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Communication of Inflation Forecast Uncertainty

The Influence of Probabilistic Communication on
Public's Perception and Economic Decision-Making

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Communication of Inflation Forecast Uncertainty: The Influence of Probabilistic Communication on Public's Perception and Economic Decision-Making

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Abstract

Effective communication of forecast uncertainty is critical for public understanding and decision-making. In macroeconomic contexts, inflation forecasts are widely communicated, yet the role of uncertainty visualisations remains underexplored in this realm. This research employed a survey-based randomised control trial to evaluate how UK public interprets probabilistic forecasts presented in various formats. Additionally, it explored, from a behavioural economics perspective, how uncertainty information is internalised and used to shape decisions. As part of a broader research agenda, the single lead-time inflation survey was first launched with 330 UK participants. Four visual formats—error bars, density strips, fan chart slices, and bell curves—were tested against a control condition. Results show that communicating uncertainty enhances interpretation accuracy relative to the control, though no format consistently outperformed others across all dimensions. Visually dense formats were perceived as harder to understand and less effective for decision-making. Thematic analysis of decision rationale reveals reliance on heuristics and exhibits loss and risk aversion. By revealing how the public interprets and utilises inflation forecast uncertainty, these findings offer valuable insights for economists, forecasters and communicators to design effective, user-centred communication tools that align public expectations and empower the public to make informed financial decisions without the risk of misinterpretation.

Keywords: Forecast Uncertainty Communication, Macroeconomics Statistics Communication, Inflation Forecasts, Uncertainty Visualisations, Decision-making under Uncertainty, Behavioural Economics

Highlights

Communication of Inflation Forecast Uncertainty: The Influence of Probabilistic Communication on Public's Perception and Economic Decision-Making

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- People generally expected high uncertainty around the central estimate. Although only a minority of respondents understood that the probability of inflation being any exact value is 0 in a statistical sense, most respondents intuited the inflation outcome being exactly the central estimate as 'very unlikely'.
- The interpretation accuracy of probabilistic forecasts (i.e. respondents' understanding of analytical uncertainty) significantly improved with visual uncertainty communication.
- In the absence of uncertainty visual aids, participants anchored their expectations on the point estimate, exhibited a stronger upward bias, and assigned lower probabilities to extreme outcomes.
- Communication tools had little influence on the forecast's cognitive (or contextual) uncertainty (i.e., how participants internalised the forecast and utilised it to support decisions).
- A clear disconnect emerged between objective probability comprehension (analytical uncertainty) and subjective, self-reported uncertainty (contextual uncertainty).
- Loss aversion is prominent in economic decision-making under macroeconomic forecast uncertainty.
- No single visual format consistently outperformed others across all metrics, suggesting selection of communication tools could take into account the prioritisation of specific key objectives.
- The perceived accuracy and certainty of the central estimate as well as the representativeness of the uncertainty showed no significant differences across groups.
- Information-dense uncertainty visualisations were perceived as more difficult to understand and less helpful for decision support.
- Uncertainty communication does not erode trust in the forecast.

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1. Introduction

Forecasts have vital applications across numerous fields, encompassing weather and climate, economics and finance, medical diagnosis, and epidemic prevention. However, in all forecasting endeavours, there is an intrinsic element of uncertainty, broadly classified into two categories: aleatoric uncertainty (i.e. inherent randomness and unpredictability in future outcomes) and epistemic uncertainty, stemming from data incompleteness, model quality, and interpretative biases. These two types of uncertainty coexist in inflation forecasts.

Forecast uncertainty is crucial to building a comprehensive view of the potential future outcomes, thereby supporting informed decision-making. Hence, reducing epistemic uncertainty and enhancing forecast quality have been central goals for forecasters. This pursuit has spurred advancements in probabilistic forecasts as well as uncertainty quantification and measurement. Nevertheless, improving quality alone is insufficient to fully realise the benefits of forecasts. Murphy (1993) highlighted three dimensions of forecast goodness: consistency, quality, and value. Among them, the forecast value—the beneficial outcomes obtained when used as an input to decision-making—carries the most significant real-world implications.

Conveying and communicating uncertainty clearly and effectively is crucial to realising the forecast value, particularly in scenarios where forecast producers and users occupy distinct roles, a common situation in practice. Effective uncertainty communication increases the reliability of predictions and enables users to comprehend and critically interpret the forecast, incorporate uncertainty considerations into their decision-making processes, and ultimately formulate robust strategies to mitigate potential risks. Empirical studies have demonstrated that uncertainty information improved decision quality and increased trust in the forecast (Joslyn and LeClerc, 2012; Möhrle et al., 2022). By contrast, poor communication practices can render uncertainty information unclear, distorted, or difficult to process. For instance, misinterpreting the probability in weather forecasts results in users taking unnecessary precautions and erodes trust and reliability (Joslyn et al., 2009).

Uncertainty communication requires multidisciplinary efforts, drawing from fields such as statistics, data visualisation, behavioural economics, psychology, and decision science. However, as emphasised by Mazzi et al. (2021), uncertainty cannot be fully understood independently of a specific context. Careful consideration of how forecasts are employed is required for developing an optimal communication strategy.

Macroeconomic forecasts, particularly inflation forecasts, hold substantial societal and economic implications for various agents (Haddow et al., 2013; Haldane, 2017; Gilbert, 2011). Central banks rely on inflation forecasts to formulate monetary policy (Aikman et al., 2011), while businesses and the public incorporate these forecasts into their inflation expectations (Coibion et al., 2019; Weber et al., 2021). These expectations subsequently inform their financial decisions today, reflecting a 'forward-looking behaviour' in economics (Hall, 2010). Empirical studies confirm that communicating uncertainty results in more accurate inflation expectations and reduces household expenditures and risky investments (Kostyshyna and Petersen, 2024; European Central Bank, 2021). Thus, current economic activities will shape actual future inflation outcomes (Aikman et al., 2011). The dynamic interplay between current economic behaviour and future inflation underscores the importance

of clear, effective, and credible forecast communication in influencing economic outcomes.

Although inflation forecasts are widely disseminated, the benefits of communicating uncertainty and the extent to which the public understands probabilistic forecasts remain contested. Much of the existing research have focused on communicating numerical and verbal summaries of distribution, leaving a gap concerning the effectiveness of visual formats in economic contexts. Multiple works (van der Bles et al., 2018; Spiegelhalter, 2024; Mazzi et al., 2021; Kostyshyna and Petersen, 2024) have highlighted gaps in understanding how users react to alternative representations of uncertainty and call for further investigation into the impact of data uncertainties on the public and experts’ use and interpretation of economics statistics, especially in the discussion of the use of fan charts by the Bank of England.

Motivated by this critical gap, this research rigorously explores visual communication tools for communicating inflation forecast uncertainty to the general public. Through survey-based, randomised controlled trials combined with hypothetical decision tasks, this study systematically evaluates how alternative visualisations of uncertainty affect individuals’ forecast interpretation and economic decision-making. The results are expected to aid economists and communicators seeking to improve public understanding of forecast uncertainty.

The paper proceeds as follows: Section 2 critically reviews the relevant theories and literature in the communication of macroeconomic statistics and decision-making under uncertainty. Section 3 details the key research questions to be answered in this research. Section 4 outlines the methodology and experimental design employed in the study. Section 5 presents key results, evaluating the effectiveness of uncertainty communication and visual formats across multiple dimensions. Finally, Section 6 summarises the key findings, synthesises the broader implications, and proposes avenues for future research.

2. Background

A comprehensive framework presented by van der Bles et al. (2018) addressed key aspects of uncertainty communication about who communicates what, in what form, to whom, and to what effect, which underpins our understanding of various stages involved in this process. Building on this foundation, this section critically reviews and discusses empirical findings and key dimensions of forecast uncertainty communication, including the target audiences, communication methods, and effects on user perceptions and decision-making, with particular focus on the context of inflation forecasts.

2.1. Forecast Audiences

Communication strategies cannot be developed without understanding the needs, goals, and capabilities of end-users. Cognitive, emotional, and behavioural responses to uncertainty vary among audiences (Spiegelhalter, 2024), shaped by factors such as their decision goals (Raftery, 2014), numeracy levels, cognitive abilities (Weber et al., 2021), and possibly socio-demographic factors (Jost, 2017). Inflation forecasts are a type of macroeconomic information widely communicated to the general public; however, explicit communication of their uncertainty is rarely seen in the media

or reaches ordinary households. Omitting probabilistic details is often justified by concerns over misunderstanding or misinterpretation (Aikman et al., 2011; Fischhoff, 2012).

The efficacy of public communication remains contested. Some studies argue that communication beyond the reading comprehension of most of the general public leads to a deficit in public understanding and public trust (McMahon, 2018) or distracts the public (Österholm et al., 2008). Therefore, straightforward communication can better align readers’ beliefs with the bank’s forecast (Haldane and McMahon, 2018). Nonetheless, some studies have found that forecast users value the appearance of uncertainty intervals and can process them when presented clearly (Greis et al., 2015; Joslyn and Savelli, 2010; Laan et al., 2015).

There is a limited body of research investigating how the general public responds to explicit uncertainty distributions and the heterogeneous interpretive ability across diverse audiences.

2.2. Communication Methods

Uncertainty may be communicated through verbal statements, numbers, or visualisations. Empirical evidence shows that perceived uncertainty is affected by visual representations (Tak et al., 2014). Although there are various ways to communicate uncertainty visually, including error bars, box plots, violin plots, and fan charts, research on how these alternative formats affect interpretation is scarce. Despite some general rules that facilitate visual communication, the most suitable approach for visualising uncertainty depends on the goal of the communicators, the context, and the audience (Spiegelhalter et al., 2011).

Central banks often deploy different communication strategies to inform and align the public’s expectations with their inflation targets. The Bank of England has been using fan charts since 1996 (Britton et al., 1998) to communicate time-series inflation forecasts and their uncertainty. A fan chart employs a two-piece normal distribution and displays the full probability distribution for the following 3 years, thereby discouraging over-focus on the central projection and fostering discussion and accountability.

However, the use of fan charts has become controversial in recent decades (Goldstein, 2024; Gilchrist, 2024; Y, 2024). The Monetary Policy Committee (MPC) has explored alternatives, such as ‘probability ribbon’ and three-dimensional fan chart, and studied visualisations used by other central banks (Aikman et al., 2011).

In a recent review, Bernanke (2024) suggested retiring fan charts in favour of clearer, more direct presentations. He also advocated de-emphasising the central projection and supplementing it with selected alternative scenarios to elucidate the rationale behind policy choices. The Bank’s response acknowledged these proposals, but stressed the need for careful design considerations of their implementation and an evaluation of whether the alternative scenarios are adequate substitutes for fan charts. Therefore, better communication strategies remain undetermined.

Empirical research using macroeconomic migration and unemployment data found that minor variations of time-series forecast representations had very few significant effects on comprehension and perception (Dryhurst et al., 2019). Another recent study of official GDP estimates concluded that quantitative visualisation of data uncertainty, via either confidence intervals, density strips, or bell curves most effec-

tively aligned users’ judgments with objective estimates of data uncertainty (Galvão and Mitchell, 2024). Echoing these findings, the present research conducts similar experiments for inflation forecasts, evaluating the effectiveness of alternative visual formats.

2.3. Level of Uncertainty

There is a trade-off in choosing how much uncertainty to display. More is not always better. Overly broad intervals offer little actionable insight. Research has found that uncertainty information elevates cognitive burden, complicating decision-making processes (Raftery, 2014). Complex or ambiguous communication degrades decision quality, triggering over-simplifications, heuristic shortcuts, and ambiguity aversion (Fox and Weber, 2002).

Conversely, communicating only deterministic or too precise information creates ‘incredible certitude’ (Manski, 2019) and overstates forecasters’ confidence (Laan et al., 2015; Aikman et al., 2011), also leading to sub-optimal decision-making.

The Bank of England presents fan charts using both ‘wide bands’ and ‘narrow bands’ (with higher specificity), yet no studies have tested how users react to these variants (van der Bles et al., 2018). Evidence from other domains suggests that task performance improved with greater specificity in uncertainty information (Bisantz et al., 2005), but more studies are needed to explore the impacts of uncertainty ranges (Tak et al., 2014). Ultimately, the aim is to present just enough uncertainty to inform without overwhelming: enough to guide action, but not so much that users ignore or misinterpret it.

2.4. Decision-Making under Uncertainty

Much of the value of uncertainty lies in how it affects decisions (Mazzi et al., 2021). Numerous studies and theories have been developed to understand how humans make decisions under uncertainty from a behavioural economics point of view. A prominent finding in forecast communication is that humans tend to seek certainty (Kaye et al., 2020; Petropoulos et al., 2020; Joslyn et al., 2009). Recent research posits that people show algorithm aversion (a reluctance to rely on predictions after observing errors) when presented with time-series forecasts (Leffrang and Müller, 2024). This may also have potential implications for the public acceptance of inflation forecast visualisations.

As noted by Aikman et al. (2011), while rational decision-makers follow the utility-maximising behaviour with efficient use of probabilistic information, it’s likely that the public is not purely rational. Instead, they take a behavioural approach, relying on availability heuristics guided by previous accessible memory or using intuitive thinking (Padilla et al., 2018), and exhibit loss aversion (Daniel Kahneman et al., 1979). Their decisions are further influenced by subjective biases, including emotions, experiences, or majority viewpoints. These shortcuts become more pronounced in high-stakes (Kunreuther et al., 2002) or extreme-probability contexts (Joslyn and Savelli, 2010).

This findings highlights the need for forecast communication to account for the cognitive and behavioural tendencies of end-users.

2.5. Trust and Credibility

Forecast communication influences perceived trust and credibility, which are paramount to communicating official statistics. Clear communication increases trust, signals transparency and honesty (van der Bles et al., 2018), and encourages decision-makers to rely on it more heavily. By contrast, poorly communicated uncertainty erodes trust, signals incompetence, and leads to scepticism (Fischhoff, 2012), causing decision-makers to ignore or discount the forecast.

Research on weather forecasts indicates that probability information increases trust in predictions (Joslyn and LeClerc, 2012; Ripberger et al., 2022). Similarly, empirical research on the communication of macroeconomic statistics (Galvão and Mitchell, 2024; Dryhurst et al., 2019) shows that uncertainty does not decrease trust in data. By building public understanding around data and forecast, central banks can strengthen their credibility and authority and foster public adherence to their policy decisions (Haldane and McMahon, 2018).

3. Research Questions

Building on the literature and identified gaps, the goal of the present research is to evaluate the effectiveness of alternative uncertainty visualisations for communicating inflation forecasts (both year-on-year forecasts and time-series forecasts) to the general public from an interdisciplinary perspective.

Here, the communication of inflation forecasts is framed as a two-stage process: interpretation and utilisation. First, whether audiences can accurately comprehend the embedded uncertainty, as Mazzi et al. (2021) highlighted, effective communication should be 'comprehensive' and 'understandable'. Second, given the central role of forecasts in decision-making, it is vital to understand how uncertainty is utilised to support and facilitate appropriate actions. Notably, an accurate interpretation does not always guarantee optimal decisions (Joslyn and LeClerc, 2013; Nadav-Greenberg and Joslyn, 2009; Daniel Kahneman et al., 1979).

This research also aims to assess the effectiveness of communication through the lens of two uncertainty types (Humpherson, 2024): specific (or analytical) uncertainty (i.e. the uncertainty involved in the production of statistics) and contextual (or cognitive) uncertainty (i.e. the inference that users draw based on their use of the statistics).

Two central questions emerge from this framework: At the interpretation stage, how well do audiences grasp the analytical uncertainty? At the utilisation stage, how effectively do they integrate that uncertainty into their contextual decision-making?

More specifically, this research aims to answer the following questions:

Probability Interpretation

- What is the extent to which individuals' interpretations of inflation forecasts are anchored or deviated from the central estimate?
- Which format leads to a high interpretation accuracy of probability?

Forecast Perception

- How do alternative formats impact the perceived uncertainty, trust, and readability of the forecast?

Decision-Making under Uncertainty

- How do various communications shape decision-making? Will inflation forecasts with uncertainty support economically rational decision-making? Does high interpretation accuracy necessarily lead to optimal decisions?
- What behavioural patterns do people exhibit when making decisions under inflation forecast uncertainty?

4. Methodology

This section details the rationale and practical implementation of the experimental design, describes the analytical techniques used to evaluate the results, and reflects on the suitability and limitations of these approaches.

4.1. Experimental Design

This study employed an online, survey-based randomised controlled trial (RCT), a common approach in relevant literature. The present research follows the methodology of a recent investigation into the communication of UK GDP data uncertainty (Galvão and Mitchell, 2024), and several survey items were adapted directly from their work. The survey was hosted on Qualtrics, and data collection was through Prolific, the same set-up used by Dryhurst et al. (2019). The experiment used a between-subjects design to avoid learning or carryover effects. The research comprises two parallel versions of surveys: one survey focuses on single lead-time inflation forecasts (year-on-year inflation), and another on time-series inflation forecasts (inflation over the following three years). Only the single lead-time survey was deployed due to time constraints and a rigorous practice of iterating between waves. The second survey was prepared but not released. Reflections on the first survey design and plans for the second wave is mentioned in Section 6.2.

The experimental stimuli (independent variable) were the inflation forecasts presented in various formats. In each version of the survey, participants were randomly assigned to one of five groups: one control group that received no uncertainty information and four treatment groups, each exposed to a distinct uncertainty visualisation format. Random assignment ensured that each group was equivalent in expectation. By comparing responses across groups, the effect on understanding, perception, and decision-making can be attributed to the communication tools used.

4.2. Participants and Data Collection

Participants were recruited through Qualtrics' built-in representative sampling feature¹, which stratifies the sample across age, sex, and ethnicity, ensuring that participants' demographics mirror the national population on key characteristics. Upon enrolment, respondents were redirected to Qualtrics to complete the survey, with each participant permitted a single submission.

¹A representative sample reflects the demographic distribution of the UK population according to census data from the UK Office of National Statistics (ONS, 2023).

While related studies have drawn samples of 1,000–10,000 participants², due to resource constraints, I aimed to recruit an initial exploratory target of approximately 300 respondents per survey. A power calculation is not typically used in similar research, possibly due to the challenge of justifying the intended effect size to detect or because the effect size itself is an exploratory metric.

Prior to launch, the survey was tested with around 20 acquaintances to review its flow and check for any typos, display, or language issues. The survey was iterated based on their feedback. The refined survey was then integrated with Prolific and pilot-tested with 10 platform users.

The data collection for the first survey was accomplished a few hours after release on the platform. An extra 30 participants were recruited to compensate for exclusions made during data cleaning. Since the survey for time-series inflation forecasts has not been conducted yet, the remainder of this report will primarily focus on the single-lead-time survey.

4.3. Ethical Considerations

Ethical considerations were duly followed, and this study was approved by the university’s ethics committee. This research was classified as low-risk.

Participants were assured of anonymity and confidentiality, informed of the study’s purpose and data-handling procedures, and reminded of their right to withdraw at any time. A consent form was presented at the start of the Qualtrics survey, which they had to explicitly agree to before proceeding to any survey questions. All participants were compensated at a rate of £9.78 per hour, in line with Prolific’s recommended range.

4.4. Forecast Uncertainty Communication Tools: Treatment Groups

Random assignment implemented in Qualtrics produced roughly 60 respondents per experimental group.³ In theory, multiple visualisations exist for representing varying degrees of uncertainty information. Visual representations in this research were selected based on the relevant literature and common practice.

The four uncertainty visualisation formats for the single-period inflation forecast were: (i) Error Bar (ii) Density Strip (Shaded Bar) (iii) Slice of Fan Chart⁴ (iv) Bell Curve. Participants in the control group saw no uncertainty visualisation.

The four uncertainty visualisation formats for the time-series inflation forecast were: (i) Fan Chart (30% interval) (ii) Fan Chart (60% interval) (iii) Fan Chart (90% interval) (iv) Time-Series Error Bars. The control group for this version saw a deterministic forecast – a line chart of the central inflation forecast over time with no uncertainty indicated. Changes in the width of the depicted interval (90% vs 60% vs

²European Central Bank. (2021) surveyed 10,000 households per wave. Galvão and Mitchell (2024) has a sample size of 3,000(4,000 in wave 2) respondents. Dryhurst et al. (2019) tested on samples of around 1,000 UK participants.

³Due to data cleaning, the final number in each group was not identical; this was duly accounted for in the data analysis process.

⁴This format enables exploring the effect of providing values of surrounding periods.

30%) allowed testing how the uncertainty range affects comprehension, perception, and decision-making.

Figure 1 displays all communication tools used in this study. The underlying forecast data were consistent under all conditions. Visuals were generated in Python (code in Appendix D). Grey was used to accommodate colour-blind users. All visual representations were accompanied by a text description. These descriptions were kept consistent in tone to avoid confounding the visuals with differing explanations. The survey questions were likewise for each experimental group, with only minimal wording adjustments to suit the visualisation.

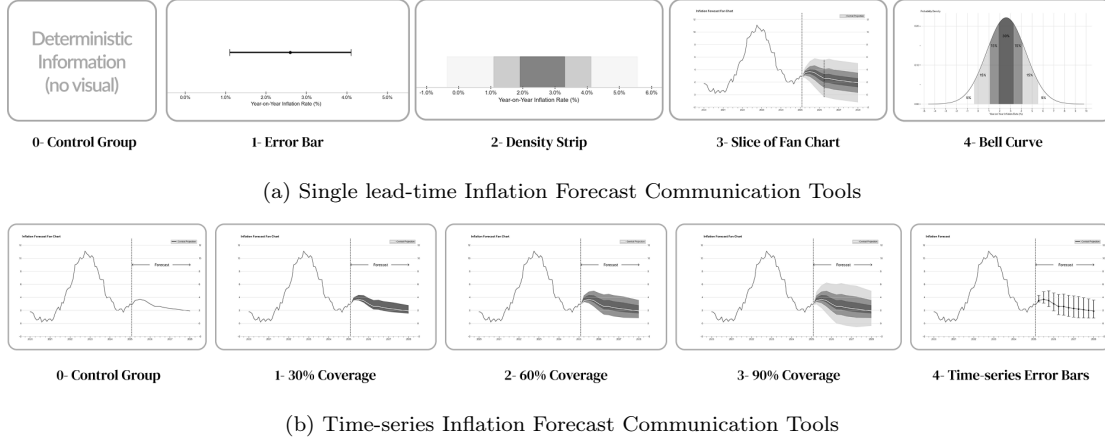


Figure 1: Communication Tools

4.5. Survey Set-up

The survey was set up in Qualtrics and optimised considering this research’s aim. A title involving uncertainty was avoided to minimize framing responses. Force response was applied to all survey questions, reflecting real-world behaviour, where forecast users infer answers when unsure. Respondents were not allowed to go back to previous questions to prevent later items from cueing earlier responses or changing how they interpreted the forecast. Throughout the survey, respondents could see their allotted inflation forecast communication tool on each and every page.

Each survey consisted of three main blocks of questions, as visually summarised in Figure 2. The full questionnaires can be found in Appendix B.⁵

4.5.1. Background Information

The first block collected information about individual characteristics, including age, highest education, and employment status. Specifically, data regarding respondents’ engagement in economic and financial activities, their attention to current and future inflation, and information sources were also gathered. Participants were asked to select the definition of inflation. Those who did not answer correctly received a brief definition and a link to the official resource for further reference.

⁵The time-series survey is subjected to modification based on the updated inflation data and learnings from the first wave.

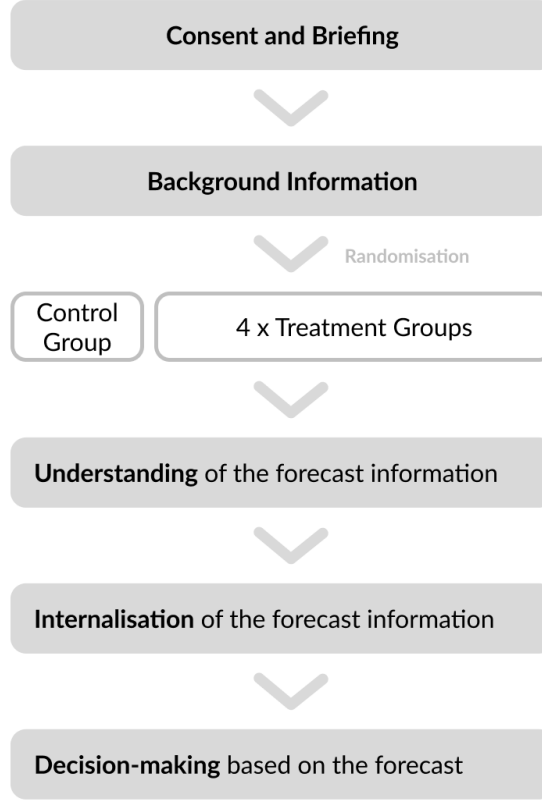


Figure 2: Survey Flow

4.5.2. Interpretation

In the following survey block, each randomised group was presented with a distinct type of inflation forecast communication tool. This block aimed to evaluate two outcomes: (i) participants' understanding of the probabilistic inflation information (ii) how they perceive and internalise the uncertainty

Participants first reported their familiarity with the communication tool and whether the forecast aligned with their prior beliefs about inflation rates. Next, respondents answer probability questions (e.g., $Q_{xx.6}$ ⁶ - '*Based on the inflation forecast provided, what do you think is the probability of inflation being exactly 2.6%?*'), as shown in Figure 3), which allowed an assessment of their understanding of the analytical uncertainty. The options were categorical in multiple-choice form, with one objectively correct value defined by the underlying uncertainty distribution. The correct answer reflected only the probabilities implied by the forecast and underlying data, not the actual inflation rate that will be realised in the future. Because the visualisations vary in the granularity of uncertain information, these survey items allowed the analysis of how communication tools affect interpretation accuracy.

⁶Since the the question number is different for each experimental group, 'xx' denotes the group-specific prefix; the digit following the decimal is the same.

Q4.6. Based on the inflation forecast provided, what do you think is the probability of inflation being **exactly** 2.6%?

- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 80%) About a 8 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 20%) About a 2 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Figure 3: Probability Question Example

Different question formats were used to draw robust conclusions. Multiple-choice items required respondents to place outcomes into predefined ranges. Accuracy on these questions is sensitive to respondents' statistical and numerical proficiency. Thus, this study introduced a novel ranking task asked participants to compare multiple potential outcomes (e.g., *Qxx.24* - 'Based on the inflation forecast provided, please rank the following inflation rates from most likely to least likely to occur: 0.8%, 2.5%, 3.9% and 6.5%.'), which provided a holistic view of how participants interpreted the overall forecast distribution regardless of their knowledge of probability.⁷

Next, participants' internalised uncertainty was evaluated by having them report forecast probabilities on a 7-point verbal likelihood scale from 'Exceptionally unlikely' to 'Virtually certain' (e.g., *Qxx.17* - 'How likely do you think it is that inflation will be exactly 2.6%?', as shown in Figure 4). To capture reasoning where relevant, respondents selected from a list of pre-defined common explanations, with an *Other*' option for additional text input.

Q4.17. How likely do you think it is that the inflation will be **exactly** 2.6%?

Exceptionally unlikely	Very unlikely	Quite unlikely	Fifty-fifty	Quite likely	Very likely	Virtually certain
☐	☐	☐	☐	☐	☐	☐

Figure 4: Likelihood Question Example

To assess forecast perception, participants were asked to subjectively rate the perceived uncertainty, the representativeness of uncertainty visualisation, the accuracy of the central estimate, the readability of the communication tool, and their trust in the forecast on a 10-point Likert scale.

To preliminarily assess respondents' understanding of skewness in uncertainty distribution, participants in treatment groups answered additional items referencing the forecast period in 2026 Q3 (featuring asymmetric uncertainty distributions, contrasting with the symmetric 2026 Q2 data used elsewhere) (e.g., *Qxx.33*).

⁷In the time series survey, participants are asked to rank time periods from the least uncertain to the most uncertain.

4.5.3. Decision Task

Drawing on relevant decision experiments (European Central Bank, 2021; Joslyn and LeClerc, 2012; Wesslen et al., 2022; Greis et al., 2016), the final survey block included a hypothetical bond-investment decision that asked respondents to choose between a nominal bond and an inflation-protected bond. Although most lay audiences (non-experts) do not rely on a full forecast distribution to make their economic and financial decisions, in order to investigate the behavioural response followed by uncertainty forecasts, the decision task was simplified to a large extent but should still reflect the reality of how each type of bond performs under different economic conditions. A pay-off table was provided to assist participants in their decisions.

The communication tools used in this block were the same as those used earlier. Participants were probed to describe their decision rationale and rate how confident they felt in their decision-making and how effective the communication tools were in supporting their decisions on a 10-point Likert scale. The economically rational decision as a benchmark is defined using underlying probability and the expected utility maximisation (EUM) strategy. For instance, based on the pay-off table, when the probability of high inflation ($>3\%$) is greater than 50%, the inflation-protected bond is optimal. To further explore behavioural patterns in decision-making, participants were asked whether gain or loss had a greater impact on their decisions, and under what conditions they may change their decisions.

4.6. Inflation Dataset

The forecast data were downloaded from the latest available Monetary Policy Report - February 2025, using the CPI inflation data conditioned on the market's expected interest rates. Visualisations were generated with the mode, uncertainty, and skew parameters, and the mode was deemed to be the central projection (Britton et al., 1998).

Historical CPI inflation data (dated back to February 2015), as deterministic values, were obtained from the Office of National Statistics (ONS) website. When constructing the visualisations, only data from 2020 onward were retained to be consistent with the graph in the Monetary Policy Reports. Inflation forecast data for 2026 Q2 was used for most questions in the year-on-year single lead-time survey. The use of any fictitious values were avoided to prevent introducing potential bias, and the participants were notified that the forecast visualisations were created using official data. Note that the forecast data published by The Bank of England assume a skewness of zero. When assessing respondents' understanding of skewness, the inflation forecast in 2026 Q3 applied a skewness of 0.8.⁸ In cases where the skewness parameter was altered, the accompanying text omitted any claim that the visual was created with official data.

⁸In the time-series survey, a skewness of 0.3 was applied to all forecasts, except those in decision-making questions.

4.7. Data Analysis

4.7.1. Data Cleaning and Preliminary Verification

The raw survey data were cleaned before the analysis to remove invalid or low-effort responses. Incomplete responses were automatically discarded in Qualtrics. The median response time was 19:56 min. As shown in Figure 5, the response duration showed high variability, with the control group finishing notably faster. Surveys completed in implausibly short times (under 10 min) were flagged for potential low effort, except for the control group as those participants did not receive items regarding skewed distribution data. The results were also rejected if rapid clicking patterns were identified or if the two ranking questions remained in default orders. Thirty additional records were randomly drawn as a further sanity check. Rejected responses were replaced by new responses from Prolific, and the verification process was repeated.

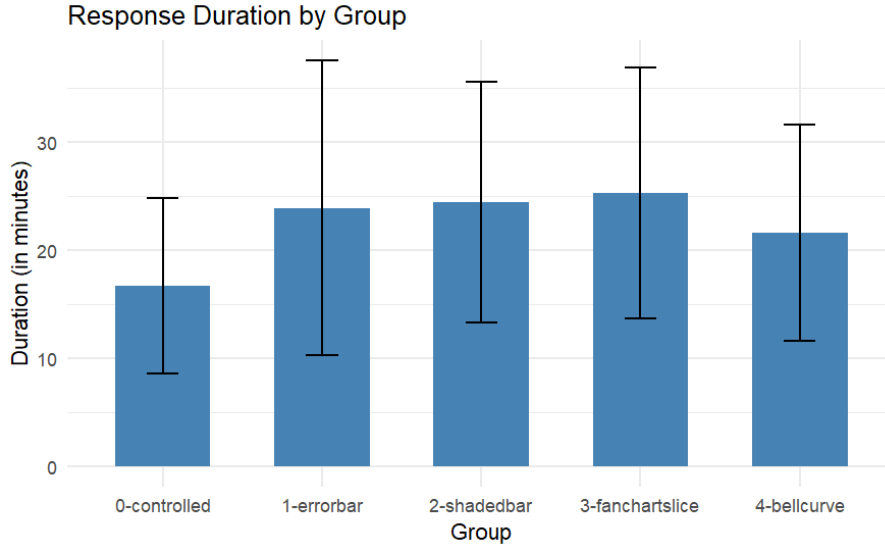


Figure 5: Response Duration Bar Plot across Groups

Differences in participant characteristics between groups are expected to be negligible on average. This was verified through preliminary demographic balance checks. The Kruskal–Wallis test (age) and chi-squared test (gender, education, and employment status) suggested there was insufficient evidence to conclude a significant differences across the five groups. Hence, the sample was comparable.

Most respondents provided substantive explanations for their bond-choice decisions, suggesting the decision scenario was at least understandable. Text inputs from 'Other' entries were recoded into existing categories where appropriate.

Data from the testing and pilot study were excluded from the final analysis. Respondents who failed to define inflation were excluded from analysis, because a baseline grasp of the concept is deemed necessary to interpret more sophisticated visual communications. These procedures ensured that the final dataset comprised attentive respondents and was suitable for subsequent analysis.

4.7.2. Analytical Approaches

Data were analysed using appropriate approaches according to the format (numeric and textual). All data analysis was performed in R (Notebook in Appendix E).

Descriptive Analysis

Each question was analysed, reported, and visualised for the full sample and by experimental groups, enabling both an overall assessment and comparisons across experimental groups.

For categorical outcome measures, the number of times each choice was selected was counted using a contingency table, and the proportion of each response was calculated. The proportion of the correct response in each group was extracted for sequential study. For Likert-scale questions, the means, median, standard deviation, and interquartile range for each group were assessed for central tendency and variability. In the case of ranking outcomes, the mean rank was calculated.

Each item's result was analysed and discussed separately to evaluate which visual format yielded the highest probabilistic interpretation accuracy. Aggregating them into a composite accuracy score was deemed inappropriate for this study.

Following the resource-constrained approach, sensitivity analysis based on the 309 sample size was used to estimate the minimum detectable effect for ordinal outcomes. The survey design afforded 80% power to detect an omnibus effect as small as Cohen's $f = 0.185$ (roughly Cohen's $d = 0.37$ for any single pair). Effects smaller than this threshold were likely undetected.

Inferential Analysis

Due to the data level and distribution, non-parametric statistical tests were used to test the hypotheses. Scores derived from Likert items (e.g., *Qxx.17*, as shown in Figure 4) were analysed using Kruskal-Wallis tests (Kruskal and Wallis, 1952) because of the ordinal nature of these scales. The test assesses whether there are statistically significant differences in the distribution of a variable across more than two independent groups. Where Kruskal-Wallis tests indicated a significant difference, post-hoc pairwise comparisons using Dunn's test (Dunn, 1964) were performed in R.

Categorical outcomes (e.g., *Qxx.6*, as shown in Figure 3) were analysed using chi-squared tests (McHugh, 2013) to assess whether two categorical variables were related or independent. When there was a significant difference, Pearson residuals identified the cells driving the χ^2 . Proportion tests were followed for items with a defined correct response. All significance tests were conducted at the 5% significance level (two-tailed), and the p-values were reported.

Decision Analysis

In terms of behavioural responses to uncertainty, the survey aims to first analyse the decision patterns through benchmarking with the decision outcome corresponding to EUM and later evaluate the decision quality after the actual inflation rate is realised.

Finally, thematic analysis (Braun and Clarke, 2006) was conducted for the open-ended responses (qualitative data). After familiarisation with all responses, initial codes were generated and subsequently merged into broader themes. Representative quotes were extracted for each theme.

4.8. Limitations

First, the sample size was relatively small compared to other macroeconomic studies, limiting the ability to detect small effects and perform granular subgroup analysis. Although participants were sampled from UK-representative populations, coverage was imperfect because elderly groups or individuals without internet access were not well represented. Moreover, participants who chose to take the survey might have been more interested in this topic.

Secondly, the perception of forecast uncertainty was measured by self-reported data which are subjective and have inherent limitations. People may show bias, such as the tendency to use extreme choices or the midpoint, on a rating scale (Paulhus, 1991).

Additionally, the decision task cannot fully simulate what happens in reality. The scenario has to be simplified to a large extent to avoid confusing and overwhelming the participants.

Finally, the treatment effects cannot be perfectly isolated. Although, in principle, all other variables are held constant, so that any outcome differences can be attributed to the communication tool used, respondents' cognitive abilities and pre-existing trust in inflation forecasts inevitably influence their answers. Despite explicit instructions to base responses on the provided inflation forecast (treatments), it is not possible to guarantee that participants do not rely on their own heuristics. Nevertheless, this limitation does not invalidate the exercise because, in practice, individuals usually combine external forecasts with their own judgment.

The methodology of this study is carefully considered to be aligned with the scope and timescale of the research and appropriate to the identified limitations.

5. Survey Results and Discussion

A selection of core findings is reported and presented below, highlighting how respondents understood probabilistic forecasts and the impacts of different communication tools, forecast perceptions, and behavioural responses to uncertainty information. The raw data, full analysis and visualisations are available in Appendix E.

5.1. Participant Characteristics

Key participant characteristics, such as their highest education, employment status, and financial activities, are summarised in Appendix C. The results should be interpreted in light of these characteristics.

5.2. Probability Interpretation

5.2.1. Central Estimate

The public often receives information through media and headline news which filter or intermediate central bank's message to a single value (Aikman et al., 2011; Haldane, 2017). Thus, effective communication requires the public to recognise that the headline value is only an estimate, and they should not expect the inflation to be exactly as headline statistical estimates.

To investigate whether people understand that the probability of inflation outcome being any exact value is 0, *Item Qxx.6* asked, 'What do you think is the probability of inflation being exactly 2.6%?' The results show that only small portion (20%) of total respondents correctly selected '<1%', whereas the modal response (30%) was '30%'. As shown in Figure 6, accuracy was lowest in the control condition and highest with the error-bar visual. However, the difference was not significant ($\chi^2(4) = 4.01, p = .40$).

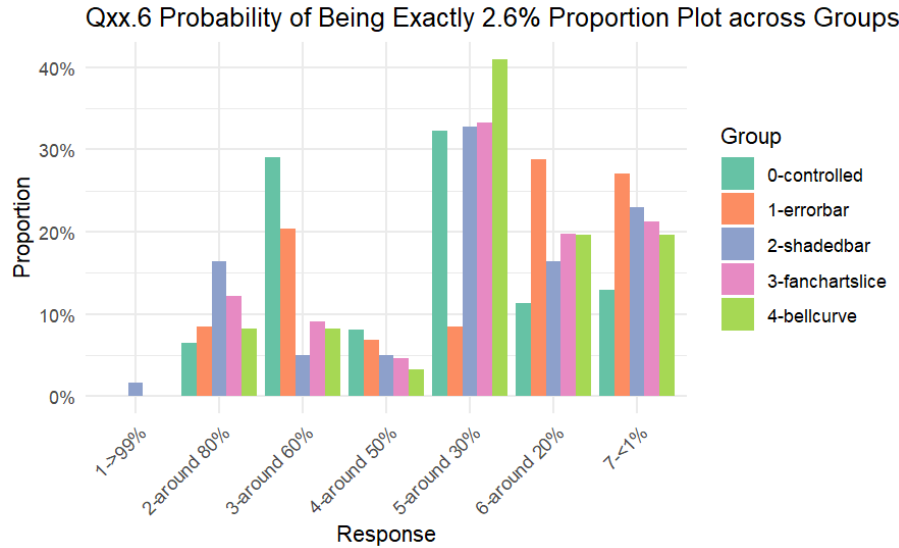


Figure 6: Item Qxx.6 Proportion Bar Plot across Groups

Later, in *item Qxx.17*, participants were asked to rate the central estimate on a seven-point verbal likelihood scale. Figure 7 shows a left-skewed distribution. Most participants (30%) think it is 'Very unlikely'. The Kruskal–Wallis test detected no group differences ($H(4) = 1.49, p = .83$), with a negligible effect size ($\epsilon^2 = -0.008$, 95% CI [-0.01, 0.03]).

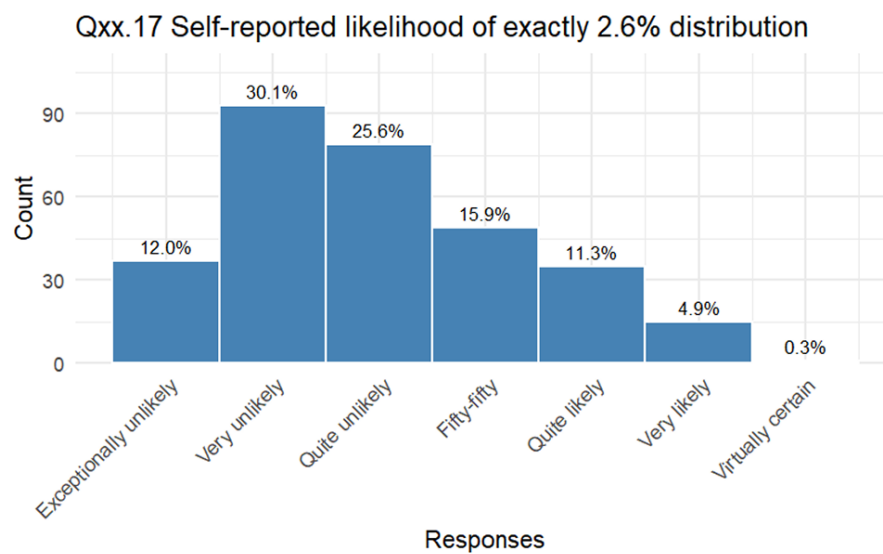


Figure 7: Item Qxx.17 Proportion Bar Plot Among all Respondents

Qualitative follow-ups revealed that less than 10% respondents selected the correct reason, which is likely to correlate more with educational attainment than with communication tools.

Overall, although not many people understood that inflation being an exact value has zero probability in a statistical sense, most participants (67.7%) intuited that the inflation being exactly the central estimate was unlikely, suggesting people do expect high uncertainty around central estimates.

5.2.2. Treatment Effects

Items *Qxx.6* to *Qxx.15* were designed to assess probabilistic understanding and investigate which visual format yields high interpretation accuracy. A higher proportion of correct responses indicates greater interpretation accuracy. Table 1 reports accuracy rates and omnibus proportion-test p-values. Five items exhibited significant group differences. The visualisations of the full distribution can be found in Appendix E.

Table 1: Understanding of Probability across Groups

Survey Item	Question	Correct Answer	0-controlled	1-errorbar	2-shadedbar	3-fanchartslice	4-bellcurve	p-value
Qxx.6*	Exactly 2.6%	(1%)	12.90%	27.12%	22.95%	21.21%	19.67%	0.40
Qxx.7*	1.9%-3.3%	(around 30%)	4.84%	30.51%	44.26%	37.88%	47.54%	1.55e-06
Qxx.8*	3.3%-5.6%	(around 30%)	22.58%	57.63%	47.54%	34.85%	45.90%	0.0012
Qxx.10	Outside of 90% coverage	(10%)	16.13%	57.63%**	44.26%	46.97%	67.21%	2.18e-07
Qxx.11	Higher than 6%	(1-5%)	20.97%	35.59%	39.34%	36.36%	47.54%	0.042
Qxx.12*	Higher than 3%	(40-50%)	24.19%	23.73%	31.15%	18.18%	21.31%	0.53
Qxx.14	Higher than 2%	(around 60%)	25.80%	61.02%	47.54%	33.33%	54.10%	0.00030
Qxx.15	Deviation from the central estimate	"equal risk"	27.42%	45.76%	59.02%	50.00%	37.70%	0.0058

Note: *Survey items marked with * were followed by a self-report likelihood question later in the survey.

**The error-bar group was asked about the probability of outcomes outside the 60% coverage rather than 90% because of the characteristics of the visualisation; thus, their results are not directly comparable.

Significant p-values are shown in bold.

The control group consistently underperformed. Although 1.9%-3.3% covers the 30% central interval, Figure 8 shows that most people (37%) in the control group judged this probability to be $\approx 80\%$. Regarding the probability of an extreme inflation rate, such as higher than 6%, respondents in the control group believed it to be '*exceptionally unlikely*' ($<1\%$), lower than the probability given by the forecast. Conversely, they overestimated the probability of inflation exceeding the Bank's 2% target, and assign probabilities above the objective 60%. Nonetheless, the reported probability of inflation higher than 3% did not differ significantly across groups. Respondents might gauge the probability in relation to the communicated reference point of 2.6%.

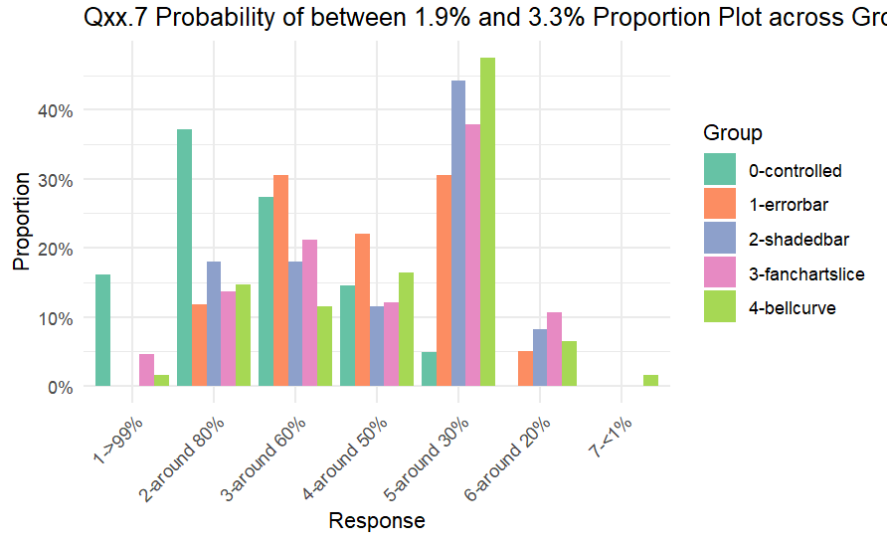


Figure 8: Item Qxx.7 Proportion Bar Plot across Groups

The underlying uncertainty distribution was symmetric, but participants in the control group believed that there was more upside risk—significantly more than the other groups. Noticeably, Figure 9 shows that this bias is attenuated but not eliminated by uncertainty visualisations. There were still more respondents in the bell-curve group who perceived a higher upside risk.

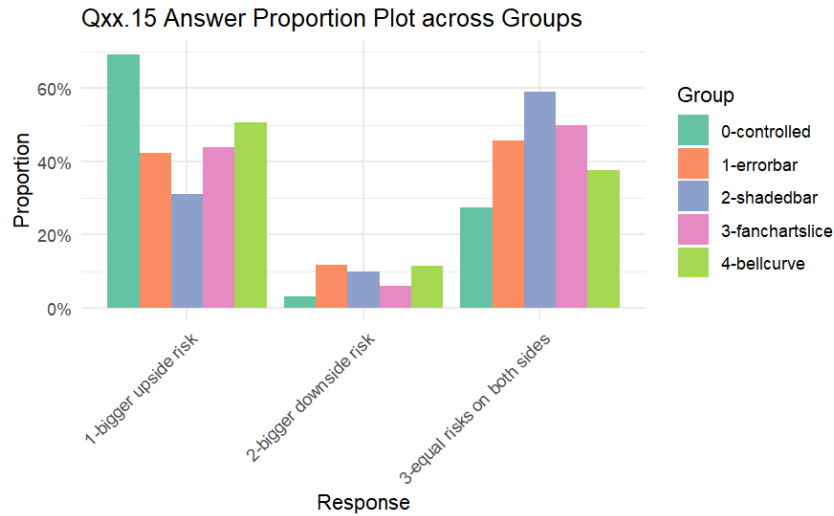


Figure 9: Item Qxx.15 Proportion Bar Plot across Groups (Risk of inflation exceeding or under-shooting the central estimate)

The ranking task asked participants to order four possible inflation outcomes by likelihood. Given the underlying distribution, the inflation rates from most to least likely are: 2.5%, 3.9%, 0.8% and 6.5%. The heatmap in Figure 10a shows that, on average, except for the control group, respondents had a correct overview of the inflation forecast. The control group again exhibited the upward bias observed in earlier items, and they achieved the lowest accuracy rate, as shown in Figure 10b. The proportion test indicates a significant difference across groups ($\chi^2(4) = 11.06$, $p = .03$).

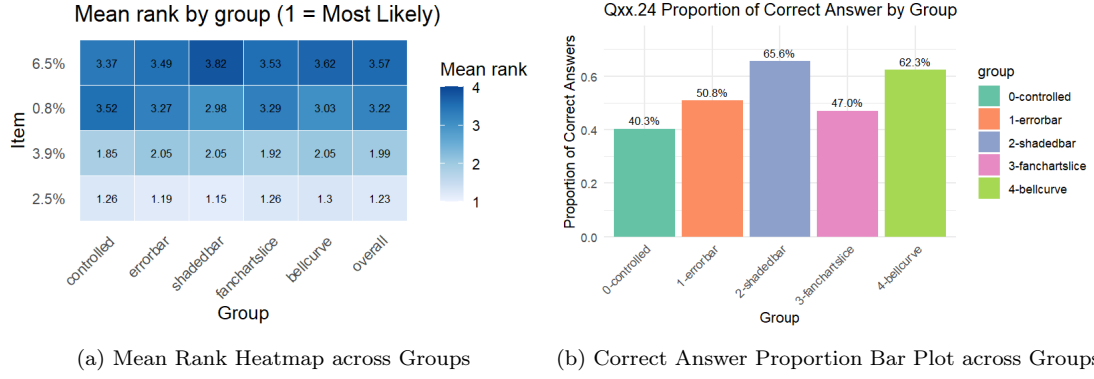


Figure 10: Qxx.24 Ranking Question Results

Overall, without visual uncertainty information, respondents tend to centre their expectations on the point estimate, exhibit a stronger upward bias, and assign lower probabilities to extreme outcomes. These results confirm that communicating uncertainty generally improves the interpretation of probabilistic information (i.e. the understanding of analytical uncertainty), and the absence of such information forces respondents to infer or guess. However, the effects among treatment groups were inconsistent. No single format was uniformly superior. Error bars, density strips, and bell curves each produced the highest accuracy for different items, suggesting that the choice of communication tools could take into account the prioritisation of specific key objectives.

Items Qxx.17 and Qxx.21–23 assessed how respondents internalised uncertainty using different communication tools. The median values were identical across groups in each of these items. Table 2 shows that the verbal likelihood people associated with different inflation outcomes (i.e. the reported cognitive uncertainty) did not differ significantly across groups, and the effects were negligible, indicating that the communication tools had little influence on how participants absorbed and internalised the forecast.

Table 2: Self-reported Likelihood across Groups

Survey Item	Question	Median	Mode	Kruskal-Wallis Test Result
Qxx.17	Exactly 2.6%	3-"Quite unlikely"	2-"Very unlikely"	$H(4) = 1.49, p = .83$ ($\epsilon^2 = -0.008$, 95% CI [-0.01, 0.03])
Qxx.21	1.9%-3.3%	5-"Quite likely"	5-"Quite likely"	$H(4) = 8.66, p = .07$ ($\epsilon^2 = 0.015$, 95% CI [-0.002, 0.08])
Qxx.22	3.3%-5.6%	3-"Quite unlikely"	3-"Quite unlikely"	$H(4) = 9.85, \mathbf{p} = .04$ ($\epsilon^2 = 0.019$, 95% CI [-0.002, 0.08])
Qxx.23	Higher than 3%	4-"Fifty-fifty"	3-"Quite unlikely"	$H(4) = 3.20, p = .53$ ($\epsilon^2 = -0.003$, 95% CI [-0.01, 0.05])

Note: Significant p-values are shown in bold.

5.3. Forecast Perception

Items Qxx.26-27 and Qxx.29-31 used 10-point sliders to gauge respondents' perceptions of the forecast. As summarised in Table 3 below, the accuracy and certainty of the central estimate and the representativeness of the uncertainty showed no significant differences across groups, and the effect sizes were negligible.

Table 3: Perception of Forecast Results across Groups

Survey Item	Question		0-controlled	1-errorbar	2-shadedbar	3-fanchartslice	4-bellcurve	KW Test Result
Qxx.26	Accuracy of central estimate	Median	7	7	7	7	6	$H(4) = 1.35, p = .85$ ($\epsilon^2 = -0.009$, 95% CI [-0.01, 0.03])
		Mean	6.15	6.41	6.64	6.38	6.54	
		SD	1.79	1.60	1.60	1.69	1.60	
Qxx.27	Certainty of central estimate	Median	6	6	6	6	6	$H(4) = 0.99, p = .91$ ($\epsilon^2 = -0.010$, 95% CI [-0.01, 0.03])
		Mean	5.85	5.69	5.95	5.91	5.77	
		SD	1.84	1.72	1.61	1.80	1.69	
Qxx.29	Representativeness of the uncertainty	Median	7	6	7	7	6	$H(4) = 5.15, p = .27$ ($\epsilon^2 = 0.003$, 95% CI [-0.008, 0.06])
		Mean	6.40	6.00	6.52	6.71	6.33	
		SD	1.82	1.90	1.22	1.40	1.83	
Qxx.30	Readability of the forecast	Median	8	7	7	6.5	6	$H(4) = 21.14, \mathbf{p} = .0003$ ($\epsilon^2 = 0.056$, 95% CI [0.02, 0.13])
		Mean	7.34	6.98	6.39	6.12	5.69	
		SD	2.18	2.17	2.00	2.17	2.55	
Qxx.31	Trust of the forecast	Median	7	6	7	6	6	$H(4) = 3.77, p = .44$ ($\epsilon^2 = -0.001$, 95% CI [-0.009, 0.05])
		Mean	6.13	6.00	6.34	6.00	5.69	
		SD	1.97	2.04	1.60	1.82	1.83	

Note: Significant p-values are shown in bold.

By contrast, readability ratings differed significantly across the groups. The control condition (no visual) was rated the easiest to understand, which was as expected because they received no visualisation aid. The error-bar format was also rated easy to interpret, likely because respondents were familiar with this type of visualisation. On the other hand, bell curves were rated the most difficult to understand, followed by fan charts, likely owing to the greater amount of uncertainty information they convey.

Van Der Bles et al. (2020) distinguish between trust in the messenger and trust in the number. *Item Qxx.31* asked participants ‘How much do you trust the inflation forecast communicated to you in this way?’, which evaluates whether the forecast and visualisation demonstrate transparency and clarity, regardless the source’s credibility. Average trust ratings differed, with the bell curve receiving the lowest score, and the density strip, the highest. However, these differences were not statistically significant, suggesting that communicating uncertainty did not erode the perceived trustworthiness in forecasts. These results are in line with prior empirical evidence (Galvão and Mitchell, 2024; Dryhurst et al., 2019).

5.4. Decision-Making under Uncertainty

The decision pattern was analysed first. The actual decision quality (whether their choice maximises utility regardless of uncertainty) will be assessed after the forecast is realised.

Overall, although the economically rational choice—given the underlying forecast distribution—is to invest in nominal bonds, only 33% of the total respondents chose this option; the remainder chose inflation-protected bonds. Figure 11 shows that bond choices did not vary by communication format ($\chi^2(4) = 2.48, p = .65$). The results also suggest that that forecast contextual uncertainty does not differ between groups but it is disconnected from its analytical uncertainty.

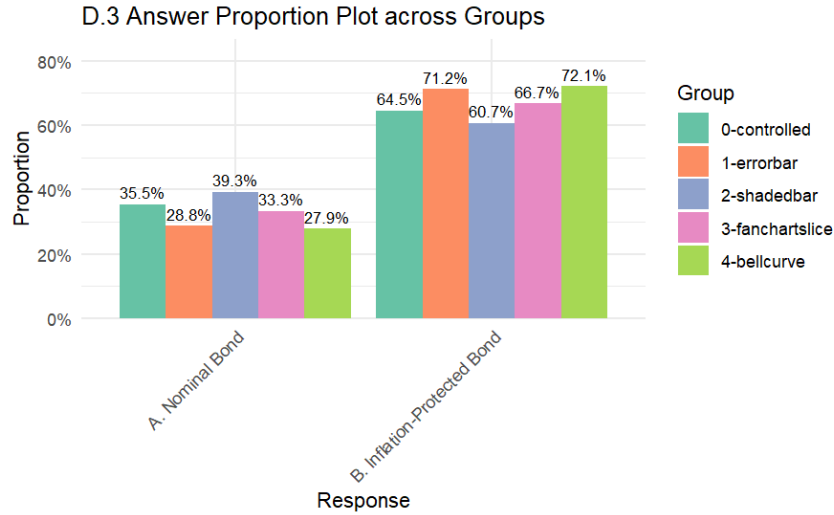


Figure 11: Item D3 (Bond Investment Decisions) Proportion Bar Plot across Groups

Table 4 indicates that respondents were generally confident in their choices, although the bell-curve format elicited the lowest mean score. Average confidence in the chosen option did not differ across groups. By contrast, ratings of the effectiveness of forecast information in assisting their decisions did vary significantly, with the bell curve and fan chart rated significantly less helpful than the other formats.

Table 4: Decision-making under Uncertainty Results across Groups

Survey Item	Question		0-controlled	1-errorbar	2-shadedbar	3-fanchartslice	4-bellcurve	KW Test Result
D4	Confidence in decision-making	Median	7	8	7	7	7	$H(4) = 7.05, p = .13$ ($\epsilon^2 = 0.01, 95\% \text{ CI } [-0.006, 0.07]$)
		Mean	7.03	7.22	7.16	6.61	6.49	
		SD	1.77	1.70	1.69	1.90	1.92	
D9	Effectiveness of the forecast	Median	7	7	7	6	7	$H(4) = 10.05, p = .03$ ($\epsilon^2 = 0.02, 95\% \text{ CI } [-0.001, 0.08]$)
		Mean	6.69	6.61	6.54	6.02	6.00	
		SD	2.00	2.07	1.75	1.89	2.07	

Note: Significant p-values are shown in bold.

More than half of the respondents thought potential loss had more impact on their decision-making and this tendency was most pronounced in the bell-curve and fan-chart group, as shown in Figure 12.

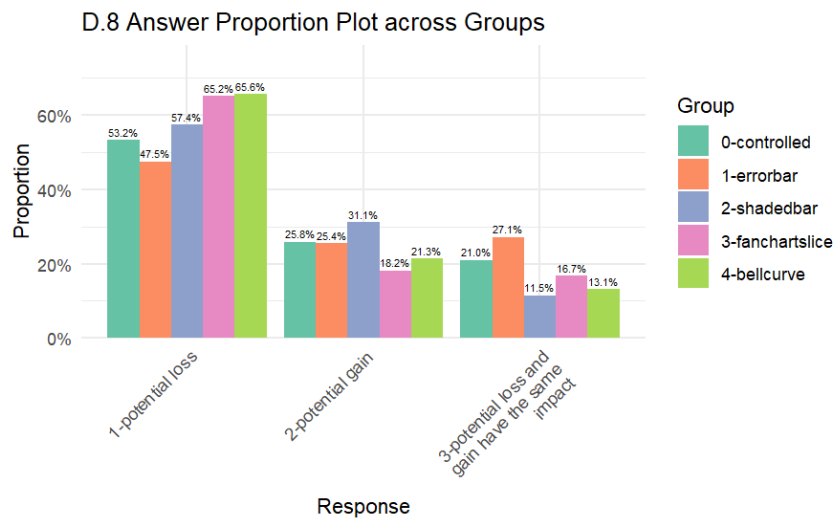


Figure 12: Item D8 (Impacts on Decision-making) Proportion Bar Plot across Groups

Considering perceived readability alongside results from decision task reveals that, as the visualisation grows more information-dense, subjective perceptions of its decision support decline and individuals feel less confidence in their decisions and become more risk-averse.

The thematic analysis of respondents' decision rationales further provides insights that confirm established decision theories. Many statements reflect heuristic substitution rather than formal probability integration. Only a minority reported explicit expected-value calculations. However, choices frequently hinged on whether inflation was expected to stay below or exceed the 3% threshold, likely because it was mentioned explicitly in the pay-off table. Loss and risk aversion were prominent among participants selecting the inflation-protected bond. They also emphasised that the economic environment is highly uncertain, unpredictable and volatile, citing geopolitical conflict, tariffs, and elections. The experiment captures a snapshot in time; thus, attitudes toward the economic outlook may vary under different macroeconomic conditions.

Conversely, nominal-bond supporters expressed they endorsed the Bank of England's projections, and often framed outcomes as proportions of the stake, thereby downplaying potential losses and tilting towards risk-seeking behaviour. Additionally, label effects and heuristic cues replaced analytic processing. The word 'protected' and the notion of a 'fixed rate' powerfully steered decisions, independent of numeric comparisons.

It is also worth noticing that the decision task was located at the end of the survey, yet many respondents cited their own inflation expectations (often diverging from the official forecast) when explaining the rationale behind their choices. This observation suggests the limitations of communication tools in anchoring expectations to the forecast provided.

6. Conclusions and Future Works

This paper concludes by synthesising the main findings, evaluating their broader implications, and proposing directions for future research in uncertainty communication.

6.1. Conclusions

This research rigorously evaluated the effectiveness of visual communication tools for conveying uncertainty in single lead-time inflation forecasts to the general public. Through survey-based controlled randomised trials, the study systematically assessed four distinct visualisation formats (error bar, density strip, fan-chart slice, and bell curve) against a control condition. Although sample-size limitations constrained the detection of smaller effects, the findings yielded valuable insights. Overall, the study successfully achieved its objectives, advancing our understanding of uncertainty communication by providing clear evidence regarding probabilistic interpretation accuracy, uncertainty internalisation, and behavioural responses to uncertainty.

The results demonstrate that people expect high uncertainty around central estimates. Communicating uncertainty through visual representations significantly

enhances probabilistic interpretation accuracy of the analytical uncertainty. Without such information, respondents anchor more towards the central estimate, underestimate tail risks and exhibit a pronounced upward bias towards higher inflation scenarios. However, self-reported likelihood ratings of perceived uncertainty (i.e. cognitive uncertainty) show no between-group variation, indicating a disconnect between objective understanding (probability) and subjective assessment (likelihood ratings).

Among the four treatment groups, no single visual format consistently outperformed others across all metrics in terms of probabilistic understanding. Instead, the results clearly demonstrate a trade-off between information depth and clarity. Detailed visualisations (e.g. bell curves and fan charts) delivered comprehensive information yet simultaneously reduced readability and perceived decision-support effectiveness. Notably, transparent visual uncertainty communication did not diminish public trust in forecasts. Although the study did not reveal a direct association between decisions and communication tools, the results provide empirical evidence aligned with established behavioural economic theories.

These findings provide valuable insights for forecasters and communicators to develop or design uncertainty communication tools and suggest considering displaying a amount of balanced uncertainty information. From an economic and social perspective, the results underscore the necessity of communicating forecast uncertainty to enhance public comprehension of macroeconomic projections. Moreover, tool selection should align with priority objectives (e.g., whether the goal is to inform or to anchor).

Looking ahead, this research rigorously established a robust methodological foundation for future exploration in forecast uncertainty communication. Specifically, it employed a decision-centred experimental design to comprehensively evaluate uncertainty communication formats and introduced a ranking assessment technique to capture respondents' holistic perceptions of uncertainty. Potential avenues for future research are detailed in the subsequent section.

6.2. Future Works

This report presents findings from an initial exploratory study on communicating single lead-time inflation forecasts, representing a first step within a broader research agenda on effectively conveying inflation forecast uncertainty. Several directions for further research and methodological refinement have been identified.

The immediate next step will be to revise the survey design and release the time-series inflation forecast survey, using the updated data. Planned improvements include incorporating attention-check questions, standardising Likert-scale anchor points, and potentially reducing the number of survey items to improve answer quality.

Additionally, while the current sample size limited subgroup analyses, future research with larger samples could rigorously explore heterogeneity in public interpretation and decision-making. With expanded sample sizes, more sophisticated parametric methods could also be employed to estimate the effects of different visualisation formats.

Regarding skewness, the Bank of England typically avoids asymmetric forecasts to prevent misinterpretation. However, preliminary evidence from this study indi-

cates that respondents can comprehend skewness information to some degree. This promising result suggests that future research should systematically evaluate public perceptions of skewed uncertainty distributions.

This research employed a survey-based randomised control trial. Future research can also explore qualitative interviews and eye-tracking techniques to explore the effects of visual factors, such as colour and icon, on forecast users' cognitive processing and emotional responses.

Within macroeconomics, a variety of decision tasks, such as saving, spending, or investment allocation can be further explored beyond fixed-income planning in current binary-choice format. The framework and methodology developed through this research should also enable future research to explore decision-centred forecast uncertainty communication experiments across other domains.

Finally, this research evaluated a limited range of visualisations, yet many other forms of uncertainty communication exist. Future studies can investigate a broader range and more distinct communication methods such as high-density region plots (Hyndman, 1996) and interactive uncertainty communication tools.

7. Project Management and Reflection

The research project followed three phases with clear objectives and timelines. In term 1, I conducted a comprehensive literature review on forecast uncertainty communication, macroeconomic statistics communication, and related areas, identifying gaps and formulating research questions.

In term 2, I refined my research questions and validated the project's direction and value by consulting the author (Professor James Mitchell) of closely related research (Galvão and Mitchell, 2024). I designed the experiment and survey in Qualtrics, sourced inflation-forecast datasets, and wrote Python code for visualizations. After iterative testing and a pilot study, I incorporated feedback to optimise flow, clarity, and wording and integrated the final survey with Prolific, deploying it to a UK-representative sample.

In term 3, I focused on data analysis, results interpretation, and compiling the report. I critically evaluated appropriate analytical approaches, systematically documented my analysis processes in R, and synthesized the findings. I also compare my results with relevant literature and report with transparency.

Throughout, I maintained rigorous organisation. I use Gantt chart (see Appendix A) to plan and track milestones, adjusting timelines each term. I use Zotero for literature management, Notion and Obsidian for notes, and GitHub for code versioning. I am grateful to receive ongoing support and guidance from my supervisor. I maintained regular communication with him through weekly meetings and email exchanges, providing progress updates, resolving queries, and clarifying the next steps, seeking and integrating feedback.

Building on my interests in statistics and user research, this project furthers my interest in the interdisciplinary area of "from statistics to decisions" and advances my technical and research capabilities. My key learnings include:

- **Technical Skills:** Survey design and deployment in Qualtrics; uncertainty visualization coding in Python; handling large-dimensional datasets in R

- Research Methodology: Systematic literature review approaches; experimental design principles; iterative development processes; transparent reporting standards.
- Professional Communication: Engaging with domain experts; professional academic communication and presentation skills; documenting processes comprehensively.

Looking ahead, I will present my work at the International Symposium on Forecasting (ISF) in Beijing this summer, which will provide additional expert feedback I can integrate to refine my approach and conduct the time-series inflation forecast survey. I also aim to further strengthen my data analytical skills and explore inferential statistical techniques and parametric tools such as cross-tabulations and correlation analyses to delve deeper into the current data and prepare for analysing the second survey.

Declaration

I acknowledge the use of ChatGPT o4-mini (OpenAI, <https://chat.openai.com/>) to improve the language of my writing and provide a foundational framework for the data analysis code, which was subsequently adapted. AI also assisted with the thematic analysis, enabling a comparison with my own analysis to identify any discrepancies.

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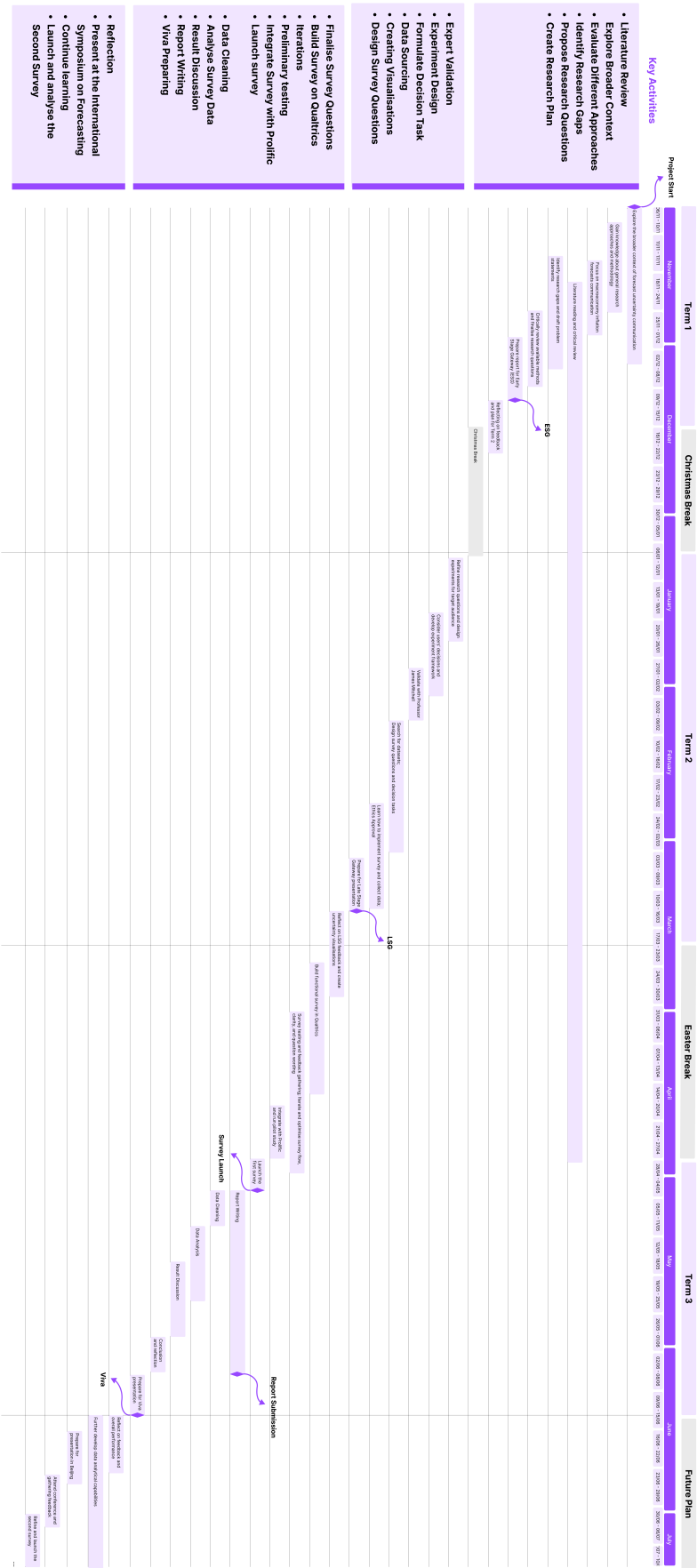
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Appendix A. Project Gantt Chart



Appendix B. Full Survey Questions

Appendix B.1. Year-on-Year Inflation Forecast Survey

IMPERIAL

Consent Form

Q1.1. Welcome!

Please read the following participant information before you can proceed to the survey.

Introduction

Imperial College London supports the practice of protecting human participants in research. The following information details why the research is being done and what is involved for you to take part in this study. *Communication of Inflation Forecast*. Your participation in this study is entirely voluntary, and you have the right to withdraw at any time without penalty or negative consequences. If you choose to withdraw, any data collected from you will be deleted and not included in the final analysis.

Summary

In this survey, you will be asked a series of questions regarding inflation forecasts in the UK.

This study will take approximately 20-30 minutes to complete. Your participation will help contribute to the understanding of effective forecast communication strategies.

Data collection and confidentiality

During this study, we will collect certain data from you without any identifiable personal information, which may include: gender, age, educational level and financial literacy data. All data collected will be stored securely and will not be disclosed to third parties or

repurposed for any unrelated usage.

Contact information

If you have any questions, concerns, or feedback related to this study, please feel free to contact:

p1921@ic.ac.uk

Dyson School of Design Engineering
Imperial College London

Statement of consent

I confirm that:

- I have read and understood the information provided in this consent form.
- I have had the opportunity to ask questions and have received satisfactory answers.
- I voluntarily agree to participate in: Communication of Inflation Forecast
- I understand that I can withdraw from the study at any time without penalty or negative consequences.

- ☐ I consent to participate.
☐ I do not consent to participate.

Prolific ID

Q2.1. What is your Prolific ID?

Please note that this response should auto-fill with the correct ID.

Screening Questions

Q3.1. Welcome to the start of the survey!

The first section will gather some background information about you. Please note that the information you provide will remain anonymous and confidential.

Once you press the 'Next' button, you will not be able to go back to the previous page.

Q3.2. What is your age? (Please only use numbers)

Q3.3. Which of the following best describes your gender identity?

- ☐ Male
- ☐ Female
- ☐ Non-binary / third gender
- ☐ Prefer not to say

Q3.4. What is your highest level of education achieved?

- ☐ No formal education
- ☐ Primary/Elementary
- ☐ Secondary School (GCSE)
- ☐ High School (A Level)
- ☐ Technical or Vocational

- ☐ Bachelors or equivalent degree level qualification (BA, BSc, etc)
- ☐ Masters or equivalent higher degree level qualification (MA, MSc, etc)
- ☐ PhD or equivalent doctoral level qualification
- ☐ Other (please specify)

Q3.5. What is your current profession or employment status?

- ☐ Student
- ☐ Employed
- ☐ Self-employed
- ☐ Retired
- ☐ Unemployed
- ☐ Other (please specify)

Q3.6. In which, if any, have you ever studied economics? (Select all that apply)

- ☐ At school
- ☐ In higher education (e.g. university, college)
- ☐ Through self-directed study (books)
- ☐ Self-motivated study (course)
- ☐ No, I have never studied economics
- ☐ Don't know / can't recall
- ☐ Other (please specify)

Q3.7. Which of the following financial activities do you actively participate in? (Select all that apply)

- ☐ Savings (e.g. regular deposits into savings accounts)
- ☐ Investments (e.g. stocks)
- ☐ Fixed Income (e.g. government bonds, corporate bonds)
- ☐ Retirement Planning (e.g. pensions)
- ☐ Trading (e.g. day trading)

- ☐ Real Estate (e.g., buying or renting properties)
- ☐ Cryptocurrencies (e.g., Bitcoin, Ethereum)
- ☐ None of the above

☐ Other (Please specify)

Q3.8. How frequently, if at all, do you read/watch/listen to news stories related to economics or the economy?

- ☐ Never
- ☐ Rarely
- ☐ Monthly
- ☐ Weekly
- ☐ Almost daily
- ☐ Every day
- ☐ Not sure

Q3.9. To the best of your knowledge, which option most accurately describes what inflation is?

- ☐ Inflation measures the difference between exports and imports
- ☐ Inflation measures the size of the economy
- ☐ Inflation measures how many people are in employment
- ☐ Inflation measures the increase in prices
- ☐ Inflation measures a government's budget balance
- ☐ I have heard about inflation but not sure what it is
- ☐ I don't have a clue what inflation is

Q3.10. Inflation is a measure of how the prices of goods and services are rising.

For further details, please refer to the link: [Inflation - Bank of England](#)

Q3.11. How much do you trust official inflation forecasts (For instance, the forecasts released by the Bank of England)?

- ☐ 1 — I do not trust them at all
- ☐ 2 — I slightly distrust them
- ☐ 3 — Neutral
- ☐ 4 — I somewhat trust them
- ☐ 5 — I fully trust them
- ☐ Unsure / no opinion

Q3.12. How often do you check the current value of inflation?

- ☐ Never
- ☐ Rarely (a few times a year)
- ☐ Sometimes (1–3 times a month)
- ☐ Often (once a week or more)

Q3.13. Which source do you rely on to inform yourself about the current value of inflation? (Choose all that apply)

- ☐ Television, radio or print media (news channels, newspapers)
- ☐ Social media (Twitter, Reddit, etc.)
- ☐ Official websites or data releases (e.g., government, central bank)
- ☐ Content platform (e.g., YouTube, blog)
- ☐ Actively search on search engines (Google, etc.)
- ☐ Family or friends
- ☐ Financial advice websites/forums
- ☐ Private banks or economic research institutes
- ☐ Other (please specify)

Q3.14. To what extent do you feel that the current value of inflation is impacting your personal finances?

- ☐ Not at all – I don't feel that current value of inflation is affecting my finances in any way.
- ☐ Slightly – Current value of inflation has a small effect on my expenses and overall financial situation.
- ☐ Moderately – Current value of inflation has some impact on my expenses and overall financial situation.
- ☐ Very significantly – Current value of inflation has a major impact on my expenses and overall financial situation.

Q3.15. How much do you adjust or take action in response to the change in current value of inflation?

- ☐ Not at all – I do not/cannot change my financial behavior based on current inflation.
- ☐ Rarely – I make minimal changes to my financial behavior in response to current inflation.
- ☐ Somewhat actively – I make occasional adjustments to my spending or saving habits due to current inflation.
- ☐ Very actively – I regularly make changes to my financial behavior or decisions based on current inflation.

Q3.16. How often do you look at inflation forecasts (that is, information about future inflation)?

- ☐ Never
- ☐ Rarely (a few times a year)
- ☐ Sometimes (1–3 times a month)
- ☐ Often (once a week or more)

Q3.17. Which source do you rely on to obtain inflation forecasts (information about future inflation)? (Choose all that apply)

- ☐ Television, radio or print media (news channels, newspapers)
- ☐ Social media (Twitter, Reddit, etc.)
- ☐ Official websites or data releases (e.g., government, central bank)

- ☐ Content platform (e.g., YouTube, blog)
- ☐ Actively search on search engines (Google, etc.)
- ☐ Family or friends
- ☐ Financial advice websites/forums
- ☐ Private banks or economic research institutes
- ☐ Other (please specify)

Q3.18. To what extent do you feel that inflation forecasts (information about future inflation) impact your personal finances?

- ☐ Not at all – I don't feel that inflation forecasts affect my finances in any way.
- ☐ Slightly – Inflation forecasts have a small effect on my expenses and overall financial situation.
- ☐ Moderately – Inflation forecasts have some impact on my expenses and overall financial situation.
- ☐ Very significantly – Inflation forecasts have a major impact on my expenses and overall financial situation.

Q3.19. How much do you adjust or take action in response to inflation forecasts (information about future inflation)?

- ☐ Not at all – I do not/cannot change my financial behavior based on inflation forecasts.
- ☐ Rarely – I make minimal changes to my financial behavior in response to inflation forecasts.
- ☐ Somewhat actively – I make occasional adjustments to my spending or saving habits due to inflation forecasts.
- ☐ Very actively – I regularly make changes to my financial behavior or decisions based on inflation forecasts.

Q3.20. In the next section of the survey, you will be presented with inflation forecasts and asked to answer a series of questions.

Please respond to all questions based on your understanding of the information provided. Please also note that many of the questions in this survey do not have a right or wrong answer. We encourage you

to respond based on your honest judgment and understanding.

Please **pay close attention** to the forecast information provided at the top of each page, as it may vary from page to page.

Understanding of Y-on-Y Forecast (Control Group)

Q4.1. The information below describes the inflation forecast in **2026Q2**. Note that the value is based on the latest data released by the Bank of England.

The most likely value for inflation in **2026Q2** is of **2.6%** (central estimate),
though the actual inflation value may deviate from this forecast.

Q4.2. Have you ever been communicated this type of information before?

- ☐ Definitely not
- ☐ Probably not
- ☐ Probably yes
- ☐ Definitely yes

Q4.3. Does the inflation forecast information provided align with your expectations?

- ☐ Strongly aligns – The inflation forecast matches exactly what I expected.
- ☐ Somewhat aligns – The inflation forecast is generally in line with my expectations.
- ☐ Neutral – The inflation forecast neither aligns nor conflicts with my expectations.

- ☐ Somewhat does not align – The inflation forecast is somewhat different from what I expected.
- ☐ Does not align at all – The inflation forecast is significantly different from what I expected.

Q4.4. How does the inflation forecast conflict with your expectations?

- ☐ The inflation forecast is higher than what I expected.
- ☐ The inflation forecast is lower than what I expected.

Q4.5.

The most likely value for inflation in **2026Q2** is of **2.6%** (central estimate),
though the actual inflation value may deviate from this forecast.

Please answer all the questions based on the information given.

Q4.6. Based on the inflation forecast provided, what do you think is the probability of inflation being **exactly** 2.6%?

- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 80%) About a 8 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 20%) About a 2 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q4.7. Based on the inflation forecast provided, what do you think is the probability of inflation being **between 1.9% and 3.3%**?

- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q4.8. Based on the inflation forecast provided, what do you think is the probability of inflation being **between 3.3% and 5.6%**?

- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q4.9.

The most likely value for inflation in 2026Q2 is of 2.6% (central estimate),
though the actual inflation value may deviate from this forecast.

Please answer all the questions based on the information given.

Q4.10. Based on the inflation forecast provided, what do you think is the probability of inflation ending up **outside** of the range from -0.4% to 5.6%?

- ☐ 40% (20% each side)
- ☐ 20% (10% each side)
- ☐ 10% (5% each side)
- ☐ 5% (2.5% each side)
- ☐ Less than 1%

Q4.11. Based on the inflation forecast provided, what do you think is the probability of inflation ending up **higher than 6%**?

- ☐ More than 20%
- ☐ 10-20%
- ☐ 5-10%
- ☐ 1-5%
- ☐ Less than 1%

Q4.12. Based on the inflation forecast provided, what do you think is the probability of inflation ending up **higher than 3%**?

- ☐ More than 60%
- ☐ 50-60%
- ☐ 40-50%
- ☐ 30-40%
- ☐ Less than 30%

Q4.13.

The most likely value for inflation in 2026Q2 is of 2.6% (central estimate),
though the actual inflation value may deviate from this forecast.

Please answer all the questions based on the information

given.

Q4.14. To keep inflation low and stable, Government sets an inflation target of 2%.

Based on the inflation forecast provided, what do you think is the probability of inflation **exceeding** 2%?

- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q4.15. Based on the inflation forecast provided, do you think there is a higher risk of inflation exceeding the central estimate or of being less than this central estimate?

- ☐ There is a bigger upside risk (that is, of inflation exceeding the central estimate).
- ☐ There is a bigger downside risk (that is, of inflation being less than the central estimate).
- ☐ The risks appear equal on both sides.

Q4.16.

The most likely value for inflation in 2026Q2 is of 2.6% (central estimate),
though the actual inflation value may deviate from this forecast.

Please answer all the questions based on the information

given.

Q4.17. How likely do you think it is that the inflation will be **exactly** 2.6%?

- | | | | | | | |
|------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Exceptionally unlikely | Very unlikely | Quite unlikely | Fifty-fifty | Quite likely | Very likely | Virtually certain |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

Q4.18.

The most likely value for inflation in 2026Q2 is of 2.6% (central estimate),
though the actual inflation value may deviate from this forecast.

Please answer all the questions based on the information given.

Q4.19. What tells you that the inflation in 2026Q2 is likely or unlikely to be **exactly** 2.6%? (Please select the option that best reflects your belief, even if other answers may also seem reasonable.)

- ☐ It is being communicated in the text that this is the most likely value.
- ☐ It is being communicated in the text that the actual value may deviate.
- ☐ There is not enough information, suggesting the central estimate is not certain.
- ☐ This is the only information given, suggesting the central estimate is quite certain.
- ☐ The probability of inflation taking any exact value is 0.
- ☐ The value seems reasonable to me. (gut feeling)
- ☐ The value seems unreasonable to me. (gut feeling)
- ☐ Other (Please specify)

Q4.20.

The most likely value for inflation in 2026Q2 is of 2.6% (central estimate),
though the actual inflation value may deviate from this forecast.

Please answer all the questions based on the information
given.

Q4.21. How likely do you think it is that the inflation will be **between
1.9% and 3.3%**?

Exceptionally unlikely	Very unlikely	Quite unlikely	Fifty-fifty	Quite likely	Very likely	Virtually certain
☐	☐	☐	☐	☐	☐	☐

Q4.22. How likely do you think it is that the inflation will be **between
3.3% and 5.6%**?

Exceptionally unlikely	Very unlikely	Quite unlikely	Fifty-fifty	Quite likely	Very likely	Virtually certain
☐	☐	☐	☐	☐	☐	☐

Q4.23. How likely do you think it is that the inflation will be **higher
than 3%**?

Exceptionally unlikely	Very unlikely	Quite unlikely	Fifty-fifty	Quite likely	Very likely	Virtually certain
☐	☐	☐	☐	☐	☐	☐

Exceptionally unlikely	Very unlikely	Quite unlikely	Fifty-fifty	Quite likely	Very likely	Virtually certain
☐	☐	☐	☐	☐	☐	☐

Q4.24. Based on the inflation forecast provided, please rank the
following inflation rates from **most likely to least likely** to occur:
0.8%, 2.5%, 3.9% and 6.5%.

0.8%
2.5%
3.9%
6.5%

Q4.25.

The most likely value for inflation in 2026Q2 is of 2.6% (central estimate),
though the actual inflation value may deviate from this forecast.

Please answer all the questions based on the information
given.

Q4.26. On a scale of 1 to 10, where 1 means not accurate at all and
10 means very accurate, how accurate do you think the central
estimate of 2.6% is?

Not accurate at all					Very accurate				
☒	2	3	4	5	6	7	8	9	10

Q4.27. On a scale of 1 to 10, where 1 means very uncertain and 10 means very certain, how certain or uncertain do you think the inflation forecast for 2026 Q2 is?

Very uncertain					Very certain				
☒	2	3	4	5	6	7	8	9	10

Q4.28. Why do you think that the forecast is more or less uncertain? (Select all that apply)

- ☐ The forecast is for a time far in the future.
- ☐ The central estimate has a low probability.
- ☐ I'm not familiar with how to interpret this type of chart.
- ☐ I believe the economic conditions are highly unpredictable.
- ☐ I believe the economic conditions are highly predictable.
- ☐ There is not enough information given, suggesting uncertainty.
- ☐ This is the only information given, suggesting certainty and confidence.
- ☐ The forecast is not as I expected.
- ☐ The forecast seems reasonable to me.
- ☐ Other (Please specify)

Q4.29. On a scale of 1 to 10, where 1 means not representative at all and 10 means very representative, how representative do you think the uncertainty being communicated is (i.e., how well the given information reflects the true variability or uncertainty of the forecast?)

Not representative at all					Very representative				
☒	2	3	4	5	6	7	8	9	10

Q4.30. On a scale of 1 to 10, where 1 means very difficult and 10 means very easy, how easy or difficult do you find the inflation forecast communication to understand?

Very difficult					Very easy				
☒	2	3	4	5	6	7	8	9	10

Q4.31. On a scale of 1 to 10, where 1 means you don't trust the inflation forecast at all and 10 means you fully trust it, how much do you trust the inflation forecast communicated to you in this way?

I don't trust at all.					I fully trust it.				
☒	2	3	4	5	6	7	8	9	10

Decision-Making using Y-on-Y Forecast (Control Group)

Q5.1. The information below describes the inflation forecast in **2026Q2**. Note that the value is based on the latest data released by the Bank of England.

The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast.

Q5.2. Assume you are an investor with £100,000 to invest in a 1-year bond. You will receive your full investment back in a single payment at the end of the 1 year.

Your goal is to maximize the real return on your investment that you will receive after 1 year.

Inflation has a direct impact on the return on investment you will receive. The following setup is simplified but it generally reflects the reality of how each type of bond responds to inflation.

There are two types of bonds available for you to choose between:

- A. Nominal Bond:** A nominal bond has a fixed rate of interest and the **real return** depends on inflation. If inflation stays low or drops ($\leq 3\%$), choosing fixed-rate nominal bonds will provide higher return. However, if inflation increases, your return will be eroded due to reduced purchasing power.
- B. Inflation-Protected Bond:** An inflation-protected bond provides stable but lower **real return** regardless of inflation. If inflation is high ($> 3\%$), choosing inflation-protected bonds will hedge against inflation by ensuring your returns are adjusted in line with inflation.

Now, you are asked to choose which bond you would like to invest your entire £100,000 in:

- A. Nominal Bond
- B. Inflation-Protected Bond

A payoff table is provided to help you understand the potential outcomes of your decision. It shows how much you could effectively

gain or lose, depending on which bond you choose to invest in, and on the actual inflation rate in 2026Q2.

	Real Return When Inflation is Low ($\leq 3\%$)	Real Return When Inflation is High ($> 3\%$)
A. Nominal Bond	+£3,000	-£1,000
B. Inflation-Protected Bond	+£1,000	+£1,000

Note that you will **not be** able to trade the bond during the next year.

Please make the decision based on the forecast information provided, but keep in mind that there is uncertainty.

Q5.3. Which bond would you choose to invest in?

- ☐ A. Nominal Bond
- ☐ B. Inflation-Protected Bond

Q5.4. On a scale of 1 to 10, where 1 means not confident at all and 10 means very confident, how confident are you in making the decision?

Not confident at all

☒ 1 2 3 4 5 6 7 8 9 10

Very confident

Q5.5. Could you explain in a few sentences how and why you decided to invest in this bond?

Q5.6. If the probability of inflation being low ($\leq 3\%$) is **greater than** the following percentages, I would consider investing in the **nominal bond**:

- ☐ 40%
- ☐ 50%
- ☐ 60%
- ☐ 70%
- ☐ 80%
- ☐ I would consider changing my decision if the inflation is perceived low, but I do not think about probability.
- ☐ I will always choose an inflation-protected bond.

Q5.7. If the probability of inflation being high ($> 3\%$) is **greater than** the following percentages, I would consider investing in the **inflation-protected bond**:

- ☐ 40%
- ☐ 50%
- ☐ 60%
- ☐ 70%
- ☐ 80%
- ☐ I would consider changing my decision if the inflation is perceived high, but I do not think about probability.
- ☐ I will always choose a nominal bond.

Q5.8. Considering the payoff for the **Nominal Bond**, which of the following has a larger impact on your decision-making?

- ☐ The potential loss if inflation turns out to be high ($> 3\%$).
- ☐ The potential gain if inflation stays low ($\leq 3\%$).
- ☐ The potential loss and the potential gain have the same impact.

Q5.9. On a scale of 1 to 10, where 1 means the forecast was not effective at all and 10 means it was very effective, how effective do you think the forecast was in assisting your decision-making?

Not effective at all

1

2

3

4

5

6

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8

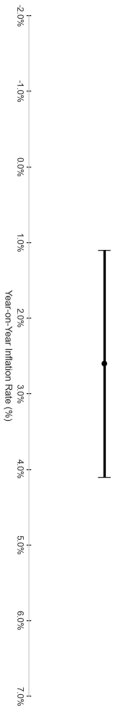
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Very effective

Understanding of Y-on-Y Forecast (Error Bar)

Q6.1. The graph below shows the inflation forecast in **2026Q2**. Note that the graph is created based on the latest data released by the Bank of England.



The most likely value for inflation in **2026Q2** is of **2.6% (central estimate)**, though the actual inflation value may deviate from this forecast. The inflation rate is quite likely to be somewhere on the black line between **1.1% and 4.1%** (a 6 in 10 chance, or 60%). And it is less likely that the inflation rate will be outside the black line (a 4 in 10 chance, or 40%).

Q6.2. Have you ever seen this type of graph before?

- ☐ Definitely not
- ☐ Probably not
- ☐ Probably yes
- ☐ Definitely yes

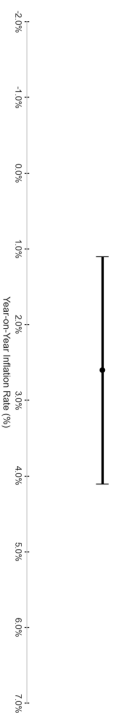
Q6.3. Does the inflation forecast information provided align with your expectations?

- ☐ Strongly aligns – The inflation forecast matches exactly what I expected.
- ☐ Somewhat aligns – The inflation forecast is generally in line with my expectations.
- ☐ Neutral – The inflation forecast neither aligns nor conflicts with my expectations.
- ☐ Somewhat does not align – The inflation forecast is somewhat different from what I expected.
- ☐ Does not align at all – The inflation forecast is significantly different from what I expected.

Q6.4. How does the inflation forecast conflict with your expectations?

- ☐ The inflation forecast is higher than what I expected.
- ☐ The inflation forecast is lower than what I expected.

Q6.5.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is quite likely to be somewhere on the black line between 1.1% and 4.1% (a 6 in 10 chance, or 60%). And it is less likely that the inflation rate will be outside the black line (a 4 in 10 chance, or 40%).

Please answer all the questions based on the information given.

Q6.6. Based on the inflation forecast provided, what do you think is the probability of inflation being **exactly** 2.6%?

- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 80%) About a 8 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 20%) About a 2 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

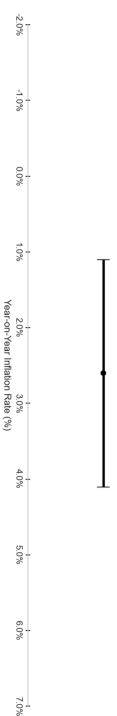
Q6.7. Based on the inflation forecast provided, what do you think is the probability of inflation being **between** 1.9% and 3.3%?

- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q6.8. Based on the inflation forecast provided, what do you think is the probability of inflation being **between 3.3% and 5.6%**?

- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q6.9.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is quite likely to be somewhere on the black line between 1.1% and 4.1% (a 6 in 10 chance, or 60%). And it is less likely that the inflation rate will be outside the black line (a 4 in 10 chance, or 40%).

Please answer all the questions based on the information given.

Q6.10. Based on the inflation forecast provided, what do you think is the probability of inflation ending up **outside** of the black line?

- ☐ 40% (20% each side)

- ☐ 20% (10% each side)
- ☐ 10% (5% each side)
- ☐ 5% (2.5% each side)
- ☐ Less than 1%

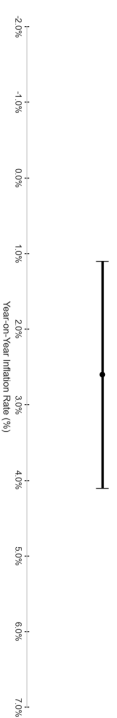
Q6.11. Based on the inflation forecast provided, what do you think is the probability of inflation ending up **higher than 6%**?

- ☐ More than 20%
- ☐ 10-20%
- ☐ 5-10%
- ☐ 1-5%
- ☐ Less than 1%

Q6.12. Based on the inflation forecast provided, what do you think is the probability of inflation ending up **higher than 3%**?

- ☐ More than 60%
- ☐ 50-60%
- ☐ 40-50%
- ☐ 30-40%
- ☐ Less than 30%

Q6.13.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is quite likely to be somewhere on the black line between 1.1% and 4.1% (a 6 in 10 chance, or 60%). And it is less likely that the inflation rate will be outside the black line (a 4 in 10 chance, or 40%).

Please answer all the questions based on the information given.

Q6.14. To keep inflation low and stable, Government sets an inflation target of 2%.

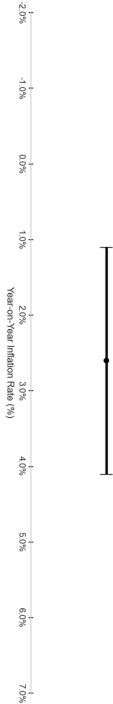
Based on the inflation forecast provided, what do you think is the probability of inflation **exceeding** 2%?

- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q6.15. Based on the inflation forecast provided, do you think there is a higher risk of inflation exceeding the central estimate or of being less than this central estimate?

- ☐ There is a bigger upside risk (that is, of inflation exceeding the central estimate).
- ☐ There is a bigger downside risk (that is, of inflation being less than the central estimate).
- ☐ The risks appear equal on both sides.

Q6.16.



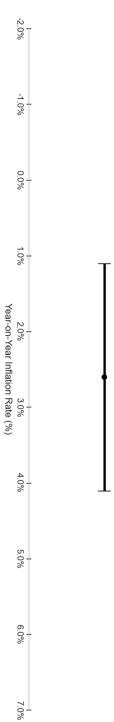
The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is quite likely to be somewhere on the black line between 1.1% and 4.1% (a 6 in 10 chance, or 60%). And it is less likely that the inflation rate will be outside the black line (a 4 in 10 chance, or 40%).

Please answer all the questions based on the information given.

Q6.17. How likely do you think it is that the inflation will be **exactly** 2.6%?

- | | | | | | | |
|------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Exceptionally unlikely | Very unlikely | Quite unlikely | Fifty-fifty | Quite likely | Very likely | Virtually certain |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

Q6.18.

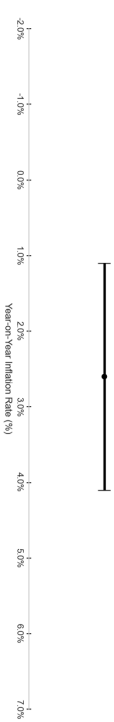


The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is quite likely to be somewhere on the black line between 1.1% and 4.1% (a 6 in 10 chance, or 60%). And it is less likely that the inflation rate will be outside the black line (a 4 in 10 chance, or 40%).

Please answer all the questions based on the information given.

- Q6.19. What tells you that the inflation in 2026Q2 is likely or unlikely to be **exactly** 2.6%? (Please select the option that best reflects your belief, even if other answers may also seem reasonable.)
- ☐ It is being communicated in the text that this is the most likely value.
 - ☐ It is being communicated in the text that the actual value may deviate.
 - ☐ The black line extends over a short distance, suggesting the central estimate somewhat likely but not certain.
 - ☐ The black line extends over a long distance, so inflation is unlikely to be around this value.
 - ☐ The centre of the black line is at 2.6%.
 - ☐ The probability of inflation taking any exact value is 0.
 - ☐ The value seems reasonable to me. (gut feeling)
 - ☐ The value seems unreasonable to me. (gut feeling)
 - ☐ Other (Please specify)

Q6.20.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is quite likely to be somewhere on the black line between 1.1% and 4.1% (a 6 in 10 chance, or 60%). And it is less likely that the inflation rate will be outside the black line (a 4 in 10 chance, or 40%).

Please answer all the questions based on the information given.

- Q6.21. How likely do you think it is that the inflation will be **between 1.9% and 3.3%**?
- | | | | | | | |
|------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Exceptionally unlikely | Very unlikely | Quite unlikely | Fifty-fifty | Quite likely | Very likely | Virtually certain |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
- Q6.22. How likely do you think it is that the inflation will be **between 3.3% and 5.6%**?
- | | | | | | | |
|------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Exceptionally unlikely | Very unlikely | Quite unlikely | Fifty-fifty | Quite likely | Very likely | Virtually certain |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

Exceptionally unlikely Very unlikely Quite unlikely Fifty-fifty Quite likely Very likely Virtually certain

Q6.23. How likely do you think it is that the inflation will be higher than 3%?

Exceptionally unlikely Very unlikely Quite unlikely Fifty-fifty Quite likely Very likely Virtually certain

Q6.24. Based on the inflation forecast provided, please rank the following inflation rates from **most likely** to **least likely** to occur: 0.8%, 2.5%, 3.9% and 6.5%.

0.8%
2.5%
3.9%
6.5%

Q6.25.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is quite likely to be somewhere on the black line between 1.1% and 4.1% (a 6 in 10 chance, or 60%). And it is less likely that the inflation rate will be outside the black line (a 4 in 10 chance, or 40%).

Please answer all the questions based on the information given.

Q6.26. On a scale of 1 to 10, where 1 means not accurate at all and 10 means very accurate, how accurate do you think the central estimate in the graph is?

Not accurate at all Very accurate

1 2 3 4 5 6 7 8 9 10

Q6.27. On a scale of 1 to 10, where 1 means very uncertain and 10 means very certain, how certain or uncertain do you think the inflation forecast for 2026 Q2 is?

Very uncertain Very certain

1 2 3 4 5 6 7 8 9 10

Q6.28. Why do you think that the forecast is more or less uncertain? (Select all that apply)

☐ The forecast is for a time far in the future.

- ☐ The central estimate has a low probability.
- ☐ I'm not familiar with how to interpret this type of chart.
- ☐ I believe the economic conditions are highly unpredictable.
- ☐ I believe the economic conditions are highly predictable.
- ☐ The error bar is widely spread.
- ☐ The error bar is concentrated.
- ☐ The forecast is not as I expected.
- ☐ The forecast seems reasonable to me.
- ☐ Other (Please specify)

Q6.29. On a scale of 1 to 10, where 1 means not representative at all and 10 means very representative, how representative do you think the uncertainty depicted in the graph is (i.e., how well the given information reflects the true variability or uncertainty of the forecast?)

Not representative at all

Very representative

1

2

3

4

5

6

7

8

9

10

Q6.30. On a scale of 1 to 10, where 1 means very difficult and 10 means very easy, how easy or difficult do you find the graph to understand?

Very difficult

Very easy

1

2

3

4

5

6

7

8

9

10

Q6.31. On a scale of 1 to 10, where 1 means you don't trust the inflation forecast at all and 10 means you fully trust it, how much do you trust the inflation forecast shown in the graph?

I don't trust at all.

I fully trust it.

1

2

3

4

5

6

7

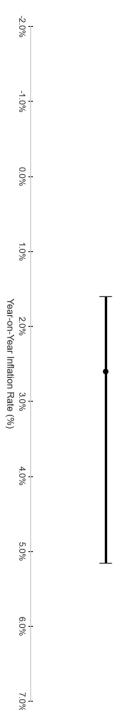
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9

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Q6.32. The graph below shows the inflation forecast in **2026Q3**.

Please note that this forecast is for a different quarter in 2026.



The most likely value for inflation in 2026Q3 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is quite likely to be somewhere on the black line between 1.6% and 5.2% (a 6 in 10 chance, or 60%). And it is less likely that the inflation rate will be outside the black line (a 4 in 10 chance, or 40%).

Please answer all the questions based on the information given.

Q6.33. Based on the inflation forecast provided, what do you think is the probability of inflation ending up **higher than 3%**?

- ☐ More than 60%
- ☐ 50-60%
- ☐ 40-50%
- ☐ 30-40%
- ☐ Less than 30%

Q6.34. How likely do you think it is that the inflation will be **higher than 3%**?

Exceptionally unlikely	Very unlikely	Quite unlikely	Fifty-fifty	Quite likely	Very likely	Virtually certain
☐	☐	☐	☐	☐	☐	☐

Q6.35. Based on the inflation forecast provided, please rank the following inflation rates from **most likely** to **least likely** to occur: 1.1%, 2.6%, 3.3% and 4.1%.

- 1.1%
- 2.6%
- 3.3%
- 4.1%

Q6.36. Based on the inflation forecast provided, do you think there is a higher risk of inflation exceeding the central estimate or of being less than this central estimate?

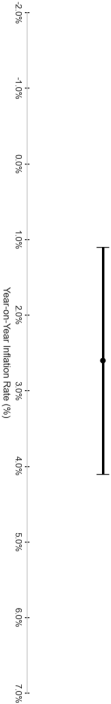
- ☐ There is a bigger upside risk (that is, of inflation exceeding the central estimate).
- ☐ There is a bigger downside risk (that is, of inflation being less than the central estimate).
- ☐ The risks appear equal on both sides.

Q6.37. If the actual inflation ends up outside the black line, which side is more likely, based on the uncertainty depicted in the graph?

- ☐ More likely to be on the upper side (where inflation is high).
- ☐ More likely to be on the lower side (where inflation is low).
- ☐ Roughly equal chances of being on either side.

Decision-Making using Y-on-Y Forecast (Error Bar)

Q7.1. The graph below shows the inflation forecast in **2026Q2**. Note that the graph is created based on the latest data released by the Bank of England.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is quite likely to be somewhere on the black line between 1.1% and 4.1% (a 6 in 10 chance, or 60%). And it is less likely that the inflation rate will be outside the black line (a 4 in 10 chance, or 40%).

Q7.2. Assume you are an investor with £100,000 to invest in a **1-year** bond. You will receive your full investment back in a single payment at the end of the 1 year.

Your goal is to maximize the real return on your investment that you will receive after 1 year.

Inflation has a direct impact on the return on investment you will receive. The following setup is simplified but it generally reflects the reality of how each type of bond responds to inflation.

There are two types of bonds available for you to choose between:

- A. Nominal Bond:** A nominal bond has a fixed rate of interest and the **real return** depends on inflation. If inflation stays low or drops ($\leq 3\%$), choosing fixed-rate nominal bonds will provide higher return. However, if inflation increases, your return will be eroded due to reduced purchasing power.
- B. Inflation-Protected Bond:** An inflation-protected bond provides stable but lower **real return** regardless of inflation. If inflation is high ($> 3\%$), choosing inflation-protected bonds will hedge against inflation by ensuring your returns are adjusted in line with inflation.

Now, you are asked to choose which bond you would like to invest your entire £100,000 in:

- A. Nominal Bond
- B. Inflation-Protected Bond

A payoff table is provided to help you understand the potential outcomes of your decision. It shows how much you could effectively gain or lose, depending on which bond you choose to invest in, and on the actual inflation rate in **2026Q2**.

	Real Return When Inflation is Low ($\leq 3\%$)	Real Return When Inflation is High ($> 3\%$)
A. Nominal Bond	+£3,000	-£1,000
B. Inflation-Protected Bond	+£1,000	+£1,000

Note that you will **not** be able to trade the bond during the next year.

Please make the decision based on the forecast information provided, but keep in mind that there is uncertainty.

Q7.3. Which bond would you choose to invest in?

- ☐ A. Nominal Bond
- ☐ B. Inflation-Protected Bond

Q7.4. On a scale of 1 to 10, where 1 means not confident at all and 10 means very confident, how confident are you in making the decision?

Not confident at all

☒ 1 2 3 4 5 6 7 8 9 10

Very confident

Q7.5. Could you explain in a few sentences how and why you decided to invest in this bond?

Q7.6. If the probability of inflation being low ($\leq 3\%$) is **greater than** the following percentages, I would consider investing in the **nominal bond**:

- ☐ 40%
- ☐ 50%
- ☐ 60%
- ☐ 70%
- ☐ 80%
- ☐ I would consider changing my decision if the inflation is perceived low, but I do not think about probability.
- ☐ I will always choose an inflation-protected bond.

Q7.7. If the probability of inflation being high (> 3%) is greater than the following percentages, I would consider investing in the inflation-protected bond:

- ☐ 40%
- ☐ 50%
- ☐ 60%
- ☐ 70%
- ☐ 80%
- ☐ I would consider changing my decision if the inflation is perceived high, but I do not think about probability.
- ☐ I will always choose a nominal bond.

Q7.8. Considering the payoff for the Nominal Bond, which of the following has a larger impact on your decision-making?

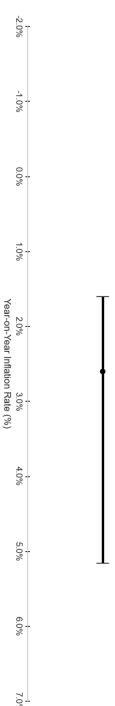
- ☐ The potential loss if inflation turns out to be high (> 3%).
- ☐ The potential gain if inflation stays low ($\leq 3\%$).
- ☐ The potential loss and the potential gain have the same impact.

Q7.9. On a scale of 1 to 10, where 1 means the forecast was not effective at all and 10 means it was very effective, how effective do you think the forecast was in assisting your decision-making?

Not effective at all Very effective

1 2 3 4 5 6 7 8 9 10

Q7.10. The following setup remains the same; however, we assume that your real return will depend on the Year-on-Year inflation rate in 2026Q3, with the forecast provided below.



The most likely value for inflation in 2026Q3 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is quite likely to be somewhere on the black line between 1.6% and 5.2% (a 6 in 10 chance, or 60%). And it is less likely that the inflation rate will be outside the black line (a 4 in 10 chance, or 40%).

Q7.11. Assume you are an investor with £100,000 to invest in a 1-year bond. You will receive your full investment back in a single payment at the end of the 1 year.

Your goal is to maximize the real return on your investment that you will receive after 1 year.

Inflation has a direct impact on the return on investment you will receive. The following setup is simplified but it generally reflects the reality of how each type of bond responds to inflation.

There are two types of bonds available for you to choose between:

A. Nominal Bond: A nominal bond has a fixed rate of interest and the **real return** depends on inflation. If inflation stays low or drops ($\leq 3\%$), choosing fixed-rate nominal bonds will provide higher return. However, if inflation increases, your return will be eroded due to reduced purchasing power.

B. Inflation-Protected Bond: An inflation-protected bond provides stable but lower **real return** regardless of inflation. If inflation is high ($> 3\%$), choosing inflation-protected bonds will hedge against inflation by ensuring your returns are adjusted in line with inflation.

Now, you are asked to choose which bond you would like to invest your entire £100,000 in:

- A. Nominal Bond
- B. Inflation-Protected Bond

A payoff table is provided to help you understand the potential outcomes of your decision. It shows how much you could effectively gain or lose, depending on which bond you choose to invest in, and on the actual inflation rate in 2026Q3.

	Real Return When Inflation is Low ($\leq 3\%$)	Real Return When Inflation is High ($> 3\%$)
A. Nominal Bond	+£3,000	-£1,000
B. Inflation-Protected Bond	+£1,000	+£1,000

Note that you will **not** be able to trade the bond during the next year.

Please make the decision based on the forecast information provided, but keep in mind that there is uncertainty.

Q7.12. Which bond would you choose to invest in?

- ☐ A. Nominal Bond
- ☐ B. Inflation-Protected Bond

Q7.13. On a scale of 1 to 10, where 1 means not confident at all and 10 means very confident, how confident are you in making the decision?

Not confident at all

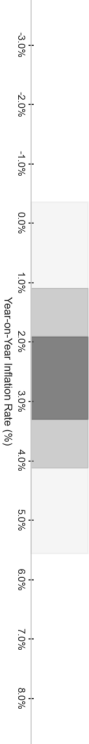
☒ 1 2 3 4 5 6 7 8 9 10

Very confident

Q7.14. Could you explain in a few sentences how and why you decided to invest in this bond?

Understanding of Y-on-Y Forecast (Shaded Stripe)

Q8.1. The graph below shows the inflation forecast in 2026Q2. Note that the graph is created based on the latest data released by the Bank of England.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Q8.2. Have you ever seen this type of graph before?

- ☐ Definitely not
- ☐ Probably not
- ☐ Probably yes
- ☐ Definitely yes

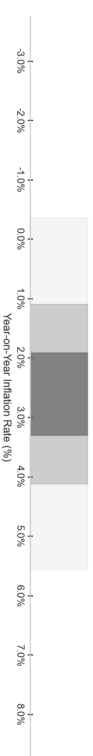
Q8.3. Does the inflation forecast information provided align with your expectations?

- ☐ Strongly aligns – The inflation forecast matches exactly what I expected.
- ☐ Somewhat aligns – The inflation forecast is generally in line with my expectations.
- ☐ Neutral – The inflation forecast neither aligns nor conflicts with my expectations.
- ☐ Somewhat does not align – The inflation forecast is somewhat different from what I expected.
- ☐ Does not align at all – The inflation forecast is significantly different from what I expected.

Q8.4. How does the inflation forecast conflict with your expectations?

- ☐ The inflation forecast is higher than what I expected.
- ☐ The inflation forecast is lower than what I expected.

Q8.5.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Please answer all the questions based on the information given.

Q8.6. Based on the inflation forecast provided, what do you think is the probability of inflation being **exactly** 2.6%?

- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 80%) About a 8 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 20%) About a 2 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

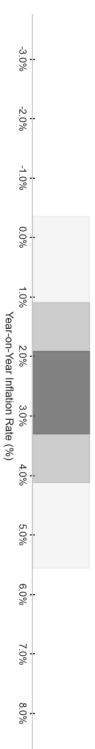
Q8.7. Based on the inflation forecast provided, what do you think is the probability of inflation being **between 1.9% and 3.3%**?

- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q8.8. Based on the inflation forecast provided, what do you think is the probability of inflation being **between 3.3% and 5.6%**?

- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q8.9.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Please answer all the questions based on the information given.

Q8.10. Based on the inflation forecast provided, what do you think is the probability of inflation ending up **outside** of the shaded area?

- ☐ 40% (20% each side)
- ☐ 20% (10% each side)
- ☐ 10% (5% each side)
- ☐ 5% (2.5% each side)
- ☐ Less than 1%

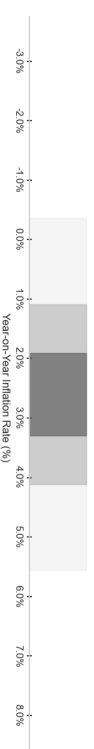
Q8.11. Based on the inflation forecast provided, what do you think is the probability of inflation ending up **higher than 6%**?

- ☐ More than 20%
- ☐ 10-20%
- ☐ 5-10%
- ☐ 1-5%
- ☐ Less than 1%

Q8.12. Based on the inflation forecast provided, what do you think is the probability of inflation ending up **higher than 3%**?

- ☐ More than 60%
- ☐ 50-60%
- ☐ 40-50%
- ☐ 30-40%
- ☐ Less than 30%

Q8.13.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Please answer all the questions based on the information given.

Q8.14. To keep inflation low and stable, Government sets an inflation target of 2%.

Based on the inflation forecast provided, what do you think is the probability of inflation **exceeding** 2%?

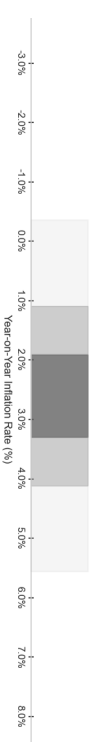
- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance

- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q8.15. Based on the inflation forecast provided, do you think there is a higher risk of inflation exceeding the central estimate or of being less than this central estimate?

- ☐ There is a bigger upside risk (that is, of inflation exceeding the central estimate).
- ☐ There is a bigger downside risk (that is, of inflation being less than the central estimate).
- ☐ The risks appear equal on both sides.

Q8.16.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Please answer all the questions based on the information given.

Q8.17. How likely do you think it is that the inflation will be **exactly** 2.6%?

- Exceptionally unlikely
- Very unlikely
- Quite unlikely
- Fifty-fifty
- Quite likely
- Very likely
- Virtually certain
- ☐
- ☐
- ☐
- ☐
- ☐
- ☐
- ☐

Q8.18.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

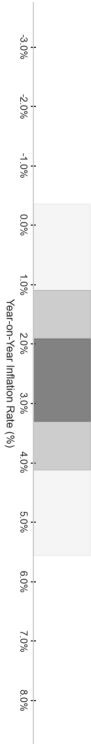
Please answer all the questions based on the information given.

Q8.19. What tells you that the inflation in 26Q2 is likely or unlikely to be **exactly** 2.6%? (Please select the option that best reflects your belief, even if other answers may also seem reasonable.)

- ☐ It is being communicated in the text that this is the most likely value.
- ☐ It is being communicated in the text that the actual value may deviate.
- ☐ The shaded bar covers a narrow range, suggesting it's somewhat likely but not certain.

- ☐ The shaded bar covers a wide range, so inflation is unlikely to be around this value.
- ☐ The centre of the shaded bar is at 2.6%.
- ☐ The probability of inflation taking any exact value is 0.
- ☐ The value seems reasonable to me. (gut feeling)
- ☐ The value seems unreasonable to me. (gut feeling)
- ☐ Other (Please specify)

Q8.20.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Please answer all the questions based on the information given.

Q8.21. How likely do you think it is that the inflation will be **between** 1.9% and 3.3%?

- Exceptionally unlikely
- Very unlikely
- Quite unlikely
- Fifty-fifty
- Quite likely
- Very likely
- Virtually certain

Exceptionally unlikely ☐ Very unlikely ☐ Quite unlikely ☐ Fifty-fifty ☐ Quite likely ☐ Very likely ☐ Virtually certain ☐

Q8.22. How likely do you think it is that the inflation will be **between 3.3% and 5.6%** ?

Exceptionally unlikely ☐ Very unlikely ☐ Quite unlikely ☐ Fifty-fifty ☐ Quite likely ☐ Very likely ☐ Virtually certain ☐

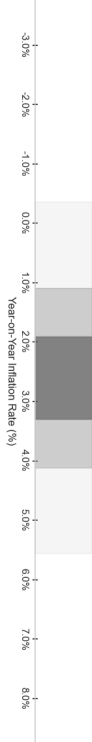
Q8.23. How likely do you think it is that the inflation will be **higher than 3%** ?

Exceptionally unlikely ☐ Very unlikely ☐ Quite unlikely ☐ Fifty-fifty ☐ Quite likely ☐ Very likely ☐ Virtually certain ☐

Q8.24. Based on the inflation forecast provided, please rank the following inflation rates from **most likely** to **least likely** to occur: 0.8%, 2.5%, 3.9% and 6.5%.

0.8%
2.5%
3.9%
6.5%

Q8.25.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Please answer all the questions based on the information given.

Q8.26. On a scale of 1 to 10, where 1 means not accurate at all and 10 means very accurate, how accurate do you think the central estimate in the graph is?

Not accurate at all ☒ 2 3 4 5 6 7 8 9 10 Very accurate

Q8.27. On a scale of 1 to 10, where 1 means very uncertain and 10 means very certain, how certain or uncertain do you think the inflation forecast for 2026 Q2 is?

Very uncertain

Very certain

1 2 3 4 5 6 7 8 9 10

Q8.28. Why do you think that the forecast is more or less uncertain?
(Select all that apply)

- ☐ The forecast is for a time far in the future.
- ☐ The central estimate has a low probability.
- ☐ I'm not familiar with how to interpret this type of chart.
- ☐ I believe the economic conditions are highly unpredictable.
- ☐ I believe the economic conditions are highly predictable.
- ☐ The shaded bar is widely spread.
- ☐ The shaded bar is concentrated.
- ☐ The forecast is not as I expected.
- ☐ The forecast seems reasonable to me.
- ☐ Other (Please specify)

Q8.29. On a scale of 1 to 10, where 1 means not representative at all and 10 means very representative, how representative do you think the uncertainty depicted in the graph is (i.e., how well the given information reflects the true variability or uncertainty of the forecast?)

Not representative at all

Very representative

1 2 3 4 5 6 7 8 9 10

Q8.30. On a scale of 1 to 10, where 1 means very difficult and 10 means very easy, how easy or difficult do you find the graph to understand?

Very difficult

Very easy

1 2 3 4 5 6 7 8 9 10

Q8.31. On a scale of 1 to 10, where 1 means you don't trust the inflation forecast at all and 10 means you fully trust it, how much do you trust the inflation forecast shown in the graph?

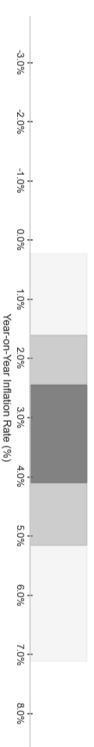
I don't trust at all.

I fully trust it.

1 2 3 4 5 6 7 8 9 10

Q8.32. The graph below shows the inflation forecast in 2026Q3.

Please note that this forecast is for a different quarter in 2026.



The most likely value for inflation in 2026Q3 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Please answer all the questions based on the information given.

Q8.33. Based on the inflation forecast provided, what do you think is the probability of inflation ending up **higher than 3%**?

- ☐ More than 60%
- ☐ 50-60%
- ☐ 40-50%
- ☐ 30-40%
- ☐ Less than 30%

Q8.34. How likely do you think it is that the inflation will be **higher than 3%**?

- | | | | | | | |
|------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Exceptionally unlikely | Very unlikely | Quite unlikely | Fifty-fifty | Quite likely | Very likely | Virtually certain |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

Q8.35. Based on the inflation forecast provided, please rank the following inflation rates from **most likely** to **least likely** to occur:

- 1.1%
- 2.6%
- 3.3%
- 4.1%

Q8.36. Based on the inflation forecast provided, do you think there is a higher risk of inflation exceeding the central estimate or of being less than this central estimate?

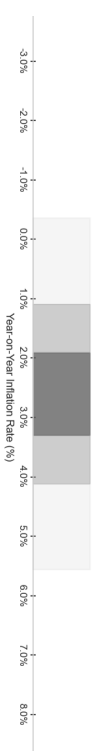
- ☐ There is a bigger upside risk (that is, of inflation exceeding the central estimate).
- ☐ There is a bigger downside risk (that is, of inflation being less than the central estimate).
- ☐ The risks appear equal on both sides.

Q8.37. If the actual inflation ends up outside the shaded area, which side is more likely, based on the uncertainty depicted in the graph?

- ☐ More likely to be on the upper side (where inflation is high).
- ☐ More likely to be on the lower side (where inflation is low).
- ☐ Roughly equal chances of being on either side.

Decision-Making using Y-on-Y Forecast (Shaded Stripe)

Q9.1. The graph below shows the inflation forecast in **2026Q2**. Note that the graph is created based on the latest data released by the Bank of England.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Q9.2. Assume you are an investor with £100,000 to invest in a 1-year bond. You will receive your full investment back in a single payment at the end of the 1 year.

Your goal is to maximize the real return on your investment that you will receive after 1 year.

Inflation has a direct impact on the return on investment you will receive. The following setup is simplified but it generally reflects the reality of how each type of bond responds to inflation.

There are two types of bonds available for you to choose between:

A. Nominal Bond: A nominal bond has a fixed rate of interest and the **real return** depends on inflation. If inflation stays low or drops ($\leq 3\%$), choosing fixed-rate nominal bonds will provide higher return. However, if inflation increases, your return will be eroded due to reduced purchasing power.

B. Inflation-Protected Bond: An inflation-protected bond provides stable but lower **real return** regardless of inflation. If inflation is high ($> 3\%$), choosing inflation-protected bonds will hedge against inflation by ensuring your returns are adjusted in line with inflation.

Now, you are asked to choose which bond you would like to invest your entire £100,000 in:

- A. Nominal Bond
- B. Inflation-Protected Bond

A payoff table is provided to help you understand the potential outcomes of your decision. It shows how much you could effectively gain or lose, depending on which bond you choose to invest in, and on the actual inflation rate in **2026Q2**.

	Real Return When Inflation Is Low ($\leq 3\%$)	Real Return When Inflation Is High ($> 3\%$)
A. Nominal Bond	+£3,000	-£1,000
B. Inflation-Protected Bond	+£1,000	+£1,000

Note that you will **not** be able to trade the bond during the next year.

Please make the decision based on the forecast information provided, but keep in mind that there is uncertainty.

Q9.3. Which bond would you choose to invest in?

- ☐ A. Nominal Bond
- ☐ B. Inflation-Protected Bond

Q9.4. On a scale of 1 to 10, where 1 means not confident at all and 10 means very confident, how confident are you in making the decision?

Not confident at all Very confident

1 2 3 4 5 6 7 8 9 10

Q9.5. Could you explain in a few sentences how and why you decided to invest in this bond?

Q9.6. If the probability of inflation being low ($\leq 3\%$) is **greater than** the following percentages, I would consider investing in the **nominal bond**:

- ☐ 40%
☐ 50%
☐ 60%
☐ 70%
☐ 80%
☐ I would consider changing my decision if the inflation is perceived low, but I do not think about probability.
☐ I will always choose an inflation-protected bond.

Q9.7. If the probability of inflation being high ($> 3\%$) is **greater than** the following percentages, I would consider investing in the **inflation-protected bond**:

- ☐ 40%
☐ 50%
☐ 60%
☐ 70%
☐ 80%
☐ I would consider changing my decision if the inflation is perceived high, but I do not think about probability.

- ☐ I will always choose a nominal bond.

Q9.8. Considering the payoff for the **Nominal Bond**, which of the following has a larger impact on your decision-making?

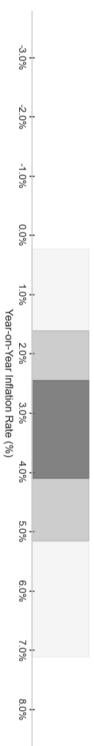
- ☐ The potential loss if inflation turns out to be high ($> 3\%$).
☐ The potential gain if inflation stays low ($\leq 3\%$).
☐ The potential loss and the potential gain have the same impact.

Q9.9. On a scale of 1 to 10, where 1 means the forecast was not effective at all and 10 means it was very effective, how effective do you think the forecast was in assisting your decision-making?

Not effective at all Very effective

1 2 3 4 5 6 7 8 9 10

Q9.10. **The following setup remains the same**; however, we assume that your real return will depend on the Year-on-Year inflation rate in **2026Q3**, with the forecast provided below.



The most likely value for inflation in 2026Q3 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Q9.11. Assume you are an investor with £100,000 to invest in a 1-year bond. You will receive your full investment back in a single payment at the end of the 1 year.

Your goal is to maximize the real return on your investment that you will receive after 1 year.

Inflation has a direct impact on the return on investment you will receive. The following setup is simplified but it generally reflects the reality of how each type of bond responds to inflation.

There are two types of bonds available for you to choose between:

A. Nominal Bond: A nominal bond has a fixed rate of interest and the **real return** depends on inflation. If inflation stays low or drops ($\leq 3\%$), choosing fixed-rate nominal bonds will provide higher return. However, if inflation increases, your return will be eroded due to reduced purchasing power.

B. Inflation-Protected Bond: An inflation-protected bond provides stable but lower **real return** regardless of

inflation. If inflation is high ($> 3\%$), choosing inflation-protected bonds will hedge against inflation by ensuring your returns are adjusted in line with inflation.

Now, you are asked to choose which bond you would like to invest your entire £100,000 in:

- A. Nominal Bond
- B. Inflation-Protected Bond

A payoff table is provided to help you understand the potential outcomes of your decision. It shows how much you could effectively gain or lose, depending on which bond you choose to invest in, and on the actual inflation rate in 2026Q3.

	Real Return When Inflation is Low ($\leq 3\%$)	Real Return When Inflation is High ($> 3\%$)
A. Nominal Bond	+£3,000	-£1,000
B. Inflation-Protected Bond	+£1,000	+£1,000

Note that you will **not** be able to trade the bond during the next year.

Please make the decision based on the forecast information provided, but keep in mind that there is uncertainty.

Q9.12. Which bond would you choose to invest in?

- ☐ A. Nominal Bond
- ☐ B. Inflation-Protected Bond

Q9.13. On a scale of 1 to 10, where 1 means not confident at all and 10 means very confident, how confident are you in making the

decision?

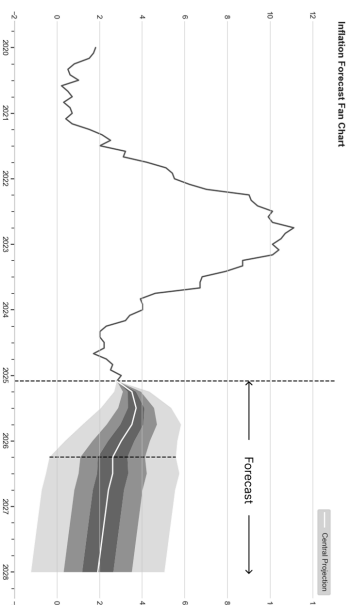
Not confident at all Very confident

1 2 3 4 5 6 7 8 9 10

Q9.14. Could you explain in a few sentences how and why you decided to invest in this bond?

Understanding of Y-on-Y Forecast (Fan Chart Slice)

Q10.1. The graph below shows the inflation forecast from 2025Q2 to 2028Q1. Note that the graph is created based on the latest data released by the Bank of England. **The forecast for 2026Q2 is of particular interest and is highlighted using a vertical dashed line on the graph.**



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Q10.2. Have you ever seen this type of graph before?

- ☐ Definitely not
- ☐ Probably not
- ☐ Probably yes
- ☐ Definitely yes

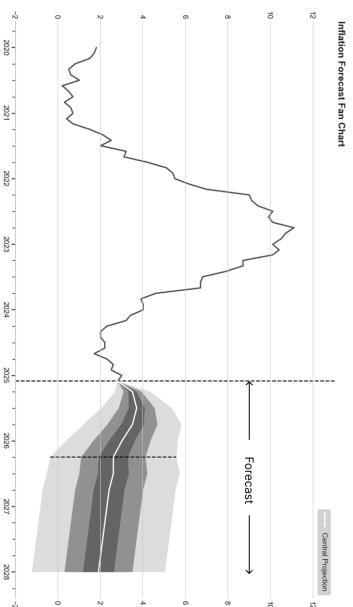
Q10.3. Does the inflation forecast in 2026Q2 align with your expectations?

- ☐ Strongly aligns – The inflation forecast matches exactly what I expected.
- ☐ Somewhat aligns – The inflation forecast is generally in line with my expectations.
- ☐ Neutral – The inflation forecast neither aligns nor conflicts with my expectations.
- ☐ Somewhat does not align – The inflation forecast is somewhat different from what I expected.
- ☐ Does not align at all – The inflation forecast is significantly different from what I expected.

Q10.4. How does the inflation forecast conflict with your expectations?

- ☐ The inflation forecast is higher than what I expected.
- ☐ The inflation forecast is lower than what I expected.

Q10.5.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Please answer all the questions based on the information given.

Q10.6. Based on the inflation forecast provided, what do you think is the probability of inflation in 2026Q2 being exactly 2.6%?

- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 80%) About a 8 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance

- ☐ (around 20%) About a 2 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

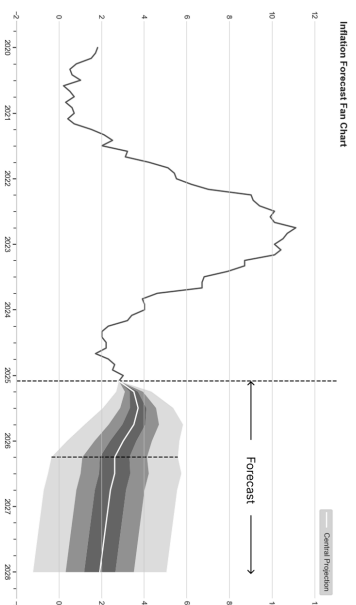
Q10.7. Based on the inflation forecast provided, what do you think is the probability of inflation in 2026Q2 being between 1.9% and 3.3%?

- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q10.8. Based on the inflation forecast provided, what do you think is the probability of inflation in 2026Q2 being between 3.3% and 5.6%?

- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q10.9.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Please answer all the questions based on the information given.

Q10.10. Based on the inflation forecast provided, what do you think is the probability of inflation in 2026Q2 ending up outside of the shaded area?

- ☐ 40% (20% each side)
- ☐ 20% (10% each side)
- ☐ 10% (5% each side)
- ☐ 5% (2.5% each side)
- ☐ Less than 1%

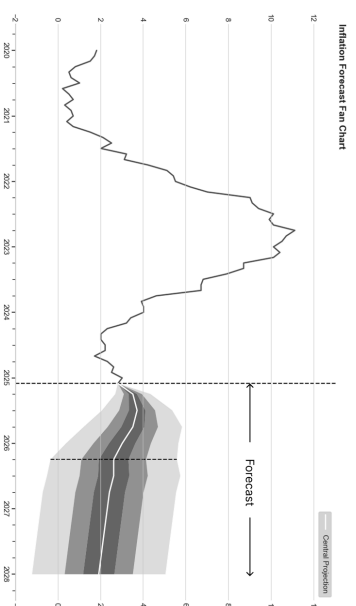
Q10.11. Based on the inflation forecast provided, what do you think is the probability of inflation in 2026Q2 ending up higher than 6%?

- ☐ More than 20%
- ☐ 10-20%
- ☐ 5-10%
- ☐ 1-5%
- ☐ Less than 1%

Q10.12. Based on the inflation forecast provided, what do you think is the probability of inflation in 2026Q2 ending up higher than 3%?

- ☐ More than 60%
- ☐ 50-60%
- ☐ 40-50%
- ☐ 30-40%
- ☐ Less than 30%

Q10.13.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Please answer all the questions based on the information given.

Q10.14. To keep inflation low and stable, Government sets an inflation target of 2%.

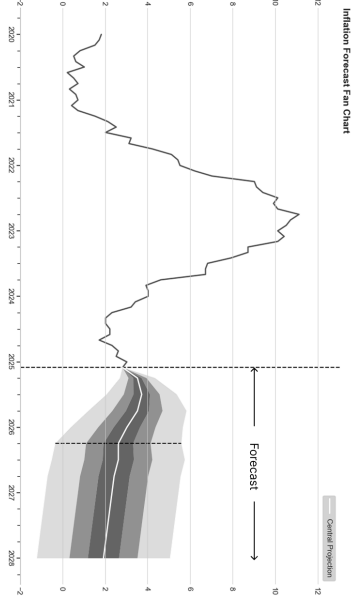
Based on the inflation forecast provided, what do you think is the probability of inflation in 2026Q2 exceeding 2%?

- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q10.15. Based on the inflation forecast provided, do you think there is a higher risk of inflation in 2026Q2 exceeding the central estimate or of being less than this central estimate?

- ☐ There is a bigger upside risk (that is, of inflation exceeding the central estimate).
- ☐ There is a bigger downside risk (that is, of inflation being less than the central estimate).
- ☐ The risks appear equal on both sides.

Q10.16.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

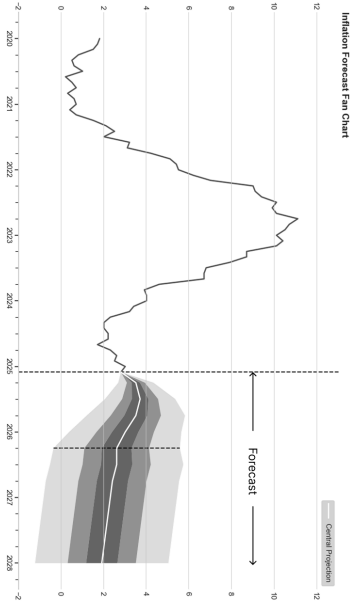
Please answer all the questions based on the information given.

Q10.17. How likely do you think it is that the inflation in 2026Q2 will be exactly 2.6%?

- Exceptionally unlikely
- Very unlikely
- Quite unlikely
- Fifty-fifty
- Quite likely
- Very likely
- Virtually certain

- Exceptionally unlikely ☐
- Very unlikely ☐
- Quite unlikely ☐
- Fifty-fifty ☐
- Quite likely ☐
- Very likely ☐
- Virtually certain ☐

Q10.18.



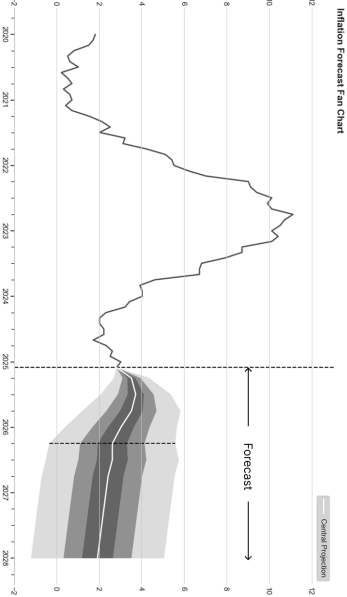
The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Please answer all the questions based on the information given.

Q10.19. What tells you that the inflation in 2026Q2 is likely or unlikely to be **exactly** 2.6%? (Please select the option that best reflects your belief, even if other answers may also seem reasonable.)

- ☐ It is being communicated in the text that this is the most likely value.
- ☐ It is being communicated in the text that the actual value may deviate.
- ☐ The bands at 2026Q2 cover a narrow range, suggesting it's somewhat likely but not certain.
- ☐ The bands at 2026Q2 cover a wide range, so inflation is unlikely to be around this value.
- ☐ The bands at 2026Q2 suddenly get wider.
- ☐ The centre of the bands is at 2.6%.
- ☐ The probability of inflation taking any exact value is 0.
- ☐ The value seems reasonable to me. (gut feeling)
- ☐ The value seems unreasonable to me. (gut feeling)
- ☐ Other (Please specify)

Q10.20.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Please answer all the questions based on the information

given.

2.5%

3.9%

6.5%

Q10.21. How likely do you think it is that the inflation in 2026Q2 will be **between 1.9% and 3.3%**?

Exceptionally unlikely	Very unlikely	Quite unlikely	Fifty-fifty	Quite likely	Very likely	Virtually certain
☐	☐	☐	☐	☐	☐	☐

Q10.22. How likely do you think it is that the inflation in 2026Q2 will be **between 3.3% and 5.6%**?

Exceptionally unlikely	Very unlikely	Quite unlikely	Fifty-fifty	Quite likely	Very likely	Virtually certain
☐	☐	☐	☐	☐	☐	☐

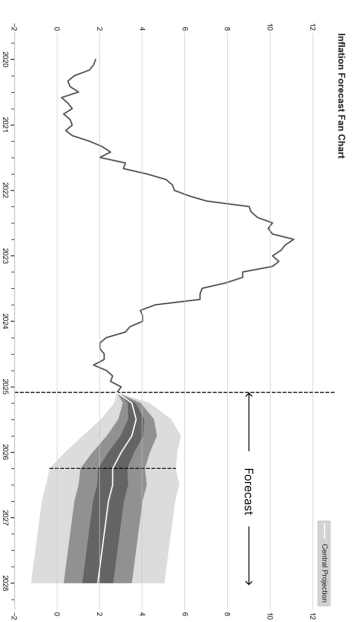
Q10.23. How likely do you think it is that the inflation in 2026Q2 will be **higher than 3%**?

Exceptionally unlikely	Very unlikely	Quite unlikely	Fifty-fifty	Quite likely	Very likely	Virtually certain
☐	☐	☐	☐	☐	☐	☐

Q10.24. Based on the inflation forecast provided, please rank the following inflation rates from **most likely** to **least likely** to occur in 2026Q2: 0.8%, 2.5%, 3.9% and 6.5%.

0.8%

Q10.25.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Please answer all the questions based on the information given.

Q10.26. On a scale of 1 to 10, where 1 means not accurate at all and 10 means very accurate, how accurate do you think the central estimate in the graph is?

Not accurate at all					Very accurate				
☒	2	3	4	5	6	7	8	9	10

Q10.27. On a scale of 1 to 10, where 1 means very uncertain and 10 means very certain, how certain or uncertain do you think the inflation forecast **for 2026 Q2** is?

Very uncertain					Very certain				
☒	2	3	4	5	6	7	8	9	10

Q10.28. Why do you think that the forecast is more or less uncertain? (Select all that apply)

- ☐ The forecast is for a time far in the future.
- ☐ The central estimate has a low probability.
- ☐ I'm not familiar with how to interpret this type of chart.
- ☐ I believe the economic conditions are highly unpredictable.
- ☐ I believe the economic conditions are highly predictable.
- ☐ The bands at 2026Q2 are widely spread.
- ☐ The bands at 2026Q2 are concentrated.
- ☐ The forecast is not as I expected.
- ☐ The forecast seems reasonable to me.
- ☐ Other (Please specify)

Q10.29. On a scale of 1 to 10, where 1 means not representative at all and 10 means very representative, how representative do you

think the uncertainty depicted in the graph is (i.e., how well the given information reflects the true variability or uncertainty of the forecast?)

Not representative at all					Very representative				
☒	2	3	4	5	6	7	8	9	10

Q10.30. On a scale of 1 to 10, where 1 means very difficult and 10 means very easy, how easy or difficult do you find the graph to understand?

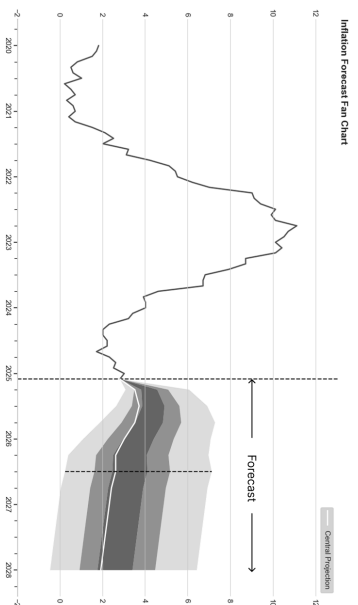
Very difficult					Very easy				
☒	2	3	4	5	6	7	8	9	10

Q10.31. On a scale of 1 to 10, where 1 means you don't trust the inflation forecast at all and 10 means you fully trust it, how much do you trust the inflation forecast shown in the graph?

I don't trust at all.					I fully trust it.				
☒	2	3	4	5	6	7	8	9	10

Q10.32. The graph below shows the inflation forecast in **2026Q3**.

Please note that this forecast is for a different quarter in 2026.



The most likely value for inflation in 2026Q3 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Please answer all the questions based on the information given.

Q10.33. Based on the inflation forecast provided, what do you think is the probability of inflation in 2026Q3 ending up higher than 3%?

- ☐ More than 60%
- ☐ 50-60%
- ☐ 40-50%
- ☐ 30-40%
- ☐ Less than 30%

Q10.34. How likely do you think it is that the inflation in 2026Q3 will be higher than 3%?

- | | | | | | | |
|------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Exceptionally unlikely | Very unlikely | Quite unlikely | Fifty-fifty | Quite likely | Very likely | Virtually certain |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

Q10.35. Based on the inflation forecast provided, please rank the following inflation rates from **most likely to least likely to occur in 2026Q3**: 1.1%, 2.6%, 3.3% and 4.1%.

- 1.1%
- 2.6%
- 3.3%
- 4.1%

Q10.36. Based on the inflation forecast provided, do you think there is a higher risk of inflation in 2026Q3 exceeding the central estimate or of being less than this central estimate?

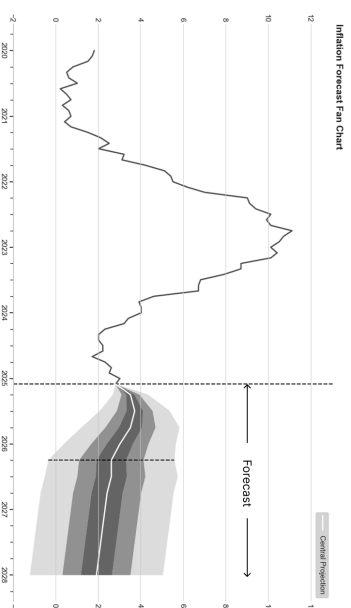
- ☐ There is a bigger upside risk (that is, of inflation exceeding the central estimate).
- ☐ There is a bigger downside risk (that is, of inflation being less than the central estimate).
- ☐ The risks appear equal on both sides.

Q10.37. If the actual inflation in 2026Q3 ends up outside the shaded area, which side is more likely, based on the uncertainty depicted in the graph?

- ☐ More likely to be on the upper side (where inflation is high).
- ☐ More likely to be on the lower side (where inflation is low).
- ☐ Roughly equal chances of being on either side.

Decision-Making using Y-on-Y Forecast (Fan Chart Slice)

Q11.1. The graph below shows the inflation forecast from 2025Q2 to 2028Q1. Note that the graph is created based on the latest data released by the Bank of England. **The forecast for 2026Q2 is of particular interest and is highlighted using a vertical dashed line on the graph.**



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Q11.2. Assume you are an investor with £100,000 to invest in a **1-year bond**. You will receive your full investment back in a single payment at the end of the 1 year.

Your goal is to maximize the real return on your investment that you will receive after 1 year.

Inflation has a direct impact on the return on investment you will receive. The following setup is simplified but it generally reflects the reality of how each type of bond responds to inflation.

There are two types of bonds available for you to choose between:

- A. Nominal Bond:** A nominal bond has a fixed rate of interest and the **real return** depends on inflation. If inflation stays low or drops ($\leq 3\%$), choosing fixed-rate nominal bonds will provide higher return. However, if inflation increases, your return will be eroded due to reduced purchasing power.
- B. Inflation-Protected Bond:** An inflation-protected bond provides stable but lower **real return** regardless of inflation. If inflation is high ($> 3\%$), choosing inflation-protected bonds will hedge against inflation by ensuring your returns are adjusted in line with inflation.

Now, you are asked to choose which bond you would like to invest your entire £100,000 in:

- A. Nominal Bond
- B. Inflation-Protected Bond

A payoff table is provided to help you understand the potential outcomes of your decision. It shows how much you could effectively gain or lose, depending on which bond you choose to invest in, and on the actual inflation rate in **2026Q2**.

	Real Return When Inflation is Low ($\leq 3\%$)	Real Return When Inflation is High ($> 3\%$)
A. Nominal Bond	+£3,000	-£1,000
B. Inflation-Protected Bond	+£1,000	+£1,000

Note that you will **not be** able to trade the bond during the next year.

Please make the decision based on the forecast information provided, but keep in mind that there is uncertainty.

Q11.3. Which bond would you choose to invest in?

- ☐ A. Nominal Bond
☐ B. Inflation-Protected Bond

Q11.4. On a scale of 1 to 10, where 1 means not confident at all and 10 means very confident, how confident are you in making the decision?

Not confident at all Very confident

☒ 1 2 3 4 5 6 7 8 9 10

Q11.5. Could you explain in a few sentences how and why you decided to invest in this bond?

Q11.6. If the probability of inflation being low ($\leq 3\%$) is **greater than** the following percentages, I would consider investing in the **nominal bond**:

- ☐ 40%
☐ 50%
☐ 60%
☐ 70%
☐ 80%
☐ I would consider changing my decision if the inflation is perceived low, but I do not think about probability.
☐ I will always choose an inflation-protected bond.

Q11.7. If the probability of inflation being high ($> 3\%$) is **greater than** the following percentages, I would consider investing in the **inflation-protected bond**:

- ☐ 40%
☐ 50%
☐ 60%
☐ 70%
☐ 80%
☐ I would consider changing my decision if the inflation is perceived high, but I do not think about probability.
☐ I will always choose a nominal bond.

Q11.8. Considering the payoff for **the Nominal Bond**, which of the following has a larger impact on your decision-making?

- ☐ The potential loss if inflation turns out to be high ($> 3\%$).
☐ The potential gain if inflation stays low ($\leq 3\%$).
☐ The potential loss and the potential gain have the same impact.

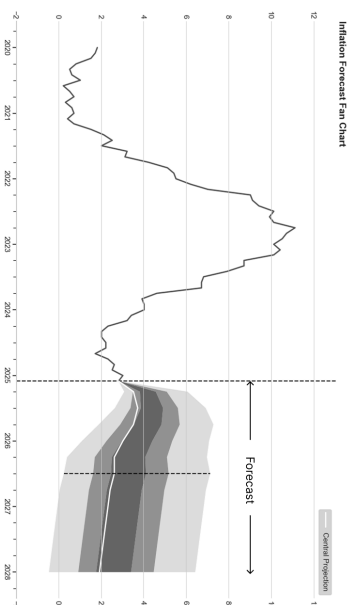
Q11.9. On a scale of 1 to 10, where 1 means the forecast was not effective at all and 10 means it was very effective, how effective do you think the forecast was in assisting your decision-making?

Not effective at all Very effective

1 2 3 4 5 6 7 8 9 10

☒ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10

Q11.10. The following setup remains the same; however, we assume that your real return will depend on the Year-on-Year inflation rate in **2026Q3**, with the forecast provided below.



The most likely value for inflation in 2026Q3 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Q11.11. Assume you are an investor with £100,000 to invest in a **1-year** bond. You will receive your full investment back in a single payment at the end of the 1 year.

Your goal is to maximize the real return on your investment that you will receive after 1 year.

Inflation has a direct impact on the return on investment you will receive. The following setup is simplified but it generally reflects the reality of how each type of bond responds to inflation.

There are two types of bonds available for you to choose between:

- A. Nominal Bond:** A nominal bond has a fixed rate of interest and the **real return** depends on inflation. If inflation stays low or drops ($\leq 3\%$), choosing fixed-rate nominal bonds will provide higher return. However, if inflation increases, your return will be eroded due to reduced purchasing power.
- B. Inflation-Protected Bond:** An inflation-protected bond provides stable but lower **real return** regardless of inflation. If inflation is high ($> 3\%$), choosing inflation-protected bonds will hedge against inflation by ensuring your returns are adjusted in line with inflation.

Now, you are asked to choose which bond you would like to invest your entire £100,000 in:

- A. Nominal Bond
- B. Inflation-Protected Bond

A payoff table is provided to help you understand the potential outcomes of your decision. It shows how much you could effectively gain or lose, depending on which bond you choose to invest in, and on the actual inflation rate in **2026Q3**.

	Real Return When Inflation is Low ($\leq 3\%$)	Real Return When Inflation is High ($> 3\%$)
A. Nominal Bond	+£3,000	-£1,000
B. Inflation-Protected Bond	+£1,000	+£1,000

Note that you will **not be** able to trade the bond during the next year.

Please make the decision based on the forecast information provided, but keep in mind that there is uncertainty.

Q11.12. Which bond would you choose to invest in?

- ☐ A. Nominal Bond
- ☐ B. Inflation-Protected Bond

Q11.13. On a scale of 1 to 10, where 1 means not confident at all and 10 means very confident, how confident are you in making the decision?

Not confident at all

Very confident

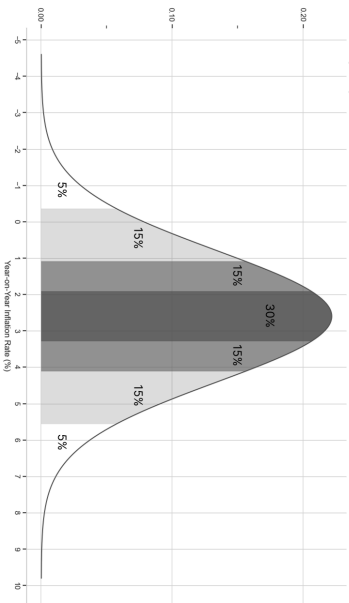
☒ 1
 2
 3
 4
 5
 6
 7
 8
 9
 10

Q11.14. Could you explain in a few sentences how and why you decided to invest in this bond?

Understanding of Y-on-Y Forecast (Bell Curve)

Q12.1. The graph below shows the inflation forecast in **2026Q2**.

Note that the graph is created based on the latest data released by the Bank of England.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Q12.2. Have you ever seen this type of graph before?

- ☐ Definitely not
- ☐ Probably not
- ☐ Probably yes
- ☐ Definitely yes

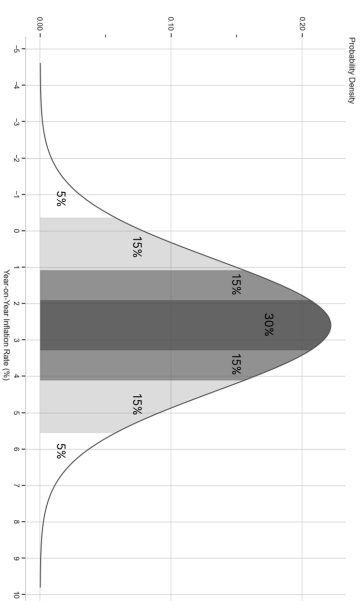
Q12.3. Does the inflation forecast information provided align with your expectations?

- ☐ Strongly aligns – The inflation forecast matches exactly what I expected.
- ☐ Somewhat aligns – The inflation forecast is generally in line with my expectations.
- ☐ Neutral – The inflation forecast neither aligns nor conflicts with my expectations.
- ☐ Somewhat does not align – The inflation forecast is somewhat different from what I expected.
- ☐ Does not align at all – The inflation forecast is significantly different from what I expected.

Q12.4. How does the inflation forecast conflict with your expectations?

- ☐ The inflation forecast is higher than what I expected.
- ☐ The inflation forecast is lower than what I expected.

Q12.5.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Please answer all the questions based on the information given.

Q12.6. Based on the inflation forecast provided, what do you think is the probability of inflation being **exactly** 2.6%?

- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 80%) About a 8 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 20%) About a 2 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

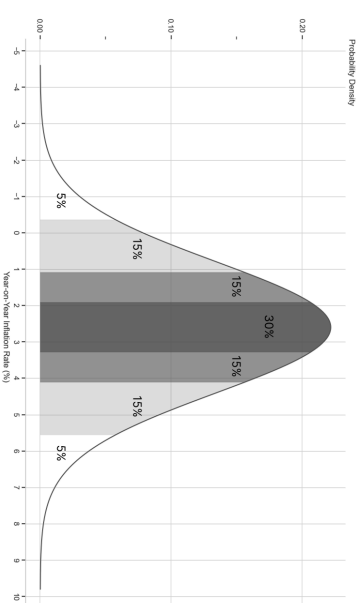
Q12.7. Based on the inflation forecast provided, what do you think is the probability of inflation being **between 1.9% and 3.3%**?

- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q12.8. Based on the inflation forecast provided, what do you think is the probability of inflation being **between 3.3% and 5.6%**?

- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q12.9.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Please answer all the questions based on the information given.

Q12.10. Based on the inflation forecast provided, what do you think is the probability of inflation ending up **outside** of the shaded area?

- ☐ 40% (20% each side)
- ☐ 20% (10% each side)
- ☐ 10% (5% each side)
- ☐ 5% (2.5% each side)
- ☐ Less than 1%

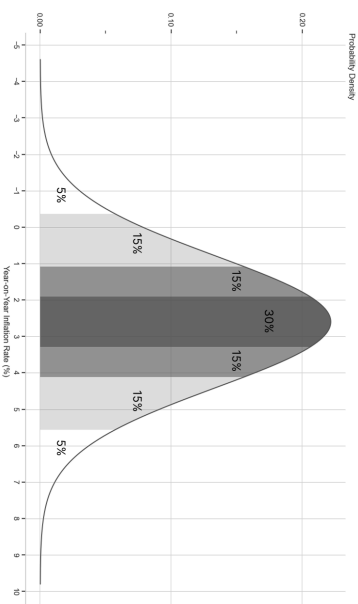
Q12.11. Based on the inflation forecast provided, what do you think is the probability of inflation ending up **higher than 6%**?

- ☐ More than 20%
- ☐ 10-20%
- ☐ 5-10%
- ☐ 1-5%
- ☐ Less than 1%

Q12.12. Based on the inflation forecast provided, what do you think is the probability of inflation ending up **higher than 3%**?

- ☐ More than 60%
- ☐ 50-60%
- ☐ 40-50%
- ☐ 30-40%
- ☐ Less than 30%

Q12.13.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Please answer all the questions based on the information given.

Q12.14. To keep inflation low and stable, Government sets an inflation target of 2%.

Based on the inflation forecast provided, what do you think is the probability of inflation **exceeding 2%**?

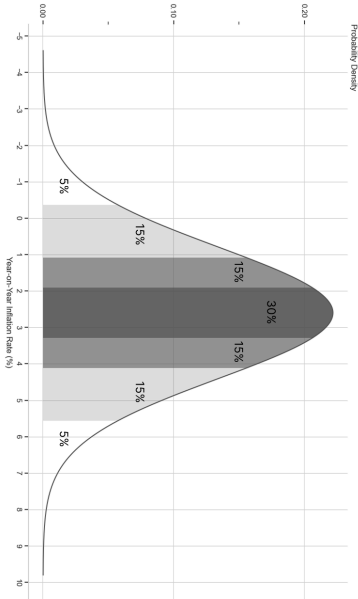
- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q12.15. Based on the inflation forecast provided, do you think there is a higher risk of inflation exceeding the central estimate or of being less than this central estimate?

- ☐ There is a bigger upside risk (that is, of inflation exceeding the central estimate).
- ☐ There is a bigger downside risk (that is, of inflation being less than the central estimate).
- ☐ The risks appear equal on both sides.

☐ Exceptionally unlikely
 ☐ Very unlikely
 ☐ Quite unlikely
 ☐ Fifty-fifty
 ☐ Quite likely
 ☐ Very likely
 ☐ Virtually certain

Q12.16.



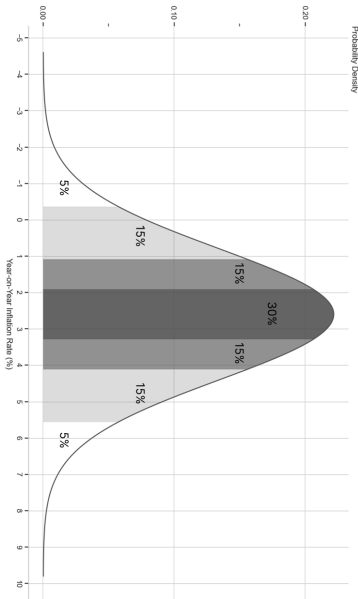
The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Please answer all the questions based on the information given.

Q12.17. How likely do you think it is that the inflation will be **exactly** 2.6%?

☐ Exceptionally unlikely
 ☐ Very unlikely
 ☐ Quite unlikely
 ☐ Fifty-fifty
 ☐ Quite likely
 ☐ Very likely
 ☐ Virtually certain

Q12.18.



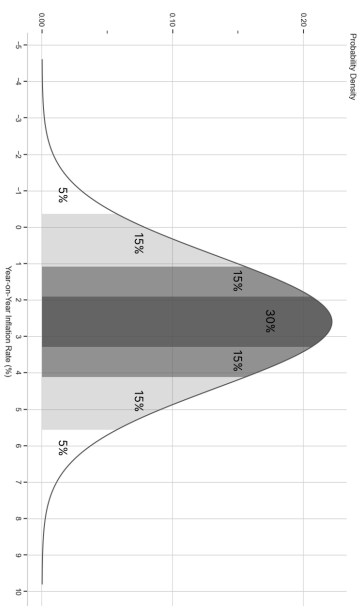
The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Please answer all the questions based on the information given.

Q12.19. What tells you that the inflation in 2026Q2 is likely or unlikely to be **exactly** 2.6%? (Please select the option that best reflects your belief, even if other answers may also seem reasonable.)

- ☐ It is being communicated in the text that this is the most likely value.
- ☐ It is being communicated in the text that the actual value may deviate.
- ☐ The probability density at 2.6% is moderate, suggesting it's somewhat likely but not certain.
- ☐ The probability density at 2.6% is low, so inflation is unlikely to be around this value.
- ☐ The peak of the curve is at 2.6%.
- ☐ The probability of inflation taking any exact value is 0.
- ☐ The value seems reasonable to me. (gut feeling)
- ☐ The value seems unreasonable to me. (gut feeling)
- ☐ Other (Please specify)

Q12.20.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Please answer all the questions based on the information given.

Q12.21. How likely do you think it is that the inflation will be between 1.9% and 3.3%?

- | | | | | | | |
|------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Exceptionally unlikely | Very unlikely | Quite unlikely | Fifty-fifty | Quite likely | Very likely | Virtually certain |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

Q12.22. How likely do you think it is that the inflation will be between 3.3% and 5.6%?

- | | | | | | | |
|------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Exceptionally unlikely | Very unlikely | Quite unlikely | Fifty-fifty | Quite likely | Very likely | Virtually certain |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

Q12.23. How likely do you think it is that the inflation will be higher than 3%?

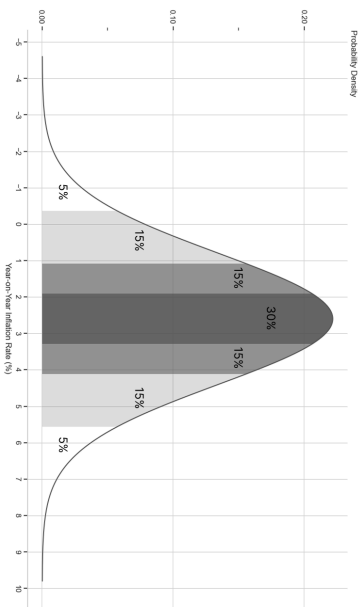
- | | | | | | | |
|------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Exceptionally unlikely | Very unlikely | Quite unlikely | Fifty-fifty | Quite likely | Very likely | Virtually certain |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

Q12.24. Based on the inflation forecast provided, please rank the following inflation rates from **most likely** to **least likely** to occur:
0.8%, 2.5%, 3.9% and 6.5%.

- 0.8%
- 2.5%
- 3.9%

6.5%

Q12.25.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Please answer all the questions based on the information given.

Q12.26. On a scale of 1 to 10, where 1 means not accurate at all and 10 means very accurate, how accurate do you think the central estimate in the graph is?

Not accurate at all

1 2 3 4 5 6 7 8 9 10

Very accurate

Q12.27. On a scale of 1 to 10, where 1 means very uncertain and 10 means very certain, how certain or uncertain do you think the inflation forecast for 2026 Q2 is?

Very uncertain

1 2 3 4 5 6 7 8 9 10

Very certain

Q12.28. Why do you think that the forecast is more or less uncertain? (Select all that apply)

- ☐ The forecast is for a time far in the future.
- ☐ The central estimate has a low probability.
- ☐ I'm not familiar with how to interpret this type of chart.
- ☐ I believe the economic conditions are highly unpredictable.
- ☐ I believe the economic conditions are highly predictable.
- ☐ The probability distribution is widely spread.
- ☐ The probability distribution is concentrated.
- ☐ The forecast is not as I expected.
- ☐ The forecast seems reasonable to me.
- ☐ Other (Please specify)

Q12.29. On a scale of 1 to 10, where 1 means not representative at all and 10 means very representative, how representative do you think the uncertainty depicted in the graph is (i.e., how well the given information reflects the true variability or uncertainty of the forecast?)

Not representative at all

Very representative

1 2 3 4 5 6 7 8 9 10

Q12.30. On a scale of 1 to 10, where 1 means very difficult and 10 means very easy, how easy or difficult do you find the graph to understand?

Very difficult

Very easy

1 2 3 4 5 6 7 8 9 10

Q12.31. On a scale of 1 to 10, where 1 means you don't trust the inflation forecast at all and 10 means you fully trust it, how much do you trust the inflation forecast shown in the graph?

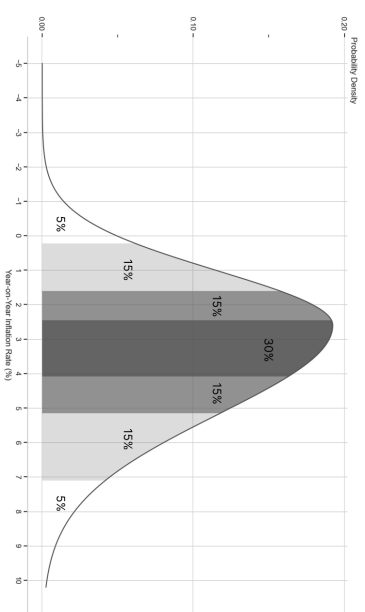
I don't trust at all.

I fully trust it.

1 2 3 4 5 6 7 8 9 10

Q12.32. The graph below shows the inflation forecast in **2026Q3**.

Please note that this forecast is for a different quarter in 2026.



The most likely value for inflation in 2026Q3 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Please answer all the questions based on the information given.

Q12.33. Based on the inflation forecast provided, what do you think is the probability of inflation ending up **higher than 3%**?

- ☐ More than 60%
- ☐ 50-60%
- ☐ 40-50%
- ☐ 30-40%
- ☐ Less than 30%

Q12.34. How likely do you think it is that the inflation will be **higher than 3%**?

Exceptionally unlikely	Very unlikely	Quite unlikely	Fifty-fifty	Quite likely	Very likely	Virtually certain
☐	☐	☐	☐	☐	☐	☐

Q12.35. Based on the inflation forecast provided, please rank the following inflation rates from **most likely** to **least likely** to occur:
1. 1%, 2. 6%, 3. 3% and 4. 1%.

- 1. 1%
- 2. 6%
- 3. 3%
- 4. 1%

Q12.36. Based on the inflation forecast provided, do you think there is a higher risk of inflation exceeding the central estimate or of being less than this central estimate?

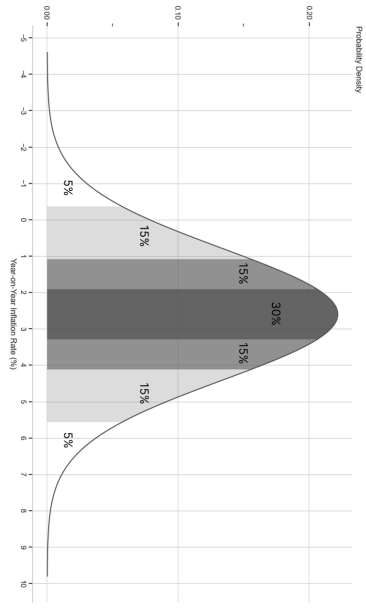
- ☐ There is a bigger upside risk (that is, of inflation exceeding the central estimate).
- ☐ There is a bigger downside risk (that is, of inflation being less than the central estimate).
- ☐ The risks appear equal on both sides.

Q12.37. If the actual inflation ends up outside the shaded area, which side is more likely, based on the uncertainty depicted in the graph?

- ☐ More likely to be on the upper side (where inflation is high).
- ☐ More likely to be on the lower side (where inflation is low).
- ☐ Roughly equal chances of being on either side.

Decision-Making using Y-on-Y Forecast (Bell Curve)

Q13.1. The graph below shows the inflation forecast in **2026Q2**. Note that the graph is created based on the latest data released by the Bank of England.



The most likely value for inflation in 2026Q2 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Q13.2. Assume you are an investor with £100,000 to invest in a 1-year bond. You will receive your full investment back in a single payment at the end of the 1 year.

Your goal is to maximize the real return on your investment that you will receive after 1 year.

Inflation has a direct impact on the return on investment you will receive. The following setup is simplified but it generally reflects the reality of how each type of bond responds to inflation.

There are two types of bonds available for you to choose between:

A. Nominal Bond: A nominal bond has a fixed rate of interest and the **real return** depends on inflation. If inflation stays low or drops ($\leq 3\%$), choosing fixed-rate nominal bonds will provide higher return. However, if inflation increases, your return will be eroded due to reduced purchasing power.

B. Inflation-Protected Bond: An inflation-protected bond provides stable but lower **real return** regardless of inflation. If inflation is high ($> 3\%$), choosing inflation-protected bonds will hedge against inflation by ensuring your returns are adjusted in line with inflation.

Now, you are asked to choose which bond you would like to invest your entire £100,000 in:

- A. Nominal Bond
- B. Inflation-Protected Bond

A payoff table is provided to help you understand the potential outcomes of your decision. It shows how much you could effectively gain or lose, depending on which bond you choose to invest in, and on the actual inflation rate in **2026Q2**.

	Real Return When Inflation is Low ($\leq 3\%$)	Real Return When Inflation is High ($> 3\%$)
A. Nominal Bond	+£3,000	-£1,000
B. Inflation-Protected Bond	+£1,000	+£1,000

Note that you will **not be** able to trade the bond during the next year.

Please make the decision based on the forecast information provided, but keep in mind that there is uncertainty.

Q13.3. Which bond would you choose to invest in?

- ☒ A. Nominal Bond
- ☐ B. Inflation-Protected Bond

Q13.4. On a scale of 1 to 10, where 1 means not confident at all and 10 means very confident, how confident are you in making the decision?

Not confident at all

☒ 2 3 4 5 6 7 8 9 10

Very confident

Q13.5. Could you explain in a few sentences how and why you decided to invest in this bond?

Q13.6. If the probability of inflation being low ($\leq 3\%$) is **greater than** the following percentages, I would consider investing in the **nominal bond**:

- ☐ 40%
- ☐ 50%
- ☐ 60%
- ☐ 70%
- ☐ 80%
- ☐ I would consider changing my decision if the inflation is perceived low, but I do not think about probability.
- ☐ I will always choose an inflation-protected bond.

Q13.7. If the probability of inflation being high ($> 3\%$) is **greater than** the following percentages, I would consider investing in the **inflation-protected bond**:

- ☐ 40%
- ☐ 50%
- ☐ 60%
- ☐ 70%
- ☐ 80%
- ☐ I would consider changing my decision if the inflation is perceived high, but I do not think about probability.
- ☐ I will always choose a nominal bond.

Q13.8. Considering the payoff for the **Nominal Bond**, which of the following has a larger impact on your decision-making?

- ☐ The potential loss if inflation turns out to be high ($> 3\%$).
- ☐ The potential gain if inflation stays low ($\leq 3\%$).
- ☐ The potential loss and the potential gain have the same impact.

Q13.9. On a scale of 1 to 10, where 1 means the forecast was not effective at all and 10 means it was very effective, how effective do you think the forecast was in assisting your decision-making?

Not effective at all

1

2

3

4

5

6

7

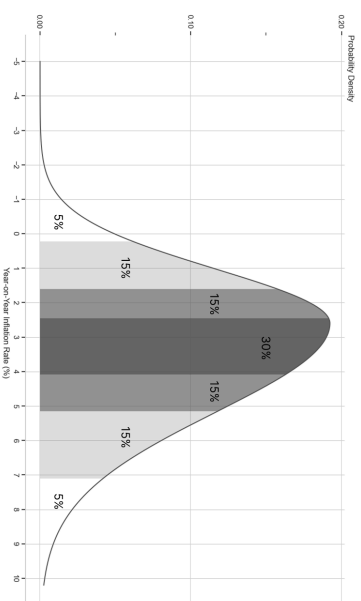
8

9

10

Very effective

Q13.10. **The following setup remains the same;** however, we assume that your real return will depend on the Year-on-Year inflation rate in **2026Q3**, with the forecast provided below.



The most likely value for inflation in 2026Q3 is of 2.6% (central estimate), though the actual inflation value may deviate from this forecast. The inflation rate is most likely to be in the dark gray area (3 out of 10 times), and within each pair of lighter gray areas on a further 3 out of 10 occasions. The total shaded area covers inflation for 9 out of 10 possible outcomes, and it is very unlikely (1 out of 10 times) that inflation will fall outside of that area.

Q13.11. Assume you are an investor with £100,000 to invest in a **1-year bond**. You will receive your full investment back in a single payment at the end of the 1 year.

Your goal is to maximize the real return on your investment that you will receive after 1 year.

Inflation has a direct impact on the return on investment you will receive. The following setup is simplified but it generally reflects the reality of how each type of bond responds to inflation.

There are two types of bonds available for you to choose between:

- A. Nominal Bond:** A nominal bond has a fixed rate of interest and the **real return** depends on inflation. If inflation stays low or drops ($\leq 3\%$), choosing fixed-rate nominal bonds will provide higher return. However, if inflation increases, your return will be eroded due to reduced purchasing power.
- B. Inflation-Protected Bond:** An inflation-protected bond provides stable but lower **real return** regardless of inflation. If inflation is high ($> 3\%$), choosing inflation-protected bonds will hedge against inflation by ensuring your returns are adjusted in line with inflation.

Now, you are asked to choose which bond you would like to invest your entire £100,000 in:

- A. Nominal Bond
- B. Inflation-Protected Bond

A payoff table is provided to help you understand the potential outcomes of your decision. It shows how much you could effectively gain or lose, depending on which bond you choose to invest in, and on the actual inflation rate in **2026Q3**.

	Real Return When Inflation is Low ($\leq 3\%$)	Real Return When Inflation is High ($> 3\%$)
A. Nominal Bond	+£3,000	-£1,000
B. Inflation-Protected Bond	+£1,000	+£1,000

Note that you will **not be** able to trade the bond during the next year.

Please make the decision based on the forecast information provided, but keep in mind that there is uncertainty.

Q13.12. Which bond would you choose to invest in?

- ☐ A. Nominal Bond
- ☐ B. Inflation-Protected Bond

Q13.13. On a scale of 1 to 10, where 1 means not confident at all and 10 means very confident, how confident are you in making the decision?

Not confident at all

2345678910

Very confident

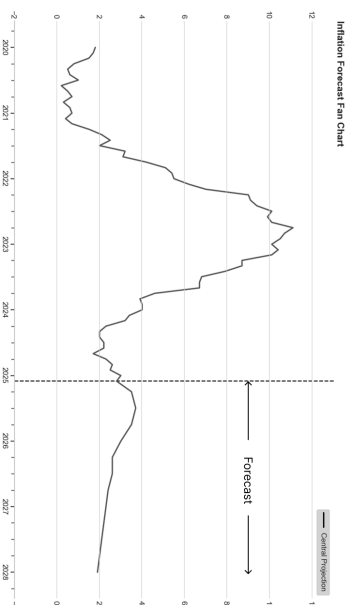
Q13.14. Could you explain in a few sentences how and why you decided to invest in this bond?

Appendix B.2. Time-series Inflation Forecast Survey

IMPERIAL

Understanding of Time-series Forecast (Deterministic)

Q8.1. The graph below shows the inflation forecast **from 2025Q2 to 2028Q1**.



The black line in the graph represents the most likely value for CPI inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast.

Please answer all the questions based on the information given.

Q8.2. Have you ever seen this type of graph before?

- ☐ Definitely not
- ☐ Probably not
- ☐ Probably yes
- ☐ Definitely yes

Q8.3. Based on the forecast, how do you think the inflation will change over the next 3 years?

- ☐ Go up
- ☐ Go down
- ☐ Increase then decrease
- ☐ Decrease then increase
- ☐ Other (please specify)

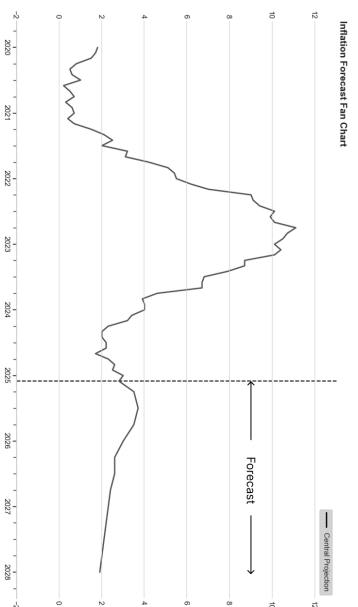
Q8.4. Does the inflation forecast information provided align with your expectations?

- ☐ Strongly aligns – The inflation forecast matches exactly what I expected.
- ☐ Somewhat aligns – The inflation forecast is generally in line with my expectations.
- ☐ Neutral – The inflation forecast neither aligns nor conflicts with my expectations.
- ☐ Somewhat does not align – The inflation forecast is somewhat different from what I expected.
- ☐ Does not align at all – The inflation forecast is significantly different from what I expected.

Q8.5. How does the inflation forecast conflict with your expectations?

- ☐ The inflation forecast is higher than what I expected.
- ☐ The inflation forecast is lower than what I expected.

Q8.6.



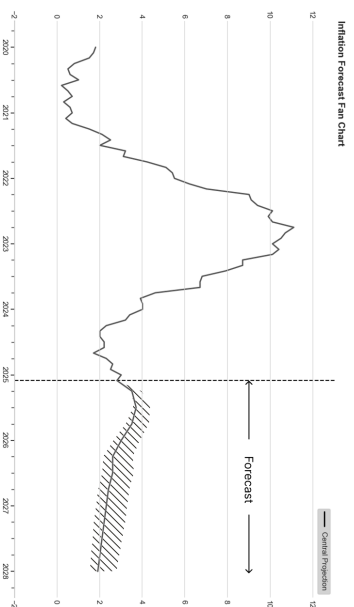
The black line in the graph represents the most likely value for CPI inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast.

Please answer all the questions based on the information given.

Q8.7. Based on the inflation forecast provided, what do you think is the probability that the inflation will follow the **central projection**?

- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 80%) About a 8 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 20%) About a 2 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

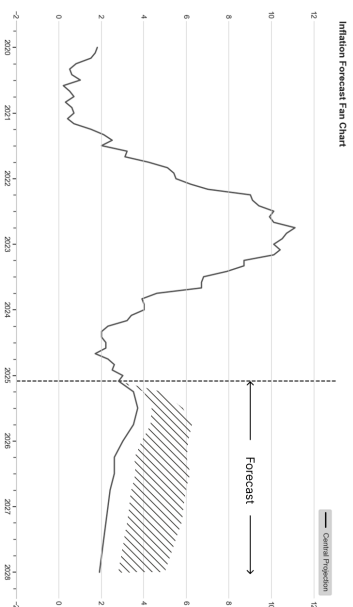
Q8.8. Based on the inflation forecast provided, what do you think is the probability that the inflation will land in the area **with stripe pattern**?



- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q8.9. Based on the inflation forecast provided, what do you think is the probability that the inflation will land in the area **with stripe**

pattern?

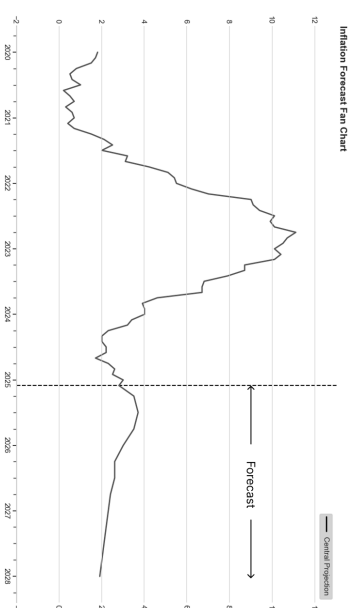


- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q8.10. Based on the inflation forecast provided, do you think there is a higher risk of inflation exceeding the central projection or of being less than this central projection?

- ☐ There is a bigger upside risk (that is, of inflation exceeding the central projection).
- ☐ There is a bigger downside risk (that is, of inflation being less than the central projection).
- ☐ The risks appear equal on both sides.

Q8.11.



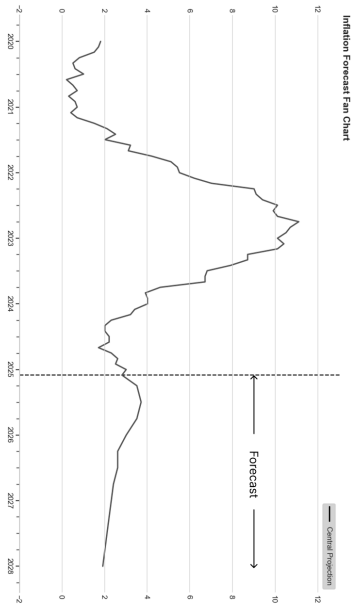
The black line in the graph represents the most likely value for CPI inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast.

Please answer all the questions based on the information given.

Q8.12. How likely do you think it is that the inflation will follow the central projection?

- ☐ Exceptionally unlikely
- ☐ Very unlikely
- ☐ Quite unlikely
- ☐ Fifty-fifty
- ☐ Quite likely
- ☐ Very likely
- ☐ Virtually certain

Q8.13.



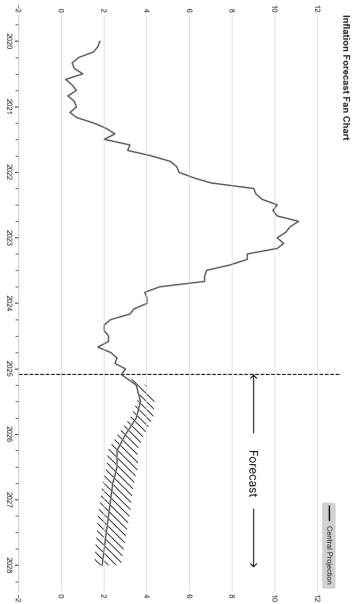
The black line in the graph represents the most likely value for CPI inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast.

Please answer all the questions based on the information given.

Q8.14. What tells you that the inflation from 2025Q2 to 2028Q1 is likely or unlikely to follow the central projection? (Please select the option that best reflects your belief, even if other answers may also seem reasonable.)

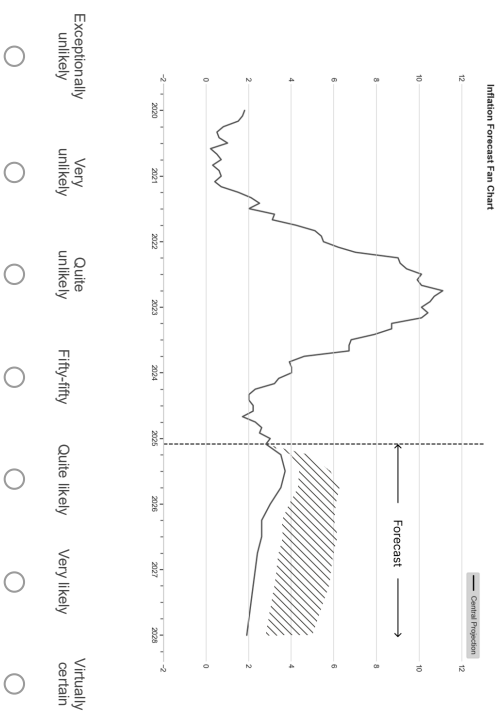
- ☐ It is being communicated in the text that these are the most likely values.
- ☐ It is being communicated in the text that the actual values may deviate.
- ☐ There is not enough information, suggesting the central projection is not certain.
- ☐ This is the only information given, suggesting the central projection is quite certain.
- ☐ The probability of inflation following any exact path is 0.
- ☐ The value seems reasonable to me. (gut feeling)
- ☐ The value seems unreasonable to me. (gut feeling)
- ☐ Other (Please specify)

Q8.15. How likely do you think it is that the inflation will land in the area with stripe pattern?

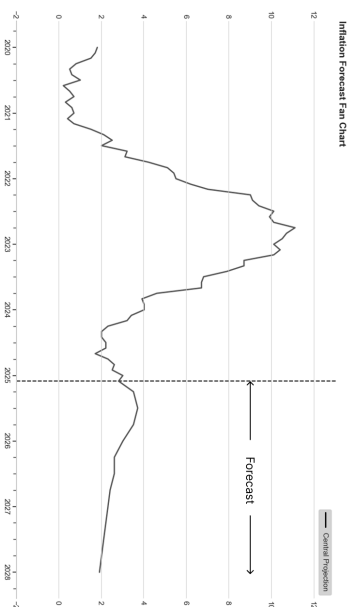


- | | | | | | | |
|------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Exceptionally unlikely | Very unlikely | Quite unlikely | Fifty-fifty | Quite likely | Very likely | Virtually certain |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

Q8.16. How likely do you think it is that the inflation will land in the area with stripe pattern?



Q8.17.



The black line in the graph represents the most likely value for CPI inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast.

Please answer all the questions based on the information given.

Q8.18. To keep inflation low and stable, Government sets an inflation target of 2%. How likely do you think it is that the inflation will consistently **exceed** 2% target?

Exceptionally unlikely Very unlikely Quite unlikely Fifty-fifty Quite likely Very likely Virtually certain

☐ ☐ ☐ ☐ ☐ ☐ ☐

Q8.19. Based on the inflation forecast provided, please rank the following time periods from **the least uncertain to the most uncertain** regarding inflation: 2025, 2026, 2027, 2028

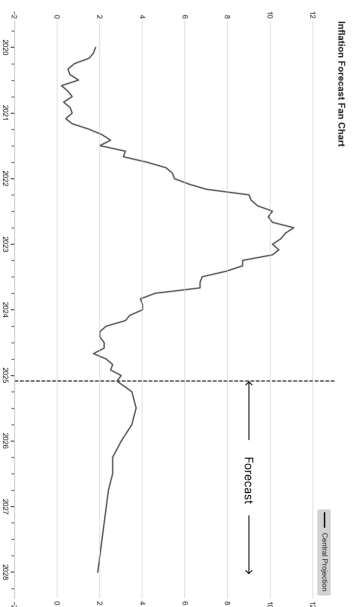
2025

2026

2027

2028

Q8.20.



The black line in the graph represents the most likely value for CPI inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast.

Please answer all the questions based on the information given.

Q8.21. What indicates that the forecasts at different time periods (2025, 2026, 2027, 2028) are more or less uncertain? (Please select the option that best reflects your belief, even if other answers may also seem reasonable.)

- ☐ It is more uncertain when the time is getting further away in the future.
- ☐ It is more uncertain when the inflation rate is high.
- ☐ It is more uncertain when the inflation rate is low.
- ☐ It is more uncertain when the central projection has an erratic value.
- ☐ Other (Please specify)

Q8.22. On a scale of 1 to 10, where 1 means not accurate at all and 10 means very accurate, how accurate do you think the central projection in the graph is?

Not accurate at all 1 2 3 4 5 6 7 8 9 10 Very accurate

Q8.23. On a scale of 1 to 10, where 1 means very uncertain and 10 means very certain, how certain or uncertain do you think the inflation forecast is?

Very uncertain 1 2 3 4 5 6 7 8 9 10 Very certain

Q8.24. Why do you think that the forecast is more or less uncertain? (Select all that apply)

- ☐ The forecast is for a time far in the future.
- ☐ The forecast fluctuates too much.
- ☐ The forecast is stable.
- ☐ I'm not familiar with how to interpret this type of chart.
- ☐ I believe the economic conditions are highly unpredictable.
- ☐ I believe the economic conditions are highly predictable.
- ☐ The forecast is not as I expected.
- ☐ The forecast seems reasonable to me.
- ☐ Other (Please specify)

Q8.25. On a scale of 1 to 10, where 1 means not representative at all and 10 means very representative, how representative do you

think the uncertainty depicted in the graph is (i.e., how well the given information reflects the true variability or uncertainty of the forecast?)

Not representative at all

Very representative

1 2 3 4 5 6 7 8 9 10

Q8.26. On a scale of 1 to 10, where 1 means very difficult and 10 means very easy, how easy or difficult do you find the graph to understand?

Very difficult

Very easy

1 2 3 4 5 6 7 8 9 10

Q8.27. On a scale of 1 to 10, where 1 means you don't trust the inflation forecast at all and 10 means you fully trust it, how much do you trust the inflation forecast shown in the graph?

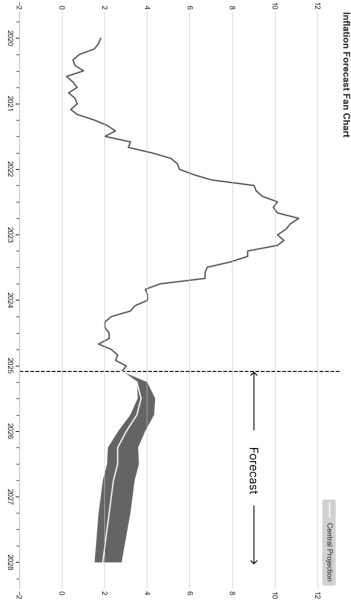
I don't trust at all.

I fully trust it.

1 2 3 4 5 6 7 8 9 10

Understanding of Time-series Forecast (Fan Chart-30)

Q9.1. The graph below shows the inflation forecast from 2025Q2 to 2028Q1.



The fan chart depicts the probability of various future outcomes for CPI inflation. The white line represents the most likely value for inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie within the darkest central band on only 30 of those occasions. And on the remaining 70 out of 100 occasions inflation can fall anywhere outside the grey area of the fan chart.

Please answer all the questions based on the information given.

Q9.2. Have you ever seen this type of graph before?

- ☐ Definitely not
- ☐ Probably not
- ☐ Probably yes
- ☐ Definitely yes

Q9.3. Based on the forecast, how do you think the inflation will change over the next 3 years?

- ☐ Go up
☐ Go down
☐ Increase then decrease
☐ Decrease then increase
☐ Other (please specify)

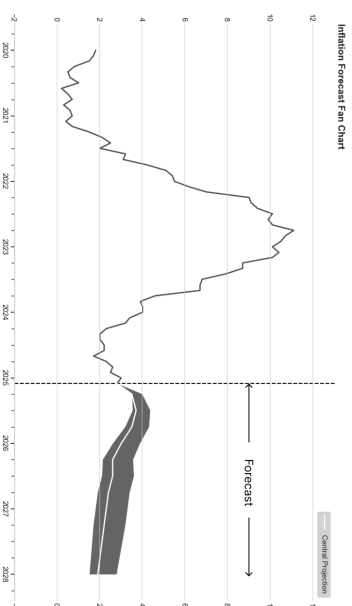
Q9.4. Does the inflation forecast information provided align with your expectations?

- ☐ Strongly aligns – The inflation forecast matches exactly what I expected.
☐ Somewhat aligns – The inflation forecast is generally in line with my expectations.
☐ Neutral – The inflation forecast neither aligns nor conflicts with my expectations.
☐ Somewhat does not align – The inflation forecast is somewhat different from what I expected.
☐ Does not align at all – The inflation forecast is significantly different from what I expected.

Q9.5. How does the inflation forecast conflict with your expectations?

- ☐ The inflation forecast is higher than what I expected.
☐ The inflation forecast is lower than what I expected.

Q9.6.



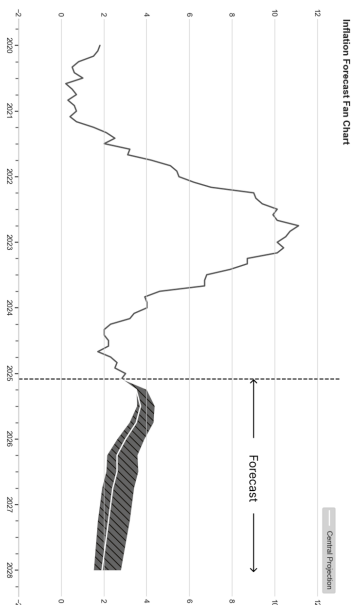
The fan chart depicts the probability of various future outcomes for CPI inflation. The white line represents the most likely value for inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie within the darkest central band on only 30 of those occasions. And on the remaining 70 out of 100 occasions inflation can fall anywhere outside the grey area of the fan chart.

Please answer all the questions based on the information given.

Q9.7. Based on the inflation forecast provided, what do you think is the probability that the inflation will follow the central projection?

- ☐ (>99%) Greater than 99 in 100 chance
☐ (around 80%) About a 8 in 10 chance
☐ (around 60%) About a 6 in 10 chance
☐ (around 50%) About a 1 in 2 chance
☐ (around 30%) About a 3 in 10 chance
☐ (around 20%) About a 2 in 10 chance
☐ (<1%) Less than 1 in 100 chance

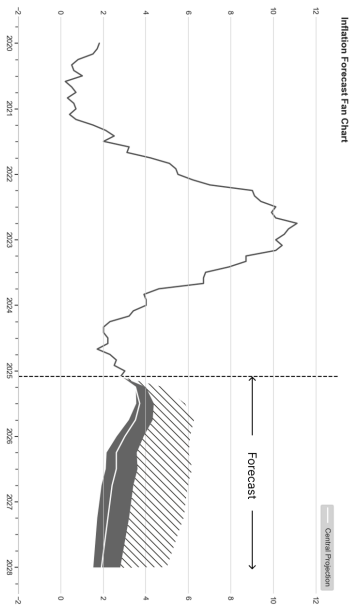
Q9.8. Based on the inflation forecast provided, what do you think is the probability that the inflation will land in the area **with stripe pattern**?



- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

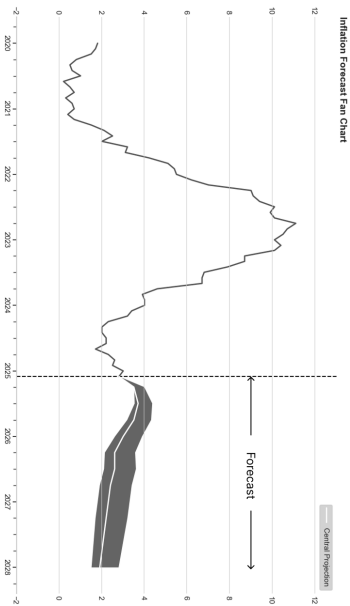
Q9.9. Based on the inflation forecast provided, what do you think is the probability that the inflation will land in the area **with stripe**

pattern?



- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q9.10.



The fan chart depicts the probability of various future outcomes for CPI inflation. The white line represents the most likely value for inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie within the darkest central band on only 30 of those occasions. And on the remaining 70 out of 100 occasions inflation can fall anywhere outside the grey area of the fan chart.

Please answer all the questions based on the information given.

Q9.11. Based on the inflation forecast provided, what do you think is the probability that inflation will fall **outside** of the shaded grey area?

- ☐ 70% (35% each side)
- ☐ 40% (20% each side)
- ☐ 20% (10% each side)
- ☐ 10% (5% each side)
- ☐ 5% (2.5% each side)
- ☐ Less than 1%

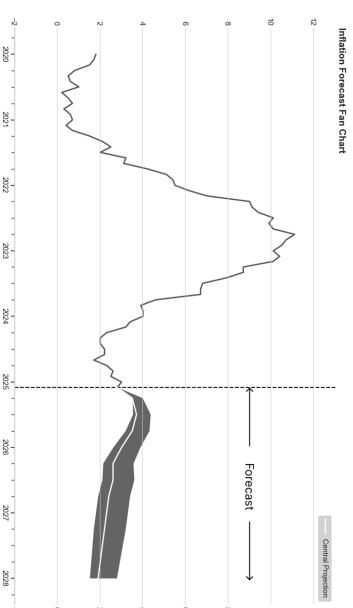
Q9.12. Based on the inflation forecast provided, do you think there is a higher risk of inflation exceeding the central projection or of being less than this central projection?

- ☐ There is a bigger upside risk (that is, of inflation exceeding the central projection).
- ☐ There is a bigger downside risk (that is, of inflation being less than the central projection).
- ☐ The risks appear equal on both sides.

Q9.13. If the actual inflation falls outside the shaded grey area, which side is more likely, based on the uncertainty depicted in the graph?

- ☐ More likely to be on the upper side (where inflation is high).
- ☐ More likely to be on the lower side (where inflation is low).
- ☐ Roughly equal chances of being on either side.

Q9.14.



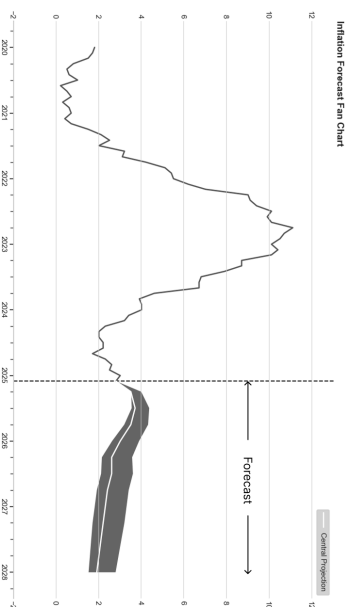
The fan chart depicts the probability of various future outcomes for CPI inflation. The white line represents the most likely value for inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie within the darkest central band on only 30 of those occasions. And on the remaining 70 out of 100 occasions inflation can fall anywhere outside the grey area of the fan chart.

Please answer all the questions based on the information given.

Q9.15. How likely do you think it is that the inflation will follow the **central projection**?

- Exceptionally unlikely Very unlikely Quite unlikely Fifty-fifty Quite likely Very likely Virtually certain
- ☐ ☐ ☐ ☐ ☐ ☐ ☐

Q9.16.



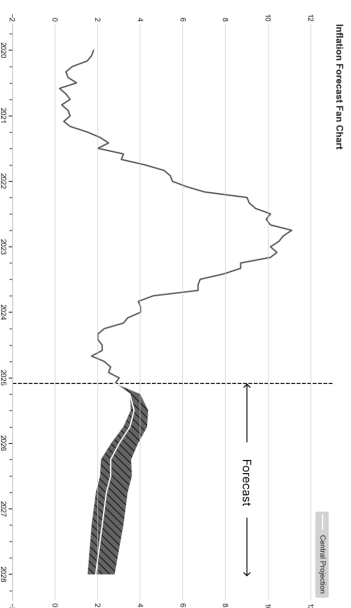
The fan chart depicts the probability of various future outcomes for CPI inflation. The white line represents the most likely value for inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie within the darkest central band on only 30 of those occasions. And on the remaining 70 out of 100 occasions inflation can fall anywhere outside the grey area of the fan chart.

Please answer all the questions based on the information given.

Q9.17. What tells you that the inflation from 2025Q2 to 2028Q1 is likely or unlikely to follow the central projection? (Please select the option that best reflects your belief, even if other answers may also seem reasonable.)

- ☐ It is being communicated in the text that these are the most likely values.
- ☐ It is being communicated in the text that the actual values may deviate.
- ☐ The fan chart is concentrated around the central projection.
- ☐ The fan chart is widely spread around the central projection.
- ☐ The central projection is not positioned in the center of the most likely band.
- ☐ The probability of inflation following any exact path is 0.
- ☐ The value seems reasonable to me. (gut feeling)
- ☐ The value seems unreasonable to me. (gut feeling)
- ☐ Other (Please specify)

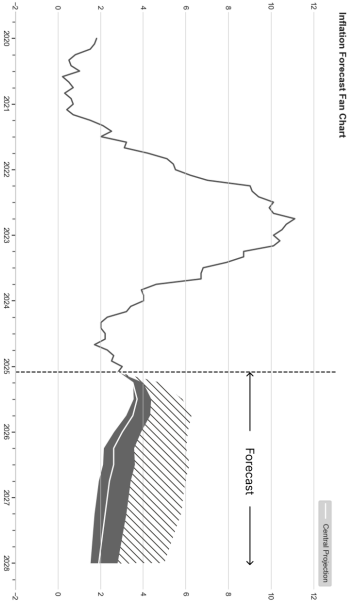
Q9.18. How likely do you think it is that the inflation will land in the area with **stripe pattern**?



- Exceptionally unlikely Very unlikely Quite unlikely Fifty-fifty Quite likely Very likely Virtually certain
- ☐ ☐ ☐ ☐ ☐ ☐ ☐

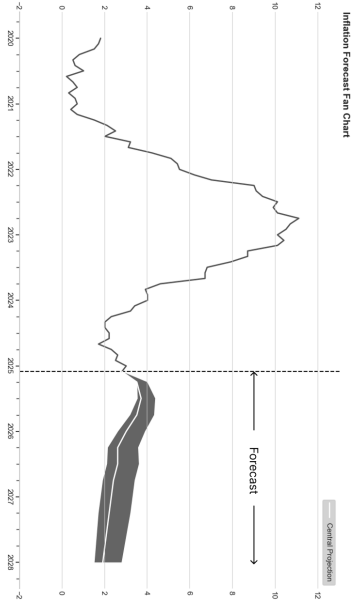
Exceptionally unlikely ☐ Very unlikely ☐ Quite unlikely ☐ Fifty-fifty ☐ Quite likely ☐ Very likely ☐ Virtually certain ☐

Q9.19. How likely do you think it is that the inflation will land in the area with stripe pattern?



Exceptionally unlikely ☐ Very unlikely ☐ Quite unlikely ☐ Fifty-fifty ☐ Quite likely ☐ Very likely ☐ Virtually certain ☐

Q9.20.



The fan chart depicts the probability of various future outcomes for CPI inflation. The white line represents the most likely value for inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie within the darkest central band on only 30 of those occasions. And on the remaining 70 out of 100 occasions inflation can fall anywhere outside the grey area of the fan chart.

Please answer all the questions based on the information given.

Q9.21. To keep inflation low and stable, Government sets an inflation target of 2%. How likely do you think it is that the inflation will consistently **exceed** 2% target?

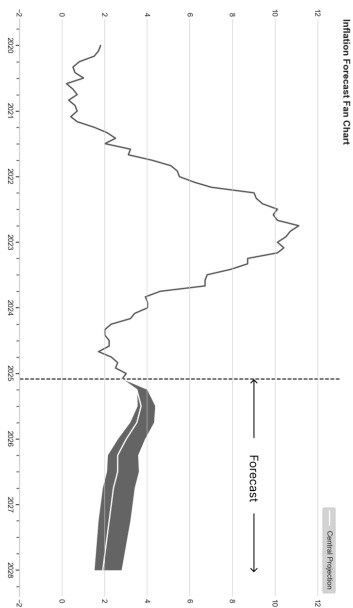
Exceptionally unlikely ☐ Very unlikely ☐ Quite unlikely ☐ Fifty-fifty ☐ Quite likely ☐ Very likely ☐ Virtually certain ☐

- Exceptionally unlikely ☐
- Very unlikely ☐
- Quite unlikely ☐
- Fifty-fifty ☐
- Quite likely ☐
- Very likely ☐
- Virtually certain ☐

Q9.22. Based on the inflation forecast provided, please rank the following time periods from **the least uncertain to the most uncertain** regarding inflation: 2025, 2026, 2027, 2028

- 2025
- 2026
- 2027
- 2028

Q9.23.



The fan chart depicts the probability of various future outcomes for CPI inflation. The white line represents the most likely value for inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie within the darkest central band on only 30 of those occasions. And on the remaining 70 out of 100 occasions inflation can fall anywhere outside the grey area of the fan chart.

Please answer all the questions based on the information given.

Q9.24. What indicates that the forecasts at different time periods (2025, 2026, 2027, 2028) are more or less uncertain? (Please select the option that best reflects your belief, even if other answers may also seem reasonable.)

- ☐ It is more uncertain when the time is getting further away in the future.
- ☐ It is more uncertain when the band is getting wider.
- ☐ It is more uncertain when the inflation rate is high.
- ☐ It is more uncertain when the inflation rate is low.
- ☐ It is more uncertain when the central projection has an erratic value.
- ☐ Other (Please specify)

Q9.25. On a scale of 1 to 10, where 1 means not accurate at all and 10 means very accurate, how accurate do you think the central projection in the graph is?

Not accurate at all

☒

2

3

4

5

6

7

8

9

10

Very accurate

Q9.26. On a scale of 1 to 10, where 1 means very uncertain and 10 means very certain, how certain or uncertain do you think the inflation forecast is?

Very uncertain

102345678910

Very certain

Q9.27. Why do you think that the forecast is more or less uncertain? (Select all that apply)

- ☐ The forecast is for a time far in the future.
- ☐ The forecast fluctuates too much.
- ☐ The forecast is stable.
- ☐ I'm not familiar with how to interpret this type of chart.
- ☐ I believe the economic conditions are highly unpredictable.
- ☐ I believe the economic conditions are highly predictable.
- ☐ The bands are widely spread.
- ☐ The bands are concentrated.
- ☐ The forecast is not as I expected.
- ☐ The forecast seems reasonable to me.
- ☐ Other (Please specify)

Q9.28. On a scale of 1 to 10, where 1 means not representative at all and 10 means very representative, how representative do you think the uncertainty depicted in the graph is (i.e., how well the given information reflects the true variability or uncertainty of the forecast?)

Not representative at all

102345678910

Very representative

Q9.29. On a scale of 1 to 10, where 1 means very difficult and 10 means very easy, how easy or difficult do you find the graph to understand?

Very difficult

102345678910

Very easy

Q9.30. On a scale of 1 to 10, where 1 means you don't trust the inflation forecast at all and 10 means you fully trust it, how much do you trust the inflation forecast shown in the graph?

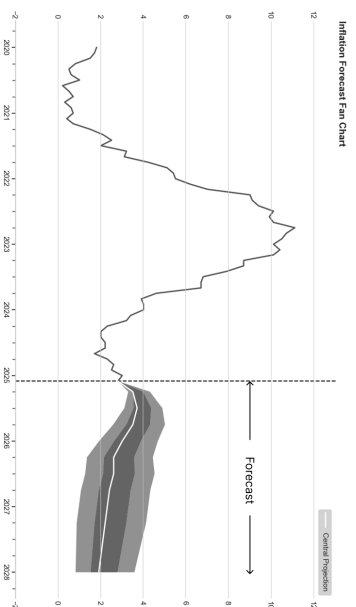
I don't trust at all.

102345678910

I fully trust it.

Understanding of Time-series Forecast (Fan Chart-60)

Q10.1. The graph below shows the inflation forecast from 2025Q2 to 2028Q1.



The fan chart depicts the probability of various future outcomes for CPI inflation. The white line represents the most likely value for inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie within the darkest central band on only 30 of those occasions. The fan chart is constructed so that outturns of inflation are also expected to lie within each pair of the lighter grey areas on 30 occasions. In any particular quarter of the forecast period, inflation is therefore expected to lie somewhere within the fans on 60 out of 100 occasions. And on the remaining 40 out of 100 occasions inflation can fall anywhere outside the grey area of the fan chart.

Please answer all the questions based on the information given.

Q10.2. Have you ever seen this type of graph before?

- ☐ Definitely not
- ☐ Probably not
- ☐ Probably yes
- ☐ Definitely yes

Q10.3. Based on the forecast, how do you think the inflation will change over the next 3 years?

- ☐ Go up
- ☐ Go down
- ☐ Increase then decrease
- ☐ Decrease then increase
- ☐ Other (please specify)

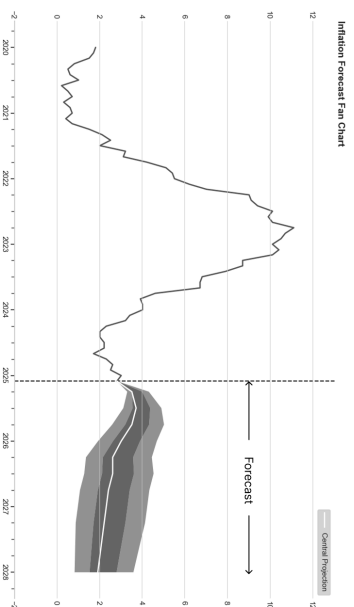
Q10.4. Does the inflation forecast information provided align with your expectations?

- ☐ Strongly aligns – The inflation forecast matches exactly what I expected.
- ☐ Somewhat aligns – The inflation forecast is generally in line with my expectations.
- ☐ Neutral – The inflation forecast neither aligns nor conflicts with my expectations.
- ☐ Somewhat does not align – The inflation forecast is somewhat different from what I expected.
- ☐ Does not align at all – The inflation forecast is significantly different from what I expected.

Q10.5. How does the inflation forecast conflict with your expectations?

- ☐ The inflation forecast is higher than what I expected.
- ☐ The inflation forecast is lower than what I expected.

Q10.6.

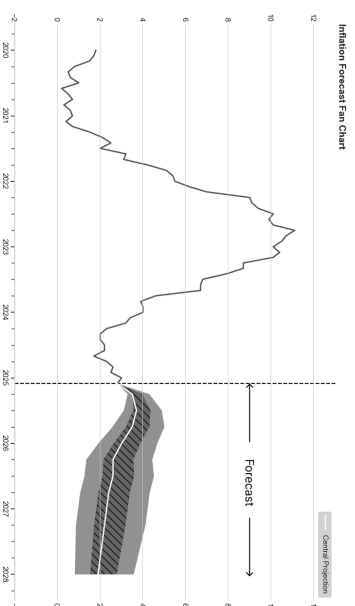


The fan chart depicts the probability of various future outcomes for CPI inflation. The white line represents the most likely value for inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie within the darkest central band on only 30 of those occasions. The fan chart is constructed so that outturns of inflation are also expected to lie within each pair of the lighter grey areas on 30 occasions. In any particular quarter of the forecast period, inflation is therefore expected to lie somewhere within the fans on 60 out of 100 occasions. And on the remaining 40 out of 100 occasions inflation can fall anywhere outside the grey area of the fan chart.

Please answer all the questions based on the information given.

- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 20%) About a 2 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q10.8. Based on the inflation forecast provided, what do you think is the probability that the inflation will land in the area **with stripe pattern**?



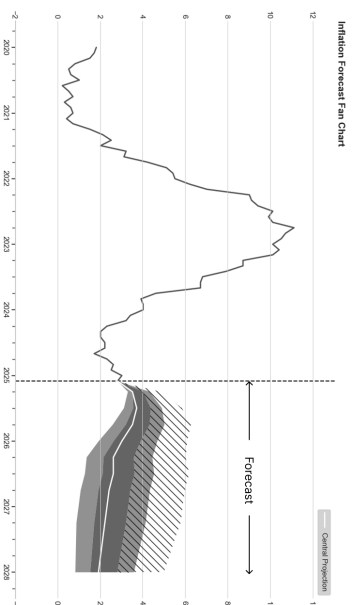
- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q10.7. Based on the inflation forecast provided, what do you think is the probability that the inflation will follow **the central projection**?

- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 80%) About a 8 in 10 chance

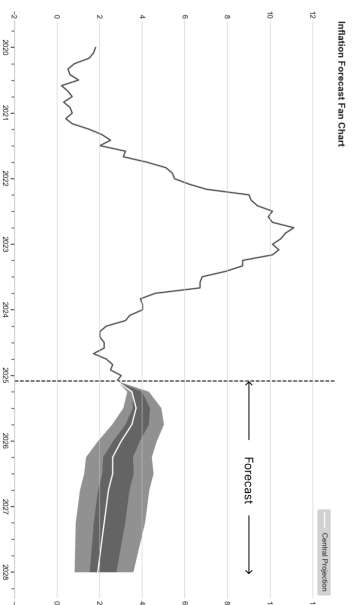
Q10.9. Based on the inflation forecast provided, what do you think is the probability that the inflation will land in the area **with stripe**

pattern?



- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q10.10.



The fan chart depicts the probability of various future outcomes for CPI inflation. The white line represents the most likely value for inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie within the darkest central band on only 30 of those occasions. The fan chart is constructed so that outturns of inflation are also expected to lie within each pair of the lighter grey areas on 30 occasions. In any particular quarter of the forecast period, inflation is therefore expected to lie somewhere within the fans on 60 out of 100 occasions. And on the remaining 40 out of 100 occasions inflation can fall anywhere outside the grey area of the fan chart.

Please answer all the questions based on the information given.

Q10.11. Based on the inflation forecast provided, what do you think is the probability that inflation will fall **outside** of the shaded grey area?

- ☐ 70% (35% each side)
- ☐ 40% (20% each side)
- ☐ 20% (10% each side)
- ☐ 10% (5% each side)
- ☐ 5% (2.5% each side)
- ☐ Less than 1%

Q10.12. Based on the inflation forecast provided, do you think there is a higher risk of inflation exceeding the central projection or of being less than this central projection?

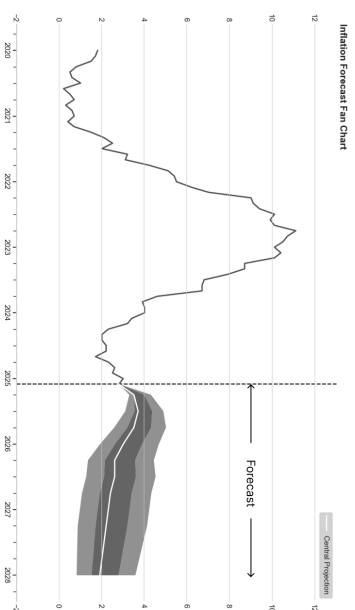
- ☐ There is a bigger upside risk (that is, of inflation exceeding the central projection).

- ☐ There is a bigger downside risk (that is, of inflation being less than the central projection).
- ☐ The risks appear equal on both sides.

Q10.13. If the actual inflation falls outside the shaded grey area, which side is more likely, based on the uncertainty depicted in the graph?

- ☐ More likely to be on the upper side (where inflation is high).
- ☐ More likely to be on the lower side (where inflation is low).
- ☐ Roughly equal chances of being on either side.

Q10.14.



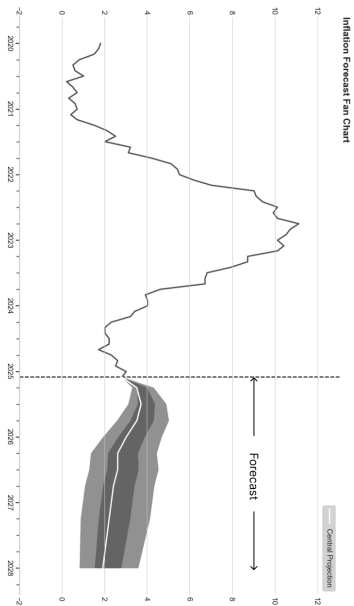
The fan chart depicts the probability of various future outcomes for CPI inflation. The white line represents the most likely values for inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie within the darkest central band on only 30 of those occasions. The fan chart is constructed so that outturns of inflation are also expected to lie within each pair of the lighter grey areas on 30 occasions. In any particular quarter of the forecast period, inflation is therefore expected to lie somewhere within the fans on 60 out of 100 occasions. And on the remaining 40 out of 100 occasions inflation can fall anywhere outside the grey area of the fan chart.

Please answer all the questions based on the information given.

Q10.15. How likely do you think it is that the inflation will follow the central projection?

- | | | | | | | | | | | | | | |
|-----------------------|------------------------|-----------------------|---------------|-----------------------|----------------|-----------------------|-------------|-----------------------|--------------|-----------------------|-------------|-----------------------|-------------------|
| ☐ | Exceptionally unlikely | ☐ | Very unlikely | ☐ | Quite unlikely | ☐ | Fifty-fifty | ☐ | Quite likely | ☐ | Very likely | ☐ | Virtually certain |
|-----------------------|------------------------|-----------------------|---------------|-----------------------|----------------|-----------------------|-------------|-----------------------|--------------|-----------------------|-------------|-----------------------|-------------------|

Q10.16.



The fan chart depicts the probability of various future outcomes for CPI inflation. The white line represents the most likely values for inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie within the darkest central band on only 30 of those occasions. The fan chart is constructed so that outturns of inflation are also expected to lie within each pair of the lighter grey areas on 30 occasions. In any particular quarter of the forecast period, inflation is therefore expected to lie somewhere within the fans on 60 out of 100 occasions. And on the remaining 40 out of 100 occasions inflation can fall anywhere outside the grey area of the fan chart.

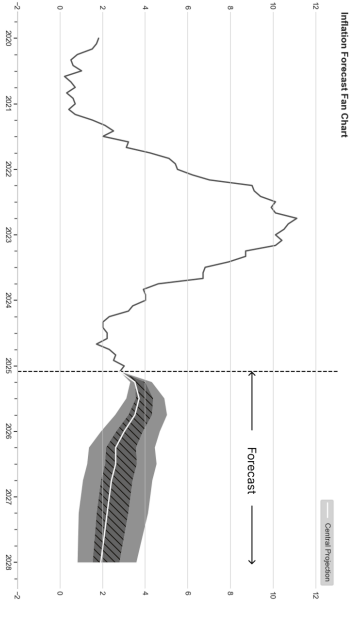
Please answer all the questions based on the information given.

Q10.17. What tells you that the inflation from 2025Q2 to 2028Q1 is likely or unlikely to follow the central projection? (Please select the

option that best reflects your belief, even if other answers may also seem reasonable.)

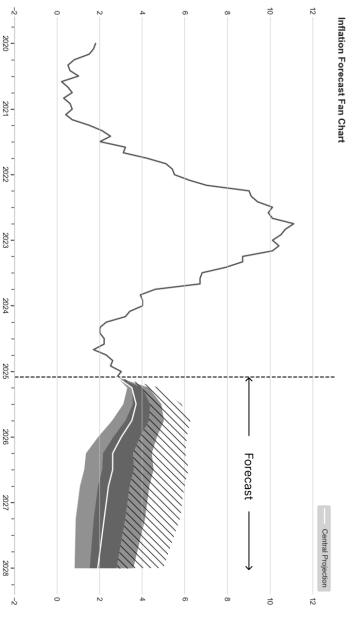
- ☐ It is being communicated in the text that these are the most likely values.
- ☐ It is being communicated in the text that the actual values may deviate.
- ☐ The fan chart is concentrated around the central projection.
- ☐ The fan chart is widely spread around the central projection.
- ☐ The central projection is not positioned in the center of the most likely band.
- ☐ The probability of inflation following any exact path is 0.
- ☐ The value seems reasonable to me. (gut feeling)
- ☐ The value seems unreasonable to me. (gut feeling)
- ☐ Other (Please specify)

Q10.18. How likely do you think it is that the inflation will land in the area with stripe pattern?



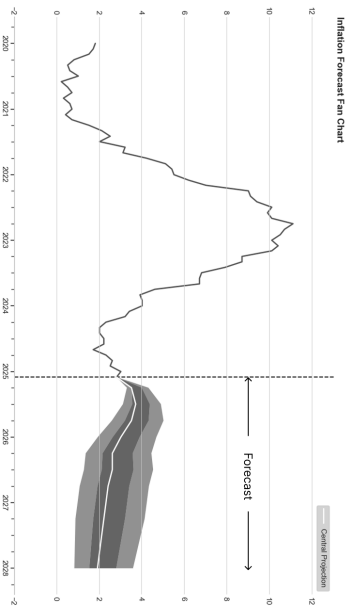
- | | | | | | | |
|------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Exceptionally unlikely | Very unlikely | Quite unlikely | Fifty-fifty | Quite likely | Very likely | Virtually certain |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

Q10.19. How likely do you think it is that the inflation will land in the area with stripe pattern?



- Exceptionally unlikely
- Very unlikely
- Quite unlikely
- Fifty-fifty
- Quite likely
- Very likely
- Virtually certain

Q10.20.



The fan chart depicts the probability of various future outcomes for CPI inflation. The white line represents the most likely values for inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie within the darkest central band on only 30 of those occasions. The fan chart is constructed so that outturns of inflation are also expected to lie within each pair of the lighter grey areas on 30 occasions. In any particular quarter of the forecast period, inflation is therefore expected to lie somewhere within the fans on 60 out of 100 occasions. And on the remaining 40 out of 100 occasions inflation can fall anywhere outside the grey area of the fan chart.

Please answer all the questions based on the information given.

Q10.21. To keep inflation low and stable, Government sets an inflation target of 2%. How likely do you think it is that the inflation will consistently **exceed** 2% target?

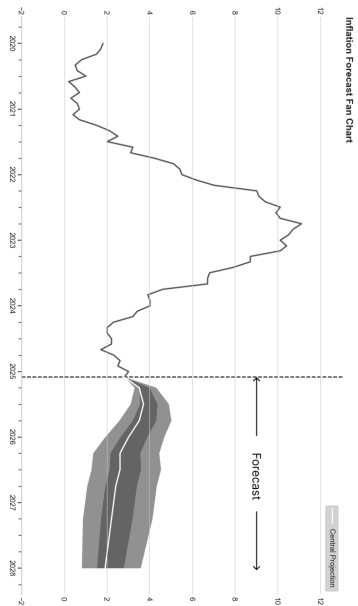
- Exceptionally unlikely
- Very unlikely
- Quite unlikely
- Fifty-fifty
- Quite likely
- Very likely
- Virtually certain

Q10.22. Based on the inflation forecast provided, please rank the following time periods from **the least uncertain** to **the most uncertain** regarding inflation: 2025, 2026, 2027, 2028

- 2025
- 2026

2027
2028

Q10.23.



The fan chart depicts the probability of various future outcomes for CPI inflation. The white line represents the most likely values for inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie within the darkest central band on only 30 of those occasions. The fan chart is constructed so that outturns of inflation are also expected to lie within each pair of the lighter grey areas on 30 occasions. In any particular quarter of the forecast period, inflation is therefore expected to lie somewhere within the fans on 60 out of 100 occasions. And on the remaining 40 out of 100 occasions inflation can fall anywhere outside the grey area of the fan chart.

Please answer all the questions based on the information given.

Q10.24. What indicates that the forecasts at different time periods (2025, 2026, 2027, 2028) are more or less uncertain? (Please select the option that best reflects your belief, even if other answers may also seem reasonable.)

- ☐ It is more uncertain when the time is getting further away in the future.
- ☐ It is more uncertain when the full band is getting wider.
- ☐ It is more uncertain when the central band is getting wider.
- ☐ It is more uncertain when the inflation rate is high.
- ☐ It is more uncertain when the inflation rate is low.
- ☐ It is more uncertain when the central projection has an erratic value.
- ☐ Other (Please specify)

Q10.25. On a scale of 1 to 10, where 1 means not accurate at all and 10 means very accurate, how accurate do you think the central projection in the graph is?

Not accurate at all Very accurate

☒ 1 2 3 4 5 6 7 8 9 10

Q10.26. On a scale of 1 to 10, where 1 means very uncertain and 10 means very certain, how certain or uncertain do you think the inflation forecast is?

Very uncertain Very certain

☒ 1 2 3 4 5 6 7 8 9 10

Q10.27. Why do you think that the forecast is more or less uncertain? (Select all that apply)

- ☐ The forecast is for a time far in the future.
- ☐ The forecast fluctuates too much.
- ☐ The forecast is stable.
- ☐ I'm not familiar with how to interpret this type of chart.
- ☐ I believe the economic conditions are highly unpredictable.
- ☐ I believe the economic conditions are highly predictable.
- ☐ The bands are widely spread.
- ☐ The bands are concentrated.
- ☐ The forecast is not as I expected.
- ☐ The forecast seems reasonable to me.
- ☐ Other (Please specify)

Q10.28. On a scale of 1 to 10, where 1 means not representative at all and 10 means very representative, how representative do you think the uncertainty depicted in the graph is (i.e., how well the given information reflects the true variability or uncertainty of the forecast?)

Not representative at all 1 2 3 4 5 6 7 8 9 10 Very representative

Q10.29. On a scale of 1 to 10, where 1 means very difficult and 10 means very easy, how easy or difficult do you find the graph to understand?

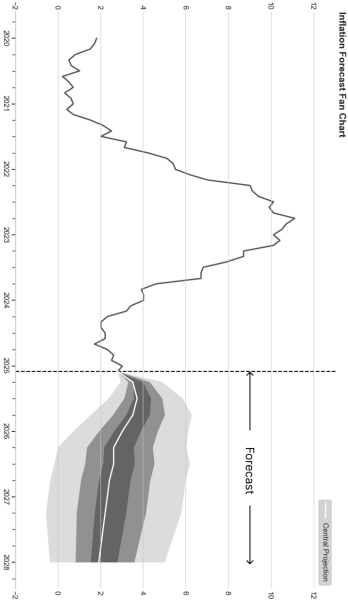
Very difficult 1 2 3 4 5 6 7 8 9 10 Very easy

Q10.30. On a scale of 1 to 10, where 1 means you don't trust the inflation forecast at all and 10 means you fully trust it, how much do you trust the inflation forecast shown in the graph?

I don't trust at all. 1 2 3 4 5 6 7 8 9 10 I fully trust it.

Understanding of Time-series Forecast (Fan Chart-90)

Q11.1. The graph below shows the inflation forecast from 2025Q2 to 2028Q1.



The fan chart depicts the probability of various future outcomes for CPI inflation. The white line represents the most likely value for inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie within the darkest central band on only 30 of those occasions. The fan chart is constructed so that outturns of inflation are also expected to lie within each pair of the lighter grey areas on 30 occasions. In any particular quarter of the forecast period, inflation is therefore expected to lie somewhere within the fans on 90 out of 100 occasions. And on the remaining 10 out of 100 occasions inflation can fall anywhere outside the grey area of the fan chart.

Please answer all the questions based on the information given.

Q11.2. Have you ever seen this type of graph before?

- ☐ Definitely not
- ☐ Probably not
- ☐ Probably yes
- ☐ Definitely yes

Q11.3. Based on the forecast, how do you think the inflation will change over the next 3 years?

- ☐ Go up
- ☐ Go down
- ☐ Increase then decrease
- ☐ Decrease then increase
- ☐ Other (please specify)

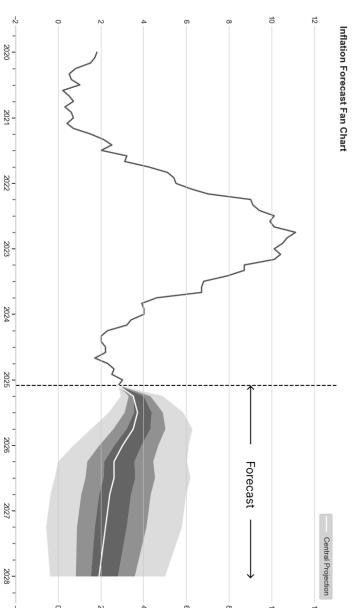
Q11.4. Does the inflation forecast information provided align with your expectations?

- ☐ Strongly aligns – The inflation forecast matches exactly what I expected.
- ☐ Somewhat aligns – The inflation forecast is generally in line with my expectations.
- ☐ Neutral – The inflation forecast neither aligns nor conflicts with my expectations.
- ☐ Somewhat does not align – The inflation forecast is somewhat different from what I expected.
- ☐ Does not align at all – The inflation forecast is significantly different from what I expected.

Q11.5. How does the inflation forecast conflict with your expectations?

- ☐ The inflation forecast is higher than what I expected.
- ☐ The inflation forecast is lower than what I expected.

Q11.6.



The fan chart depicts the probability of various future outcomes for CPI inflation. The white line represents the most likely value for inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie within the darkest central band on only 30 of those occasions. The fan chart is constructed so that outturns of inflation are also expected to lie within each pair of the lighter grey areas on 30 occasions. In any particular quarter of the forecast period, inflation is therefore expected to lie somewhere within the fans on 90 out of 100 occasions. And on the remaining 10 out of 100 occasions inflation can fall anywhere outside the grey area of the fan chart.

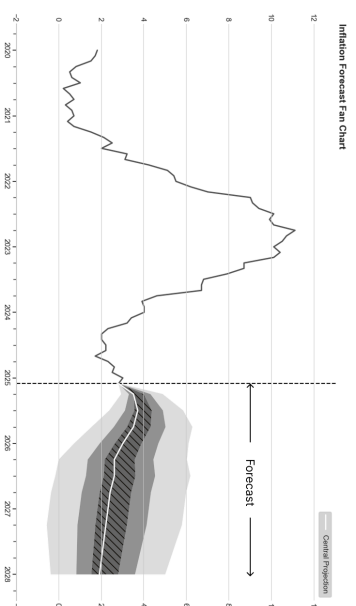
Please answer all the questions based on the information given.

Q11.7. Based on the inflation forecast provided, what do you think is the probability that the inflation will follow **the central projection**?

- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 80%) About a 8 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 20%) About a 2 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q11.8. Based on the inflation forecast provided, what do you think is the probability that the inflation will land in the area **with stripe**

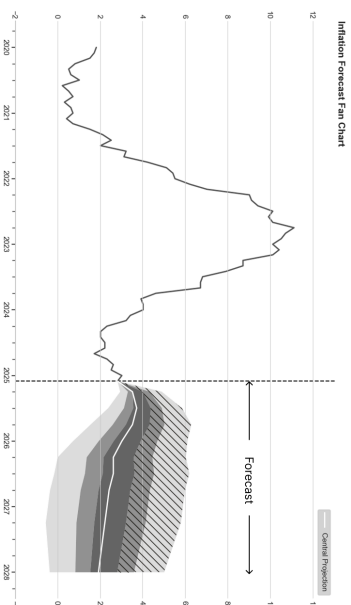
pattern?



- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

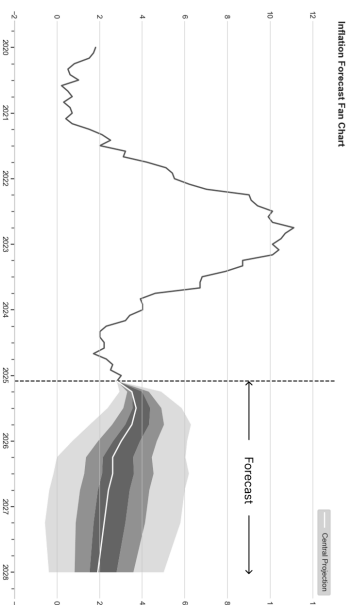
Q11.9. Based on the inflation forecast provided, what do you think is the probability that the inflation will land in the area **with stripe**

pattern?



- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q11.10.



The fan chart depicts the probability of various future outcomes for CPI inflation. The white line represents the most likely values for inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie within the darkest central band on only 30 of those occasions. The fan chart is constructed so that outturns of inflation are also expected to lie within each pair of the lighter grey areas on 30 occasions. In any particular quarter of the forecast period, inflation is therefore expected to lie somewhere within the fans on 90 out of 100 occasions. And on the remaining 10 out of 100 occasions inflation can fall anywhere outside the grey area of the fan chart.

Please answer all the questions based on the information given.

Q11.11. Based on the inflation forecast provided, what do you think is the probability that inflation will fall **outside** of the shaded grey area?

- ☐ 70% (35% each side)
- ☐ 40% (20% each side)
- ☐ 20% (10% each side)
- ☐ 10% (5% each side)
- ☐ 5% (2.5% each side)
- ☐ Less than 1%

Q11.12. Based on the inflation forecast provided, do you think there is a higher risk of inflation exceeding the central projection or of being less than this central projection?

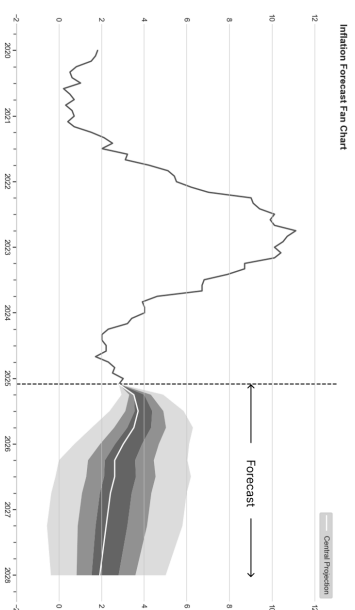
- ☐ There is a bigger upside risk (that is, of inflation exceeding the central projection).

- ☐ There is a bigger downside risk (that is, of inflation being less than the central projection).
- ☐ The risks appear equal on both sides.

Q11.13. If the actual inflation falls outside the shaded grey area, which side is more likely, based on the uncertainty depicted in the graph?

- ☐ More likely to be on the upper side (where inflation is high).
- ☐ More likely to be on the lower side (where inflation is low).
- ☐ Roughly equal chances of being on either side.

Q11.14.



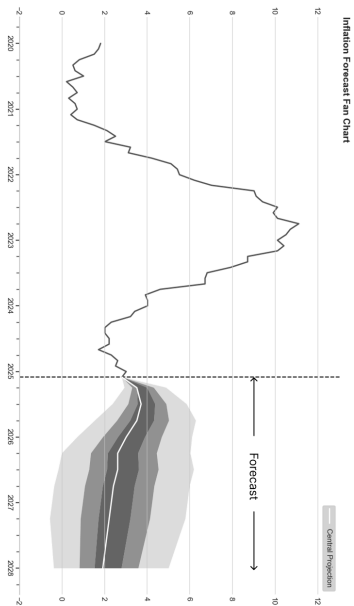
The fan chart depicts the probability of various future outcomes for CPI inflation. The white line represents the most likely values for inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie within the darkest central band on only 30 of those occasions. The fan chart is constructed so that outturns of inflation are also expected to lie within each pair of the lighter grey areas on 30 occasions. In any particular quarter of the forecast period, inflation is therefore expected to lie somewhere within the fans on 90 out of 100 occasions. And on the remaining 10 out of 100 occasions inflation can fall anywhere outside the grey area of the fan chart.

Please answer all the questions based on the information given.

Q11.15. How likely do you think it is that the inflation will follow the central projection?

- ☐ Exceptionally unlikely
- ☐ Very unlikely
- ☐ Quite unlikely
- ☐ Fifty-fifty
- ☐ Quite likely
- ☐ Very likely
- ☐ Virtually certain

Q11.16.



The fan chart depicts the probability of various future outcomes for CPI inflation. The white line represents the most likely values for inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie within the darkest central band on only 30 of those occasions. The fan chart is constructed so that outturns of inflation are also expected to lie within each pair of the lighter grey areas on 30 occasions. In any particular quarter of the forecast period, inflation is therefore expected to lie somewhere within the fans on 90 out of 100 occasions. And on the remaining 10 out of 100 occasions inflation can fall anywhere outside the grey area of the fan chart.

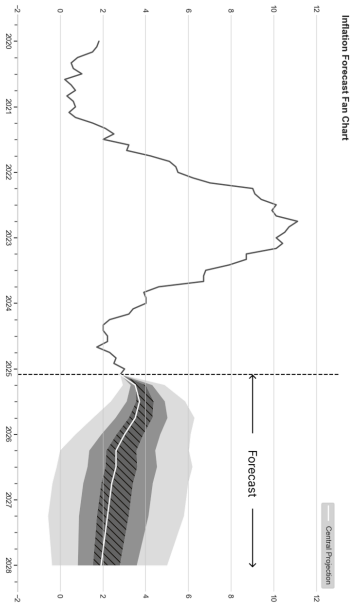
Please answer all the questions based on the information given.

Q11.17. What tells you that the inflation from 2025Q2 to 2028Q1 is likely or unlikely to follow the central projection? (Please select the

option that best reflects your belief, even if other answers may also seem reasonable.)

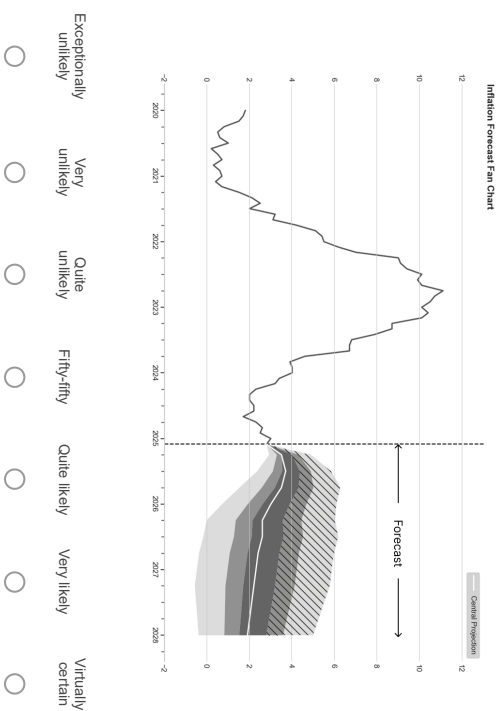
- ☐ It is being communicated in the text that these are the most likely values.
- ☐ It is being communicated in the text that the actual values may deviate.
- ☐ The fan chart is concentrated around the central projection.
- ☐ The fan chart is widely spread around the central projection.
- ☐ The central projection is not positioned in the center of the most likely band.
- ☐ The probability of inflation following any exact path is 0.
- ☐ The value seems reasonable to me. (gut feeling)
- ☐ The value seems unreasonable to me. (gut feeling)
- ☐ Other (Please specify)

Q11.18. How likely do you think it is that the inflation will land in the area with stripe pattern?

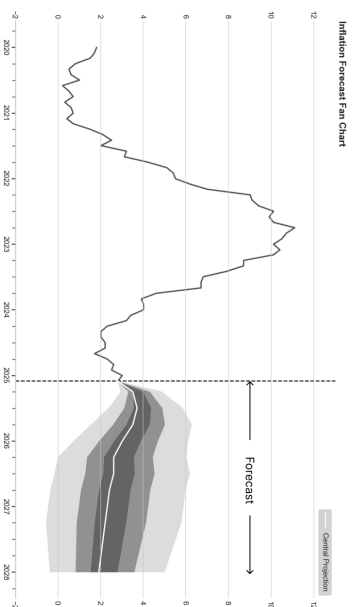


- | | | | | | | |
|------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Exceptionally unlikely | Very unlikely | Quite unlikely | Fifty-fifty | Quite likely | Very likely | Virtually certain |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

Q11.19. How likely do you think it is that the inflation will land in the area with stripe pattern?



Q11.20.



The fan chart depicts the probability of various future outcomes for CPI inflation. The white line represents the most likely values for inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie within the darkest central band on only 30 of those occasions. The fan chart is constructed so that outturns of inflation are also expected to lie within each pair of the lighter grey areas on 30 occasions. In any particular quarter of the forecast period, inflation is therefore expected to lie somewhere within the fans on 90 out of 100 occasions. And on the remaining 10 out of 100 occasions inflation can fall anywhere outside the grey area of the fan chart.

Please answer all the questions based on the information given.

Q11.21. To keep inflation low and stable, Government sets an inflation target of 2%. How likely do you think it is that the inflation will consistently **exceed** 2% target?

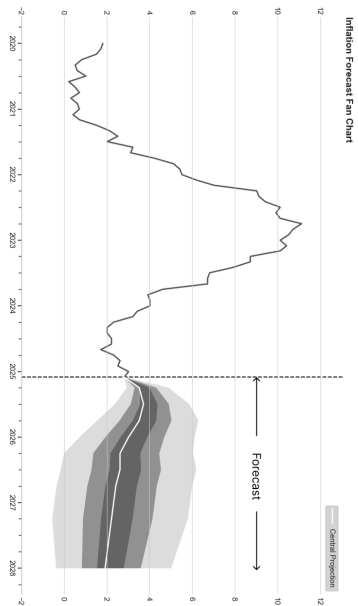
- Exceptionally unlikely Very unlikely Quite unlikely Fifty-fifty Quite likely Very likely Virtually certain
- ☐ ☐ ☐ ☐ ☐ ☐ ☐

Q11.22. Based on the inflation forecast provided, please rank the following time periods from **the least uncertain** to **the most uncertain** regarding inflation: 2025, 2026, 2027, 2028

- 2025
- 2026

2027
2028

Q11.23.



The fan chart depicts the probability of various future outcomes for CPI inflation. The white line represents the most likely values for inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie within the darkest central band on only 30 of those occasions. The fan chart is constructed so that outturns of inflation are also expected to lie within each pair of the lighter grey areas on 30 occasions. In any particular quarter of the forecast period, inflation is therefore expected to lie somewhere within the fans on 90 out of 100 occasions. And on the remaining 10 out of 100 occasions inflation can fall anywhere outside the grey area of the fan chart.

Please answer all the questions based on the information given.

Q11.24. What indicates that the forecasts at different time periods (2025, 2026, 2027, 2028) are more or less uncertain? (Please select the option that best reflects your belief, even if other answers may also seem reasonable.)

- ☐ It is more uncertain when the time is getting further away in the future.
- ☐ It is more uncertain when the full band is getting wider.
- ☐ It is more uncertain when the central band is getting wider.
- ☐ It is more uncertain when the inflation rate is high.
- ☐ It is more uncertain when the inflation rate is low.
- ☐ It is more uncertain when the central projection has an erratic value.
- ☐ Other (Please specify)

Q11.25. On a scale of 1 to 10, where 1 means not accurate at all and 10 means very accurate, how accurate do you think the central projection in the graph is?

Not accurate at all Very accurate

☒ 1 2 3 4 5 6 7 8 9 10

Q11.26. On a scale of 1 to 10, where 1 means very uncertain and 10 means very certain, how certain or uncertain do you think the inflation forecast is?

Very uncertain Very certain

☒ 1 2 3 4 5 6 7 8 9 10

Q11.27. Why do you think that the forecast is more or less uncertain? (Select all that apply)

- ☐ The forecast is for a time far in the future.
- ☐ The forecast fluctuates too much.
- ☐ The forecast is stable.
- ☐ I'm not familiar with how to interpret this type of chart.
- ☐ I believe the economic conditions are highly unpredictable.
- ☐ I believe the economic conditions are highly predictable.
- ☐ The bands are widely spread.
- ☐ The bands are concentrated.
- ☐ The forecast is not as I expected.
- ☐ The forecast seems reasonable to me.
- ☐ Other (Please specify)

Q11.28. On a scale of 1 to 10, where 1 means not representative at all and 10 means very representative, how representative do you think the uncertainty depicted in the graph is (i.e., how well the given information reflects the true variability or uncertainty of the forecast?)

Not representative at all	1	2	3	4	5	6	7	8	9	10	Very representative
	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	

Q11.29. On a scale of 1 to 10, where 1 means very difficult and 10 means very easy, how easy or difficult do you find the graph to understand?

Very difficult	1	2	3	4	5	6	7	8	9	10	Very easy
	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	

Q11.30. On a scale of 1 to 10, where 1 means you don't trust the inflation forecast at all and 10 means you fully trust it, how much do you trust the inflation forecast shown in the graph?

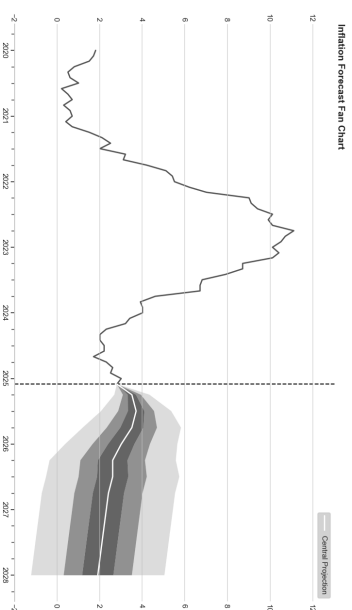
I don't trust at all.	1	2	3	4	5	6	7	8	9	10	I fully trust it.
	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	

Decision-making using Time-series Forecast (Fan Chart-90)

Q12.1. The graph below shows the inflation forecast from 2025Q2 to 2028Q1.

Please note that the graph is created based on the latest data released by the Bank of England and that it differs slightly from the one you have seen before.

Please answer the following questions using this updated information.



The fan chart depicts the probability of various future outcomes for CPI inflation. The white line represents the most likely value for inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie within the darkest central band on only 30 of those occasions. The fan chart is constructed so that outcomes of inflation are also expected to lie within each pair of the lighter gray areas on 30 occasions. In any particular quarter of the forecast period, inflation is therefore expected to lie somewhere within the fans on 90 out of 100 occasions. And on the remaining 10 out of 100 occasions inflation can fall anywhere outside the gray area of the fan chart.

Q12.2. Assume you are an investor with £100,000 to invest in a 3-year bond. You will receive returns at the end of each year.

Your goal is to maximize the cumulative return on your investment from this bond over the next 3 years.

Inflation has a direct impact on the return on investment you will

receive. The following setup is simplified but it generally reflects the reality of how each type of bond responds to inflation.

There are two types of bonds available for you to choose between:

A. Nominal Bond: A nominal bond has a fixed rate of interest and the **real returns** depend on inflation. If inflation stays low or drops ($\leq 3\%$), choosing fixed-rate nominal bonds will provide higher returns. However, if inflation increases, your returns will be eroded due to reduced purchasing power.

B. Inflation-Protected Bond: An inflation-protected bond provides stable but lower **real returns** regardless of inflation. If inflation is high ($> 3\%$), choosing inflation-protected bonds will hedge against inflation by ensuring your returns are adjusted in line with inflation.

Now, you are asked to choose which bond you would like to invest your entire £100,000 in:

- A. Nominal Bond
- B. Inflation-Protected Bond

A payoff table is provided to help you understand the potential outcomes of your decision. It shows how much you could effectively gain or lose **at the end of each year**, depending on which bond you choose to invest in, and on the actual inflation rate for that year.

	Real Return When Inflation is Low ($\leq 3\%$)	Real Return When Inflation is High ($> 3\%$)
A. Nominal Bond	+£3,000	-£1,000
B. Inflation-Protected Bond	+£1,000	+£1,000

Note that you will **not be** able to trade the bond during these 3 years.

Please make the decision based on the forecast information provided, but keep in mind that there is uncertainty.

Q12.3. Which bond would you choose to invest in?

- ☐ A. Nominal Bond
- ☐ B. Inflation-Protected Bond

Q12.4. On a scale of 1 to 10, where 1 means not confident at all and 10 means very confident, how confident are you in making the decision?

Not confident at all

☒ 1 2 3 4 5 6 7 8 9 10

Very confident

Q12.5. Could you explain in a few sentences how and why you decided to invest in this bond?

Q12.6. Considering the payoff for the **Nominal Bond**, which of the following has a larger impact on your real return?

- ☐ The potential loss if inflation turns out to be high ($> 3\%$).
- ☐ The potential gain if inflation stays low ($\leq 3\%$).
- ☐ The potential loss and the potential gain have the same impact.

Q12.7. On a scale of 1 to 10, where 1 means the forecast was not effective at all and 10 means it was very effective, how effective do you think the forecast was in assisting your decision-making?

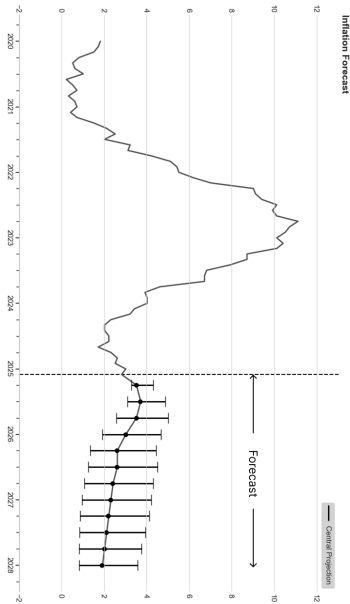
Not effective at all

☒ 1 2 3 4 5 6 7 8 9 10

Very effective

Understanding of Time-series Forecast (error bar)

Q13.1. The graph below shows the inflation forecast from 2025Q2 to 2028Q1.



The black line represents the most likely value for CPI inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie somewhere on the black error bar on 60 of those occasions. And on the remaining 40 out of 100 occasions inflation can fall anywhere outside the error bar.

Please answer all the questions based on the information given.

Q13.2. Have you ever seen this type of graph before?

- ☐ Definitely not
- ☐ Probably not
- ☐ Probably yes
- ☐ Definitely yes

Q13.3. Based on the forecast, how do you think the inflation will change over the next 3 years?

- ☐ Go up
- ☐ Go down
- ☐ Increase then decrease
- ☐ Decrease then increase
- ☐ Other (please specify)

Q13.4. Does the inflation forecast information provided align with your expectations?

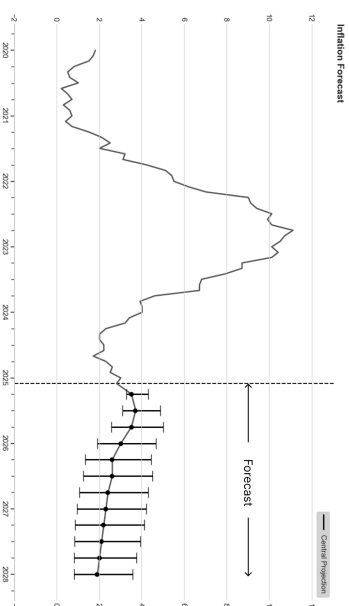
- ☐ Strongly aligns – The inflation forecast matches exactly what I expected.

- ☐ Somewhat aligns – The inflation forecast is generally in line with my expectations.
- ☐ Neutral – The inflation forecast neither aligns nor conflicts with my expectations.
- ☐ Somewhat does not align – The inflation forecast is somewhat different from what I expected.
- ☐ Does not align at all – The inflation forecast is significantly different from what I expected.

Q13.5. How does the inflation forecast conflict with your expectations?

- ☐ The inflation forecast is higher than what I expected.
- ☐ The inflation forecast is lower than what I expected.

Q13.6.



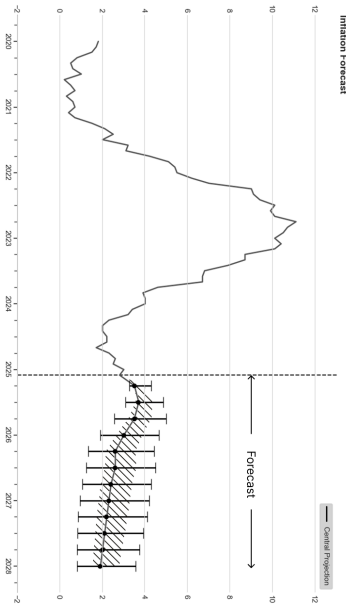
The black line represents the most likely value for CPI inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie somewhere on the black error bar on 60 of those occasions. And on the remaining 40 out of 100 occasions inflation can fall anywhere outside the error bar.

Please answer all the questions based on the information given.

Q13.7. Based on the inflation forecast provided, what do you think is the probability that the inflation will follow the **central projection**?

- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 80%) About a 8 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 20%) About a 2 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

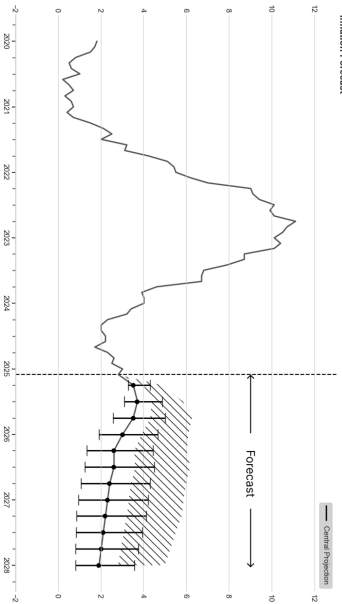
Q13.8. Based on the inflation forecast provided, what do you think is the probability that the inflation will land in the area **with stripe pattern**?



- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance

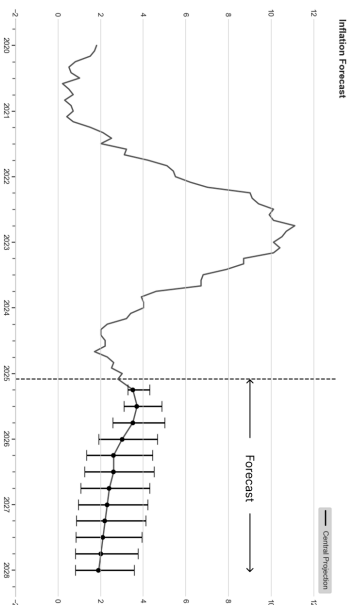
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q13.9. Based on the inflation forecast provided, what do you think is the probability that the inflation will land in the area **with stripe pattern**?



- ☐ (>99%) Greater than 99 in 100 chance
- ☐ (around 90%) About a 9 in 10 chance
- ☐ (around 60%) About a 6 in 10 chance
- ☐ (around 50%) About a 1 in 2 chance
- ☐ (around 30%) About a 3 in 10 chance
- ☐ (around 10%) About a 1 in 10 chance
- ☐ (<1%) Less than 1 in 100 chance

Q13.10. The graph below shows the inflation forecast **from 2025Q2 to 2028Q1**.



The black line represents the most likely value for CPI inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie somewhere on the black error bar on 60 of those occasions. And on the remaining 40 out of 100 occasions inflation can fall anywhere outside the error bar.

Please answer all the questions based on the information given.

Q13.11. Based on the inflation forecast provided, what do you think is the probability of inflation ending up **outside** of the areas covered by error bars?

- ☐ 70% (35% each side)
- ☐ 40% (20% each side)
- ☐ 20% (10% each side)
- ☐ 10% (5% each side)
- ☐ 5% (2.5% each side)
- ☐ Less than 1%

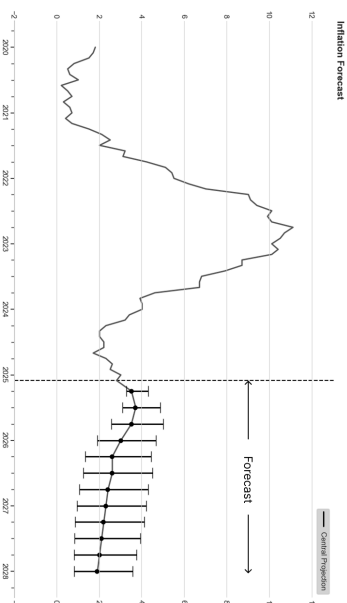
Q13.12. Based on the information provided, do you think there is a higher risk of inflation exceeding the central projection or of being less than this central projection?

- ☐ There is a bigger upside risk (that is, of inflation exceeding the central projection).
- ☐ There is a bigger downside risk (that is, of inflation being less than the central projection).
- ☐ The risks appear equal on both sides.

Q13.13. If the actual inflation ends up outside the error bar, which side is more likely, based on the uncertainty depicted in the graph?

- ☐ More likely to be on the upper side (where inflation is high).
- ☐ More likely to be on the lower side (where inflation is low).
- ☐ Roughly equal chances of being on either side.

Q13.14. The graph below shows the inflation forecast from 2025Q2 to 2028Q1.



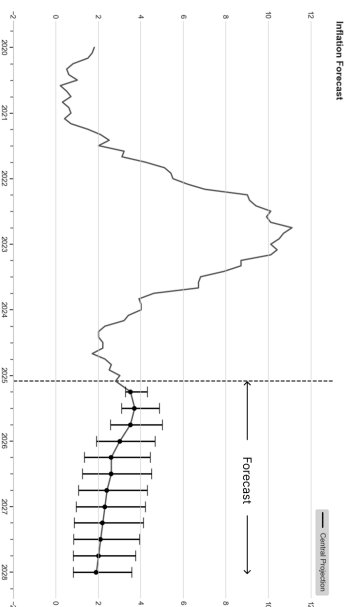
The black line represents the most likely value for CPI inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie somewhere on the black error bar on 60 of those occasions. And on the remaining 40 out of 100 occasions inflation can fall anywhere outside the error bar.

Please answer all the questions based on the information given.

Q13.15. How likely do you think it is that the inflation will follow the central projection?

- ☐ Exceptionally unlikely
 ☐ Very unlikely
 ☐ Quite unlikely
 ☐ Fifty-fifty
 ☐ Quite likely
 ☐ Very likely
 ☐ Virtually certain

Q13.16. The graph below shows the inflation forecast from 2025Q2 to 2028Q1.



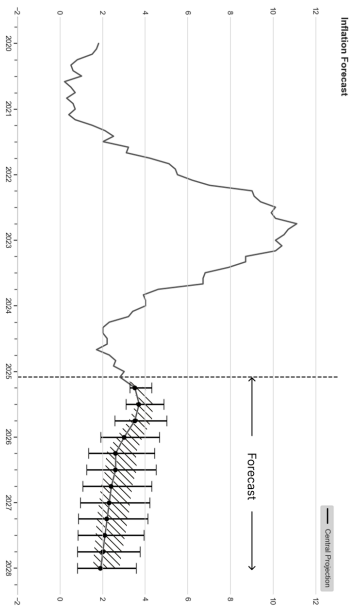
The black line represents the most likely value for CPI inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie somewhere on the black error bar on 60 of those occasions. And on the remaining 40 out of 100 occasions inflation can fall anywhere outside the error bar.

Please answer all the questions based on the information given.

Q13.17. What tells you that the inflation from 2025Q2 to 2028Q1 is likely or unlikely to follow the central projection? (Please select the option that best reflects your belief, even if other answers may also seem reasonable.)

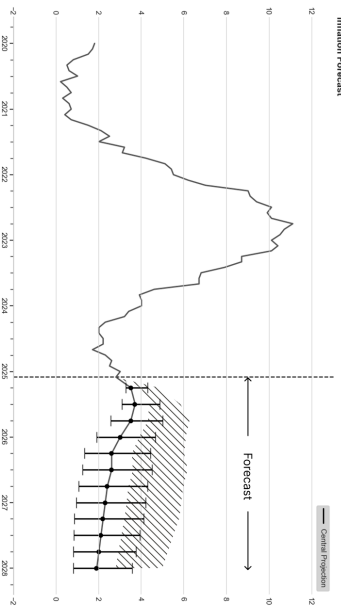
- ☐ It is being communicated in the text that these are the most likely values.
☐ It is being communicated in the text that the actual values may deviate.
☐ The black error bar extends over a short distance, suggesting the central estimate somewhat likely but not certain.
☐ The black error bar extends over a long distance, so inflation is unlikely to be around this value.
☐ The central projection is not positioned in the center of the error bar.
☐ The probability of inflation following any exact path is 0.
☐ The value seems reasonable to me. (gut feeling)
☐ The value seems unreasonable to me. (gut feeling)
☐ Other (Please specify)

Q13.18. How likely do you think it is that the inflation will land in the area with stripe pattern?



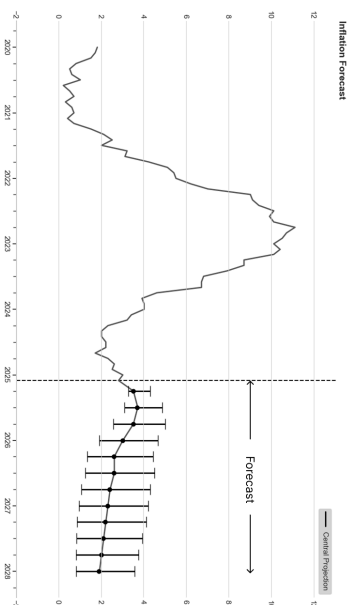
Exceptionally unlikely Very unlikely Quite unlikely Fifty-fifty Quite likely Very likely Virtually certain

Q13.19. How likely do you think it is that the inflation will land in the area with stripe pattern?



Exceptionally unlikely Very unlikely Quite unlikely Fifty-fifty Quite likely Very likely Virtually certain

Q13.20. The graph below shows the inflation forecast from 2025Q2 to 2028Q1.



The black line represents the most likely value for CPI inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie somewhere on the black error bar on 60 of those occasions. And on the remaining 40 out of 100 occasions inflation can fall anywhere outside the error bar.

Please answer all the questions based on the information given.

Q13.21. To keep inflation low and stable, Government sets an inflation target of 2%. How likely do you think it is that the inflation will consistently **exceed** 2% target?

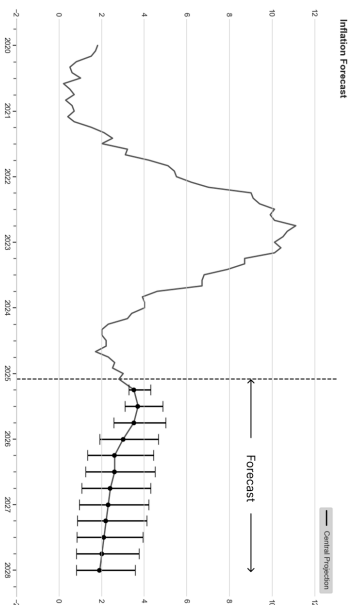
Exceptionally unlikely Very unlikely Quite unlikely Fifty-fifty Quite likely Very likely Virtually certain

- Exceptionally unlikely ☐
- Very unlikely ☐
- Quite unlikely ☐
- Fifty-fifty ☐
- Quite likely ☐
- Very likely ☐
- Virtually certain ☐

Q13.22. Based on the inflation forecast provided, please rank the following time periods from **the least uncertain to the most uncertain** regarding inflation: 2025, 2026, 2027, 2028

- 2025
- 2026
- 2027
- 2028

Q13.23. The graph below shows the inflation forecast **from 2025Q2 to 2028Q1**.



The black line represents the most likely value for CPI inflation from 2025Q2 to 2028Q1 (central projection), though the actual inflation value may deviate from this forecast. If economic circumstances identical to today's were to prevail on 100 occasions, the MPC's judgement is that inflation in any particular quarter would lie somewhere on the black error bar on 60 of those occasions. And on the remaining 40 out of 100 occasions inflation can fall anywhere outside the error bar.

Please answer all the questions based on the information given.

Q13.24. What indicates that the forecasts at different time periods (2025, 2026, 2027, 2028) are more or less uncertain? (Please select the option that best reflects your belief, even if other answers may also seem reasonable.)

- ☐ It is more uncertain when the time is getting further away in the future.
- ☐ It is more uncertain when the error bar is extending over a longer distance.
- ☐ It is more uncertain when the inflation rate is high.
- ☐ It is more uncertain when the inflation rate is low.
- ☐ It is more uncertain when the central projection has an erratic value.
- ☐ Other (Please specify)

Q13.25. On a scale of 1 to 10, where 1 means not accurate at all and 10 means very accurate, how accurate do you think the central projection in the graph is?

Not accurate at all

☒ 1
 2
 3
 4
 5
 6
 7
 8
 9
 10
 Very accurate

Q13.26. On a scale of 1 to 10, where 1 means very uncertain and 10 means very certain, how certain or uncertain do you think the inflation forecast is?

Very uncertain

10

2

3

4

5

6

7

8

9

10

Very certain

Q13.27. Why do you think that the forecast is more or less uncertain? (Select all that apply)

- ☐ The forecast is for a time far in the future.
- ☐ The forecast fluctuates too much.
- ☐ The forecast is stable.
- ☐ I'm not familiar with how to interpret this type of chart.
- ☐ I believe the economic conditions are highly unpredictable.
- ☐ I believe the economic conditions are highly predictable.
- ☐ The error bars are widely extended.
- ☐ The error bars are concentrated.
- ☐ The forecast is not as I expected.
- ☐ The forecast seems reasonable to me.
- ☐ Other (Please specify)

Q13.28. On a scale of 1 to 10, where 1 means not representative at all and 10 means very representative, how representative do you think the uncertainty depicted in the graph is (i.e., how well the given information reflects the true variability or uncertainty of the forecast?)

Not representative at all

10

2

3

4

5

6

7

8

9

10

Very representative

Q13.29. On a scale of 1 to 10, where 1 means very difficult and 10 means very easy, how easy or difficult do you find the graph to understand?

Very difficult

10

2

3

4

5

6

7

8

9

10

Very easy

Q13.30. On a scale of 1 to 10, where 1 means you don't trust the inflation forecast at all and 10 means you fully trust it, how much do you trust the inflation forecast shown in the graph?

I don't trust at all.

10

2

3

4

5

6

7

8

9

10

I fully trust it.

Appendix C. Participant Characteristics

Of the 309 respondents retained for analysis, over half of the respondents (55.01%) held at least a bachelor's degree and 60.84% of respondents were employed. Nearly half (48.9%) had never studied economics. Most reported some financial activity: 90% of respondents have savings and around half of them are involved in investments or retirement planning. The decision task later simulates fixed-income planning, but it is worth noticing that only 16.2% of participants engage with fixed-income.

Among all 330 responses, 21 cases did not correctly identify the definition of inflation, and 14 of them mistakenly equated it with the size of the economy. Their data were excluded from subsequent analyses. On a 5-point Likert scale, most respondents rate they somewhat trust the official inflation forecast (Mean = 3.7, Median = 4).

The distribution in Figure C.13 shows that current inflation has more impact on people's personal finances than inflation forecasts. Table C.5 indicates that when exposed to the communication visualisations, more people think they have seen bell curves and error bars before, with the least proportion of people who have seen density strip before (30% of that group).

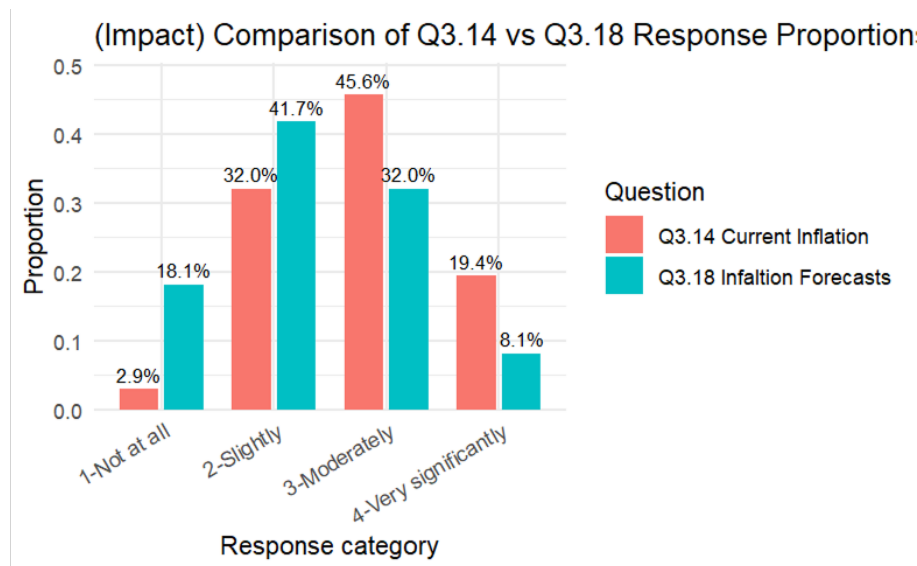


Figure C.13: Impact of Current Inflation versus Inflation Forecasts on Personal Finances Comparison

Table C.5: Prior Familiarity with the Communication Visualisation

Seen before	1-errorbar	2-shadedbar	3-fanchartslice	4-bellcurve
Yes	45.76%	29.51%	39.39%	50.82%
No	54.24%	70.49%	60.61%	49.18%

Regarding prior expectations of inflation, 55.02% of total respondents think the forecast is somewhat aligned with their expectations and 31.07% hold neutral opinions. Among 30 people who judged the forecast misaligned, 90% of them believed that the forecast was lower than their expectations. People seem to hold an upward bias in inflation rates.

Appendix D. Forecast Visualisation Code

[Link to Visualisation Generation GitHub Repository](#)

Appendix E. Data Analysis

Appendix E.1. Code in R Notebook

[Link to R Notebook Webpage](#)

[Link to Data Analysis GitHub Repository \(including the raw data\)](#)

Appendix E.2. Analysis Results

Nora Master Project Data Analysis Notebook

Nora Luo

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1. Notation
2. Load and split the data
3. Screening Questions
4. Actual Survey Analysis
5. Statistical Testing
6. Skewed Data
7. Decision-making

README

- run the Load and split the data chunk first
- results was filtered based on Q3.9 - whether respondents understand the concept of inflation
- !: We reject the null hypothesis.

Load and split the data

```
library(readr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##   filter, lag
```

```
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(ggplot2)
library(tidyr)
library(FSA)
```

```
## ## FSA v0.10.0. See citation('FSA') if used in publication.
## ## Run fishR() for related website and fishR('IFAR') for related book.
```

```
library(car)
```

```
## Warning: package 'car' was built under R version 4.4.3
```

```
## Loading required package: carData
```

```
## Warning: package 'carData' was built under R version 4.4.3
```

```
## Registered S3 methods overwritten by 'car':
##   method      from
##   hist.boot    FSA
##   confint.boot FSA
```

```
##
## Attaching package: 'car'
```

```
## The following object is masked from 'package:FSA':
##
##   bootCase
```

```
## The following object is masked from 'package:dplyr':
##
##   recode
```

```
library(rstatix)
```

```
## Warning: package 'rstatix' was built under R version 4.4.3
```

```
##  
## Attaching package: 'rstatix'
```

```
## The following object is masked from 'package:stats':  
##  
## filter
```

```
library(effectsiz)
```

```
## Warning: package 'effectsiz' was built under R version 4.4.3
```

```
##  
## Attaching package: 'effectsiz'
```

```
## The following objects are masked from 'package:rstatix':  
##  
## cohens_d, eta_squared
```

```
exported_data <- read_csv("final_rawdata_value_1605.csv")
```

```
## Rows: 332 Columns: 262
```

```
## — Column specification —————  
## Delimiter: ","  
## chr (262): StartDate, EndDate, Status, IPAddress, Progress, Duration (in sec...  
##  
## i Use `spec()` to retrieve the full column specification for this data.  
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```

exported_data <- subset(exported_data, select = -c(StartDate,EndDate,Status,IPAddress,Progress,Finished,RecordedDate, RecipientLastName,RecipientFirstName,
                                                    RecipientEmail,ExternalReference,LocationLatitude,LocationLongitude,DistributionChannel,UserLanguage))
rawdata_value <- subset(exported_data, select =-c(ResponseId,Q1.1,Q2.1))
rawdata_value <- rawdata_value[3:nrow(rawdata_value), ]

# Label the group
rawdata_value <- rawdata_value %>%
  mutate(group = case_when(
    !is.na(Q4.2) ~ '0-controlled',
    !is.na(Q6.2) ~ '1-errorbar',
    !is.na(Q8.2) ~ '2-shadedbar',
    !is.na(Q10.2) ~ '3-fanchartslice',
    !is.na(Q12.2) ~ '4-bellcurve'
  ),
  treatment = case_when(
    group %in% c("1-errorbar", "2-shadedbar", "3-fanchartslice", "4-bellcurve")
    ~ "treatment_group",
    TRUE
    ~ as.character(group) # or another fallback
  ))
# create a copy of the data for Q3.9 analysis
rawdata_q3_9 <- rawdata_value
# filter participants who do not understand the concept of inflation
rawdata_value <- rawdata_value[rawdata_value$Q3.9 == "4", ]

# rawdata_value <- rawdata_value %>%
#   mutate(group = case_when(
#     !is.na(Q4.2) ~ 0,
#     !is.na(Q6.2) ~ 1,
#     !is.na(Q8.2) ~ 2,
#     !is.na(Q10.2) ~ 3,
#     !is.na(Q12.2) ~ 4
#   ))

# meta-data
total_respondents <- nrow(rawdata_value)

group_total <- rawdata_value %>%
  group_by(group) %>%
  summarise(n_respondents = n())

total_treatment_respondents <- total_respondents - group_total$n_respondents[1]

# graph preparation

likelihood_labels = c(
  "9" = "Exceptionally unlikely",
  "10" = "Very unlikely",
  "11" = "Quite unlikely",
  "12" = "Fifty-fifty",
  "13" = "Quite likely",
  "14" = "Very likely",
  "15" = "Virtually certain"
)

# filter treatment groups
controlled_group <- rawdata_value[!is.na(rawdata_value$Q4.2), ]
error_bar <-rawdata_value[!is.na(rawdata_value$Q6.2), ]
shaded_bar <-rawdata_value[!is.na(rawdata_value$Q8.2), ]
fanchart_slice <-rawdata_value[!is.na(rawdata_value$Q10.2), ]
bell_curve <-rawdata_value[!is.na(rawdata_value$Q12.2), ]

# adding labels to the groups
# controlled_group$group <- 1
# error_bar$group <- 2
# shaded_bar$group <- 3
# fanchart_slice$group <- 4
# bell_curve$group <- 5

# remove unnecessary columns
controlled_group = controlled_group[,grep1("^(Q3|Q4|Q5)",names(controlled_group))|names(controlled_group) %in% c("Duration (in seconds)", "group", "treatment")]
error_bar = error_bar[,grep1("^(Q3|Q6|Q7)",names(error_bar))| names(error_bar) %in% c("Duration (in seconds)", "group", "treatment")]
shaded_bar = shaded_bar[,grep1("^(Q3|Q8|Q9)",names(shaded_bar))| names(shaded_bar)%in% c("Duration (in seconds)", "group", "treatment")]
fanchart_slice = fanchart_slice[,grep1("^(Q3|Q10|Q11)",names(fanchart_slice))| names(fanchart_slice) %in% c("Duration (in seconds)", "group", "treatment")]
bell_curve = bell_curve[,grep1("^(Q3|Q12|Q13)",names(bell_curve))| names(bell_curve) %in% c("Duration (in seconds)", "group", "treatment")]

```

```
# controlled_group = controlled_group[,!sapply(controlled_group, function(x) mean(is.na(x)))>0.9]

group_list <- list(
  controlled = controlled_group,
  errorbar = error_bar,
  shadedbar = shaded_bar,
  fanchartslice = fanchart_slice,
  bellcurve = bell_curve
)

skewed_group_list <- group_list[-1]
```

Screening Questions

! Time Duration

Variable: continuous

Method: check normality + kruskal-wallis test + post-hoc test

Result: $p < 0.05 \rightarrow$ we can reject the hypothesis

controlled group is significantly different from (error bar; shaded bar; fanchart slice; BUT NOT fanchartslice)

```
time_duration_table <- rawdata_value[,c("Duration (in seconds)", "group")]
time_duration_table$`Duration (in mins)` <- as.numeric(time_duration_table$`Duration (in seconds)`)/60

# report quantile information
summary(time_duration_table$`Duration (in mins)`)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  4.117  14.233  19.817  22.397  27.133  69.617
```

```
# test for normality --> It's definitely not normal distribution
# shapiro.test(time_duration_table$`Duration (in mins)`)

# run the kruskal-wallis test
kruskal_test <- kruskal.test(`Duration (in mins)` ~ group, data = time_duration_table)
print(kruskal_test)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  Duration (in mins) by group
## Kruskal-Wallis chi-squared = 30.533, df = 4, p-value = 3.811e-06
```

```
# run post-hoc test
dunn_test <- dunnTest(`Duration (in mins)` ~ group, data = time_duration_table, method = "bonferroni")
```

```
## Warning: group was coerced to a factor.
```

```
print(dunn_test)
```

```
## Dunn (1964) Kruskal-Wallis multiple comparison
```

```
##      p-values adjusted with the Bonferroni method.
```

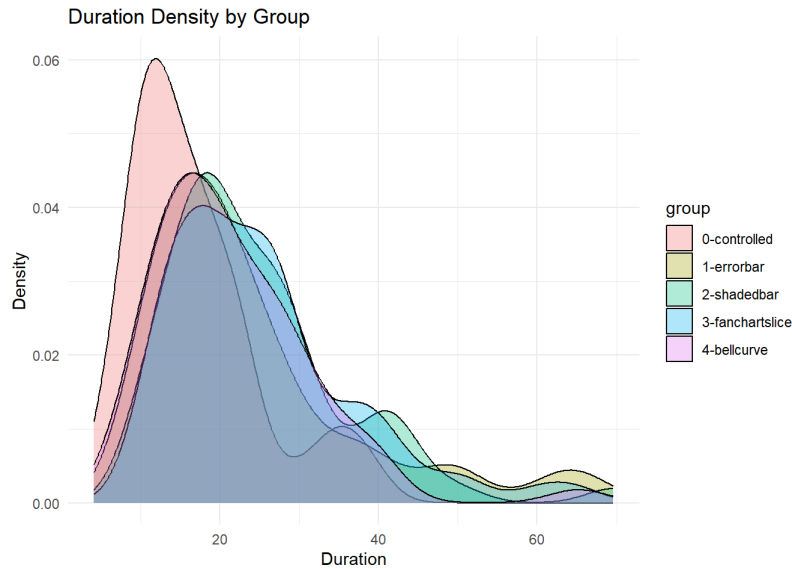
```
##           Comparison      Z      P.unadj      P.adj
## 1      0-controlled - 1-errorbar -3.5097005 4.486117e-04 4.486117e-03
## 2      0-controlled - 2-shadedbar -4.5684263 4.913997e-06 4.913997e-05
## 3      1-errorbar - 2-shadedbar -1.0161356 3.095648e-01 1.000000e+00
## 4      0-controlled - 3-fanchartslice -4.9558935 7.199866e-07 7.199866e-06
## 5      1-errorbar - 3-fanchartslice -1.3294386 1.837033e-01 1.000000e+00
## 6      2-shadedbar - 3-fanchartslice -0.2964099 7.669170e-01 1.000000e+00
## 7      0-controlled - 4-bellcurve -3.0646454 2.179282e-03 2.179282e-02
## 8      1-errorbar - 4-bellcurve  0.4690361 6.390438e-01 1.000000e+00
## 9      2-shadedbar - 4-bellcurve  1.4977050 1.342099e-01 1.000000e+00
## 10     3-fanchartslice - 4-bellcurve  1.8233127 6.825604e-02 6.825604e-01
```

Time duration summary table

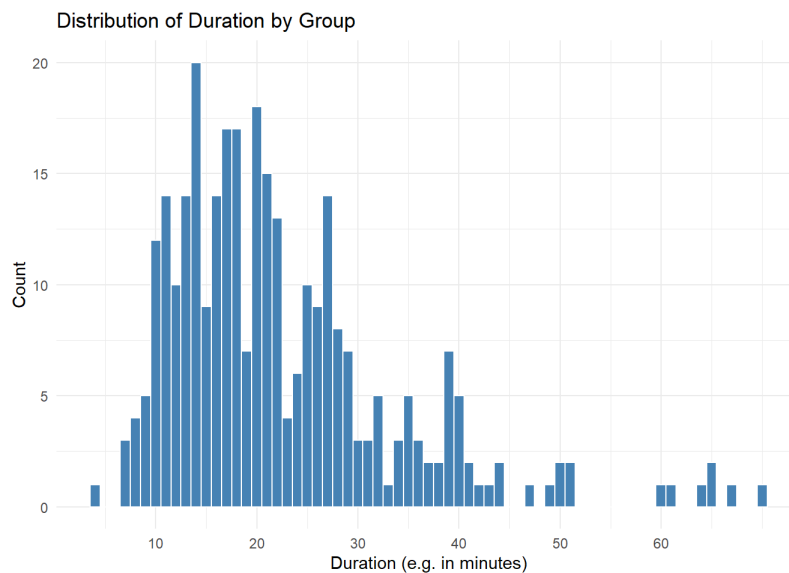
```
time_duration_summary <- time_duration_table %>%
  group_by(group) %>%
  summarise(mean_duration = mean(`Duration (in mins)`, na.rm = TRUE),
            sd_duration = sd(`Duration (in mins)`, na.rm = TRUE),
            .groups = 'drop')
```

Plot the distribution of different groups

```
# density plot
ggplot(time_duration_table, aes(x = `Duration (in mins)`, fill = group)) +
  geom_density(alpha = 0.3) +
  # facet_wrap(~ group, scales = "free") +
  labs(title = "Duration Density by Group",
       x = "Duration",
       y = "Density") +
  theme_minimal()
```

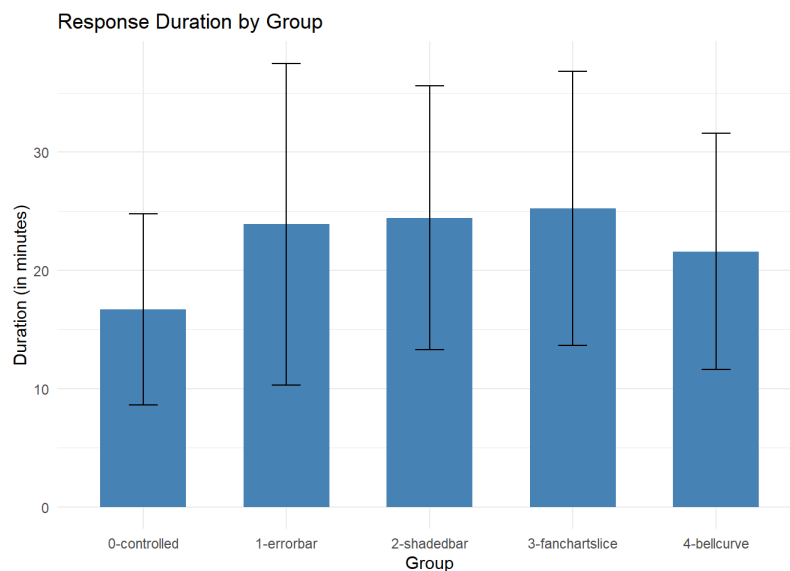


```
# histogram
ggplot(time_duration_table, aes(x = `Duration (in mins)`) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  # facet_wrap(~ group) +
  labs(title = "Distribution of Duration by Group",
       x = "Duration (e.g. in minutes)",
       y = "Count") +
  scale_x_continuous(breaks = seq(0, max(time_duration_table$`Duration (in mins)`), na.rm = TRUE), by = 10)) +
  theme_minimal()
```



Plot time duration bar chart. It's not valid as the data is not normal distribution.

```
ggplot(time_duration_summary, aes(x = group, y = mean_duration)) +
  geom_bar(stat = "identity", fill = "steelblue", width = 0.6) +
  geom_errorbar(aes(ymin = mean_duration - sd_duration, ymax = mean_duration + sd_duration), width = 0.2) +
  labs(title = "Response Duration by Group",
       y = "Duration (in minutes)",
       x = "Group") +
  theme_minimal()
```



Q3.2 Age

What is your age?

Variable: continuous

Method: kruskal-wallis test

Result: Fail to reject the null hypothesis

```
age_table <- rawdata_value[, c("Q3.2", "group")]
age_table$`Q3.2` <- as.numeric(gsub("[^0-9.]", "", age_table$`Q3.2`))

# test for normality
# shapiro.test(age_table$`Q3.2`)
# bartlett.test(`Q3.2` ~ group, data = age_table)

# run the kruskal-wallis test
kruskal_test <- kruskal.test(`Q3.2` ~ group, data = age_table)
print(kruskal_test)
```

```
##
## Kruskal-Wallis rank sum test
##
## data: Q3.2 by group
## Kruskal-Wallis chi-squared = 3.0501, df = 4, p-value = 0.5495
```

```
age_summary <- age_table %>%
  group_by(group) %>%
  summarise(mean_age = mean(`Q3.2`, na.rm = TRUE),
            sd_age = sd(`Q3.2`, na.rm = TRUE),
            .groups = 'drop')
```

Q3.3 Gender

Which of the following best describes your gender identity?

Variable: categorical

Method: contingency Table + chi-squared test

Result: Fail to reject the null hypothesis

```
q3_3_table <- rawdata_value[, c("Q3.3", "group")]
# q3_3_table$Q3.3 <- recode(q3_3_table$Q3.3,
#                           "1" = "Male",
#                           "2" = "Female",
#                           .default = "Other")
q3_3_contingency <- table(q3_3_table$Q3.3, q3_3_table$group)
print(q3_3_contingency)
```

```
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1           34         28         29           35          28
## 2            28         29          31           31          33
## 3              0          1          1            0           0
## 4              0          1          0            0           0
```

```
# Chi-squared test
chisq.test(q3_3_contingency)
```

```
## Warning in chisq.test(q3_3_contingency): Chi-squared approximation may be
## incorrect
```

```
##
## Pearson's Chi-squared test
##
## data: q3_3_contingency
## X-squared = 8.7585, df = 12, p-value = 0.7234
```

Calculate the gender percentage

```
# percentage of male
male_per <- q3_3_contingency[1,]
gender_group_total <- colSums(q3_3_contingency)
male_prop <- male_per/gender_group_total
```

Q3.4 Highest Education

What is your highest level of education achieved?

Variable: categorical with written insights from choice 9 Method: contingency Table + chi-squared test

Result: Fail to reject the null hypothesis

```
q3_4_table <- rawdata_value[rawdata_value$`Q3.4` != "9", c("Q3.4", "group")]

q3_4_contingency <- table(q3_4_table$Q3.4, q3_4_table$group)
# Chi-squared test
chisq.test(q3_4_contingency)
```

```
## Warning in chisq.test(q3_4_contingency): Chi-squared approximation may be
## incorrect
```

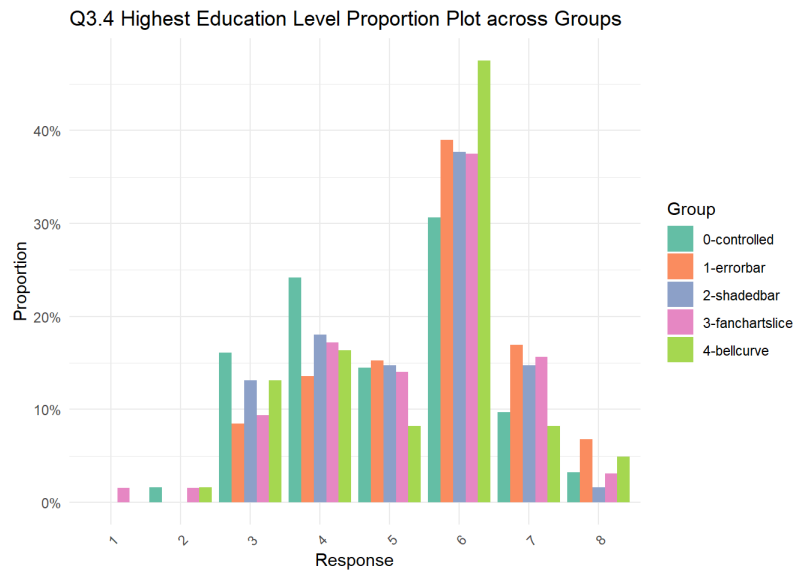
```
##
## Pearson's Chi-squared test
##
## data: q3_4_contingency
## X-squared = 18.836, df = 28, p-value = 0.9031
```

Highest Education Level proportion bar plot across different groups

```
q3_4_prop <- as.data.frame(q3_4_contingency)
# rename columns
names(q3_4_prop) <- c("Q3.4", "Group", "Count")

# calculate proportions within each group
q3_4_prop <- q3_4_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))

ggplot(q3_4_prop, aes(x = as.factor(`Q3.4`), y = Proportion, fill = Group)) +
  geom_bar(stat = "identity", position = "dodge") +
  # facet_wrap(~ Group) +
  labs(
    title = "Q3.4 Highest Education Level Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  # scale_x_discrete(labels = c(
  #   "1" = "No formal education",
  #   "2" = "Primary/Elementary",
  #   "3" = "Secondary School (GCSE)",
  #   "4" = "High School (A Level)",
  #   "5" = "Technical or Vocational",
  #   "6" = "Bachelors or equivalent degree level qualification",
  #   "7" = "Masters or equivalent higher degree level qualification",
  #   "8" = "PhD or equivalent doctoral level qualification"))+
  scale_fill_brewer(palette = "Set2")+
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



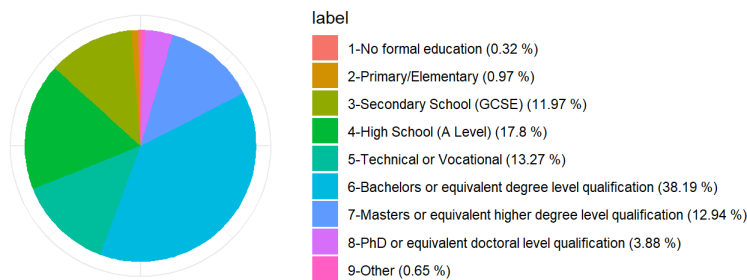
Plot pie chart including all participants

```
q3_4_table <- rawdata_value[, c("Q3.4", "group")]

# Q3.4 summary table
q3_4_summary <- q3_4_table %>%
  group_by(`Q3.4`) %>%
  summarise(count = n()) %>%
  mutate(proportion = count / sum(count),
         label =
           case_when(
             Q3.4 == "1" ~ "1-No formal education",
             Q3.4 == "2" ~ "2-Primary/Elementary",
             Q3.4 == "3" ~ "3-Secondary School (GCSE)",
             Q3.4 == "4" ~ "4-High School (A Level)",
             Q3.4 == "5" ~ "5-Technical or Vocational",
             Q3.4 == "6" ~ "6-Bachelors or equivalent degree level qualification",
             Q3.4 == "7" ~ "7-Masters or equivalent higher degree level qualification",
             Q3.4 == "8" ~ "8-PhD or equivalent doctoral level qualification",
             Q3.4 == "9" ~ "9-Other"
           ),
         label = paste0(label, " (", round(proportion * 100, 2), "%)")
  )

ggplot(q3_4_summary, aes(x = "", y = proportion, fill = label)) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  labs(title = "Q3.4 Highest Education Level Summary Pie Chart",
       x = "",
       y = "") +
  theme_minimal() +
  theme(axis.text.x = element_blank())
```


Q3.4 Highest Education Level Summary Pie Chart



Q3.5 Employee Status

What is your current profession or employment status?

Variable: categorical

Method: contingency Table + chi-squared test

Result: Fail to reject the null hypothesis

```
q3_5_table <- rawdata_value[rawdata_value$`Q3.5` != "6", c("Q3.5", "group")]
q3_5_contingency <- table(q3_5_table$Q3.5, q3_5_table$group)
# print(q3_5_contingency)

# Chi-squared test
chisq.test(q3_5_contingency)
```

```
## Warning in chisq.test(q3_5_contingency): Chi-squared approximation may be
## incorrect
```

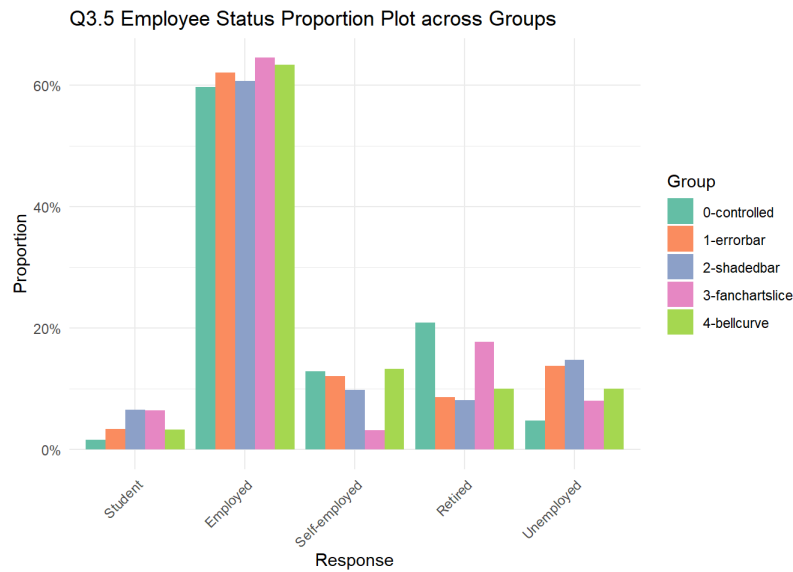
```
##
## Pearson's Chi-squared test
##
## data:  q3_5_contingency
## X-squared = 17.324, df = 16, p-value = 0.3649
```

Employee status proportion bar plot across different groups

```
q3_5_prop <- as.data.frame(q3_5_contingency)
# rename columns
names(q3_5_prop) <- c("Q3.5", "Group", "Count")

# calculate proportions within each group
q3_5_prop <- q3_5_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))

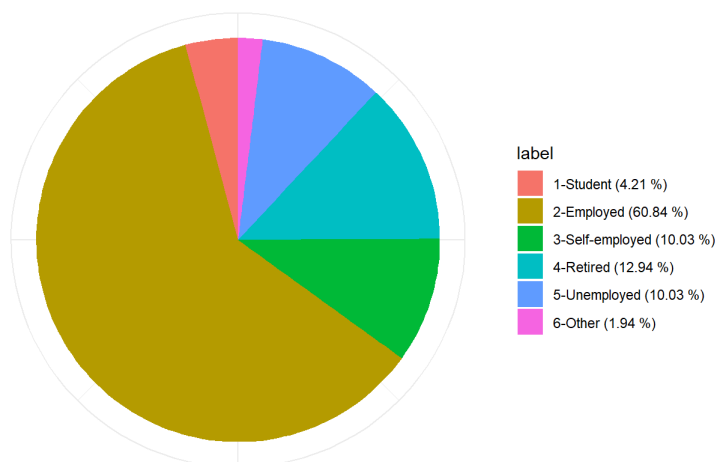
ggplot(q3_5_prop, aes(x = as.factor(`Q3.5`), y = Proportion, fill = Group)) +
  geom_bar(stat = "identity", position = "dodge") +
  # facet_wrap(~ Group) +
  labs(
    title = "Q3.5 Employee Status Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_x_discrete(labels = c(
    "1" = "Student",
    "2" = "Employed",
    "3" = "Self-employed",
    "4" = "Retired",
    "5" = "Unemployed"
  )) +
  scale_fill_brewer(palette = "Set2") +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Plot pie chart including all participants

```
q3_5_table <- rawdata_value[, c("Q3.5", "group")]
q3_5_summary <- q3_5_table %>%
  group_by(`Q3.5`) %>%
  summarise(count = n()) %>%
  mutate(proportion = count / sum(count),
         label =
           case_when(
             Q3.5 == "1" ~ "1-Student",
             Q3.5 == "2" ~ "2-Employed",
             Q3.5 == "3" ~ "3-Self-employed",
             Q3.5 == "4" ~ "4-Retired",
             Q3.5 == "5" ~ "5-Unemployed",
             Q3.5 == "6" ~ "6-Other"
           ),
         label = paste0(label, " (", round(proportion * 100, 2), "%)")
  )
ggplot(q3_5_summary, aes(x = "", y = proportion, fill = label)) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  labs(title = "Q3.5 Employee Status Summary Pie Chart",
       x = "",
       y = "") +
  theme_minimal() +
  theme(axis.text.x = element_blank())
```

Q3.5 Employee Status Summary Pie Chart



Q3.6 Economics Education

In which, if any, have you ever studied economics? (Select all that apply)

Variable: categorical

Method: convert to list and count + chi-squared test

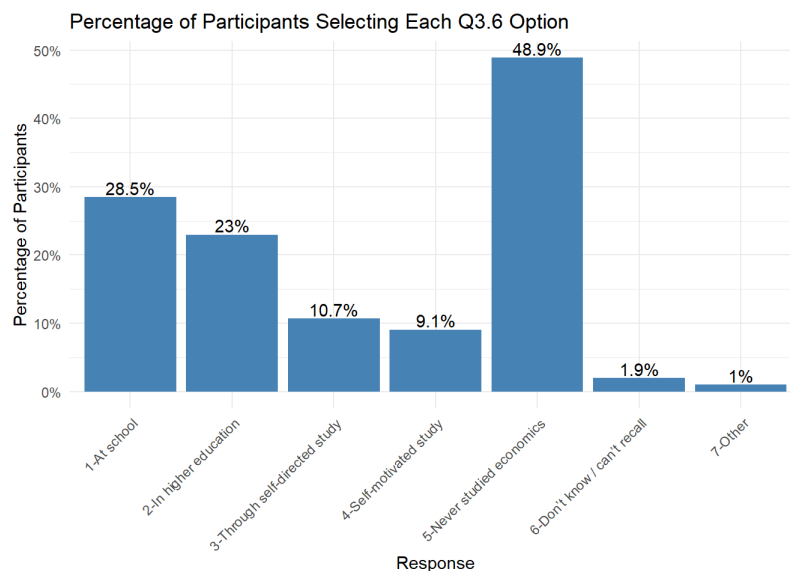
Result: Fail to reject the null hypothesis

Plot percentage bar chart including all participants

```
q3_6_table <- rawdata_value[, c("Q3.6", "group")]
q3_6_table$Q3.6 <- lapply(q3_6_table$Q3.6, function(x) as.list(as.numeric(strsplit(x, ",")[[1]])))

# count the number of occurrences of each value
q3_6_summary <- q3_6_table %>%
  unnest(Q3.6) %>%
  group_by(`Q3.6`) %>%
  summarise(count = n()) %>%
  mutate(proportion = count / total_respondents,
         label =
           case_when(
             Q3.6 == "1" ~ "1-At school",
             Q3.6 == "2" ~ "2-In higher education",
             Q3.6 == "3" ~ "3-Through self-directed study",
             Q3.6 == "4" ~ "4-Self-motivated study",
             Q3.6 == "5" ~ "5-Never studied economics",
             Q3.6 == "6" ~ "6-Don't know / can't recall",
             Q3.6 == "7" ~ "7-Other"
           ))

# q3_6_summary$Q3.6 <- as.numeric(q3_6_summary$Q3.6)
ggplot(q3_6_summary, aes(x = factor(label), y = proportion)) +
  geom_col(fill = "steelblue") +
  geom_text(aes(label = paste0(round(proportion * 100, 1), "%"),
                 vjust = -0.2)) +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  labs(
    title = "Percentage of Participants Selecting Each Q3.6 Option",
    x = "Response",
    y = "Percentage of Participants"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
q3_6_table <- rawdata_value[, c("Q3.6", "group")]
q3_6_table$Q3.6 <- lapply(q3_6_table$Q3.6, function(x) as.list(as.numeric(strsplit(x, ",")[[1]])))
# count the number of occurrences of each value
q3_6_prop <- q3_6_table %>%
  unnest(Q3.6) %>%
  group_by(group, Q3.6) %>%
  summarise(count = n(), .groups = 'drop')
q3_6_prop$Q3.6 <- as.numeric(q3_6_prop$Q3.6)

# create a contingency table
q3_6_contingency <- xtabs(count ~ Q3.6 + group, data = q3_6_prop)
print(q3_6_contingency)
```

```
##      group
## Q3.6 0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1      18      19      13      17      21
## 2       8      14      17      20      12
## 3       4       9       8       7       5
## 4       4       7       5       8       4
## 5      37      27      29      30      28
## 6       2       0       1       1       2
## 7       0       2       0       0       1
```

```
# Chi-squared test
chisq.test(q3_6_contingency)
```

```
## Warning in chisq.test(q3_6_contingency): Chi-squared approximation may be
## incorrect
```

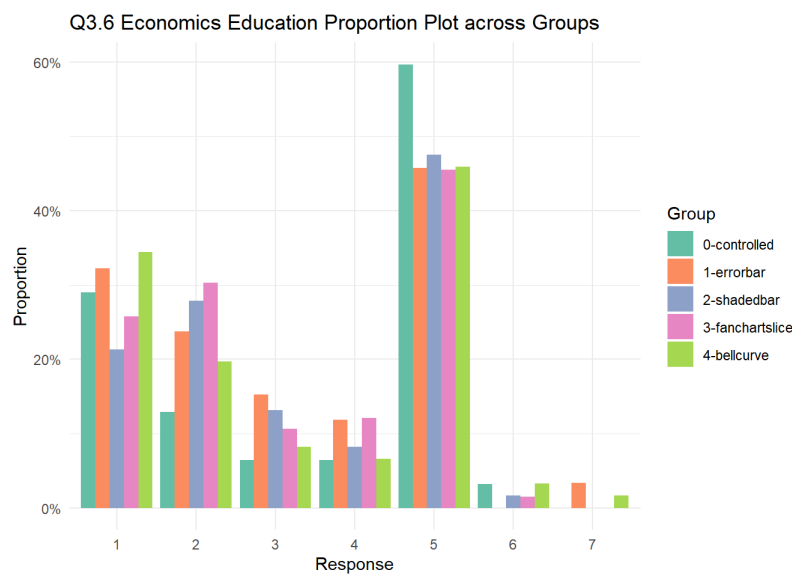
```
##
## Pearson's Chi-squared test
##
## data:  q3_6_contingency
## X-squared = 21.68, df = 24, p-value = 0.5983
```

```
# q3_6_test <- q3_6_prop %>%
#   group_by(`Q3.6`) %>%
#   summarise(total_count = sum(count), .groups = "drop")
```

Economics Education proportion bar plot across different groups

```
# convert contingency table to data frame # count 0 as well
q3_6_prop <- as.data.frame(q3_6_contingency)
# calculate proportions within each group
q3_6_prop <- q3_6_prop %>%
  left_join(group_total, by = "group") %>%
  mutate(Proportion = Freq / n_respondents)

ggplot(q3_6_prop, aes(x = factor(Q3.6), y = Proportion, fill = factor(group))) +
  geom_col(position = "dodge") +
  labs(
    title = "Q3.6 Economics Education Proportion Plot across Groups",
    x = "Response",
    y = "Proportion",
    fill = "Group"
  ) +
  scale_fill_brewer(palette = "Set2") +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal()
```



Q3.7 Financial Activities

Which of the following financial activities do you actively participate in? (Select all that apply) - Selected Choice Fixed Income (e.g., government bonds, corporate bonds)

Variable: categorical

Method: convert to list and count + chi-squared test

Result: Fail to reject the null hypothesis

```
q3_7_table <- rawdata_value[, c("Q3.7", "group")]
q3_7_table$Q3.7 <- lapply(q3_7_table$Q3.7, function(x) as.list(as.numeric(strsplit(x, ",")[[1]])))
# count the number of occurrences of each value in each group
q3_7_prop <- q3_7_table %>%
  unnest(Q3.7) %>%
  group_by(group, `Q3.7`) %>%
  summarise(count = n(), .groups = 'drop')
q3_7_prop$Q3.7 <- as.numeric(q3_7_prop$Q3.7)

# create a contingency table
q3_7_contingency <- xtabs(count ~ `Q3.7` + group, data = q3_7_prop)
print(q3_7_contingency)
```

```
##      group
## Q3.7 0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 11          54          54          54          63          53
## 12          29          33          27          36          25
## 13           9          17           8           6          10
## 14          32          34          30          27          32
## 15           3          12           5           9           4
## 16           1           7           3           7           8
## 17          11          15          13          12           9
## 18           5           5           4           2           6
## 19           2           2           0           0           0
```

```
# Chi-squared test
chisq.test(q3_7_contingency)
```

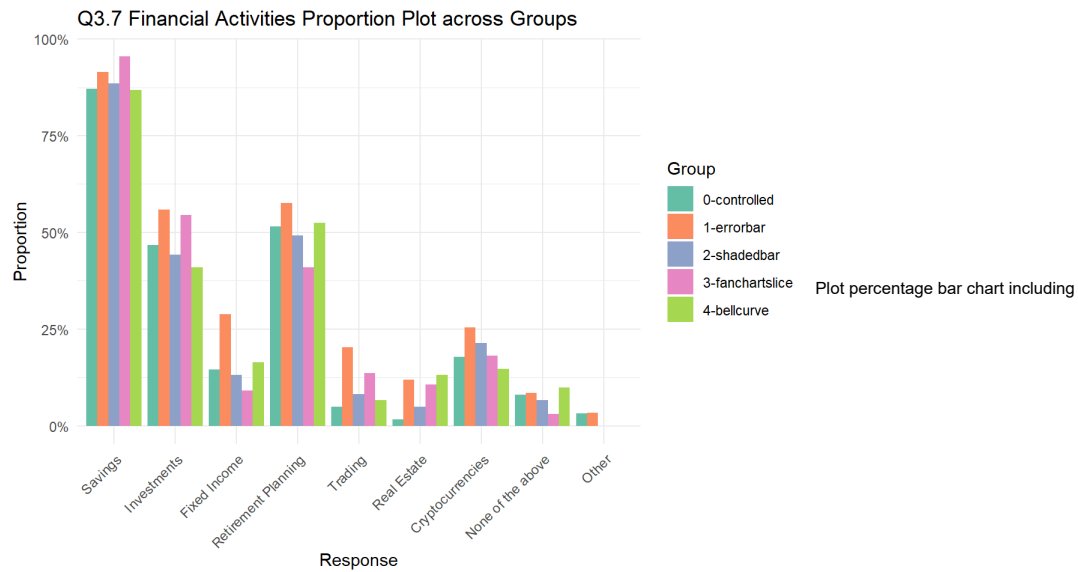
```
## Warning in chisq.test(q3_7_contingency): Chi-squared approximation may be
## incorrect
```

```
##
## Pearson's Chi-squared test
##
## data:  q3_7_contingency
## X-squared = 30.965, df = 32, p-value = 0.5188
```

Plot financial activities proportion bar plot across different groups

```
# convert contingency table to data frame
q3_7_prop <- as.data.frame(q3_7_contingency) %>%
  left_join(group_total, by = "group") %>%
  mutate(Proportion = Freq / n_respondents)

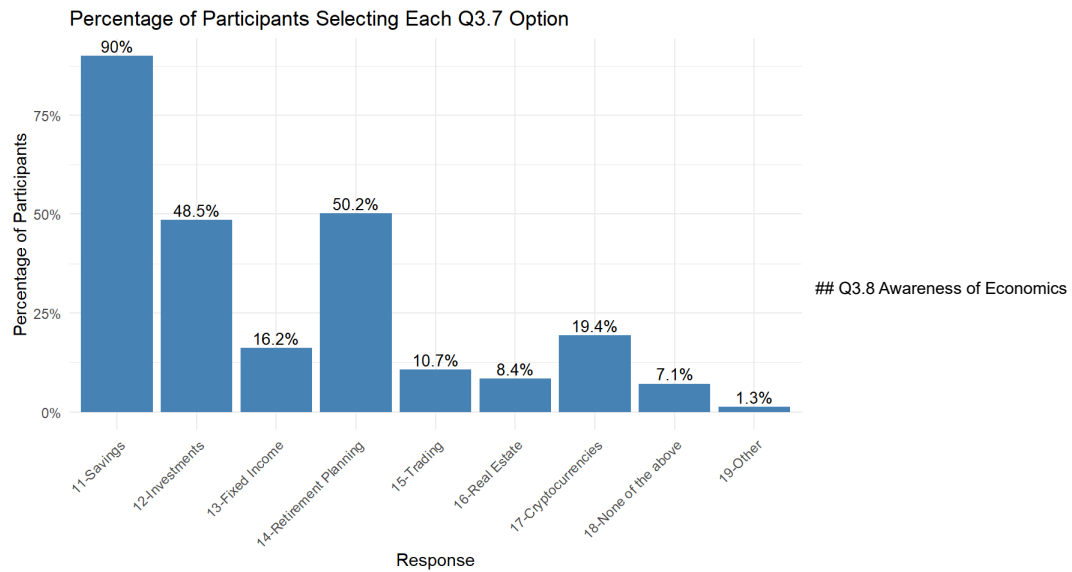
ggplot(q3_7_prop, aes(x = factor(`Q3.7`), y = Proportion, fill = factor(group))) +
  geom_col(position = "dodge") +
  labs(
    title = "Q3.7 Financial Activities Proportion Plot across Groups",
    x = "Response",
    y = "Proportion",
    fill = "Group"
  ) +
  scale_x_discrete(labels = c(
    "11" = "Savings",
    "12" = "Investments",
    "13" = "Fixed Income",
    "14" = "Retirement Planning",
    "15" = "Trading",
    "16" = "Real Estate",
    "17" = "Cryptocurrencies",
    "18" = "None of the above",
    "19" = "Other"
  )) +
  scale_fill_brewer(palette = "Set2") +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



all participants

```
q3_7_summary <- q3_7_prop %>%
  group_by(`Q3.7`) %>%
  summarise(total_count = sum(Freq), .groups = "drop") %>%
  mutate(proportion = total_count / total_respondents,
    label =
      case_when(
        `Q3.7` == "11" ~ "11-Savings",
        `Q3.7` == "12" ~ "12-Investments",
        `Q3.7` == "13" ~ "13-Fixed Income",
        `Q3.7` == "14" ~ "14-Retirement Planning",
        `Q3.7` == "15" ~ "15-Trading",
        `Q3.7` == "16" ~ "16-Real Estate",
        `Q3.7` == "17" ~ "17-Cryptocurrencies",
        `Q3.7` == "18" ~ "18-None of the above",
        `Q3.7` == "19" ~ "19-Other"
      )
  )

ggplot(q3_7_summary, aes(x = factor(label), y = proportion)) +
  geom_col(fill = "steelblue") +
  geom_text(aes(label = paste0(round(proportion * 100, 1), "%")),
    vjust = -0.3, size = 3.5) +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  labs(
    title = "Percentage of Participants Selecting Each Q3.7 Option",
    x = "Response",
    y = "Percentage of Participants"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



How frequently, if at all, do you read/watch/listen to news stories related to economics or the economy?

Variable: categorical

Method: contingency table + chi-squared test

Result: Fail to reject the null hypothesis

```
q3_8_table <- rawdata_value[, c("Q3.8", "group")]
```

```
q3_8_contingency <- table(q3_8_table$Q3.8, q3_8_table$group)
print(q3_8_contingency)
```

```
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1           2           0           3           0           4
## 2           11          11          10          10          8
## 3           9           9           7          12         10
## 4           24          15          16          16         23
## 5           14          18          19          18         11
## 6           2           6           6           9           5
## 7           0           0           0           1           0
```

```
# Chi-squared test
# chisq.test(q3_8_contingency)
chisq.test(q3_8_contingency[-nrow(q3_8_contingency), ])
```

```
## Warning in chisq.test(q3_8_contingency[-nrow(q3_8_contingency), ]): Chi-squared
## approximation may be incorrect
```

```
##
## Pearson's Chi-squared test
##
## data:  q3_8_contingency[-nrow(q3_8_contingency), ]
## X-squared = 20.071, df = 20, p-value = 0.4535
```

Awareness of Economics proportion bar plot across different groups

```

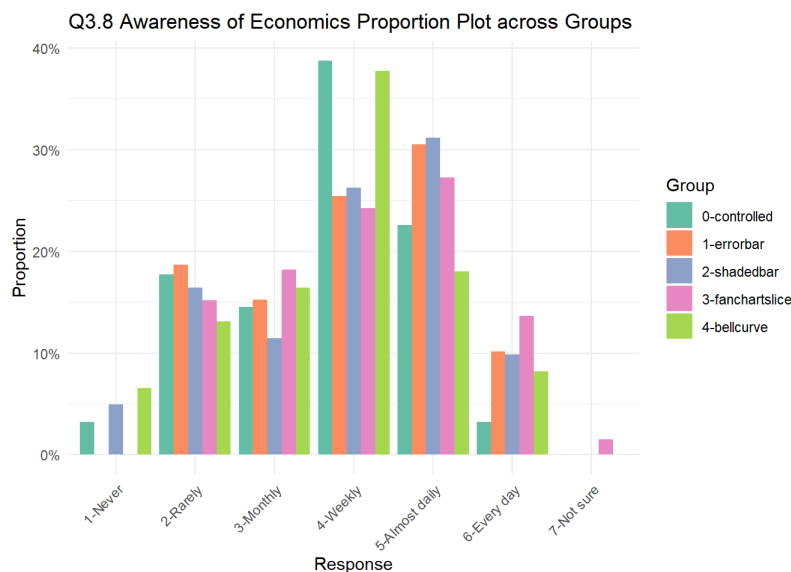
q3_8_prop <- as.data.frame(q3_8_contingency)
# rename columns
names(q3_8_prop) <- c("Q3.8", "Group", "Count")

# add a labelled column
q3_8_prop <- q3_8_prop %>%
  mutate(Q3.8_label = case_when(
    Q3.8 == "1" ~ "1-Never",
    Q3.8 == "2" ~ "2-Rarely",
    Q3.8 == "3" ~ "3-Monthly",
    Q3.8 == "4" ~ "4-Weekly",
    Q3.8 == "5" ~ "5-Almost daily",
    Q3.8 == "6" ~ "6-Every day",
    Q3.8 == "7" ~ "7-Not sure"
  ))

# calculate proportions within each group
q3_8_prop <- q3_8_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))

ggplot(q3_8_prop, aes(x = as.factor(`Q3.8_label`), y = Proportion, fill = Group)) +
  geom_col(position = "dodge") +
  # facet_wrap(~ Group) +
  labs(
    title = "Q3.8 Awareness of Economics Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2") +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```



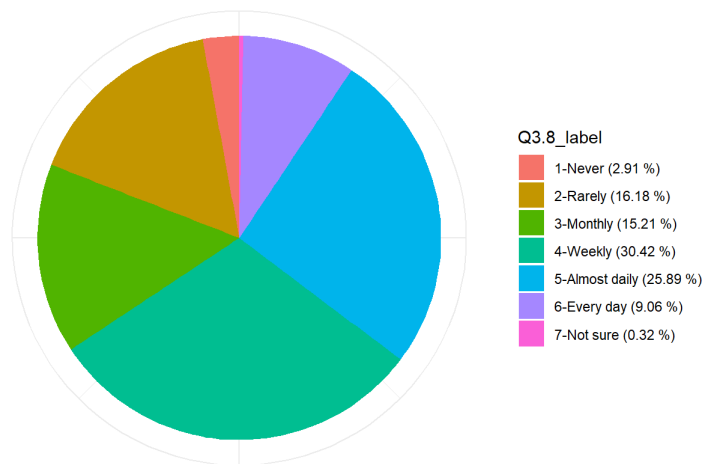
Plot pie chart including all participants

```

q3_8_summary <- q3_8_prop %>%
  group_by(`Q3.8_label`) %>%
  summarise(total_count = sum(Count), .groups = "drop") %>%
  mutate(proportion = total_count / sum(total_count),
    `Q3.8_label` = paste0(`Q3.8_label`, " (", round(proportion * 100, 2), "%)")
  )
ggplot(q3_8_summary, aes(x = "", y = proportion, fill = `Q3.8_label`)) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  labs(title = "Q3.8 Awareness of Economics Summary Pie Chart",
    x = "",
    y = "") +
  theme_minimal() +
  theme(axis.text.x = element_blank())

```


Q3.8 Awareness of Economics Summary Pie Chart



Q3.9 Inflation Knowledge

To the best of your knowledge, which option most accurately describes what inflation is?

Definitive correct answer

Variable: categorical

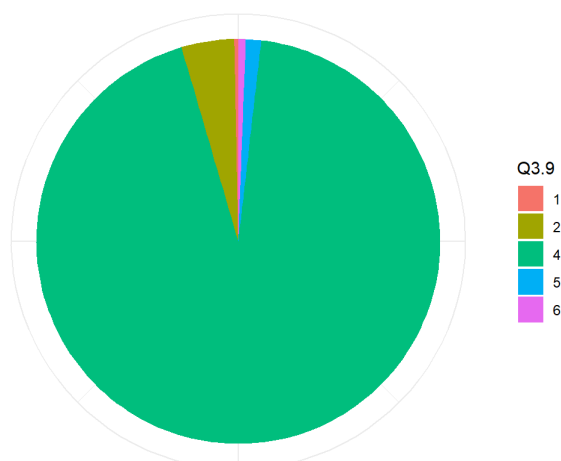
Method: descriptive analysis

Respondents were filtered based on whether if they can understand what inflation is.

```
q3_9_table <- rawdata_q3_9[, c("Q3.9", "group")]
q3_9_summary <- q3_9_table %>%
  group_by(`Q3.9`) %>%
  summarise(count = n()) %>%
  mutate(proportion = count / sum(count))

ggplot(q3_9_summary, aes(x = "", y = proportion, fill = `Q3.9`)) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  labs(title = "Q3.9 Inflation Knowledge Summary Pie Chart",
       x = "",
       y = "") +
  theme_minimal() +
  theme(axis.text.x = element_blank())
```

Q3.9 Inflation Knowledge Summary Pie Chart



Q3.11 Trust of Official Data

How much do you trust official inflation forecasts (For instance, the forecasts released by the Bank of England)?

ascending value-increasing trust + exclude 6- Unsure / no opinion

Variable: treat as continuous

Method: Kruskal-Wallis test

Result: Fail to reject the null hypothesis

```
q3_11_table <- rawdata_value[rawdata_value$`Q3.11` != "6", c("Q3.11", "group")]
q3_11_table$Q3.11 <- as.numeric(q3_11_table$`Q3.11`)
# check normality --> Not normally distributed
# shapiro.test(q3_11_table$Q3.11)
# bartlett.test(`Q3.11` ~ group, data = q3_11_table)

# run kruskal test
kruskal_test <- kruskal.test(Q3.11 ~ group, data = q3_11_table)
print(kruskal_test)
```

```
##
## Kruskal-Wallis rank sum test
##
## data: Q3.11 by group
## Kruskal-Wallis chi-squared = 3.4585, df = 4, p-value = 0.4842
```

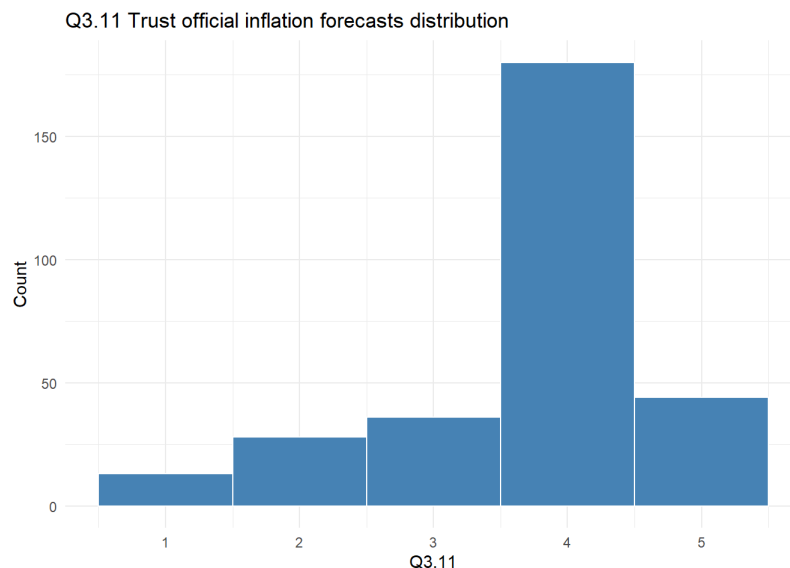
```
q3_11_summary <- q3_11_table %>%
  group_by(group) %>%
  summarise(q3_11_mean = mean(as.numeric(Q3.11), na.rm = TRUE),
            q3_11_sd = sd(as.numeric(Q3.11), na.rm = TRUE),
            .groups = 'drop')

# report quantile information
summary(q3_11_table$Q3.11)
```

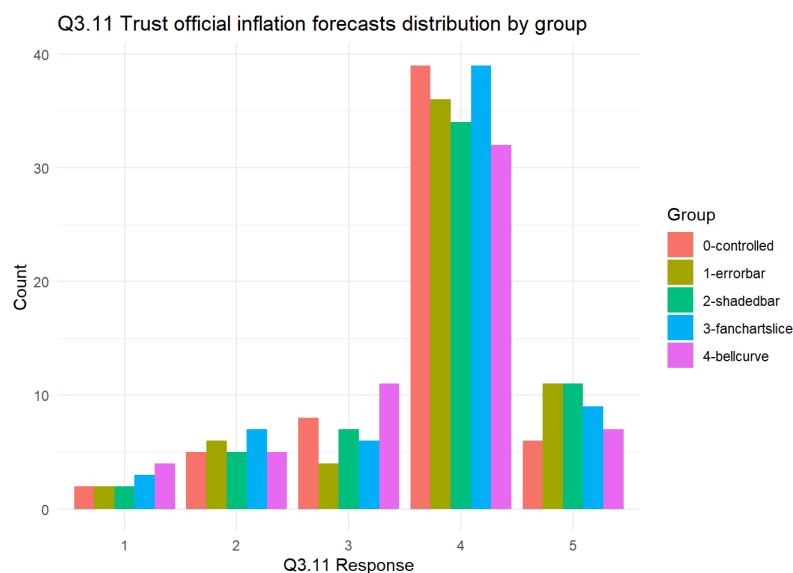
```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  1.000   3.000   4.000   3.711   4.000   5.000
```

Plot the distribution of different groups

```
# histogram including all participants
ggplot(q3_11_table, aes(x = Q3.11)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  # facet_wrap(~ group) +
  labs(title = "Q3.11 Trust official inflation forecasts distribution",
       x = "Q3.11",
       y = "Count") +
  scale_x_continuous(breaks = seq(0, max(q3_11_table$Q3.11, na.rm = TRUE), by = 1)) +
  theme_minimal()
```



```
# Grouped bar-style histogram
ggplot(q3_11_table, aes(x = factor(Q3.11), fill = factor(group))) +
  geom_bar(position = "dodge") +
  labs(
    title = "Q3.11 Trust official inflation forecasts distribution by group",
    x = "Q3.11 Response",
    y = "Count",
    fill = "Group"
  ) +
  theme_minimal()
```



Q3.12 Frequency of Checking Current Inflation

How often do you check the current value of inflation? Variable: categorical

Method: contingency table + chi-squared test

Result: Fail to reject the null hypothesis

```
# Initial Check
summary(as.numeric(rawdata_value$Q3.12))
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1.000  2.000   2.000   2.417  3.000   4.000
```

```
q3_12_table <- rawdata_value[, c("Q3.12", "group")]
q3_12_contingency <- table(q3_12_table$Q3.12, q3_12_table$group)
print(q3_12_contingency)
```

```
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
##      1           5         4           8           7           6
##      2           35        25          30          21          28
##      3           17        25          21          32          26
##      4            5         5           2           6           1
```

```
# Chi-squared test
chisq.test(q3_12_contingency)
```

```
## Warning in chisq.test(q3_12_contingency): Chi-squared approximation may be
## incorrect
```

```
##
##      Pearson's Chi-squared test
##
## data:  q3_12_contingency
## X-squared = 15.166, df = 12, p-value = 0.2325
```

Frequency of Checking Current Inflation proportion bar plot across different groups

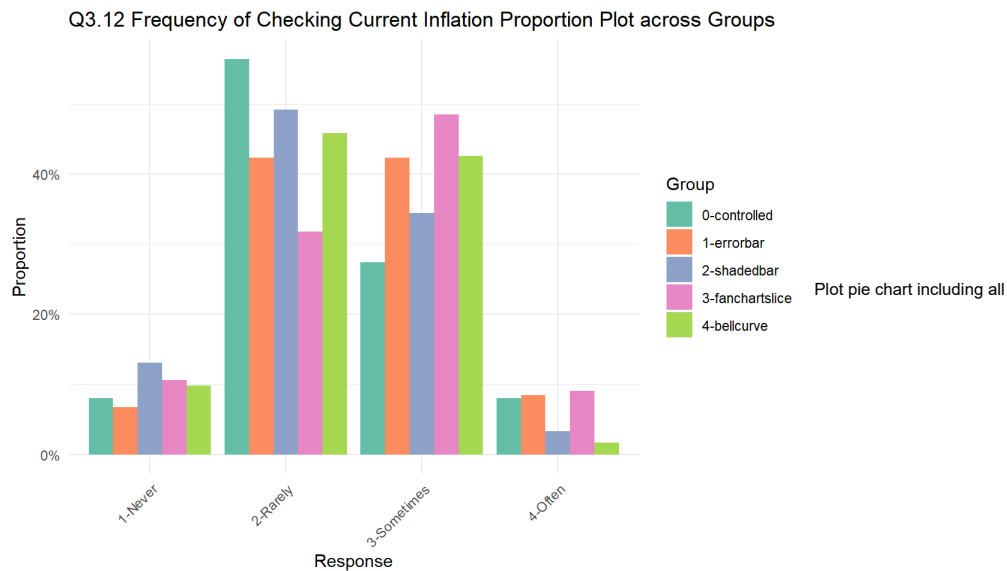
```

q3_12_prop <- as.data.frame(q3_12_contingency)
# rename columns
names(q3_12_prop) <- c("Q3.12", "Group", "Count")

# add a labelled column
q3_12_prop <- q3_12_prop %>%
  mutate(Q3.12_label = case_when(
    Q3.12 == "1" ~ "1-Never",
    Q3.12 == "2" ~ "2-Rarely",
    Q3.12 == "3" ~ "3-Sometimes",
    Q3.12 == "4" ~ "4-Often"
  ))
# calculate proportions within each group
q3_12_prop <- q3_12_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))

ggplot(q3_12_prop, aes(x = as.factor(`Q3.12_label`), y = Proportion, fill = Group)) + geom_col(position = "dodge") +
  labs(
    title = "Q3.12 Frequency of Checking Current Inflation Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2") +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```



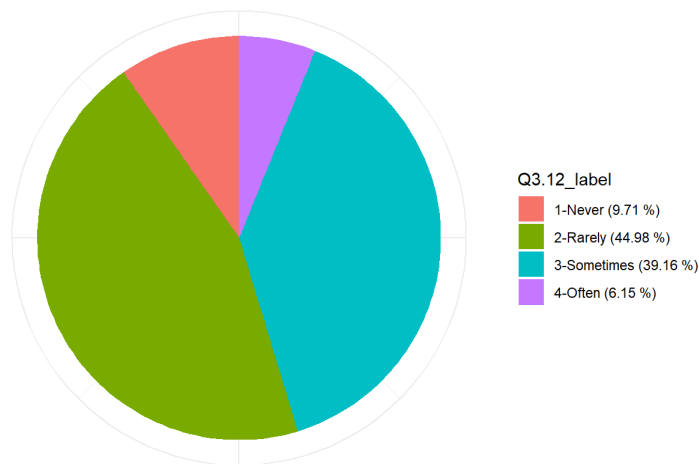
participants

```

q3_12_summary <- q3_12_prop %>%
  group_by(`Q3.12_label`) %>%
  summarise(total_count = sum(Count), .groups = "drop") %>%
  mutate(proportion = total_count / sum(total_count),
    `Q3.12_label` = paste0(`Q3.12_label`, " (", round(proportion * 100, 2), "%)"))
ggplot(q3_12_summary, aes(x = "", y = proportion, fill = `Q3.12_label`)) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  labs(title = "Q3.12 Frequency of Checking Current Inflation Summary Pie Chart",
    x = "",
    y = "") +
  theme_minimal() +
  theme(axis.text.x = element_blank())

```

Q3.12 Frequency of Checking Current Inflation Summary Pie Chart



Q3.13 Source of Current Inflation

Which source do you rely on to inform yourself about the current value of inflation?

Variable: categorical

Method: convert to list and count + chi-squared test

Result: Fail to reject the null hypothesis

```
q3_13_table <- rawdata_value[, c("Q3.13", "group")]
q3_13_table$Q3.13 <- lapply(q3_13_table$Q3.13, function(x) as.list(as.numeric(strsplit(x, ",")[[1]])))
# count the number of occurrences of each value
q3_13_prop <- q3_13_table %>%
  unnest(Q3.13) %>%
  group_by(group, `Q3.13`) %>%
  summarise(count = n(), .groups = 'drop')
q3_13_prop$Q3.13 <- as.numeric(q3_13_prop$Q3.13)
# create a contingency table
q3_13_contingency <- xtabs(count ~ `Q3.13` + group, data = q3_13_prop)
print(q3_13_contingency)
```

```
##      group
## Q3.13 0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1          43          41          33          44          36
## 2          16          16          15          13          16
## 3          24          28          26          29          21
## 4           5           7           8           4          10
## 5          15          15          22          21          17
## 6           6           6           7           4          11
## 7          15          13          15          19          13
## 8           5           8           5           5           1
## 9           2           0           0           0           0
```

```
# Chi-squared test
chisq.test(q3_13_contingency)
```

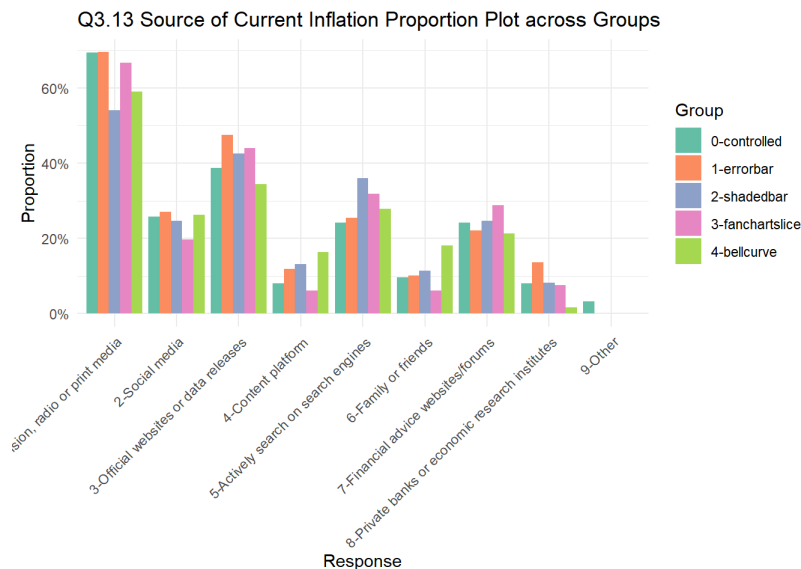
```
## Warning in chisq.test(q3_13_contingency): Chi-squared approximation may be
## incorrect
```

```
##
## Pearson's Chi-squared test
##
## data:  q3_13_contingency
## X-squared = 28.283, df = 32, p-value = 0.6553
```

Plot source of current inflation proportion bar plot across different groups

```
# convert contingency table to data frame
q3_13_prop <- as.data.frame(q3_13_contingency) %>%
  left_join(group_total, by = "group") %>%
  mutate(Proportion = Freq / n_respondents)

# add a Labelled column
q3_13_prop <- q3_13_prop %>%
  mutate(Q3.13_label = case_when(
    `Q3.13` == "1" ~ "1-Television, radio or print media",
    `Q3.13` == "2" ~ "2-Social media",
    `Q3.13` == "3" ~ "3-Official websites or data releases",
    `Q3.13` == "4" ~ "4-Content platform",
    `Q3.13` == "5" ~ "5-Actively search on search engines",
    `Q3.13` == "6" ~ "6-Family or friends",
    `Q3.13` == "7" ~ "7-Financial advice websites/forums",
    `Q3.13` == "8" ~ "8-Private banks or economic research institutes",
    `Q3.13` == "9" ~ "9-Other"
  ))
ggplot(q3_13_prop, aes(x = factor(`Q3.13_label`), y = Proportion, fill = group)) +
  geom_col(position = "dodge") +
  labs(
    title = "Q3.13 Source of Current Inflation Proportion Plot across Groups",
    x = "Response",
    y = "Proportion",
    fill = "Group"
  ) +
  scale_fill_brewer(palette = "Set2") +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

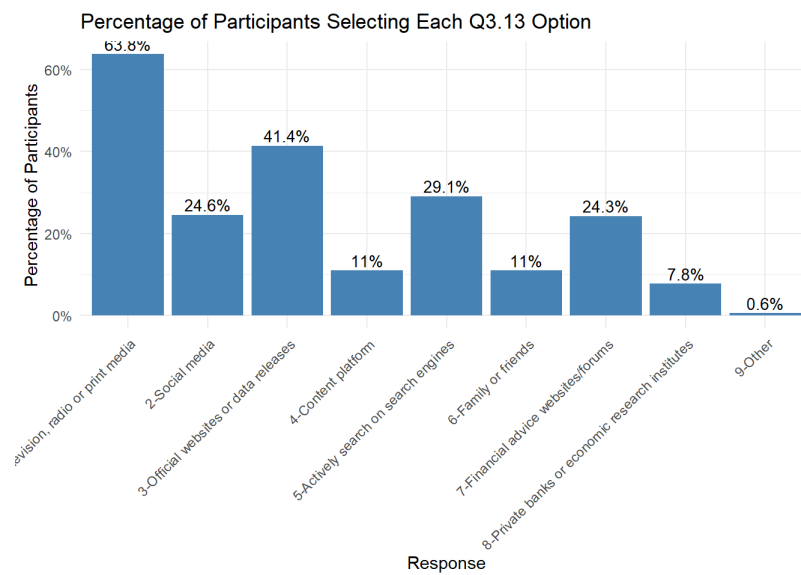


Plot percentage bar chart including

all participants

```
q3_13_summary <- q3_13_prop %>%
  group_by(`Q3.13_label`) %>%
  summarise(total_count = sum(Freq), .groups = "drop") %>%
  mutate(proportion = total_count / total_respondents)

ggplot(q3_13_summary, aes(x = factor(Q3.13_label), y = proportion)) +
  geom_col(fill = "steelblue") +
  geom_text(aes(label = paste0(round(proportion * 100, 1), "%"),
    vjust = -0.3, size = 3.5) +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  labs(
    title = "Percentage of Participants Selecting Each Q3.13 Option",
    x = "Response",
    y = "Percentage of Participants"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Q3.14 Impact of Current Inflation

To what extent do you feel that the current value of inflation is impacting your personal finances? Variable: categorical

Method: contingency table + chi-squared test

Result: Fail to reject the null hypothesis

```
# Initial Check
summary(as.numeric(rawdata_value$Q3.14))
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  1.000   2.000   3.000   2.816   3.000   4.000
```

```
q3_14_table <- rawdata_value[, c("Q3.14", "group")]
q3_14_contingency <- table(q3_14_table$Q3.14, q3_14_table$group)
print(q3_14_contingency)
```

```
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
##  1           3           0           4           1           1
##  2           23          17          18          21          20
##  3           25          32          30          29          25
##  4           11          10           9          15          15
```

```
# Chi-squared test
chisq.test(q3_14_contingency)
```

```
## Warning in chisq.test(q3_14_contingency): Chi-squared approximation may be
## incorrect
```

```
##
##      Pearson's Chi-squared test
##
## data:  q3_14_contingency
## X-squared = 10.903, df = 12, p-value = 0.5373
```

Impact of Current Inflation proportion bar plot across different groups

```

q3_14_prop <- as.data.frame(q3_14_contingency)
# rename columns
names(q3_14_prop) <- c("Q3.14", "Group", "Count")
# add a labelled column
q3_14_prop <- q3_14_prop %>%
  mutate(Q3.14_label = case_when(
    Q3.14 == "1" ~ "1-Not at all",
    Q3.14 == "2" ~ "2-Slightly",
    Q3.14 == "3" ~ "3-Moderately",
    Q3.14 == "4" ~ "4-Very significantly"
  ))
# calculate proportions within each group
q3_14_prop <- q3_14_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))
ggplot(q3_14_prop, aes(x = as.factor(`Q3.14_label`), y = Proportion, fill = Group)) + geom_col(position = "dodge") +
  labs(
    title = "Q3.14 Impact of Current Inflation Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2") +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```



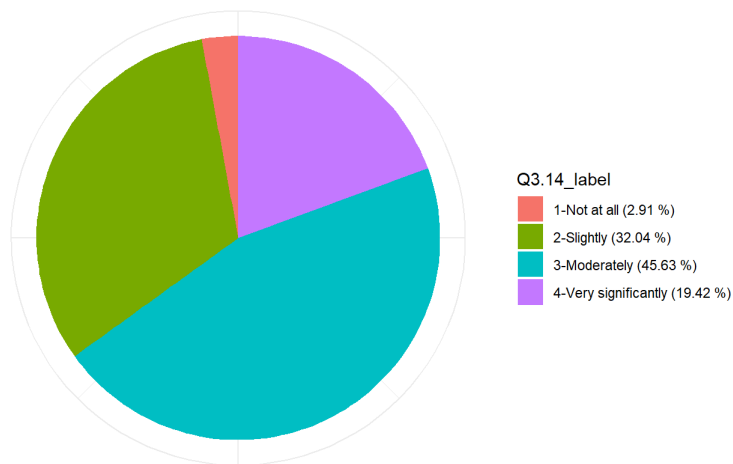
participants

```

q3_14_summary <- q3_14_prop %>%
  group_by(`Q3.14_label`) %>%
  summarise(total_count = sum(Count), .groups = "drop") %>%
  mutate(proportion = total_count / sum(total_count),
    `Q3.14_label` = paste0(`Q3.14_label`, " (", round(proportion * 100, 2), "%)")
ggplot(q3_14_summary, aes(x = "", y = proportion, fill = `Q3.14_label`)) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  labs(title = "Q3.14 Impact of Current Inflation Summary Pie Chart",
    x = "",
    y = "") +
  theme_minimal() +
  theme(axis.text.x = element_blank())

```


Q3.14 Impact of Current Inflation Summary Pie Chart



Q3.15 Response to Current Inflation

How much do you adjust or take action in response to the change in current value of inflation?

Variable: categorical

Method: contingency table + chi-squared test

Result: Fail to reject the null hypothesis

```
# Initial Check
summary(as.numeric(rawdata_value$Q3.15))
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1.00   2.00   3.00   2.57   3.00   4.00
```

```
q3_15_table <- rawdata_value[, c("Q3.15", "group")]
q3_15_contingency <- table(q3_15_table$Q3.15, q3_15_table$group)
print(q3_15_contingency)
```

```
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1           7           2           7           10           6
## 2          24          20          22          15          19
## 3          25          28          25          36          32
## 4           6           9           7           5           4
```

```
# Chi-squared test
chisq.test(q3_15_contingency)
```

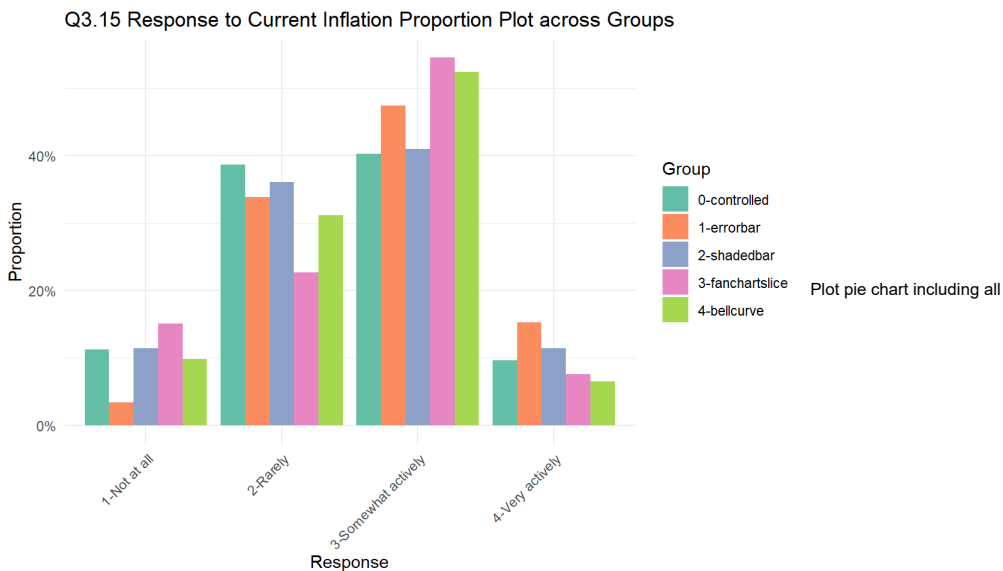
```
##
## Pearson's Chi-squared test
##
## data:  q3_15_contingency
## X-squared = 12.467, df = 12, p-value = 0.4089
```

Response to Current Inflation proportion bar plot across different groups

```

q3_15_prop <- as.data.frame(q3_15_contingency)
# rename columns
names(q3_15_prop) <- c("Q3.15", "Group", "Count")
# add a labelled column
q3_15_prop <- q3_15_prop %>%
  mutate(Q3.15_label = case_when(
    Q3.15 == "1" ~ "1-Not at all",
    Q3.15 == "2" ~ "2-Rarely",
    Q3.15 == "3" ~ "3-Somewhat actively",
    Q3.15 == "4" ~ "4-Very actively"
  ))
# calculate proportions within each group
q3_15_prop <- q3_15_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))
ggplot(q3_15_prop, aes(x = as.factor(`Q3.15_label`), y = Proportion, fill = Group)) + geom_col(position = "dodge") +
  labs(
    title = "Q3.15 Response to Current Inflation Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2")+
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal()+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```



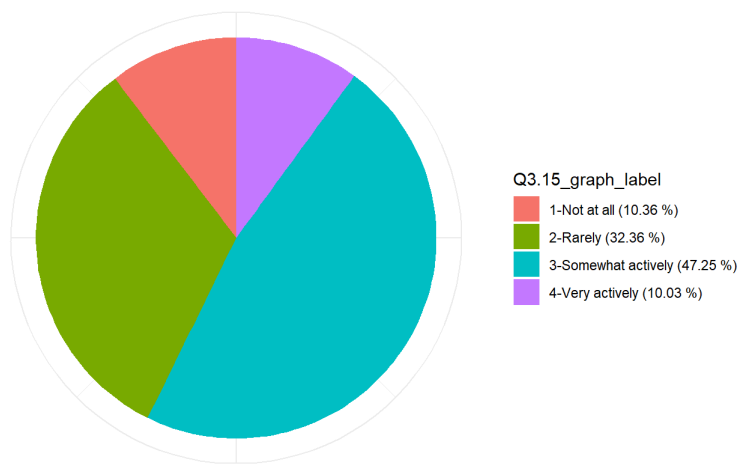
participants

```

q3_15_summary <- q3_15_prop %>%
  group_by(`Q3.15_label`) %>%
  summarise(total_count = sum(Count), .groups = "drop") %>%
  mutate(proportion = total_count / sum(total_count),
    `Q3.15_graph_label` = paste0(`Q3.15_label`, " (", round(proportion * 100, 2), "%)")
ggplot(q3_15_summary, aes(x = "", y = proportion, fill = `Q3.15_graph_label`)) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  labs(title = "Q3.15 Response to Current Inflation Summary Pie Chart",
    x = "",
    y = "") +
  theme_minimal() +
  theme(axis.text.x = element_blank())

```

Q3.15 Response to Current Inflation Summary Pie Chart



Q3.16 Frequency of Checking Inflation Forecasts

How often do you look at inflation forecasts (that is, information about future inflation)?

Variable: categorical

Method: contingency table + chi-squared test

Result: Fail to reject the null hypothesis

```
# Initial Check
summary(as.numeric(rawdata_value$Q3.16))
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1.000   2.000   2.000   2.181   3.000   4.000
```

```
q3_16_table <- rawdata_value[, c("Q3.16", "group")]
q3_16_contingency <- table(q3_16_table$Q3.16, q3_16_table$group)
print(q3_16_contingency)
```

```
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1           9         14          13          11          11
## 2          39         23          26          31          31
## 3          10         19          20          20          19
## 4           4          3           2           4           0
```

```
# Chi-squared test
chisq.test(q3_16_contingency)
```

```
## Warning in chisq.test(q3_16_contingency): Chi-squared approximation may be
## incorrect
```

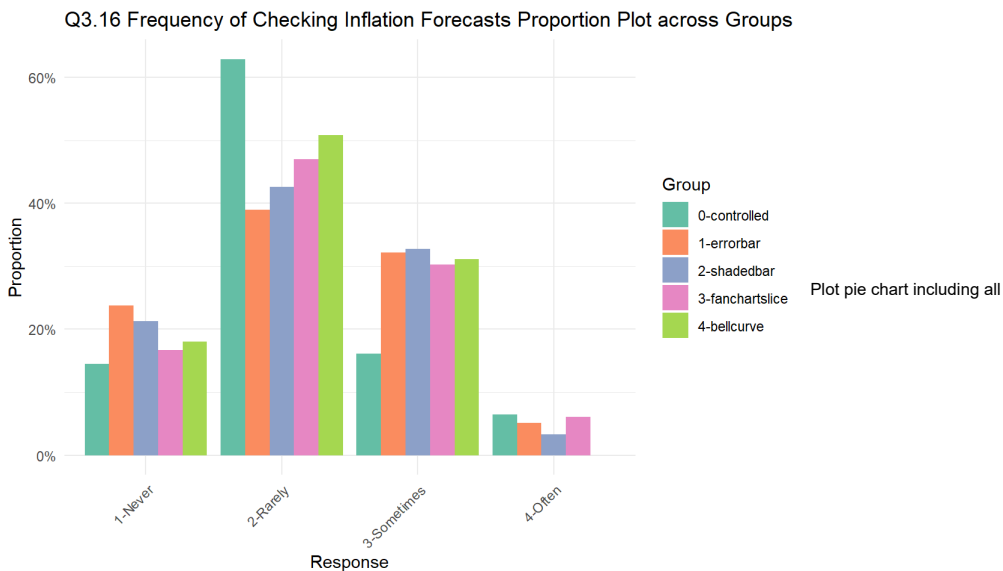
```
##
## Pearson's Chi-squared test
##
## data:  q3_16_contingency
## X-squared = 14.353, df = 12, p-value = 0.2787
```

Frequency of Checking Inflation Forecasts proportion bar plot across different groups

```

q3_16_prop <- as.data.frame(q3_16_contingency)
# rename columns
names(q3_16_prop) <- c("Q3.16", "Group", "Count")
# add a labelled column
q3_16_prop <- q3_16_prop %>%
  mutate(Q3.16_label = case_when(
    Q3.16 == "1" ~ "1-Never",
    Q3.16 == "2" ~ "2-Rarely",
    Q3.16 == "3" ~ "3-Sometimes",
    Q3.16 == "4" ~ "4-Often"
  ))
# calculate proportions within each group
q3_16_prop <- q3_16_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))
ggplot(q3_16_prop, aes(x = as.factor(`Q3.16_label`), y = Proportion, fill = Group)) + geom_col(position = "dodge") +
  labs(
    title = "Q3.16 Frequency of Checking Inflation Forecasts Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2")+
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal()+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```



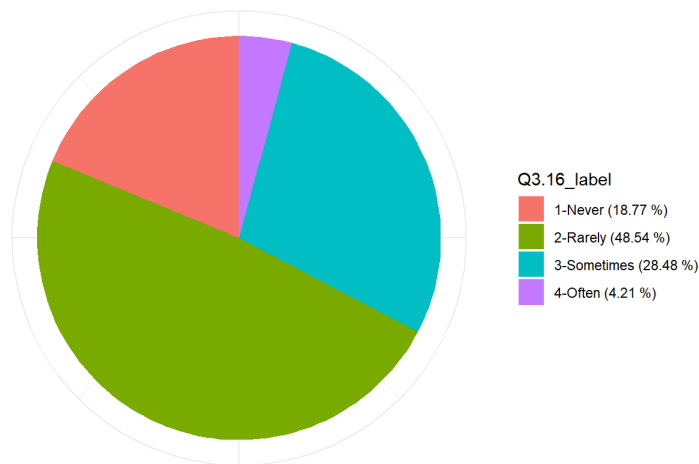
participants

```

q3_16_summary <- q3_16_prop %>%
  group_by(`Q3.16_label`) %>%
  summarise(total_count = sum(Count), .groups = "drop") %>%
  mutate(proportion = total_count / sum(total_count),
    `Q3.16_label` = paste0(`Q3.16_label`, " (", round(proportion * 100, 2), "%)"))
ggplot(q3_16_summary, aes(x = "", y = proportion, fill = `Q3.16_label`)) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  labs(title = "Q3.16 Frequency of Checking Inflation Forecasts Summary Pie Chart",
    x = "",
    y = "") +
  theme_minimal() +
  theme(axis.text.x = element_blank())

```

Q3.16 Frequency of Checking Inflation Forecasts Summary Pie Chart



Q3.17 Source of Inflation Forecasts

Which source do you rely on to obtain inflation forecasts (information about future inflation)?

Variable: categorical

Method: convert to list and count + chi-squared test

Result: Fail to reject the null hypothesis

```
q3_17_table <- rawdata_value[, c("Q3.17", "group")]
q3_17_table$Q3.17 <- lapply(q3_17_table$Q3.17, function(x) as.list(as.numeric(strsplit(x, ",")[[1]])))
# count the number of occurrences of each value
q3_17_prop <- q3_17_table %>%
  unnest(Q3.17) %>%
  group_by(group, `Q3.17`) %>%
  summarise(count = n(), .groups = 'drop')
q3_17_prop$Q3.17 <- as.numeric(q3_17_prop$Q3.17)
# create a contingency table
q3_17_contingency <- xtabs(count ~ `Q3.17` + group, data = q3_17_prop)
print(q3_17_contingency)
```

```
##      group
## Q3.17 0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1          38          30          33          35          33
## 2          17          14          14          11          13
## 3          16          22          23          29          22
## 4           5           9           8           4           7
## 5          14          16          18          17          13
## 6           4           4           7           4           8
## 7          15          16          16          20          16
## 8           5           9           4           6           2
## 9           2           0           0           0           0
```

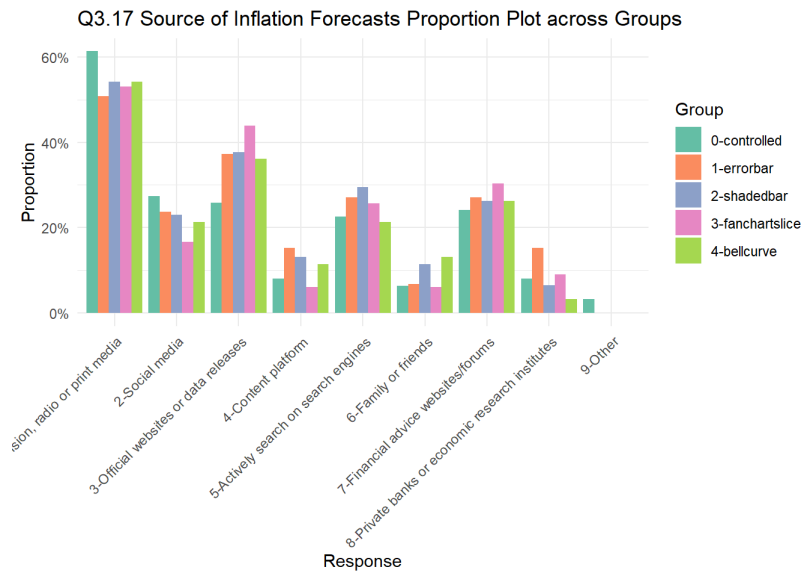
```
# Chi-squared test
chisq.test(q3_17_contingency)
```

```
## Warning in chisq.test(q3_17_contingency): Chi-squared approximation may be
## incorrect
```

```
##
## Pearson's Chi-squared test
##
## data: q3_17_contingency
## X-squared = 26.222, df = 32, p-value = 0.7537
```

Plot source of inflation forecasts proportion bar plot across different groups

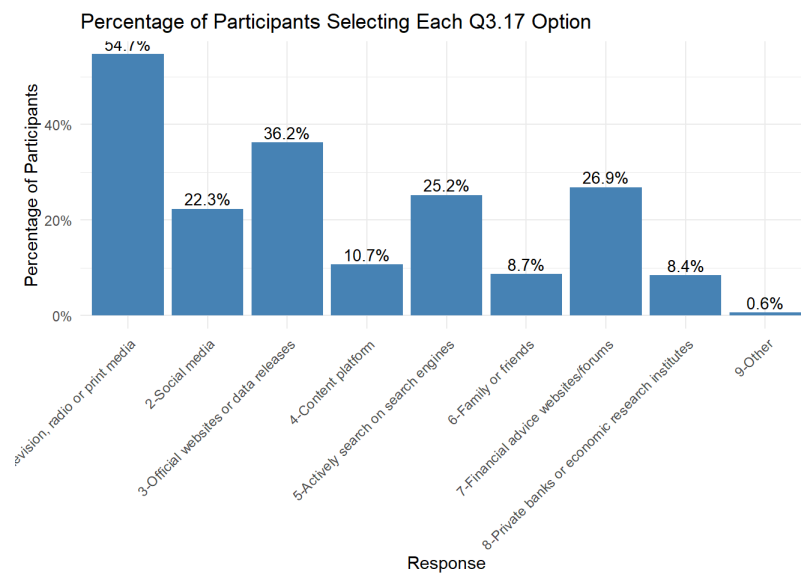
```
# convert contingency table to data frame
q3_17_prop <- as.data.frame(q3_17_contingency) %>%
  left_join(group_total, by = "group") %>%
  mutate(Proportion = Freq / n_respondents)
# add a labelled column
q3_17_prop <- q3_17_prop %>%
  mutate(Q3.17_label = case_when(
    `Q3.17` == "1" ~ "1-Television, radio or print media",
    `Q3.17` == "2" ~ "2-Social media",
    `Q3.17` == "3" ~ "3-Official websites or data releases",
    `Q3.17` == "4" ~ "4-Content platform",
    `Q3.17` == "5" ~ "5-Actively search on search engines",
    `Q3.17` == "6" ~ "6-Family or friends",
    `Q3.17` == "7" ~ "7-Financial advice websites/forums",
    `Q3.17` == "8" ~ "8-Private banks or economic research institutes",
    `Q3.17` == "9" ~ "9-Other"
  ))
ggplot(q3_17_prop, aes(x = factor(`Q3.17_label`), y = Proportion, fill = group)) +
  geom_col(position = "dodge") +
  labs(
    title = "Q3.17 Source of Inflation Forecasts Proportion Plot across Groups",
    x = "Response",
    y = "Proportion",
    fill = "Group"
  ) +
  scale_fill_brewer(palette = "Set2") +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Plot percentage bar chart including

all participants

```
q3_17_summary <- q3_17_prop %>%
  group_by(`Q3.17_label`) %>%
  summarise(total_count = sum(Freq), .groups = "drop") %>%
  mutate(proportion = total_count / total_respondents)
ggplot(q3_17_summary, aes(x = factor(Q3.17_label), y = proportion)) +
  geom_col(fill = "steelblue") +
  geom_text(aes(label = paste0(round(proportion * 100, 1), "%"),
    vjust = -0.3, size = 3.5) +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  labs(
    title = "Percentage of Participants Selecting Each Q3.17 Option",
    x = "Response",
    y = "Percentage of Participants"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Q3.18 Impact of Inflation Forecasts

To what extent do you feel that inflation forecasts (information about future inflation) impact your personal finances?

Variable: categorical

Method: contingency table + chi-squared test

Result: Fail to reject the null hypothesis

```
# Initial Check
summary(as.numeric(rawdata_value$Q3.18))
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 1.000   2.000   2.000   2.301   3.000   4.000
```

```
q3_18_table <- rawdata_value[, c("Q3.18", "group")]
q3_18_contingency <- table(q3_18_table$Q3.18, q3_18_table$group)
print(q3_18_contingency)
```

```
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1          13          10          15          9          9
## 2          25          28          20          30          26
## 3          21          15          21          21          21
## 4           3           6           5           6           5
```

```
# Chi-squared test
chisq.test(q3_18_contingency)
```

```
## Warning in chisq.test(q3_18_contingency): Chi-squared approximation may be
## incorrect
```

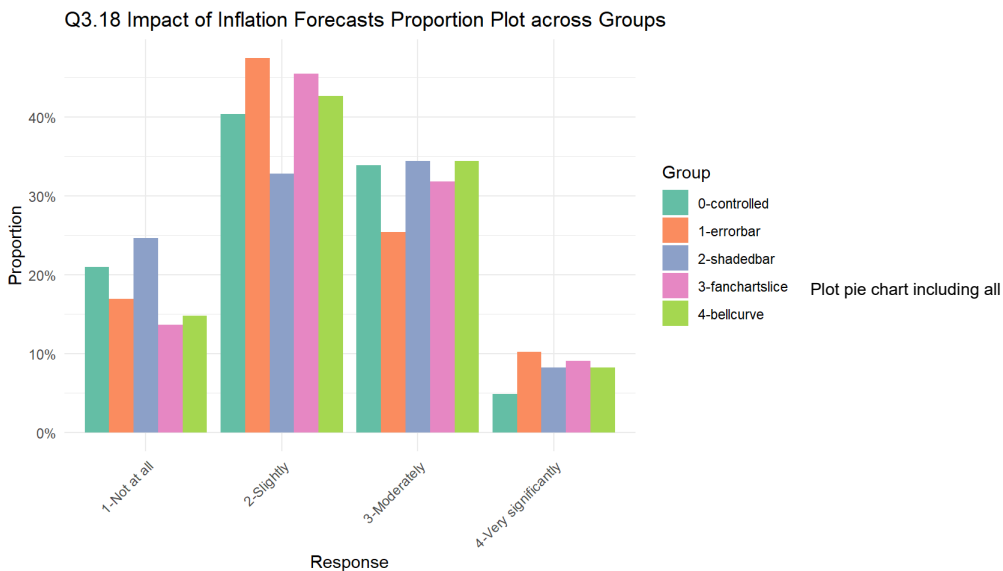
```
##
## Pearson's Chi-squared test
##
## data:  q3_18_contingency
## X-squared = 7.0349, df = 12, p-value = 0.8553
```

Impact of Inflation Forecasts proportion bar plot across different groups

```

q3_18_prop <- as.data.frame(q3_18_contingency)
# rename columns
names(q3_18_prop) <- c("Q3.18", "Group", "Count")
# add a labelled column
q3_18_prop <- q3_18_prop %>%
  mutate(Q3.18_label = case_when(
    Q3.18 == "1" ~ "1-Not at all",
    Q3.18 == "2" ~ "2-Slightly",
    Q3.18 == "3" ~ "3-Moderately",
    Q3.18 == "4" ~ "4-Very significantly"
  ))
# calculate proportions within each group
q3_18_prop <- q3_18_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))
ggplot(q3_18_prop, aes(x = as.factor(`Q3.18_label`), y = Proportion, fill = Group)) + geom_col(position = "dodge") +
  labs(
    title = "Q3.18 Impact of Inflation Forecasts Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2")+
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal()+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```



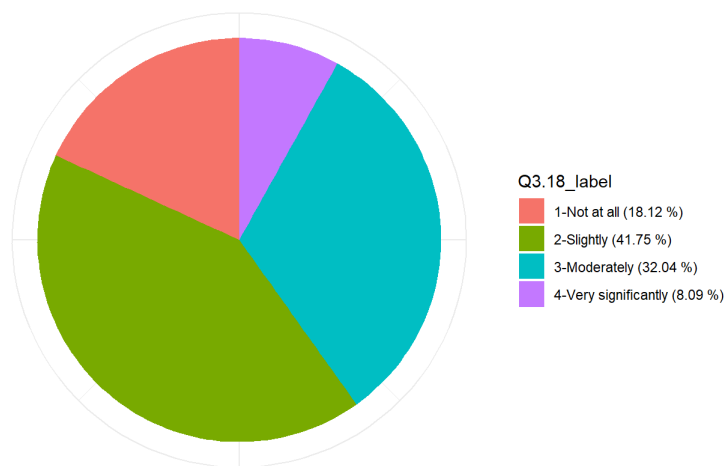
participants

```

q3_18_summary <- q3_18_prop %>%
  group_by(`Q3.18_label`) %>%
  summarise(total_count = sum(Count), .groups = "drop") %>%
  mutate(proportion = total_count / sum(total_count),
    `Q3.18_label` = paste0(`Q3.18_label`, " (", round(proportion * 100, 2), "%)")
ggplot(q3_18_summary, aes(x = "", y = proportion, fill = `Q3.18_label`)) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  labs(title = "Q3.18 Impact of Inflation Forecasts Summary Pie Chart",
    x = "",
    y = "") +
  theme_minimal() +
  theme(axis.text.x = element_blank())

```


Q3.18 Impact of Inflation Forecasts Summary Pie Chart



Q3.19 Response to Inflation Forecasts

How much do you adjust or take action in response to inflation forecasts (information about future inflation)?

Variable: categorical

Method: contingency table + chi-squared test

Result: Fail to reject the null hypothesis

```
# Initial Check
summary(as.numeric(rawdata_value$Q3.19))
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  1.000   2.000   2.000   2.236   3.000   4.000
```

```
q3_19_table <- rawdata_value[, c("Q3.19", "group")]
q3_19_contingency <- table(q3_19_table$Q3.19, q3_19_table$group)
print(q3_19_contingency)
```

```
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
##  1             16          11           17             19           12
##  2              21          23           21             20           19
##  3              23          19           19             23           28
##  4               2           6            4              4            2
```

```
# Chi-squared test
chisq.test(q3_19_contingency)
```

```
## Warning in chisq.test(q3_19_contingency): Chi-squared approximation may be
## incorrect
```

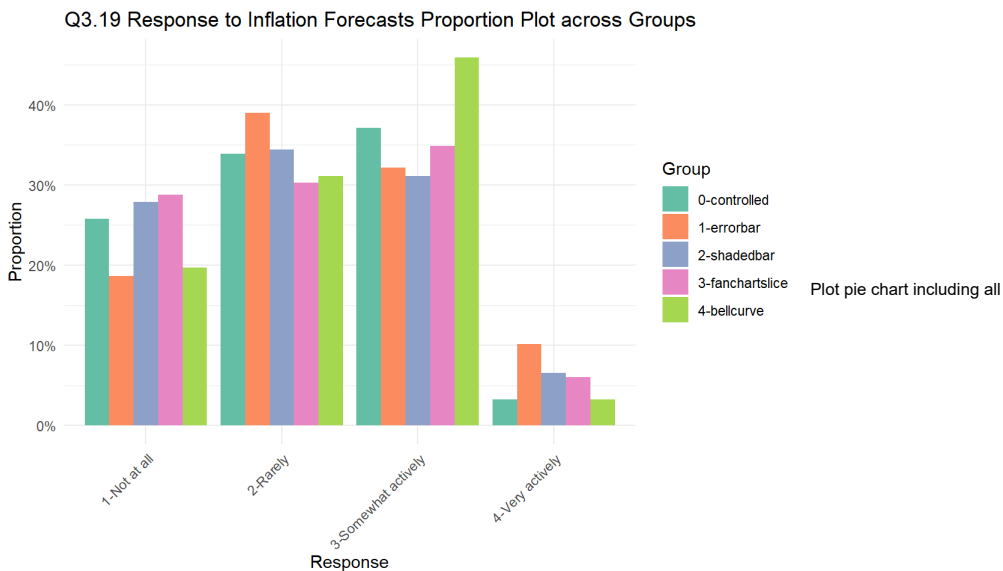
```
##
##  Pearson's Chi-squared test
##
## data:  q3_19_contingency
## X-squared = 8.7777, df = 12, p-value = 0.7218
```

Response to Inflation Forecasts proportion bar plot across different groups

```

q3_19_prop <- as.data.frame(q3_19_contingency)
# rename columns
names(q3_19_prop) <- c("Q3.19", "Group", "Count")
# add a labelled column
q3_19_prop <- q3_19_prop %>%
  mutate(Q3.19_label = case_when(
    Q3.19 == "1" ~ "1-Not at all",
    Q3.19 == "2" ~ "2-Rarely",
    Q3.19 == "3" ~ "3-Somewhat actively",
    Q3.19 == "4" ~ "4-Very actively"
  ))
# calculate proportions within each group
q3_19_prop <- q3_19_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))
ggplot(q3_19_prop, aes(x = as.factor(`Q3.19_label`), y = Proportion, fill = Group)) + geom_col(position = "dodge") +
  labs(
    title = "Q3.19 Response to Inflation Forecasts Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2")+
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal()+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```



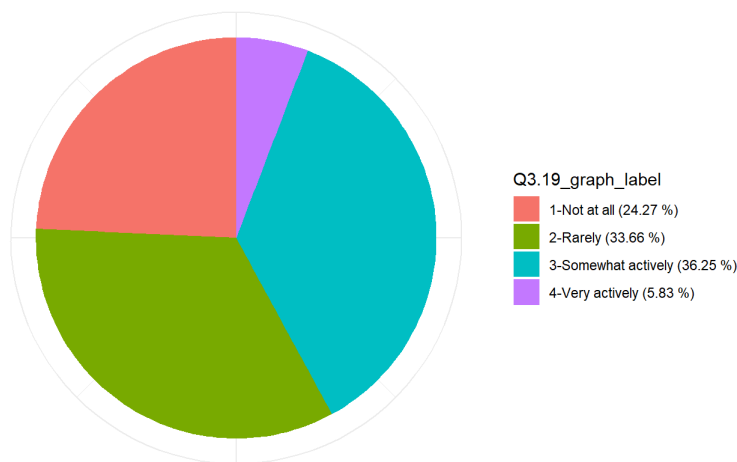
participants

```

q3_19_summary <- q3_19_prop %>%
  group_by(`Q3.19_label`) %>%
  summarise(total_count = sum(Count), .groups = "drop") %>%
  mutate(proportion = total_count / sum(total_count),
    `Q3.19_graph_label` = paste0(`Q3.19_label`, " (", round(proportion * 100, 2), "%)")
ggplot(q3_19_summary, aes(x = "", y = proportion, fill = `Q3.19_graph_label`)) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  labs(title = "Q3.19 Response to Inflation Forecasts Summary Pie Chart",
    x = "",
    y = "") +
  theme_minimal() +
  theme(axis.text.x = element_blank())

```

Q3.19 Response to Inflation Forecasts Summary Pie Chart



Comparison of current inflation and inflation forecast

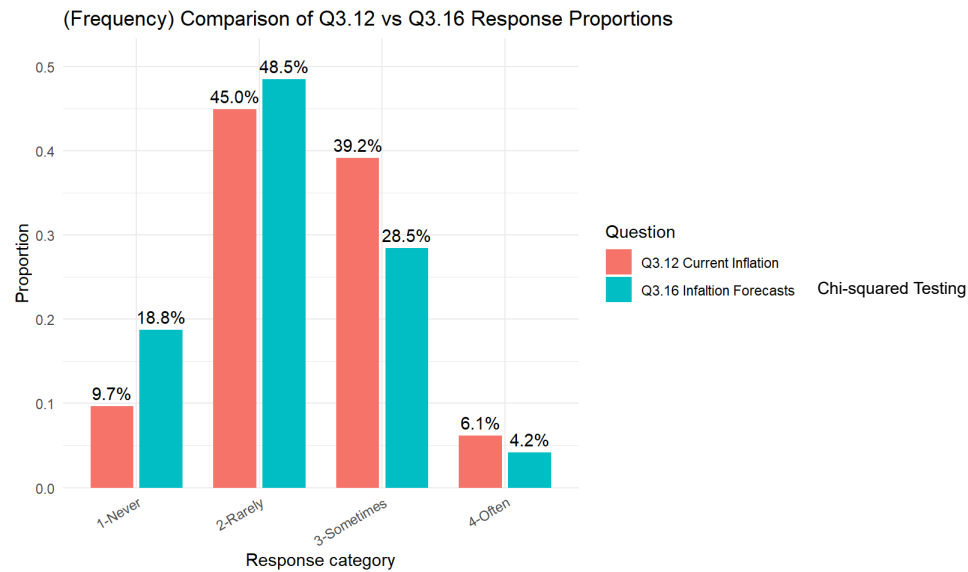
Frequency

```
q3_12_long <- q3_12_prop %>%
group_by(`Q3.12_label`) %>%
  rename(choice = `Q3.12_label`) %>%
  summarise(total_count = sum(Count), .groups = "drop") %>%
  mutate(proportion = total_count / sum(total_count),
         question = "Q3.12 Current Inflation")

q3_16_long <- q3_16_prop %>%
group_by(`Q3.16_label`) %>%
  rename(choice = `Q3.16_label`) %>%
  summarise(total_count = sum(Count), .groups = "drop") %>%
  mutate(proportion = total_count / sum(total_count),
         question = "Q3.16 Infaltion Forecasts")

# bind them together
q3_frequency <- bind_rows(q3_12_long, q3_16_long)

ggplot(q3_frequency, aes(x = choice, y = proportion, fill = question)) +
  geom_col(position = position_dodge(width = 0.8), width = 0.7) +
  geom_text(aes(label = scales::percent(proportion, accuracy = 0.1)),
            position = position_dodge(width = 0.8), vjust = -0.5) +
  scale_y_continuous(
    expand = expansion(mult = c(0, 0.1))
  ) +
  labs(
    title = "(Frequency) Comparison of Q3.12 vs Q3.16 Response Proportions",
    x = "Response category",
    y = "Proportion",
    fill = "Question"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 30, hjust = 1)
  )
```



```
q3_12_t <- q3_12_table %>%
  rename(choice = `Q3.12`) %>%
  mutate(question = "Q3.12")
q3_16_t <- q3_16_table %>%
  rename(choice = `Q3.16`) %>%
  mutate(question = "Q3.16")
# bind them together
q3_frequency_t <- bind_rows(q3_12_t, q3_16_t)
q3_frequency_table <- table(q3_frequency_t$choice, q3_frequency_t$question)
print(q3_frequency_table)
```

```
##
##      Q3.12 Q3.16
## 1      30    58
## 2     139   150
## 3     121    88
## 4      19    13
```

```
chisq.test(q3_frequency_table)
```

```
##
## Pearson's Chi-squared test
##
## data:  q3_frequency_table
## X-squared = 15.663, df = 3, p-value = 0.001329
```

```
chisq_result <- chisq.test(q3_frequency_table)
std_resid <- chisq_result$stdres
print(std_resid)
```

```
##
##      Q3.12      Q3.16
## 1 -3.2230942  3.2230942
## 2 -0.8868293  0.8868293
## 3  2.8059077 -2.8059077
## 4  1.0892353 -1.0892353
```

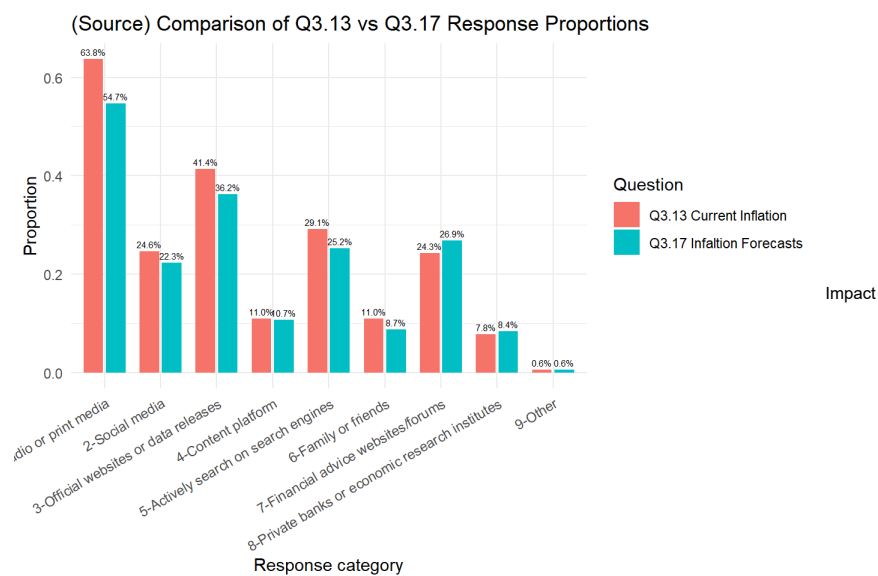
```
# cramerV(q3_frequency_table)
```

Source

```

q3_13_long <- q3_13_summary %>%
  rename(choice = `Q3.13_label`) %>%
  mutate(question = "Q3.13 Current Inflation")
q3_17_long <- q3_17_summary %>%
  rename(choice = `Q3.17_label`) %>%
  mutate(question = "Q3.17 Inflation Forecasts")
# bind them together
q3_source <- bind_rows(q3_13_long, q3_17_long)
# plot
ggplot(q3_source, aes(x = choice, y = proportion, fill = question)) +
  geom_col(position = position_dodge(width = 0.8), width = 0.7) +
  geom_text(aes(label = scales::percent(proportion, accuracy = 0.1)),
            position = position_dodge(width = 0.8), vjust = -0.5, size = 2) +
  labs(
    title = "(Source) Comparison of Q3.13 vs Q3.17 Response Proportions",
    x = "Response category",
    y = "Proportion",
    fill = "Question"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 30, hjust = 1)
  )

```



```

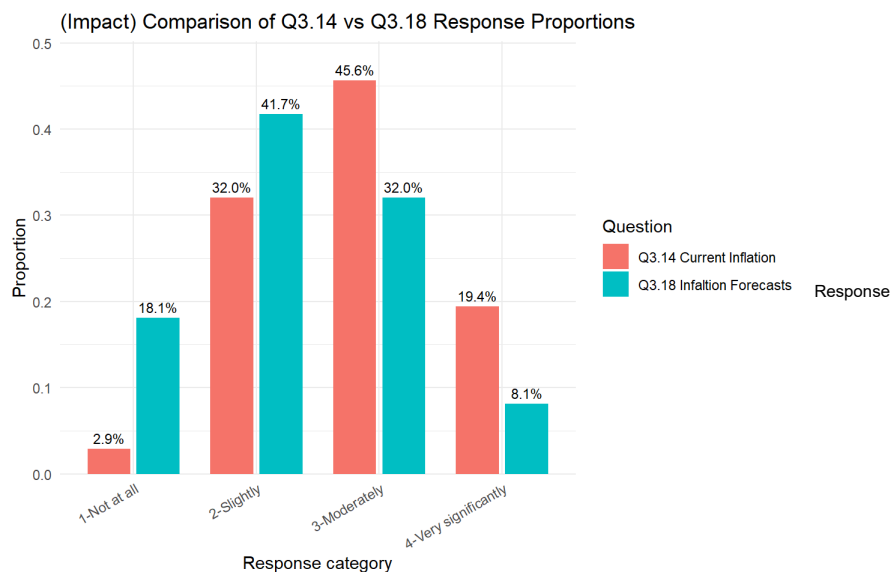
q3_14_long <- q3_14_prop %>%
group_by(`Q3.14_label`) %>%
  rename(choice = `Q3.14_label`) %>%
  summarise(total_count = sum(Count), .groups = "drop") %>%
  mutate(proportion = total_count / sum(total_count),
         question = "Q3.14 Current Inflation")

q3_18_long <- q3_18_prop %>%
  group_by(`Q3.18_label`) %>%
  rename(choice = `Q3.18_label`) %>%
  summarise(total_count = sum(Count), .groups = "drop") %>%
  mutate(proportion = total_count / sum(total_count),
         question = "Q3.18 Infaltion Forecasts")

# bind them together
q3_impact <- bind_rows(q3_14_long, q3_18_long)

ggplot(q3_impact, aes(x = choice, y = proportion, fill = question)) +
  # facet_wrap(~ question) +
  geom_col(position = position_dodge(width = 0.8), width = 0.7) +
  geom_text(aes(label = scales::percent(proportion, accuracy = 0.1)),
            position = position_dodge(width = 0.8), vjust = -0.5, size = 3) +
  scale_y_continuous(
    expand = expansion(mult = c(0, 0.1))
  ) +
  labs(
    title = "(Impact) Comparison of Q3.14 vs Q3.18 Response Proportions",
    x = "Response category",
    y = "Proportion",
    fill = "Question"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 30, hjust = 1)
  )

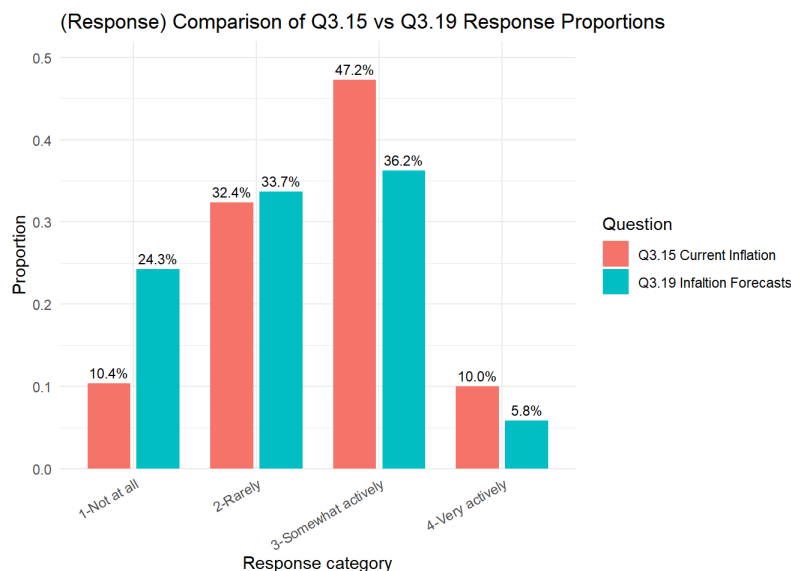
```



```

q3_15_long <- q3_15_prop %>%
group_by(`Q3.15_label`) %>%
  rename(choic = `Q3.15_label`) %>%
  summarise(total_count = sum(Count), .groups = "drop") %>%
  mutate(proportion = total_count / sum(total_count),
         question = "Q3.15 Current Inflation")
q3_19_long <- q3_19_prop %>%
group_by(`Q3.19_label`) %>%
  rename(choic = `Q3.19_label`) %>%
  summarise(total_count = sum(Count), .groups = "drop") %>%
  mutate(proportion = total_count / sum(total_count),
         question = "Q3.19 Inflation Forecasts")
# bind them together
q3_response <- bind_rows(q3_15_long, q3_19_long)
ggplot(q3_response, aes(x = choic, y = proportion, fill = question)) +
  geom_col(position = position_dodge(width = 0.8), width = 0.7) +
  geom_text(aes(label = scales::percent(proportion, accuracy = 0.1)),
            position = position_dodge(width = 0.8), vjust = -0.5, size = 3) +
  scale_y_continuous(
    expand = expansion(mult = c(0, 0.1))
  ) +
  labs(
    title = "(Response) Comparison of Q3.15 vs Q3.19 Response Proportions",
    x = "Response category",
    y = "Proportion",
    fill = "Question"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 30, hjust = 1)
  )

```



Actual Survey Analysis

Qxx.2 Graph Familiarity

Have you ever been communicated this type of information before?

Variable: categorical

Method: Contingency Table + Chi-squared test

recode to yes and no

Result: We can reject the null hypothesis, meaning people are more or less familiar with certain type of visualisations.

```
# combine data from different groups
survey_q2 <- data.frame()
survey_q2_list <- list("Q4.2", "Q6.2", "Q8.2", "Q10.2", "Q12.2")

for (i in seq_along(group_list)) {
  df <- group_list[[i]]
  col <- survey_q2_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("q2_value", "group")
  survey_q2 <- rbind(survey_q2, temp_table)
}

# recode the category
survey_q2 <- survey_q2 %>%
  mutate(q2_recode = case_when(
    q2_value %in% c("1", "2") ~ "N",
    q2_value %in% c("3", "4") ~ "Y"
  ))

survey_q2_contingency <- table(survey_q2$q2_recode, survey_q2$group)
# print the contingency table
print(survey_q2_contingency)
```

```
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
##      N           31          32          43             40          30
##      Y           31          27          18             26          31
```

```
# Chi-squared test
chisq.test(survey_q2_contingency)
```

```
##
##      Pearson's Chi-squared test
##
## data:  survey_q2_contingency
## X-squared = 7.8234, df = 4, p-value = 0.09827
```

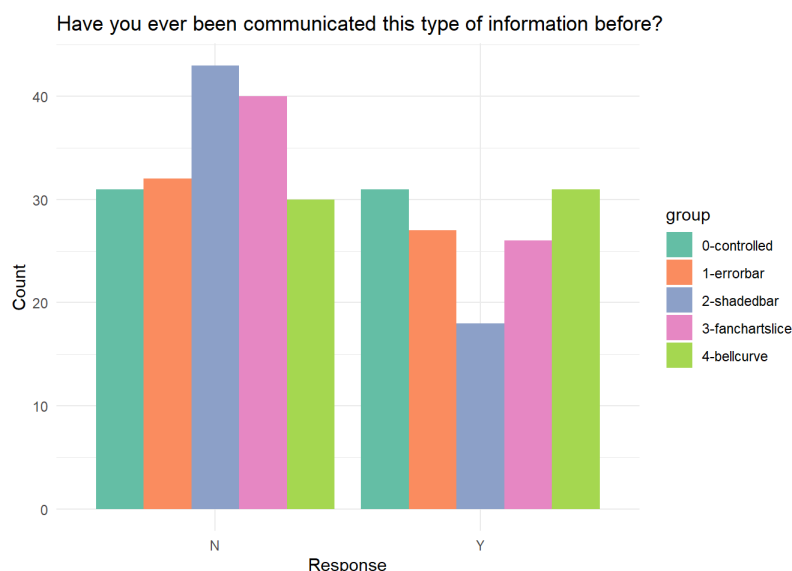
```
# chisq_result <- chisq.test(survey_q2_contingency)
# chisq_result$stdres
# chisq_result$expected #(If many values are < 5, your Chi-squared test may be less reliable.)
# Post-hoc test for N --> no single pairwise difference is strong
# pairwise.prop.test(
#   x = survey_q2_contingency["N", ],
#   n = colSums(survey_q2_contingency),
#   p.adjust.method = "holm"
# )
```

Summary Table of proportion of yes and no

```
yes_per <- survey_q2_contingency[2,]
no_per <- survey_q2_contingency[1,]
familiarity_group_total <- colSums(survey_q2_contingency)
yes_prop <- yes_per/familiarity_group_total
no_prop <- no_per/familiarity_group_total
# convert to data frame
survey_q2_summary <- data.frame(yes_prop, no_prop)
```

Plot the graph

```
ggplot(survey_q2, aes(x = q2_recode, fill = group)) +
  geom_bar(position = "dodge") +
  labs(title = "Have you ever been communicated this type of information before?",
       x = "Response",
       y = "Count") +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal()
```

Qxx.3 Prior Expectation

Does the inflation forecast information provided align with your expectations?

Variable: categorical

Method: Contingency Table + Chi-squared test

Result: Fail to reject the null hypothesis

```
# combine data from different groups
survey_q3 <- data.frame()
survey_q3_list <- list("Q4.3", "Q6.3", "Q8.3", "Q10.3", "Q12.3")
for (i in seq_along(group_list)) {
  df <- group_list[[i]]
  col <- survey_q3_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("q3_value", "group")
  survey_q3 <- rbind(survey_q3, temp_table)
}

# Initial Check
summary(as.numeric(survey_q3$q3_value))
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  1.000   2.000   2.000   2.476   3.000   5.000
```

Test the differences between groups

```
survey_q3_contingency <- table(survey_q3$q3_value, survey_q3$group)
# print the contingency table
print(survey_q3_contingency)
```

```
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
##  1           4           3           1           1           4
##  2          33          39          33          34          31
##  3          19          15          21          21          20
##  4           6           1           6          10           3
##  5           0           1           0           0           3
```

```
# Chi-squared test
chisq.test(survey_q3_contingency)
```

```
## Warning in chisq.test(survey_q3_contingency): Chi-squared approximation may be
## incorrect
```

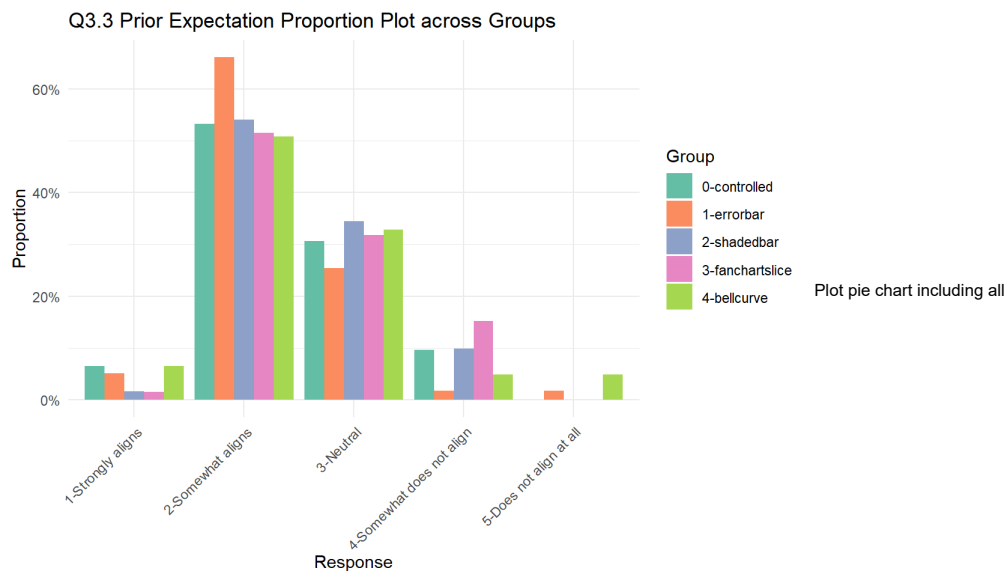
```
##
##      Pearson's Chi-squared test
##
## data:  survey_q3_contingency
## X-squared = 22.933, df = 16, p-value = 0.1155
```

Plot prior expectation proportion bar plot across different groups

```

survey_q3_prop <- as.data.frame(survey_q3_contingency)
# rename columns
names(survey_q3_prop) <- c("Q3.3", "Group", "Count")
# add a labelled column
survey_q3_prop <- survey_q3_prop %>%
  mutate(Q3.3_label = case_when(
    Q3.3 == "1" ~ "1-Strongly aligns",
    Q3.3 == "2" ~ "2-Somewhat aligns",
    Q3.3 == "3" ~ "3-Neutral",
    Q3.3 == "4" ~ "4-Somewhat does not align",
    Q3.3 == "5" ~ "5-Does not align at all",
  ))
# calculate proportions within each group
survey_q3_prop <- survey_q3_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))
ggplot(survey_q3_prop, aes(x = as.factor(`Q3.3_label`), y = Proportion, fill = Group)) + geom_col(position = "dodge") +
  labs(
    title = "Q3.3 Prior Expectation Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2") +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```



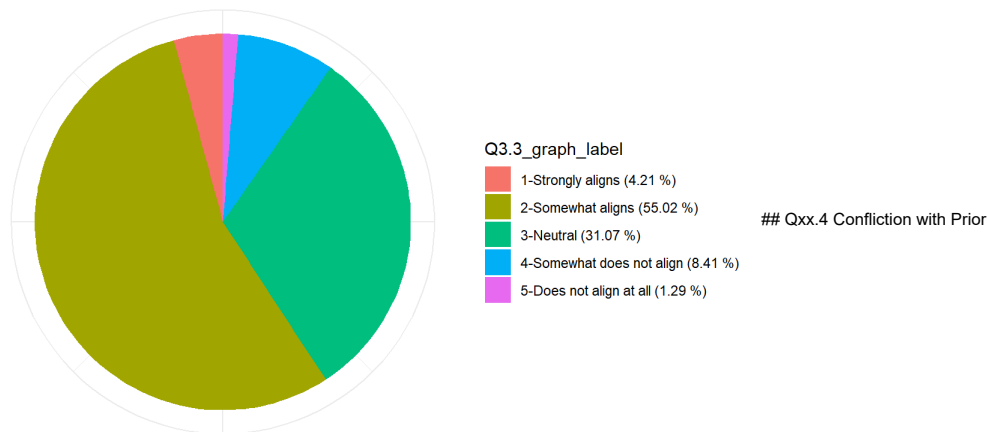
participants

```

survey_q3_summary <- survey_q3_prop %>%
  group_by(`Q3.3_label`) %>%
  summarise(total_count = sum(Count), .groups = "drop") %>%
  mutate(proportion = total_count / sum(total_count),
    `Q3.3_graph_label` = paste0(`Q3.3_label`, " (", round(proportion * 100, 2), "%)"))
ggplot(survey_q3_summary, aes(x = "", y = proportion, fill = `Q3.3_graph_label`)) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  labs(title = "Q3.3 Prior Expectation Summary Pie Chart",
    x = "",
    y = "") +
  theme_minimal() +
  theme(axis.text.x = element_blank())

```

Q3.3 Prior Expectation Summary Pie Chart



Expectation How does the inflation forecast conflict with your expectations?

Variable: categorical

Method: calculate total number of people who think the forecast is not aligned with their expectation

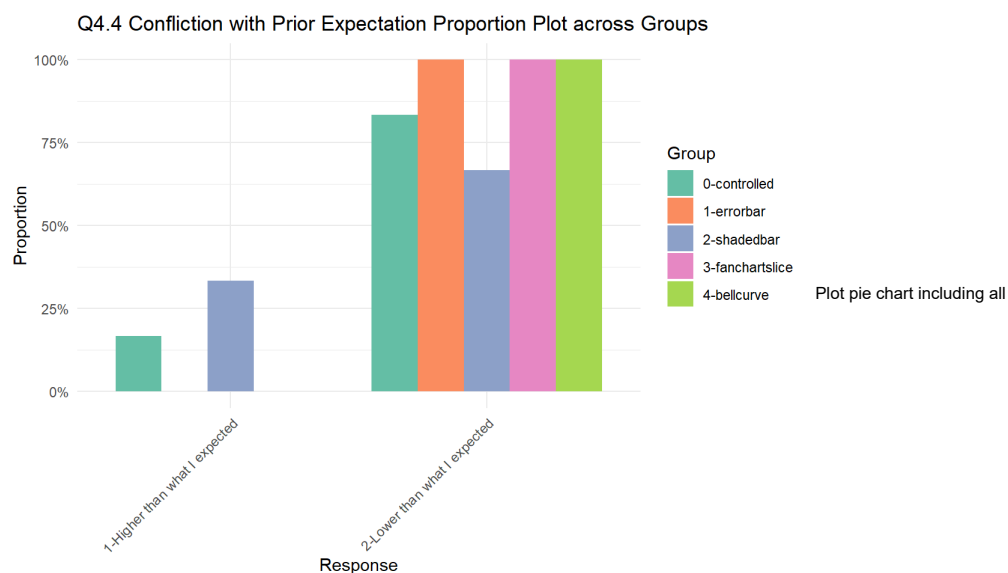
Result: Among people who think the forecast is not aligned with their expectation, 90% percent of respondents thinks the forecast is lower than what they expected.

```
# Check the number of people answered this question
n_not_align <- sum(survey_q3_summary$total_count[4:5])
# Combine data from different groups
survey_q4 <- data.frame()
survey_q4_list <- list("Q4.4", "Q6.4", "Q8.4", "Q10.4", "Q12.4")
for (i in seq_along(group_list)) {
  df <- group_list[[i]]
  col <- survey_q4_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("q4_value", "group")
  survey_q4 <- rbind(survey_q4, temp_table)
}
survey_q4 <- survey_q4[!is.na(survey_q4$q4_value), ]
survey_q4_contingency <- table(survey_q4$q4_value, survey_q4$group)
# print the contingency table
print(survey_q4_contingency)
```

```
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
##      1           1           0           2           0           0
##      2           5           2           4           10          6
```

Plot confliction with prior expectation proportion bar plot across different groups

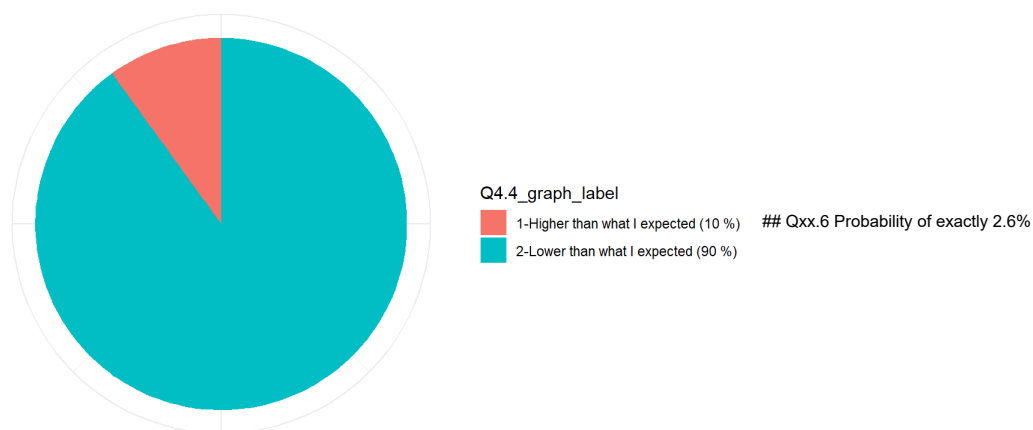
```
survey_q4_prop <- as.data.frame(survey_q4_contingency)
# rename columns
names(survey_q4_prop) <- c("Q4.4", "Group", "Count")
# add a labelled column
survey_q4_prop <- survey_q4_prop %>%
  mutate(Q4.4_label = case_when(
    Q4.4 == "1" ~ "1-Higher than what I expected",
    Q4.4 == "2" ~ "2-Lower than what I expected"
  ))
# calculate proportions within each group
survey_q4_prop <- survey_q4_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))
ggplot(survey_q4_prop, aes(x = as.factor(`Q4.4_label`), y = Proportion, fill = Group)) + geom_col(position = "dodge") +
  labs(
    title = "Q4.4 Conflicition with Prior Expectation Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2") +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



participants

```
survey_q4_summary <- survey_q4_prop %>%
  group_by(`Q4.4_label`) %>%
  summarise(total_count = sum(Count), .groups = "drop") %>%
  mutate(proportion = total_count / sum(total_count),
         `Q4.4_graph_label` = paste0(`Q4.4_label`, " (", round(proportion * 100, 2), "%)"))
ggplot(survey_q4_summary, aes(x = "", y = proportion, fill = `Q4.4_graph_label`)) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  labs(title = "Q4.4 Confliction with Prior Expectation Summary Pie Chart",
       x = "",
       y = "") +
  theme_minimal() +
  theme(axis.text.x = element_blank())
```

Q4.4 Confliction with Prior Expectation Summary Pie Chart



what do you think is the probability of inflation being exactly 2.6%?

Definitive Correct Answer

descending order

Variable: categorical

Method: contingency table & proportion test

Result: There is no statistical difference in terms of correctly answering the probability of inflation being exactly 2.6%.

```
# combine data from different groups
survey_q6 <- data.frame()
survey_q6_list <- list("Q4.6","Q6.6", "Q8.6", "Q10.6", "Q12.6")
for (i in seq_along(group_list)) {
  df <- group_list[[i]]
  col <- survey_q6_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("q6_value", "group")
  survey_q6 <- rbind(survey_q6, temp_table)
}
survey_q6$q6_value <- as.numeric(survey_q6$q6_value)

survey_q6_contingency <- table(survey_q6$q6_value, survey_q6$group)
print(survey_q6_contingency)
```

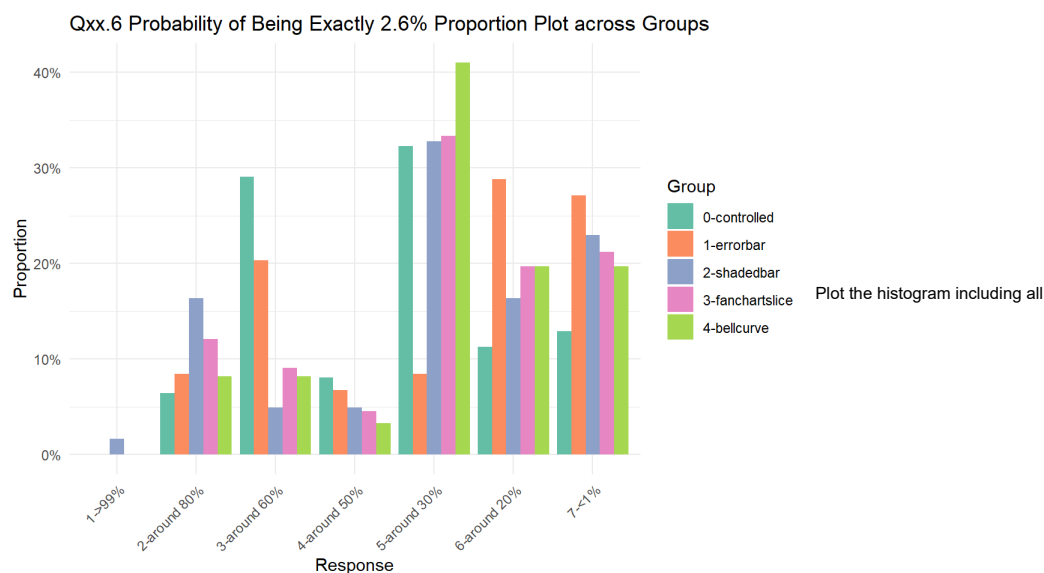
```
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1          0          0          1          0          0
## 2          4          5         10          8          5
## 3         18         12          3          6          5
## 4          5          4          3          3          2
## 5         20          5         20         22         25
## 6          7         17         10         13         12
## 7          8         16         14         14         12
```

```
# whether one proportion differs across groups
less_than_1_per <- survey_q6_contingency[7,]
survey_q6_group_total <- colSums(survey_q6_contingency)
prop.test(less_than_1_per, survey_q6_group_total)
```

```
##
## 5-sample test for equality of proportions without continuity correction
##
## data: less_than_1_per out of survey_q6_group_total
## X-squared = 4.0132, df = 4, p-value = 0.4042
## alternative hypothesis: two.sided
## sample estimates:
## prop 1 prop 2 prop 3 prop 4 prop 5
## 0.1290323 0.2711864 0.2295082 0.2121212 0.1967213
```

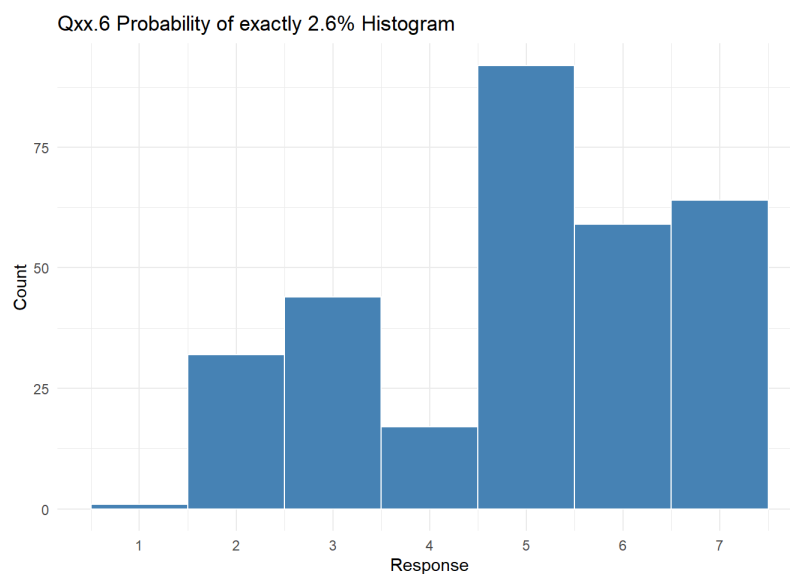
Plot probability of exactly 2.6% proportion bar plot across different groups

```
survey_q6_prop <- as.data.frame(survey_q6_contingency)
# rename columns
names(survey_q6_prop) <- c("Qxx.6", "Group", "Count")
# add a labelled column
survey_q6_prop <- survey_q6_prop %>%
  mutate(Qxx.6_label = case_when(
    Qxx.6 == "1" ~ "1->99%",
    Qxx.6 == "2" ~ "2-around 80%",
    Qxx.6 == "3" ~ "3-around 60%",
    Qxx.6 == "4" ~ "4-around 50%",
    Qxx.6 == "5" ~ "5-around 30%",
    Qxx.6 == "6" ~ "6-around 20%",
    Qxx.6 == "7" ~ "7-<1%"
  ))
# calculate proportions within each group
survey_q6_prop <- survey_q6_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))
ggplot(survey_q6_prop, aes(x = as.factor(`Qxx.6_label`), y = Proportion, fill = Group)) + geom_col(position = "dodge") +
  # facet_wrap(~Group)+
  labs(
    title = "Qxx.6 Probability of Being Exactly 2.6% Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2")+
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal()+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



participants

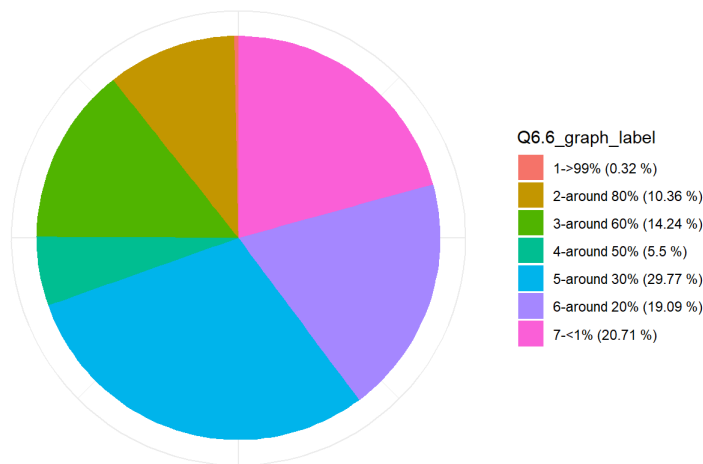
```
# histogram including all participants
ggplot(survey_q6, aes(x = q6_value)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  labs(title = "Qxx.6 Probability of exactly 2.6% Histogram",
        x = "Response",
        y = "Count") +
  scale_x_continuous(breaks = seq(0, max(survey_q6$q6_value, na.rm = TRUE), by = 1)) +
  theme_minimal()
```



Plot pie chart including all participants

```
survey_q6_summary <- survey_q6_prop %>%
  group_by(`Qxx.6_label`) %>%
  summarise(total_count = sum(count), .groups = "drop") %>%
  mutate(proportion = total_count / sum(total_count),
         `Q6.6_graph_label` = paste0(`Qxx.6_label`, " (", round(proportion * 100, 2), "%)"))
ggplot(survey_q6_summary, aes(x = "", y = proportion, fill = `Q6.6_graph_label`)) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  labs(title = "Qxx.6 Probability of exactly 2.6% Summary Pie Chart",
        x = "",
        y = "") +
  theme_minimal() +
  theme(axis.text.x = element_blank())
```

Qxx.6 Probability of exactly 2.6% Summary Pie Chart



! Qxx.7 Probability of between 1.9% and 3.3%

what do you think is the probability of inflation being between 1.9% and 3.3%?

Definitive Correct Answer

descending order

Method: contingency table & proportion test

Result: There is statistical difference in control group in terms of the proportion of the correct answer of how people think the probability of inflation being between 1.9% and 3.3% is.

```
# combine data from different groups
survey_q7 <- data.frame()
survey_q7_list <- list("Q4.7", "Q6.7", "Q8.7", "Q10.7", "Q12.7")
for (i in seq_along(group_list)) {
  df <- group_list[[i]]
  col <- survey_q7_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("q7_value", "group")
  survey_q7 <- rbind(survey_q7, temp_table)
}
survey_q7$q7_value <- as.numeric(survey_q7$q7_value)
```

Test the proportion of correct answer

```
survey_q7_contingency <- table(survey_q7$q7_value, survey_q7$group)
# print the contingency table
print(survey_q7_contingency)
```

```
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1           10           0           0           3           1
## 2           23           7          11           9           9
## 3           17          18          11          14           7
## 4            9          13           7           8          10
## 5            3          18          27          25          29
## 6            0           3           5           7           4
## 7            0           0           0           0           1
```

```
# correct answer
survey_q7_correct <- survey_q7_contingency[5,]
survey_q7_group_total <- colSums(survey_q7_contingency)
prop.test(survey_q7_correct, survey_q7_group_total)
```

```
##
## 5-sample test for equality of proportions without continuity correction
##
## data: survey_q7_correct out of survey_q7_group_total
## X-squared = 32.443, df = 4, p-value = 1.553e-06
## alternative hypothesis: two.sided
## sample estimates:
##   prop 1   prop 2   prop 3   prop 4   prop 5
## 0.0483871 0.3050847 0.4426230 0.3787879 0.4754098
```

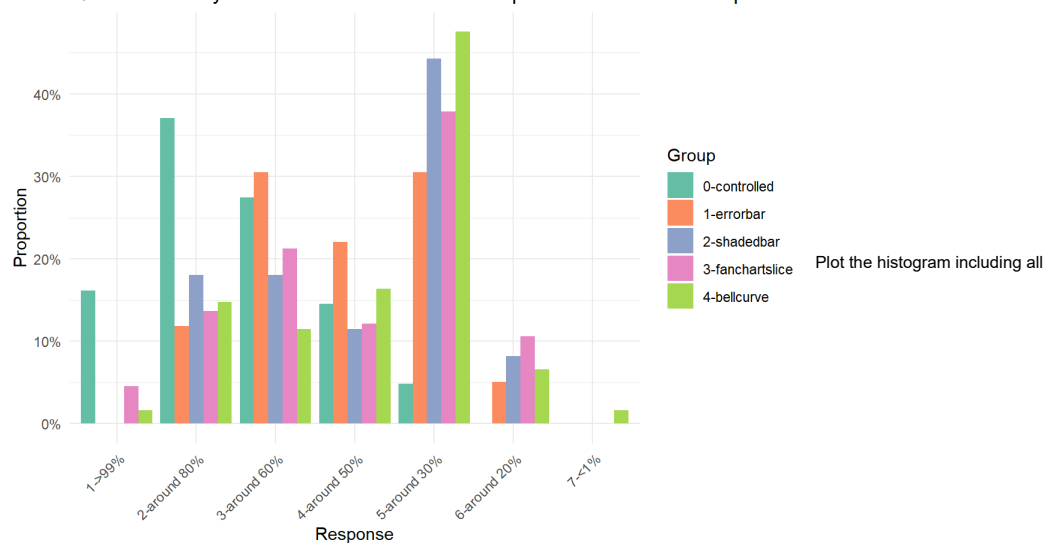
```
# pairwise proportion test
pairwise.prop.test(
  x = survey_q7_contingency[, ],
  n = colSums(survey_q7_contingency),
  p.adjust.method = "bonferroni" # or "holm", "BH"
)
```

```
##
## Pairwise comparisons using Pairwise comparison of proportions
##
## data: survey_q7_contingency[, ] out of colSums(survey_q7_contingency)
##
##          0-controlled 1-errorbar 2-shadedbar 3-fanchartslice
## 1-errorbar 0.00489      -          -          -
## 2-shadedbar 1.1e-05      1.00000      -          -
## 3-fanchartslice 0.00017  1.00000      1.00000      -
## 4-bellcurve  2.1e-06      0.84724      1.00000      1.00000
##
## P value adjustment method: bonferroni
```

Plot probability of between 1.9% and 3.3% proportion bar plot across different groups

```
survey_q7_prop <- as.data.frame(survey_q7_contingency)
# rename columns
names(survey_q7_prop) <- c("Qxx.7", "Group", "Count")
# add a labelled column
survey_q7_prop <- survey_q7_prop %>%
  mutate(Qxx.7_label = case_when(
    Qxx.7 == "1" ~ "1->99%",
    Qxx.7 == "2" ~ "2-around 80%",
    Qxx.7 == "3" ~ "3-around 60%",
    Qxx.7 == "4" ~ "4-around 50%",
    Qxx.7 == "5" ~ "5-around 30%",
    Qxx.7 == "6" ~ "6-around 20%",
    Qxx.7 == "7" ~ "7-<1%"
  ))
# calculate proportions within each group
survey_q7_prop <- survey_q7_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))
ggplot(survey_q7_prop, aes(x = as.factor(`Qxx.7_label`), y = Proportion, fill = Group)) + geom_col(position = "dodge") +
  # facet_wrap(~Group)+
  labs(
    title = "Qxx.7 Probability of between 1.9% and 3.3% Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2")+
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal()+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

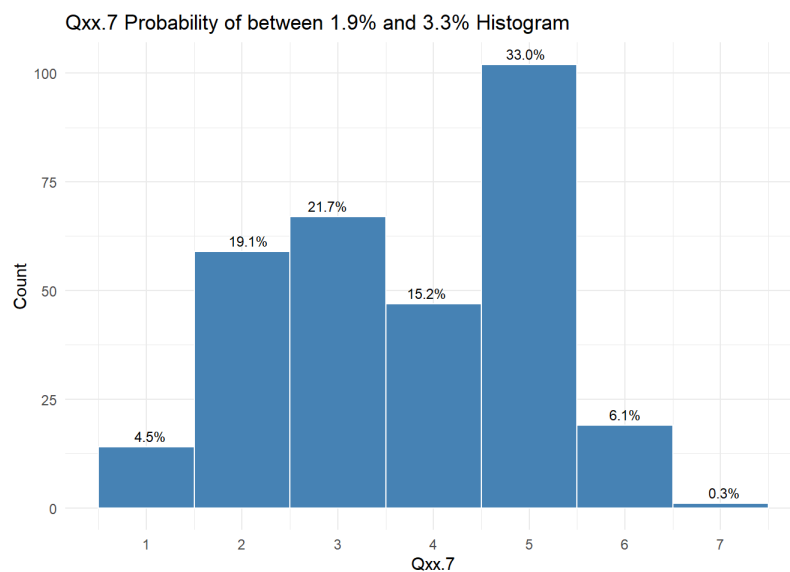
Qxx.7 Probability of between 1.9% and 3.3% Proportion Plot across Groups



participants


```
# histogram including all participants
ggplot(survey_q7, aes(x = q7_value)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  geom_text(
    stat = "bin",
    aes(
      label = after_stat(
        ifelse(
          count == 0,
          "",
          scales::percent(count / total_respondents, accuracy = 0.1)
        )
      ), vjust = -0.5, size = 3) +
  # facet_wrap(~ group) +
  labs(title = "Qxx.7 Probability of between 1.9% and 3.3% Histogram",
    x = "Qxx.7",
    y = "Count") +
  scale_x_continuous(breaks = seq(0, max(survey_q7$q7_value, na.rm = TRUE), by = 1)) +
  theme_minimal()
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



Qxx.8 Probability of between 3.3% and 5.6%

what do you think is the probability of inflation being between 3.3% and 5.6%?

Definitive Correct Answer

descending order

Method: contingency table & proportion test

Result: The control group and the error bar group has shown statistical difference.

```
# combine data from different groups
survey_q8 <- data.frame()
survey_q8_list <- list("Q4.8", "Q6.8", "Q8.8", "Q10.8", "Q12.8")
for (i in seq_along(survey_q8_list)) {
  df <- group_list[[i]]
  col <- survey_q8_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("q8_value", "group")
  survey_q8 <- rbind(survey_q8, temp_table)
}
survey_q8$q8_value <- as.numeric(survey_q8$q8_value)
```

contingency table

```
survey_q8_contingency <- table(survey_q8$q8_value, survey_q8$group)
# print the contingency table
print(survey_q8_contingency)
```

```
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1          1          0          1          1          1
## 2          3          2          3          2          7
## 3          13         4          5          10         5
## 4          5          5          9          6          2
## 5          14         34         29         23         28
## 6          18         13         11         22         12
## 7          8          1          3          2          6
```

```
# correct answer
survey_q8_correct <- survey_q8_contingency[5,]
survey_q8_group_total <- colSums(survey_q8_contingency)
prop.test(survey_q8_correct, survey_q8_group_total)
```

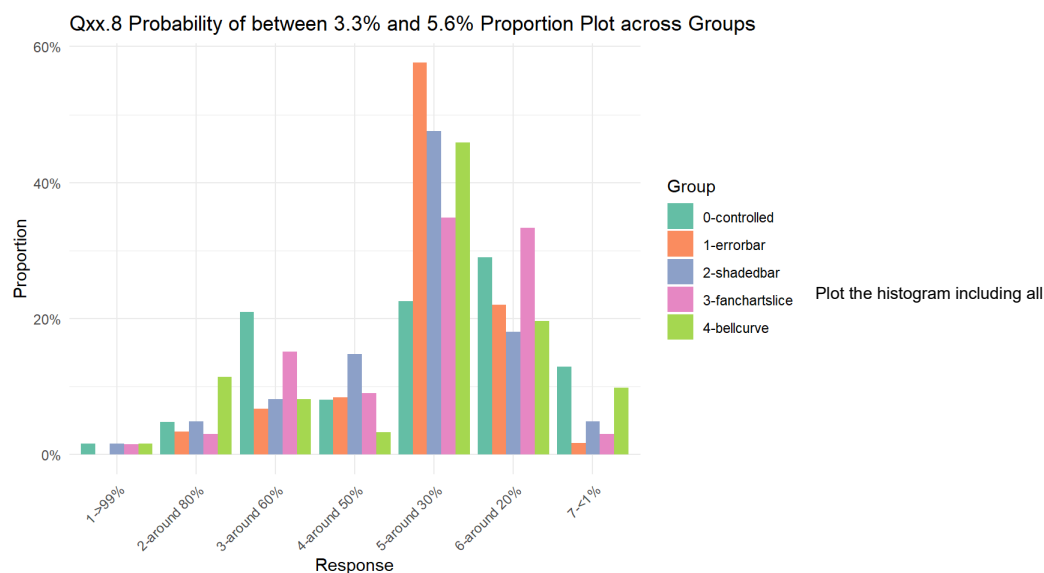
```
##
## 5-sample test for equality of proportions without continuity correction
##
## data: survey_q8_correct out of survey_q8_group_total
## X-squared = 18.077, df = 4, p-value = 0.001192
## alternative hypothesis: two.sided
## sample estimates:
## prop 1 prop 2 prop 3 prop 4 prop 5
## 0.2258065 0.5762712 0.4754098 0.3484848 0.4590164
```

```
# pairwise proportion test
pairwise.prop.test(
  x = survey_q8_contingency[5, ],
  n = colSums(survey_q8_contingency),
  p.adjust.method = "bonferroni" # or "holm", "BH"
)
```

```
##
## Pairwise comparisons using Pairwise comparison of proportions
##
## data: survey_q8_contingency[5, ] out of colSums(survey_q8_contingency)
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice
## 1-errorbar      0.0017      -      -      -
## 2-shadedbar      0.0666      1.0000      -      -
## 3-fanchartslice 1.0000      0.1766      1.0000      -
## 4-bellcurve      0.1118      1.0000      1.0000      1.0000
##
## P value adjustment method: bonferroni
```

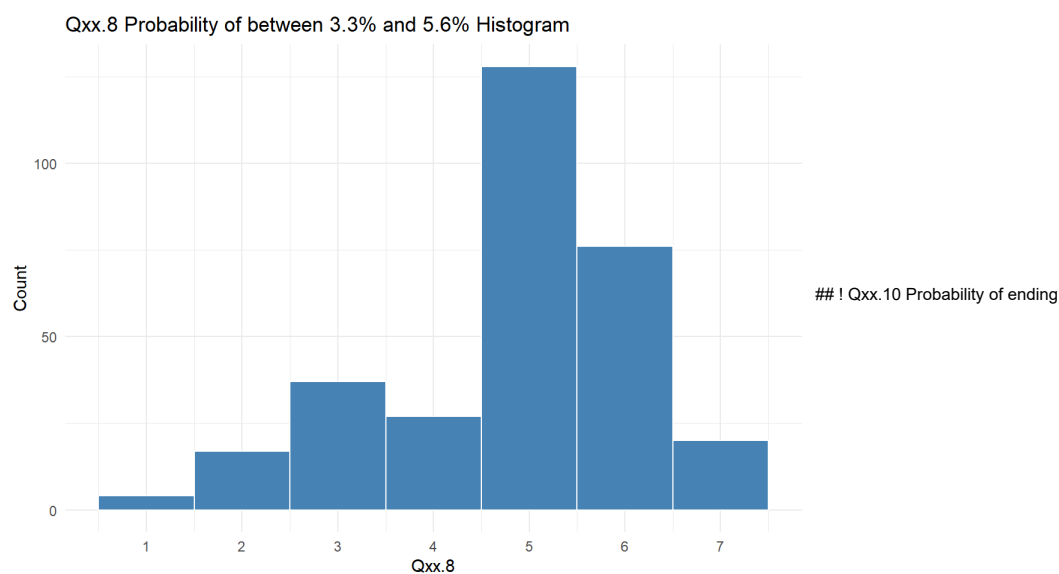
Plot probability of between 3.3% and 5.6% proportion bar plot across different groups

```
survey_q8_prop <- as.data.frame(survey_q8_contingency)
# rename columns
names(survey_q8_prop) <- c("Qxx.8", "Group", "Count")
# add a labelled column
survey_q8_prop <- survey_q8_prop %>%
  mutate(Qxx.8_label = case_when(
    Qxx.8 == "1" ~ "1->99%",
    Qxx.8 == "2" ~ "2-around 80%",
    Qxx.8 == "3" ~ "3-around 60%",
    Qxx.8 == "4" ~ "4-around 50%",
    Qxx.8 == "5" ~ "5-around 30%",
    Qxx.8 == "6" ~ "6-around 20%",
    Qxx.8 == "7" ~ "7-<1%"
  ))
# calculate proportions within each group
survey_q8_prop <- survey_q8_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))
ggplot(survey_q8_prop, aes(x = as.factor(`Qxx.8_label`), y = Proportion, fill = Group)) + geom_col(position = "dodge") +
  # facet_wrap(~Group)+
  labs(
    title = "Qxx.8 Probability of between 3.3% and 5.6% Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2")+
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal()+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



participants

```
# histogram including all participants
ggplot(survey_q8, aes(x = q8_value)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  # facet_wrap(~ group) +
  labs(title = "Qxx.8 Probability of between 3.3% and 5.6% Histogram",
        x = "Qxx.8",
        y = "Count") +
  scale_x_continuous(breaks = seq(0, max(survey_q8$q8_value, na.rm = TRUE), by = 1)) +
  theme_minimal()
```



outside the range Based on the inflation forecast provided, what do you think is the probability of inflation ending up outside of the range from -0.4% to 5.6%?

Definitive Correct Answer (varied between group)

descending order

Variable: categorical

Method: calculate the proportion for each group + proportion test

Result: Control group perform significant worse than other groups.

```

survey_q10 <- data.frame(
  group = character(0),
  correct = integer(0),
  total = integer(0),
  prop = numeric(0)
)
survey_q10_list <- list("Q4.10", "Q6.10", "Q8.10", "Q10.10", "Q12.10")
survey_q10_correct <- c(8,12,14,14,14) # 10%, 40%, 10%, 10%, 10%

for (i in seq_along(group_list)) {
  df <- group_list[[i]]
  col <- survey_q10_list[[i]]
  q10_correct <- survey_q10_correct[i]

  # count total number of people in each group
  total_count <- nrow(df)
  # count number of people who answered correctly
  correct_count <- sum(df[, col] == q10_correct, na.rm = TRUE)
  # compute the proportion
  prop <- correct_count / total_count
  # capture the group name
  grp_label <- names(group_list[i])
  # row-bind into survey_q10
  survey_q10 <- rbind(survey_q10, data.frame(
    group = grp_label,
    correct = correct_count,
    total = total_count,
    prop = prop))
}

# proportion test --> There is significant difference
prop.test(x = survey_q10$correct,
  n = survey_q10$total)

```

```

##
## 5-sample test for equality of proportions without continuity correction
##
## data: survey_q10$correct out of survey_q10$total
## X-squared = 36.59, df = 4, p-value = 2.187e-07
## alternative hypothesis: two.sided
## sample estimates:
## prop 1 prop 2 prop 3 prop 4 prop 5
## 0.1612903 0.5762712 0.4426230 0.4696970 0.6721311

```

```

# pairwise proportion test
pairwise.prop.test(
  x = setNames(survey_q10$correct, survey_q10$group),
  n = setNames(survey_q10$total, survey_q10$group),
  p.adjust.method = "bonferroni" # or "holm", "BH"
)

```

```

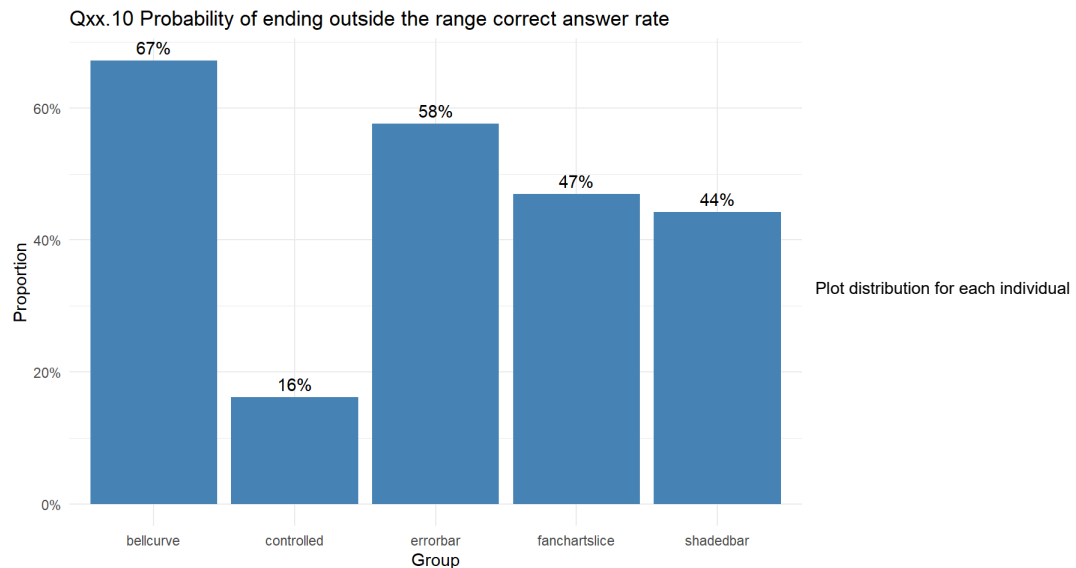
##
## Pairwise comparisons using Pairwise comparison of proportions
##
## data: setNames(survey_q10$correct, survey_q10$group) out of setNames(survey_q10$total, survey_q10$group)
##
## controlled errorbar shadedbar fanchartslice
## errorbar 5.3e-05 - - -
## shadedbar 0.0135 1.0000 - -
## fanchartslice 0.0039 1.0000 1.0000 -
## bellcurve 2.6e-07 1.0000 0.1781 0.3392
##
## P value adjustment method: bonferroni

```

```

ggplot(survey_q10, aes(x = group, y = prop)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  geom_text(aes(label = scales::percent(prop, accuracy = 1)), vjust = -0.5) +
  labs(title = "Qxx.10 Probability of ending outside the range correct answer rate",
    x = "Group",
    y = "Proportion") +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal()

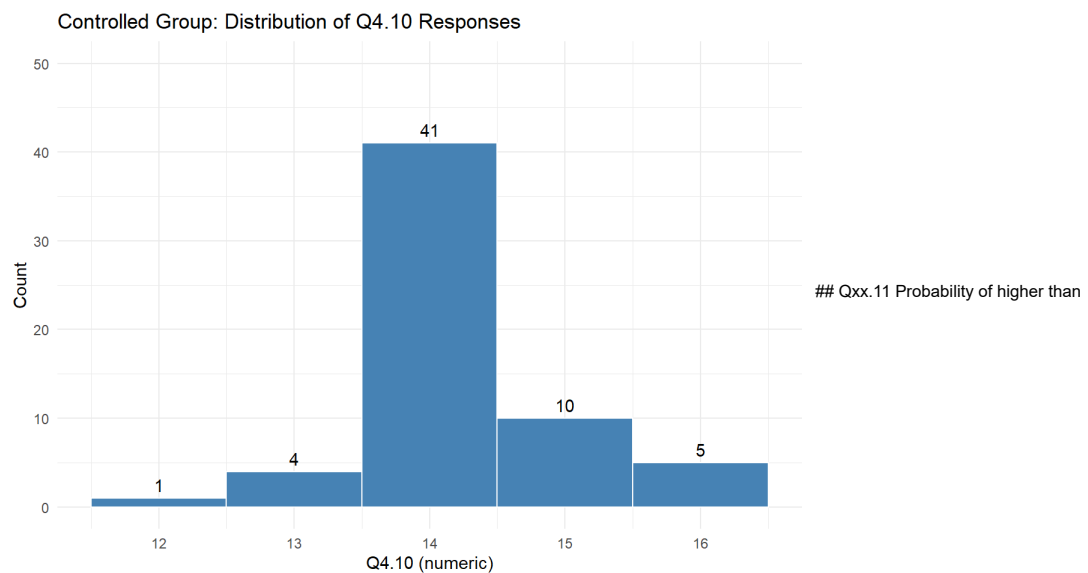
```



group (change the group and question number manually)

```
ggplot(bell_curve, aes(x = as.numeric(as.character(Q12.10)))) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  scale_y_continuous(
    limits = c(0, 50),      # force y from 0 up to 60
    breaks = seq(0, 50, by=10) # ticks at 0, 10, 20, ..., 60
  ) +
  geom_text(aes(label = ..count..), stat = "count", vjust = -0.5) +
  labs(
    title = "Controlled Group: Distribution of Q4.10 Responses",
    x = "Q4.10 (numeric)",
    y = "Count"
  ) +
  theme_minimal()
```

```
## Warning: The dot-dot notation (`..count..`) was deprecated in ggplot2 3.4.0.
## i Please use `after_stat(count)` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



6% Based on the inflation forecast provided, what do you think is the probability of inflation ending up higher than 6%?

Definitive Correct Answer - 3

variable: categorical

Method: contingency table & proportion test

Result: The bell curve and the control group has shown significant difference. The majority people think the probability is extremely low (less than 1%).

```
# combine data from different groups
survey_q11 <- data.frame()
survey_q11_list <- list("Q4.11", "Q6.11", "Q8.11", "Q10.11", "Q12.11")
for (i in seq_along(group_list)) {
  df <- group_list[[i]]
  col <- survey_q11_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("q11_value", "group")
  survey_q11 <- rbind(survey_q11, temp_table)
}
survey_q11$q11_value <- as.numeric(survey_q11$q11_value)
```

Test the proportion of correct answer

```
survey_q11_contingency <- table(survey_q11$q11_value, survey_q11$group)
# print the contingency table
print(survey_q11_contingency)
```

```
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1           1           4           1           0           4
## 2           13          12           9          10           9
## 3           13          21          24          24          29
## 4           33          15          24          30          16
## 5            2           7           3           2           3
```

```
survey_q11_correct <- survey_q11_contingency[, ]
survey_q11_group_total <- colSums(survey_q11_contingency)
prop.test(survey_q11_correct, survey_q11_group_total)
```

```
##
## 5-sample test for equality of proportions without continuity correction
##
## data: survey_q11_correct out of survey_q11_group_total
## X-squared = 9.9199, df = 4, p-value = 0.0418
## alternative hypothesis: two.sided
## sample estimates:
## prop 1 prop 2 prop 3 prop 4 prop 5
## 0.2096774 0.3559322 0.3934426 0.3636364 0.4754098
```

Post-hoc test

```
pairwise.prop.test(
  x = survey_q11_contingency[, ],
  n = colSums(survey_q11_contingency),
  p.adjust.method = "bonferroni" # or "holm", "BH"
)
```

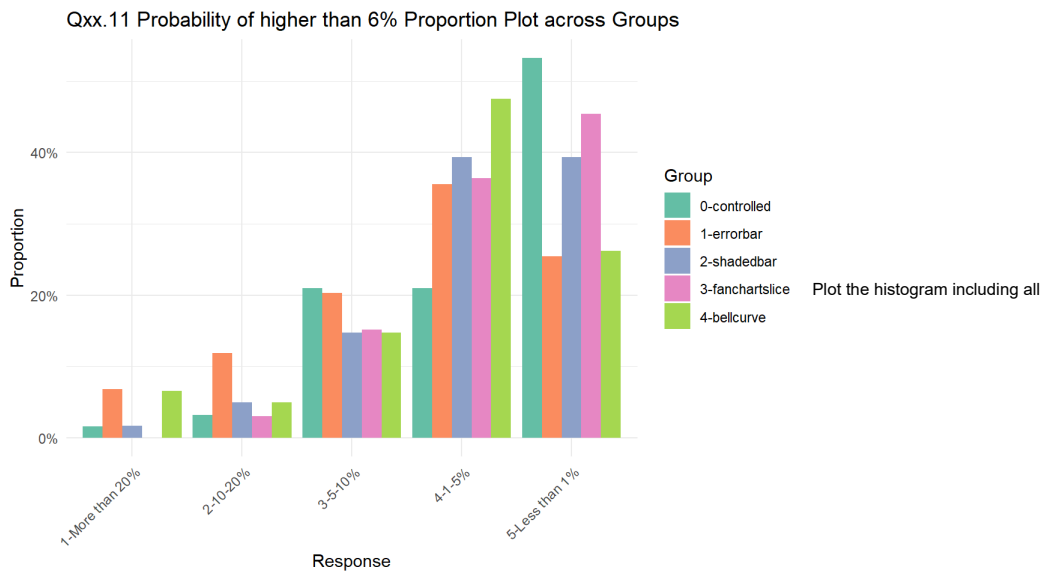
```
##
## Pairwise comparisons using Pairwise comparison of proportions
##
## data: survey_q11_contingency[, ] out of colSums(survey_q11_contingency)
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice
## 1-errorbar      1.000      -          -          -
## 2-shadedbar      0.428      1.000      -          -
## 3-fanchartslice 0.845      1.000      1.000      -
## 4-bellcurve      0.035      1.000      1.000      1.000
##
## P value adjustment method: bonferroni
```

Plot probability of higher than 6% proportion bar plot across different groups

```

survey_q11_prop <- as.data.frame(survey_q11_contingency)
# rename columns
names(survey_q11_prop) <- c("Qxx.11", "Group", "Count")
# add a labelled column
survey_q11_prop <- survey_q11_prop %>%
  mutate(Qxx.11_label = case_when(
    Qxx.11 == "1" ~ "1-More than 20%",
    Qxx.11 == "2" ~ "3-5-10%",
    Qxx.11 == "3" ~ "4-1-5%",
    Qxx.11 == "4" ~ "5-Less than 1%",
    Qxx.11 == "5" ~ "2-10-20%"
  ))
# calculate proportions within each group
survey_q11_prop <- survey_q11_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))
ggplot(survey_q11_prop, aes(x = as.factor(`Qxx.11_label`), y = Proportion, fill = Group)) + geom_col(position = "dodge") +
  # facet_wrap(~Group)+
  labs(
    title = "Qxx.11 Probability of higher than 6% Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2")+
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal()+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```

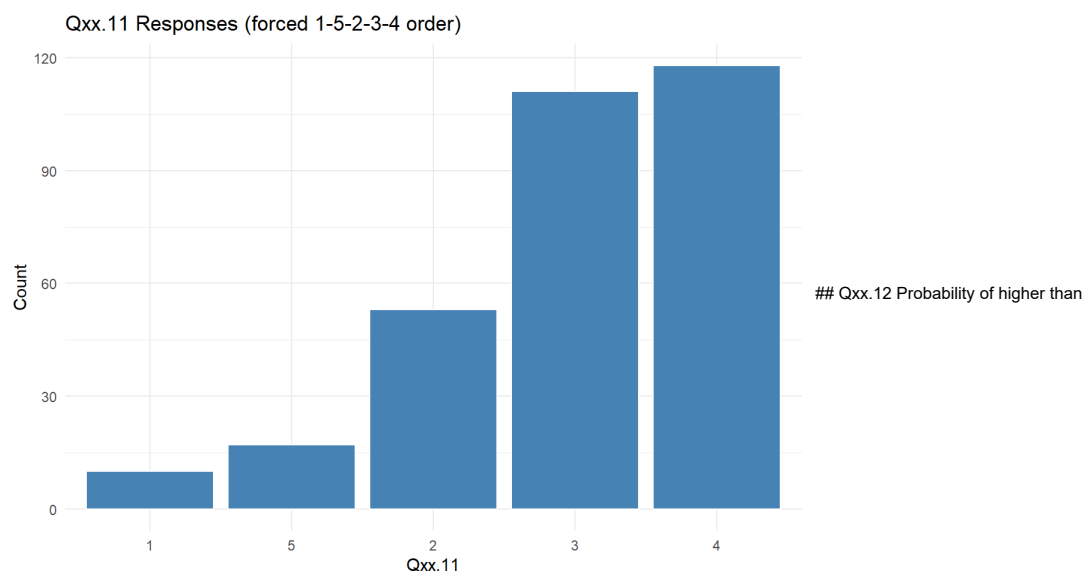


participants

```

# force order for graph
survey_q11 <- survey_q11 %>%
  mutate(
    q11_value_re = factor(
      q11_value,
      levels = c(1, 5, 2, 3, 4)
    )
  )
# histogram including all participants
ggplot(survey_q11, aes(x = q11_value_re)) +
  geom_bar(fill = "steelblue", color = "white") +
  labs(
    title = "Qxx.11 Responses (forced 1-5-2-3-4 order)",
    x = "Qxx.11",
    y = "Count"
  ) +
  theme_minimal()

```



3% Based on the inflation forecast provided, what do you think is the probability of inflation ending up higher than 3%?

Definitive Correct Answer - 1

variable: categorical

Method: contingency table & proportion test

Result: There is no statistical difference and the majority people didn't get the answer right.

```
# combine data from different groups
survey_q12 <- data.frame()
survey_q12_list <- list("Q4.12", "Q6.12", "Q8.12", "Q10.12", "Q12.12")
for (i in seq_along(group_list)) {
  df <- group_list[[i]]
  col <- survey_q12_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("q12_value", "group")
  survey_q12 <- rbind(survey_q12, temp_table)
}
survey_q12$q12_value <- as.numeric(survey_q12$q12_value)
```

Test the proportion of correct answer

```
survey_q12_contingency <- table(survey_q12$q12_value, survey_q12$group)
# print the contingency table
print(survey_q12_contingency)
```

```
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1           3           3           2           5           4
## 2          15          14          19          12          13
## 3          19          18          26          29          22
## 4          11           7          11          12          13
## 5          14          17           3           8           9
```

```
survey_q12_correct <- survey_q12_contingency[,2,]
survey_q12_group_total <- colSums(survey_q12_contingency)
prop.test(survey_q12_correct, survey_q12_group_total)
```

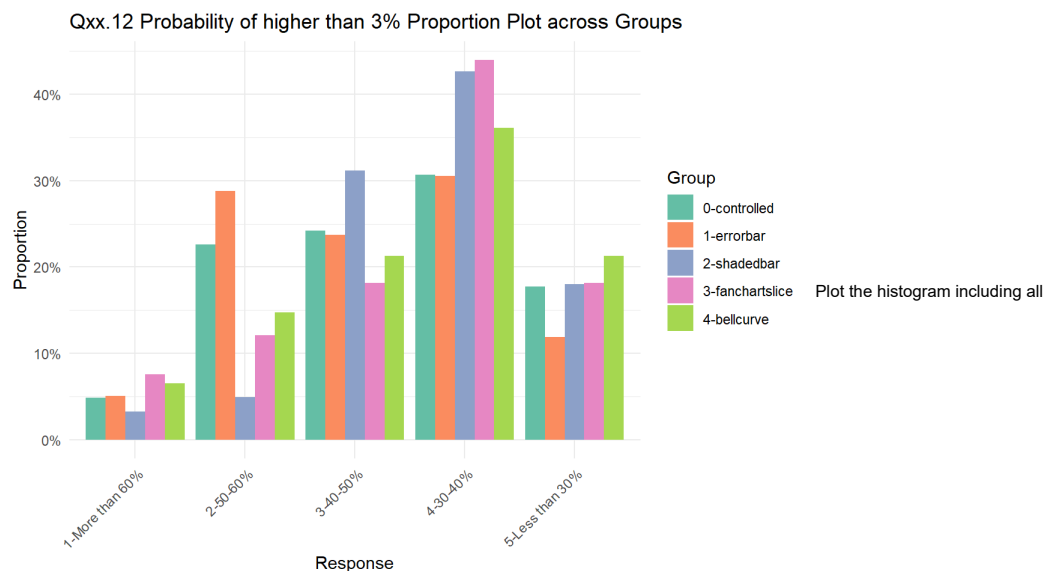
```
##
## 5-sample test for equality of proportions without continuity correction
##
## data: survey_q12_correct out of survey_q12_group_total
## X-squared = 3.1893, df = 4, p-value = 0.5267
## alternative hypothesis: two.sided
## sample estimates:
## prop 1 prop 2 prop 3 prop 4 prop 5
## 0.2419355 0.2372881 0.3114754 0.1818182 0.2131148
```

Plot probability of higher than 3% proportion bar plot across different groups


```

survey_q12_prop <- as.data.frame(survey_q12_contingency)
# rename columns
names(survey_q12_prop) <- c("Qxx.12", "Group", "Count")
# add a labelled column
survey_q12_prop <- survey_q12_prop %>%
  mutate(Qxx.12_label = case_when(
    Qxx.12 == "1" ~ "1-More than 60%",
    Qxx.12 == "2" ~ "3-40-50%",
    Qxx.12 == "3" ~ "4-30-40%",
    Qxx.12 == "4" ~ "5-Less than 30%",
    Qxx.12 == "5" ~ "2-50-60%"
  ))
# calculate proportions within each group
survey_q12_prop <- survey_q12_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))
ggplot(survey_q12_prop, aes(x = as.factor(`Qxx.12_label`), y = Proportion, fill = Group)) + geom_col(position = "dodge") +
  # facet_wrap(~Group)+
  labs(
    title = "Qxx.12 Probability of higher than 3% Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2")+
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal()+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```

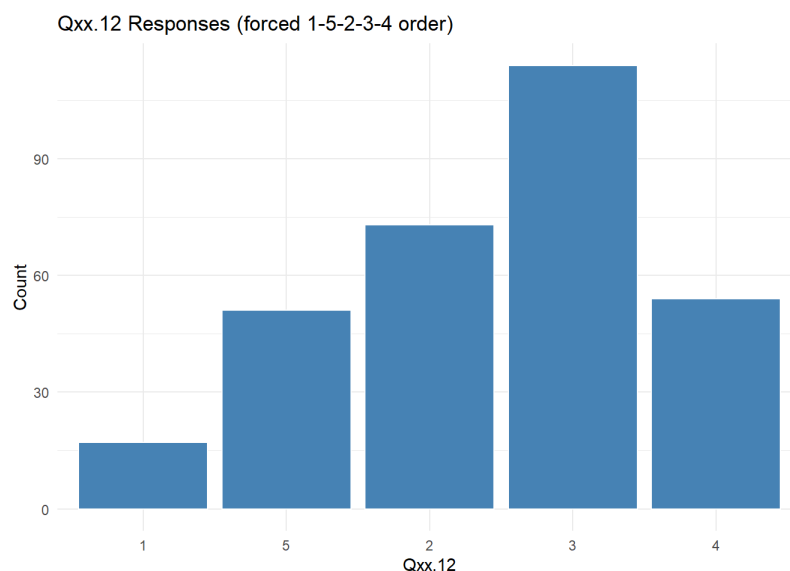


participants

```

# force order for graph
survey_q12 <- survey_q12 %>%
  mutate(
    q12_value_re = factor(
      q12_value,
      levels = c(1, 5, 2, 3, 4)
    )
  )
# histogram including all participants
ggplot(survey_q12, aes(x = q12_value_re)) +
  geom_bar(fill = "steelblue", color = "white") +
  labs(
    title = "Qxx.12 Responses (forced 1-5-2-3-4 order)",
    x = "Qxx.12",
    y = "Count"
  ) +
  theme_minimal()

```



! Qxx.14 Probability of higher than 2% target

To keep inflation low and stable, Government sets an inflation target of 2%. Based on the inflation forecast provided, what do you think is the probability of inflation exceeding 2%?

Definitive Correct Answer - 3

variable: categorical

Method: contingency table & proportion test

Result: Control group is significant different from error Bar group and bell curve group; Error bar group has shown statistical difference compared to fan chart slice. 44% of total respondents can understand the probability of inflation being higher than 6% correctly.

```
# combine data from different groups
survey_q14 <- data.frame()
survey_q14_list <- list("Q4.14", "Q6.14", "Q8.14", "Q10.14", "Q12.14")
for (i in seq_along(group_list)) {
  df <- group_list[[i]]
  col <- survey_q14_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("q14_value", "group")
  survey_q14 <- rbind(survey_q14, temp_table)
}
survey_q14$q14_value <- as.numeric(survey_q14$q14_value)
```

Test the proportion of correct answer

```
survey_q14_contingency <- table(survey_q14$q14_value, survey_q14$group)
# print the contingency table
print(survey_q14_contingency)
```

```
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1          10           1           1           1           4
## 2          21           3           7          14           7
## 3          16          36          29          22          33
## 4           9          10           8          18           2
## 5           3           7          13           8          12
## 6           2           1           3           3           3
## 7           1           1           0           0           0
```

```
survey_q14_correct <- survey_q14_contingency[,3]
survey_q14_group_total <- colSums(survey_q14_contingency)
prop.test(survey_q14_correct, survey_q14_group_total)
```

```
##
##      5-sample test for equality of proportions without continuity correction
##
## data: survey_q14_correct out of survey_q14_group_total
## X-squared = 21.144, df = 4, p-value = 0.0002965
## alternative hypothesis: two.sided
## sample estimates:
##      prop 1      prop 2      prop 3      prop 4      prop 5
## 0.2580645 0.6101695 0.4754098 0.3333333 0.5409836
```

Pairwise proportion test

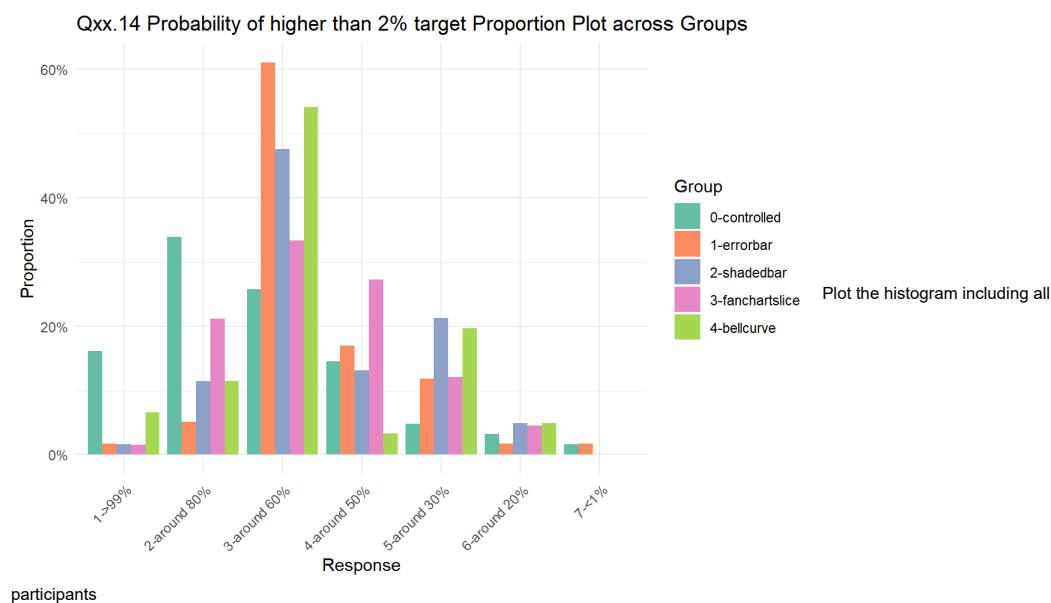
```
pairwise.prop.test(
  x = survey_q14_contingency[, ],
  n = colSums(survey_q14_contingency),
  p.adjust.method = "bonferroni" # or "holm", "BH"
)
```

```
##
## Pairwise comparisons using Pairwise comparison of proportions
##
## data: survey_q14_contingency[, ] out of colSums(survey_q14_contingency)
##
##          0-controlled 1-errorbar 2-shadedbar 3-fanchartslice
## 1-errorbar 0.0019      -          -          -
## 2-shadedbar 0.2062     1.0000     -          -
## 3-fanchartslice 1.0000    0.0352    1.0000    -
## 4-bellcurve 0.0253     1.0000     1.0000    0.2924
##
## P value adjustment method: bonferroni
```

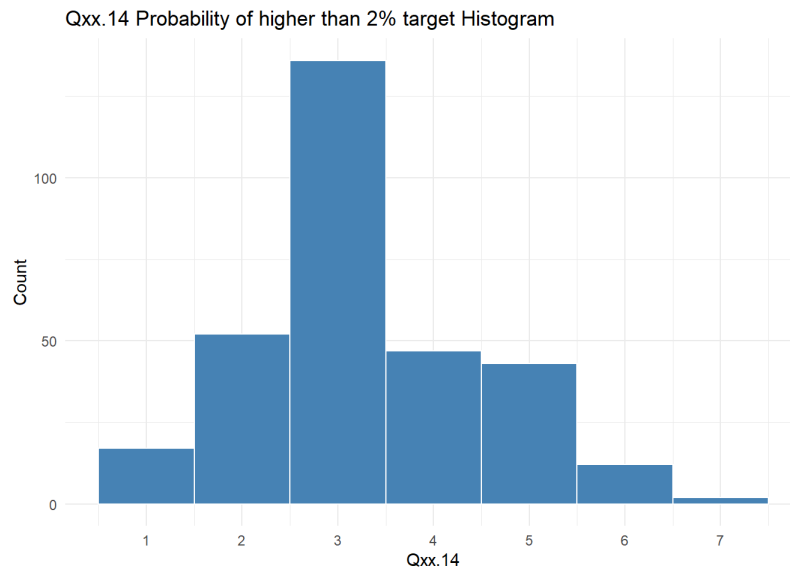
Plot probability of higher than 2% target proportion bar plot across different groups

```
survey_q14_prop <- as.data.frame(survey_q14_contingency)
# rename columns
names(survey_q14_prop) <- c("Qxx.14", "Group", "Count")
# add a labelled column
survey_q14_prop <- survey_q14_prop %>%
  mutate(Qxx.14_label = case_when(
    Qxx.14 == "1" ~ "1->99%",
    Qxx.14 == "2" ~ "2-around 80%",
    Qxx.14 == "3" ~ "3-around 60%",
    Qxx.14 == "4" ~ "4-around 50%",
    Qxx.14 == "5" ~ "5-around 30%",
    Qxx.14 == "6" ~ "6-around 20%",
    Qxx.14 == "7" ~ "7-<1%"
  ))
# calculate proportions within each group
survey_q14_prop <- survey_q14_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))

ggplot(survey_q14_prop, aes(x = as.factor(`Qxx.14_label`), y = Proportion, fill = Group)) + geom_col(position = "dodge") +
  # facet_wrap(~Group)+
  labs(
    title = "Qxx.14 Probability of higher than 2% target Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2")+
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal()+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
# histogram including all participants
ggplot(survey_q14, aes(x = q14_value)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  # facet_wrap(~ group) +
  labs(title = "Qxx.14 Probability of higher than 2% target Histogram",
        x = "Qxx.14",
        y = "Count") +
  scale_x_continuous(breaks = seq(0, max(survey_q14$q14_value, na.rm = TRUE), by = 1)) +
  theme_minimal()
```



Qxx.15 Deviation from central estimate

Based on the inflation forecast provided, do you think there is a higher risk of inflation in 2026Q2 exceeding the central estimate or of being less than this central estimate?

Definitive Correct Answer - 3

variable: categorical

Method: contingency table & proportion test

Result: The control group shows significant difference compared to the shaded bar group in terms of the correctly choosing understanding the risk is equal. Among all participants, only around 8% respondents think there is downside risk. The largest proportion (48%) of participants believe there is upside risk.

```
# combine data from different groups
survey_q15 <- data.frame()
survey_q15_list <- list("Q4.15", "Q6.15", "Q8.15", "Q10.15", "Q12.15")
for (i in seq_along(survey_q15_list)) {
  df <- survey_q15_list[[i]]
  col <- survey_q15_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("q15_value", "group")
  survey_q15 <- rbind(survey_q15, temp_table)
}
survey_q15$q15_value <- as.numeric(survey_q15$q15_value)
```

Contingency table

```
survey_q15_contingency <- table(survey_q15$q15_value, survey_q15$group)
# print the contingency table
print(survey_q15_contingency)
```

```
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1             43          25          19             29          31
## 2              2           7           6              4           7
## 3             17          27          36             33          23
```

```
# chi-squared test
chisq_result <- chisq.test(survey_q15_contingency)
```

```
## Warning in chisq.test(survey_q15_contingency): Chi-squared approximation may be
## incorrect
```

```
std_resid <- chisq_result$stdres
print(std_resid)
```

```
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1      3.8412018 -0.8891507 -2.8673008      -0.6665092  0.5667920
## 2      -1.6460439  1.0613387  0.4465225      -0.7767218  0.9613564
## 3      -2.9439670  0.3010103  2.6349682      1.1049110 -1.1078420
```

```
# correct answer
survey_q15_correct <- survey_q15_contingency[,3]
survey_q15_group_total <- colSums(survey_q15_contingency)
prop.test(survey_q15_correct, survey_q15_group_total)
```

```
##
## 5-sample test for equality of proportions without continuity correction
##
## data: survey_q15_correct out of survey_q15_group_total
## X-squared = 14.519, df = 4, p-value = 0.005811
## alternative hypothesis: two.sided
## sample estimates:
## prop 1 prop 2 prop 3 prop 4 prop 5
## 0.2741935 0.4576271 0.5901639 0.5000000 0.3770492
```

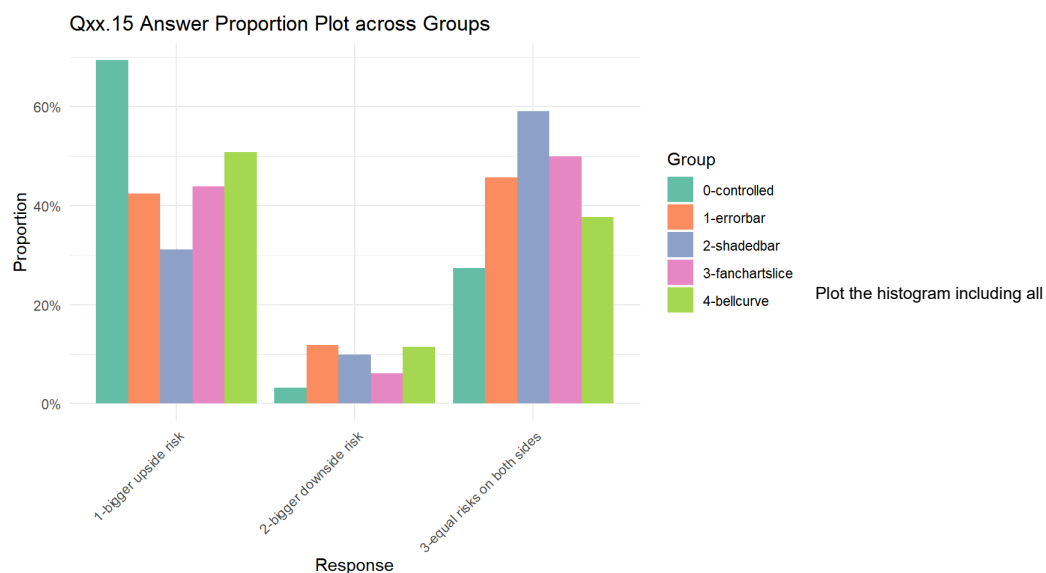
Pairwise proportion test

```
pairwise.prop.test(
  x = survey_q15_contingency[,3],
  n = colSums(survey_q15_contingency),
  p.adjust.method = "bonferroni" # or "holm", "BH"
)
```

```
##
## Pairwise comparisons using Pairwise comparison of proportions
##
## data: survey_q15_contingency[,3] out of colSums(survey_q15_contingency)
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice
## 1-errorbar      0.5644      -      -      -
## 2-shadedbar      0.0079      1.0000      -      -
## 3-fanchartslice 0.1487      1.0000      1.0000      -
## 4-bellcurve      1.0000      1.0000      0.2970      1.0000
##
## P value adjustment method: bonferroni
```

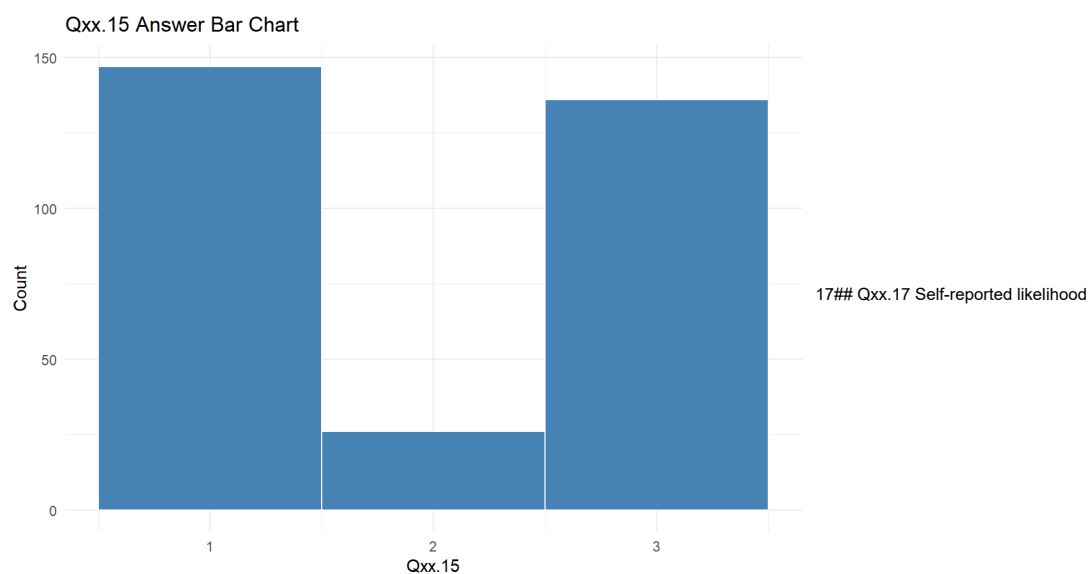
Plot proportion bar plot across different groups

```
survey_q15_prop <- as.data.frame(survey_q15_contingency)
# rename columns
names(survey_q15_prop) <- c("Qxx.15", "Group", "Count")
# add a labelled column
survey_q15_prop <- survey_q15_prop %>%
  mutate(Qxx.15_label = case_when(
    Qxx.15 == "1" ~ "1-bigger upside risk",
    Qxx.15 == "2" ~ "2-bigger downside risk",
    Qxx.15 == "3" ~ "3-equal risks on both sides"
  ))
# calculate proportions within each group
survey_q15_prop <- survey_q15_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))
ggplot(survey_q15_prop, aes(x = as.factor(`Qxx.15_label`), y = Proportion, fill = Group)) + geom_col(position = "dodge") +
  # facet_wrap(~Group)+
  labs(
    title = "Qxx.15 Answer Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2")+
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal()+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



participants

```
# histogram including all participants
ggplot(survey_q15, aes(x = q15_value)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  # facet_wrap(~ group) +
  labs(title = "Qxx.15 Answer Bar Chart",
        x = "Qxx.15",
        y = "Count") +
  scale_x_continuous(breaks = seq(0, max(survey_q15$q15_value, na.rm = TRUE), by = 1)) +
  theme_minimal()
```



of exactly 2.6% How likely do you think it is that the inflation will be exactly 2.6%?

ascending value-increasing likelihood

Variable: treat as continuous

Method: Kruskal-Wallis test

Result: Fail to reject the null hypothesis. The overall distribution is skewed towards unlikely.

```
# combine data from different groups
survey_q17 <- data.frame()
survey_q17_list <- list("Q4.17_1", "Q6.17_1", "Q8.17_1", "Q10.17_1", "Q12.17_1")
for (i in seq_along(group_list)) {
  df <- group_list[[i]]
  col <- survey_q17_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("q17_value", "group")
  survey_q17 <- rbind(survey_q17, temp_table)
}
survey_q17$q17_value <- as.numeric(survey_q17$q17_value)

#initial check
summary(survey_q17$q17_value)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##         9      10       11      11      12      15
```

Summary Table

```
survey_q17_summary <- survey_q17 %>%
  group_by(group) %>%
  summarise(
    median = median(q17_value, na.rm = TRUE),
    IQR     = IQR(q17_value, na.rm = TRUE), # same as Q3 - Q1
    Q1      = quantile(q17_value, 0.25, na.rm = TRUE),
    Q3      = quantile(q17_value, 0.75, na.rm = TRUE)
  )
print(survey_q17_summary)
```

```
## # A tibble: 5 x 5
##   group      median  IQR   Q1   Q3
##   <chr>      <dbl> <dbl> <dbl> <dbl>
## 1 0-controlled      11     2    10    12
## 2 1-errorbar       11     2    10    12
## 3 2-shadedbar      11     2    10    12
## 4 3-fanchartslice  11     2    10    12
## 5 4-bellcurve      11     2    10    12
```

KW test

```
# check normality --> the data is not normally distributed
# shapiro.test(survey_q17$q17_value)

# run Kruskal-Wallis test
kruskal_test <- kruskal.test(q17_value ~ group, data = survey_q17)
print(kruskal_test)
```

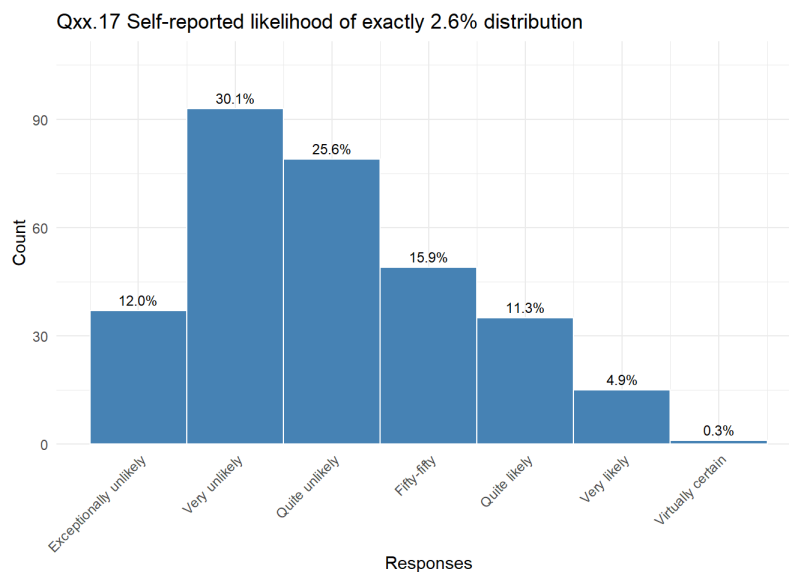
```
##
##  Kruskal-Wallis rank sum test
##
## data:  q17_value by group
## Kruskal-Wallis chi-squared = 1.4881, df = 4, p-value = 0.8287
```

```
# Effect size
kw_es <- kruskal_effsize(q17_value ~ group, data = survey_q17, ci = TRUE)
print(kw_es)
```

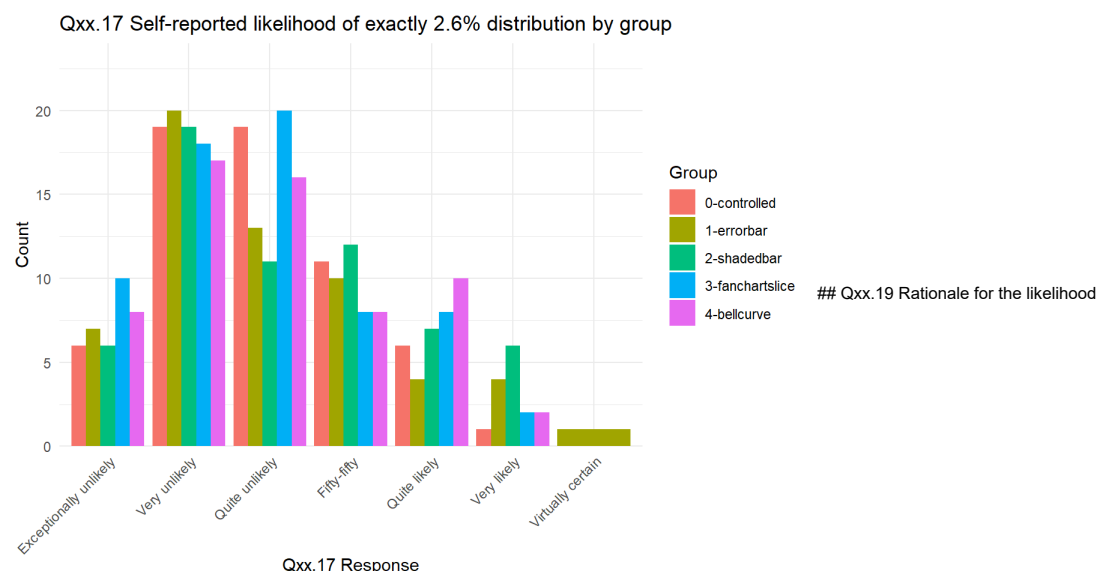
```
## # A tibble: 1 x 7
##   .y.      n effsize conf.low conf.high method magnitude
## * <chr>  <int>  <dbl>   <dbl>   <dbl> <chr>   <ord>
## 1 q17_value  309 -0.00826  -0.01    0.04 eta2[H] small
```

Plot the distribution of different groups

```
# histogram including all participants
ggplot(survey_q17, aes(x = q17_value)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  geom_text(
    stat = "count",
    aes(label = scales::percent(after_stat(count / total_respondents), accuracy = 0.1)),
    position = position_dodge(width = 0.9),
    vjust = -0.5,
    size = 3
  ) +
  # facet_wrap(~ group) +
  scale_y_continuous(
    expand = expansion(mult = c(0, 0.2))) +
  labs(title = "Qxx.17 Self-reported likelihood of exactly 2.6% distribution",
    x = "Responses",
    y = "Count") +
  scale_x_continuous(breaks = as.numeric(names(likelihood_labels)),
    labels = likelihood_labels) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
# Grouped bar-style histogram
ggplot(survey_q17, aes(x = factor(q17_value), fill = factor(group))) +
  geom_bar(position = "dodge") +
  # facet_wrap(~ group) +
  scale_y_continuous(
    expand = expansion(mult = c(0, 0.2))) +
  # add labels
  scale_x_discrete(labels=likelihood_labels) +
  labs(
    title = "Qxx.17 Self-reported likelihood of exactly 2.6% distribution by group",
    x = "Qxx.17 Response",
    y = "Count",
    fill = "Group"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

of inflation being exactly 2.6% What tells you that the inflation in 2026Q2 is likely or unlikely to be exactly 2.6%? (Please select the option that best reflects your belief, even if other answers may also seem reasonable.) - Selected Choice

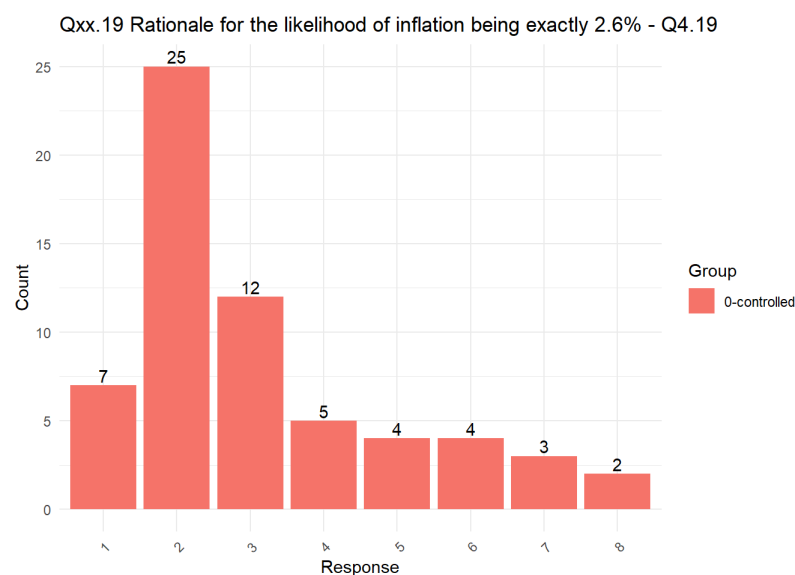
Variable: categorical

Method: contingency table + Not comparable across group

Function:

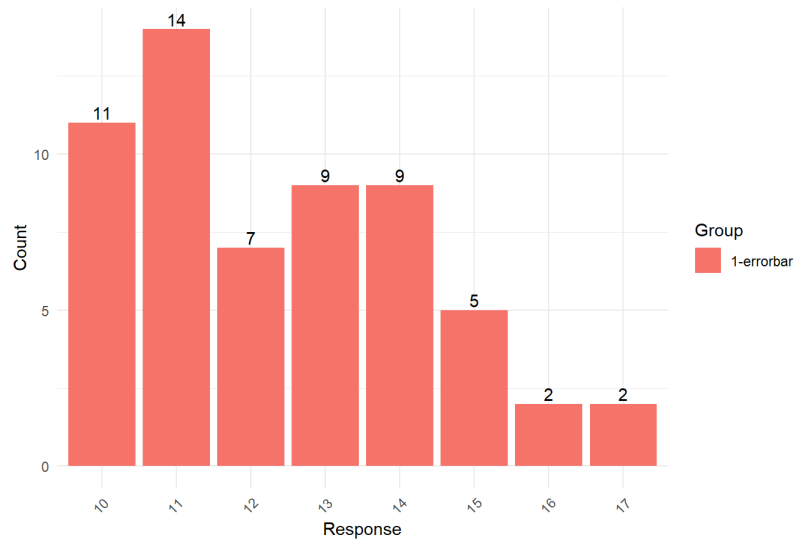
```
plot_survey_q19 <- function(survey_data, question_header){
  survey_q19 <- survey_data[, c(question_header, "group")]
  survey_q19_contingency <- table(survey_q19[[question_header]], survey_q19$group)

  # visualise the contingency table as a bar plot
  ggplot(as.data.frame(survey_q19_contingency), aes(x = Var1, y = Freq, fill = Var2)) + geom_text(aes(label = Freq),
    vjust = -0.25) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = paste("Qxx.19 Rationale for the likelihood of inflation being exactly 2.6% -", question_header),
    x = "Response",
    y = "Count",
    fill = "Group") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
}
plot_survey_q19(controlled_group, "Q4.19")
```



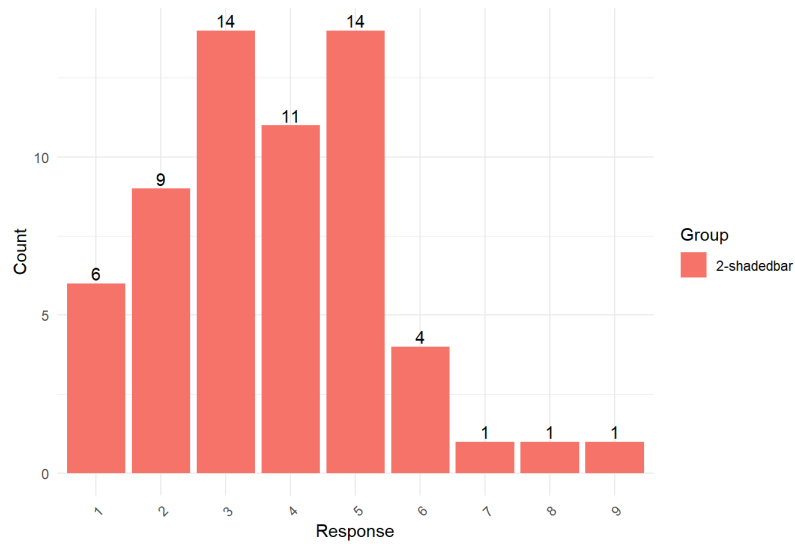
```
plot_survey_q19(error_bar, "Q6.19")
```

Qxx.19 Rationale for the likelihood of inflation being exactly 2.6% - Q6.19



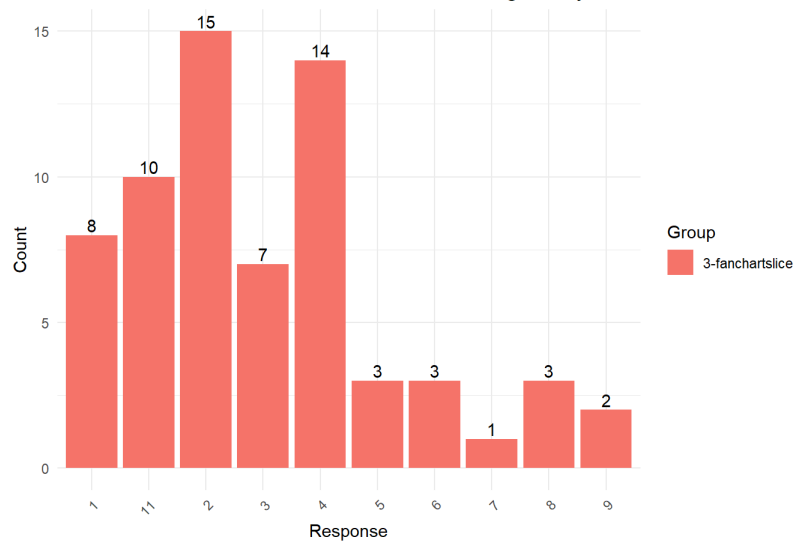
```
plot_survey_q19(shaded_bar, "Q8.19")
```

Qxx.19 Rationale for the likelihood of inflation being exactly 2.6% - Q8.19

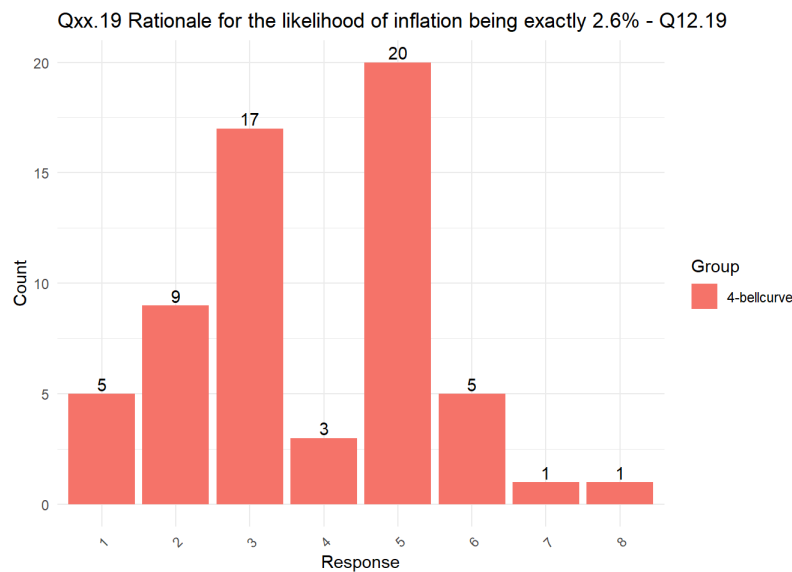


```
plot_survey_q19(fanchart_slice, "Q10.19")
```

Qxx.19 Rationale for the likelihood of inflation being exactly 2.6% - Q10.19



```
plot_survey_q19(bell_curve, "Q12.19")
```



Qxx.21 Self-reported likelihood of inflation being between 1.9% and 3.3%

How likely do you think it is that the inflation will be between 1.9% and 3.3%?

ascending value-increasing likelihood

Variable: treat as continuous

Method: Kruskal-Wallis test

Result: Fail to reject the null hypothesis. The majority of respondents think it's quite likely.

```
# combine data from different groups
survey_q21 <- data.frame()
survey_q21_list <- list("Q4.21_1", "Q6.21_1", "Q8.21_1", "Q10.21_1", "Q12.21_1")
for (i in seq_along(survey_q21_list)) {
  df <- survey_q21_list[[i]]
  col <- survey_q21_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("q21_value", "group")
  survey_q21 <- rbind(survey_q21, temp_table)
}
survey_q21$q21_value <- as.numeric(survey_q21$q21_value)
```

Summary Table

```
survey_q21_summary <- survey_q21 %>%
  group_by(group) %>%
  summarise(
    median = median(q21_value, na.rm = TRUE),
    IQR = IQR(q21_value, na.rm = TRUE), # same as Q3 - Q1
    Q1 = quantile(q21_value, 0.25, na.rm = TRUE),
    Q3 = quantile(q21_value, 0.75, na.rm = TRUE)
  )
print(survey_q21_summary)
```

```
## # A tibble: 5 x 5
##   group      median IQR   Q1   Q3
##   <chr>      <dbl> <dbl> <dbl> <dbl>
## 1 0-controlled      13     1    13    14
## 2 1-errorbar       13     1    12    13
## 3 2-shadedbar      13     2    12    14
## 4 3-fanchartslice  13     2    12    14
## 5 4-bellcurve      13     1    12    13
```

KW test

```
# summary(survey_q21)
# check normality --> the data is not normally distributed
# shapiro.test(survey_q21$q21_value)

# run Kruskal-Wallis test
kruskal_test <- kruskal.test(q21_value ~ group, data = survey_q21)
print(kruskal_test)
```

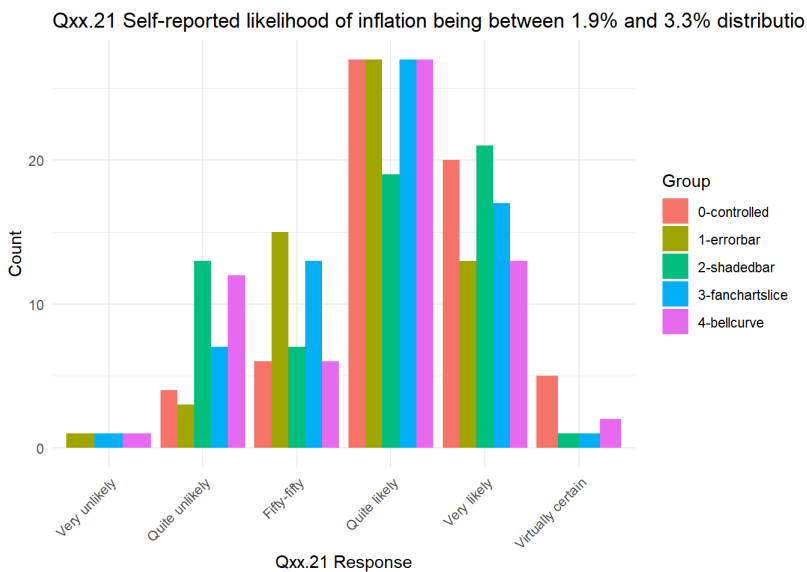
```
##
## Kruskal-Wallis rank sum test
##
## data: q21_value by group
## Kruskal-Wallis chi-squared = 8.6617, df = 4, p-value = 0.07013
```

```
# Effect size
kw_es <- kruskal_effsize(q21_value ~ group, data = survey_q21, ci = TRUE)
print(kw_es)
```

```
## # A tibble: 1 x 7
##   .y.      n effsize conf.low conf.high method magnitude
## * <chr>   <int>   <dbl>   <dbl>   <dbl> <chr>   <ord>
## 1 q21_value   309  0.0153   -0.004    0.07 eta2[H] small
```

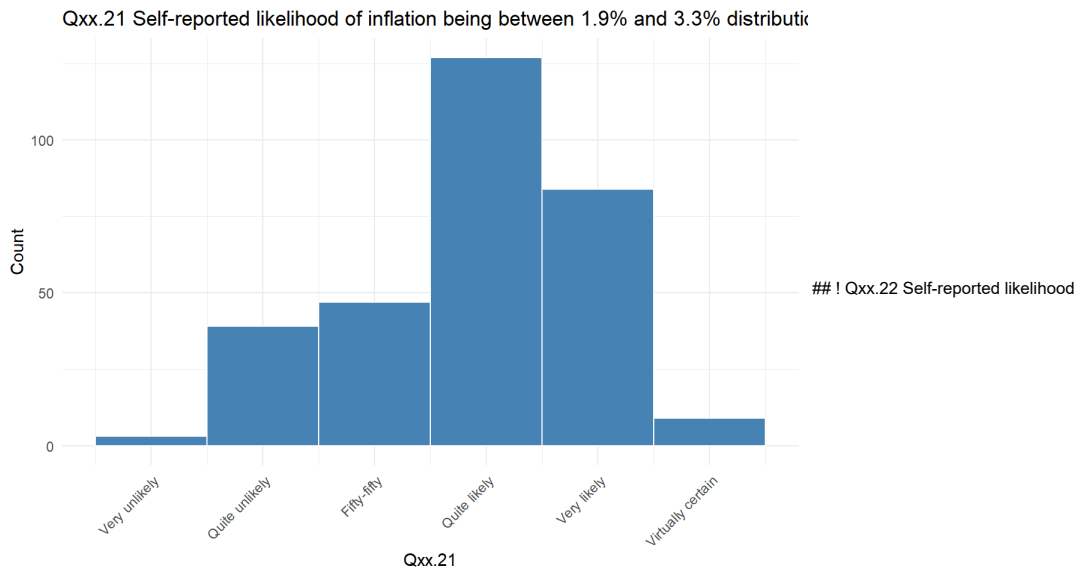
Plot the distribution of different groups

```
# Grouped bar-style histogram
ggplot(survey_q21, aes(x = factor(q21_value), fill = factor(group))) +
  geom_bar(position = "dodge") +
  # facet_wrap(~ group) +
  # add labels
  scale_x_discrete(labels=likelihood_labels) +
  labs(
    title = "Qxx.21 Self-reported likelihood of inflation being between 1.9% and 3.3% distribution by group",
    x = "Qxx.21 Response",
    y = "Count",
    fill = "Group"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Plot bar chart including all participants

```
# histogram including all participants
ggplot(survey_q21, aes(x = q21_value)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  # facet_wrap(~ group) +
  # add labels
  labs(title = "Qxx.21 Self-reported likelihood of inflation being between 1.9% and 3.3% distribution",
    x = "Qxx.21",
    y = "Count") +
  scale_x_continuous(breaks = as.numeric(names(likelihood_labels)),
    labels = likelihood_labels) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



of inflation being between 3.3% and 5.6% How likely do you think it is that the inflation will be between 3.3% and 5.6%?

ascending value-increasing likelihood

Variable: treat as continuous

Method: Kruskal-Wallis test

Result: The distribution of shaded bar group and bell curve show statistical difference. Near half of the respondents think it's quite unlikely.

```
# combine data from different groups
survey_q22 <- data.frame()
survey_q22_list <- list("Q4.22_1", "Q6.22_1", "Q8.22_1", "Q10.22_1", "Q12.22_1")
for (i in seq_along(group_list)) {
  df <- group_list[[i]]
  col <- survey_q22_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("q22_value", "group")
  survey_q22 <- rbind(survey_q22, temp_table)
}
survey_q22$q22_value <- as.numeric(survey_q22$q22_value)
summary(survey_q22$q22_value)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      9.00  11.00   11.00   11.36  12.00   15.00
```

Summary Table

```
survey_q22_summary <- survey_q22 %>%
  group_by(group) %>%
  summarise(
    median = median(q22_value, na.rm = TRUE),
    IQR     = IQR(q22_value, na.rm = TRUE), # same as Q3 - Q1
    Q1      = quantile(q22_value, 0.25, na.rm = TRUE),
    Q3      = quantile(q22_value, 0.75, na.rm = TRUE),
    mean    = mean(q22_value, na.rm = TRUE),
  )
print(survey_q22_summary)
```

```
## # A tibble: 5 × 6
##   group      median  IQR    Q1    Q3  mean
##   <chr>      <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 0-controlled    11     1    11    12  11.3
## 2 1-errorbar     11     1    11    12  11.3
## 3 2-shadedbar    11     0    11    11  11.1
## 4 3-fanchartslice 11     1    11    12  11.4
## 5 4-bellcurve    11     2    11    13  11.7
```

KW test

```
# check normality --> the data is not normally distributed
# shapiro.test(survey_q22$q22_value)

# run Kruskal-Wallis test
kruskal_test <- kruskal.test(q22_value ~ group, data = survey_q22)
print(kruskal_test)
```

```
##
## Kruskal-Wallis rank sum test
##
## data: q22_value by group
## Kruskal-Wallis chi-squared = 9.8534, df = 4, p-value = 0.04297
```

```
# Effect size
kw_es <- kruskal_effsize(q22_value ~ group, data = survey_q22, ci = TRUE)
print(kw_es)
```

```
## # A tibble: 1 x 7
##   .y.      n effsize conf.low conf.high method magnitude
## * <chr>   <int>   <dbl>   <dbl>   <dbl> <chr>   <ord>
## 1 q22_value    309  0.0193  -0.0038    0.08 eta2[H] small
```

```
# Post-hoc test
dunnTest(q22_value ~ group, data = survey_q22, method = "bonferroni")
```

```
## Warning: group was coerced to a factor.
```

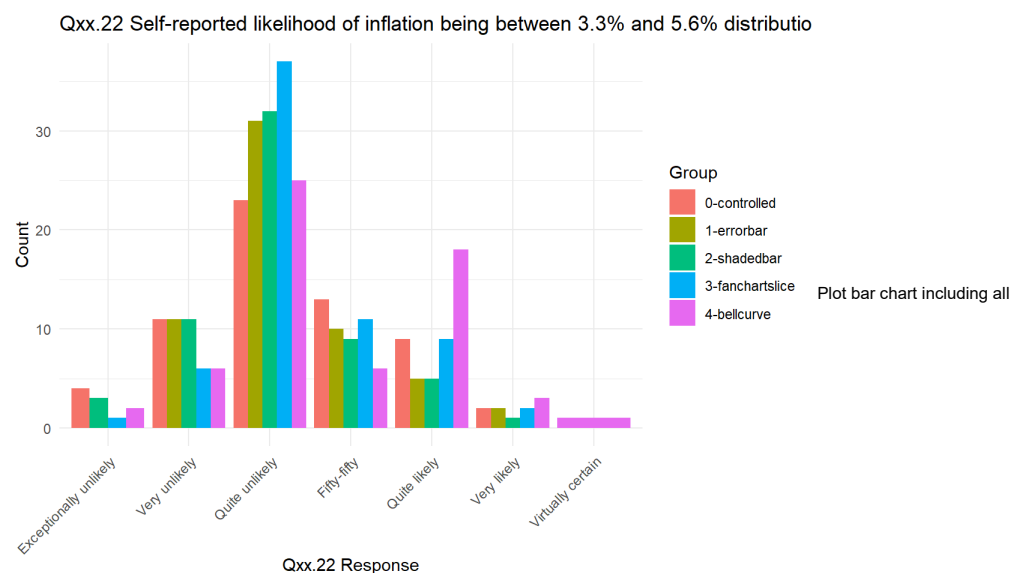
```
## Dunn (1964) Kruskal-Wallis multiple comparison
```

```
## p-values adjusted with the Bonferroni method.
```

```
##           Comparison      Z    P.unadj    P.adj
## 1 0-controlled - 1-errorbar 0.3519613 0.724867319 1.0000000
## 2 0-controlled - 2-shadedbar 1.0909284 0.275304410 1.0000000
## 3 1-errorbar - 2-shadedbar 0.7268652 0.467308520 1.0000000
## 4 0-controlled - 3-fanchartslice -0.5808237 0.561359281 1.0000000
## 5 1-errorbar - 3-fanchartslice -0.9306361 0.352041833 1.0000000
## 6 2-shadedbar - 3-fanchartslice -1.6860866 0.091779118 0.91779118
## 7 0-controlled - 4-bellcurve -1.8775914 0.060437079 0.60437079
## 8 1-errorbar - 4-bellcurve -2.2049193 0.027459761 0.27459761
## 9 2-shadedbar - 4-bellcurve -2.9565257 0.003111263 0.03111263
## 10 3-fanchartslice - 4-bellcurve -1.3280766 0.184152801 1.0000000
```

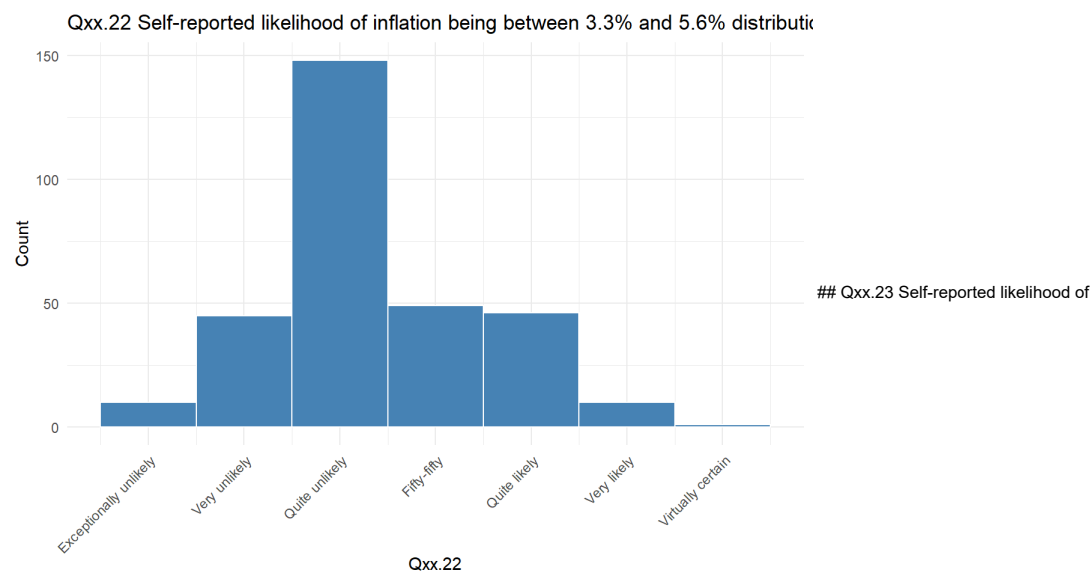
Plot the distribution of different groups

```
# Grouped bar-style histogram
ggplot(survey_q22, aes(x = factor(q22_value), fill = factor(group))) +
  geom_bar(position = "dodge") +
  # facet_wrap(~ group) +
  # add labels
scale_x_discrete(labels=likelihood_labels) +
labs(
  title = "Qxx.22 Self-reported likelihood of inflation being between 3.3% and 5.6% distribution by group",
  x = "Qxx.22 Response",
  y = "Count",
  fill = "Group"
) +
theme_minimal() +
theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



participants

```
# histogram including all participants
ggplot(survey_q22, aes(x = q22_value)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  # facet_wrap(~ group) +
  labs(title = "Qxx.22 Self-reported likelihood of inflation being between 3.3% and 5.6% distribution",
        x = "Qxx.22",
        y = "Count") +
  scale_x_continuous(breaks = as.numeric(names(likelihood_labels)),
                    labels = likelihood_labels) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



inflation being higher than 3% How likely do you think it is that the inflation will be higher than 3%?

ascending value-increasing likelihood

Variable: treat as continuous

Method: Kruskal-Wallis test

Result: Fail to reject the null hypothesis. The majority of respondents think it's quite unlikely, but there is no significant difference.

```
# combine data from different groups
survey_q23 <- data.frame()
survey_q23_list <- list("Q4.23_1", "Q6.23_1", "Q8.23_1", "Q10.23_1", "Q12.23_1")
for (i in seq_along(group_list)) {
  df <- group_list[[i]]
  col <- survey_q23_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("q23_value", "group")
  survey_q23 <- rbind(survey_q23, temp_table)
}
survey_q23$q23_value <- as.numeric(survey_q23$q23_value)
summary(survey_q23$q23_value)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      9.0    11.0    12.0    11.9    13.0    15.0
```

Summary Table

```
survey_q23_summary <- survey_q23 %>%
  group_by(group) %>%
  summarise(
    median = median(q23_value, na.rm = TRUE),
    IQR    = IQR(q23_value, na.rm = TRUE), # same as Q3 - Q1
    Q1     = quantile(q23_value, 0.25, na.rm = TRUE),
    Q3     = quantile(q23_value, 0.75, na.rm = TRUE)
  )
print(survey_q23_summary)
```

```
## # A tibble: 5 × 5
##   group      median  IQR   Q1   Q3
##   <chr>      <dbl> <dbl> <dbl> <dbl>
## 1 0-controlled      12     2    11    13
## 2 1-errorbar       12     2    11    13
## 3 2-shadedbar      12     1    11    12
## 4 3-fanchartslice  12     2    11    13
## 5 4-bellcurve      12     2    11    13
```

KW test

```
# check normality --> the data is not normally distributed
# shapiro.test(survey_q23$q23_value)

# initial check
summary(survey_q23)
```

```
##      q23_value      group
##      Min.   : 9.0   Length:309
##      1st Qu.:11.0   Class :character
##      Median :12.0   Mode  :character
##      Mean   :11.9
##      3rd Qu.:13.0
##      Max.   :15.0
```

```
# run Kruskal-Wallis test
kruskal_test <- kruskal.test(q23_value ~ group, data = survey_q23)
print(kruskal_test)
```

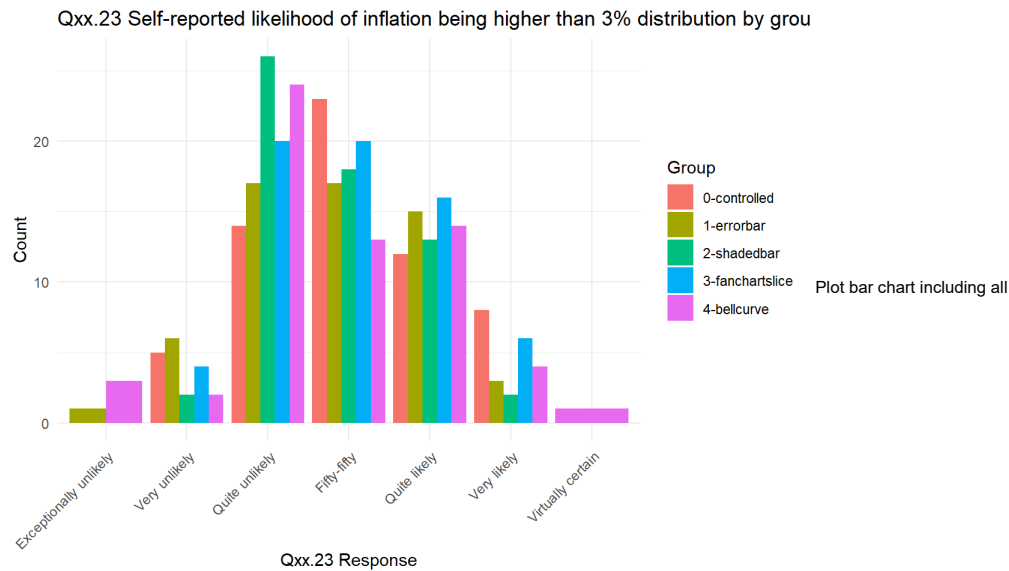
```
##
##  Kruskal-Wallis rank sum test
##
## data:  q23_value by group
## Kruskal-Wallis chi-squared = 3.1999, df = 4, p-value = 0.525
```

```
# Effect size
kw_es <- kruskal_effsize(q23_value ~ group, data = survey_q23, ci = TRUE)
print(kw_es)
```

```
## # A tibble: 1 × 7
##   .y.      n effsize conf.low conf.high method magnitude
## * <chr> <int> <dbl> <dbl> <dbl> <chr> <ord>
## 1 q23_value 309 -0.00263 -0.0098 0.05 eta2[H] small
```

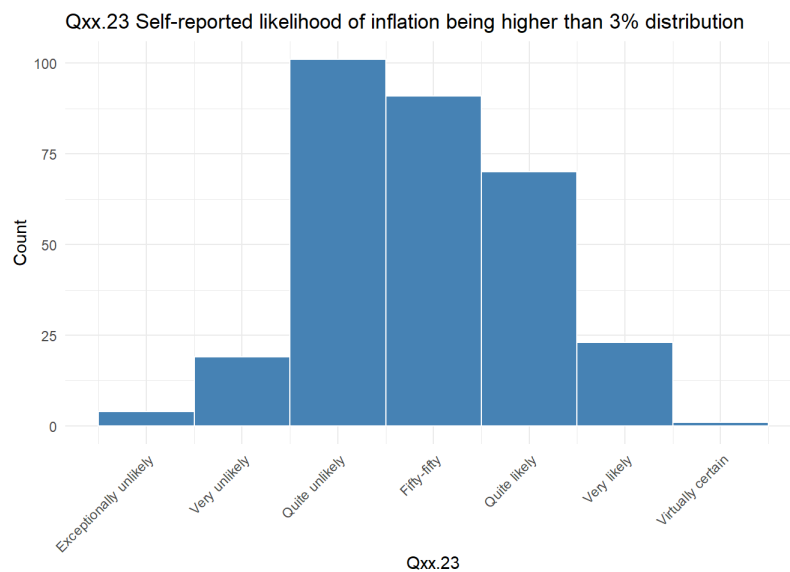
Plot the distribution of different groups


```
# Grouped bar-style histogram
ggplot(survey_q23, aes(x = factor(q23_value), fill = factor(group))) +
  geom_bar(position = "dodge") +
  # facet_wrap(~ group) +
  # add labels
  scale_x_discrete(labels=likelihood_labels) +
  labs(
    title = "Qxx.23 Self-reported likelihood of inflation being higher than 3% distribution by group",
    x = "Qxx.23 Response",
    y = "Count",
    fill = "Group"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



participants

```
# histogram including all participants
ggplot(survey_q23, aes(x = q23_value)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  # facet_wrap(~ group) +
  labs(title = "Qxx.23 Self-reported likelihood of inflation being higher than 3% distribution",
    x = "Qxx.23",
    y = "Count") +
  scale_x_continuous(breaks = as.numeric(names(likelihood_labels)),
    labels = likelihood_labels) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



! Qxx.24 Ranking

Based on the inflation forecast provided, please rank the following inflation rates from most likely to least likely to occur: 0.8%, 2.5%, 3.9% and 6.5%. - 0.8%

Lower mean rank = *more preferred*

Method: Mean Rank

```
# combine data from different groups
survey_q24 <- controlled_group[, c("Q4.24_1", "Q4.24_2", "Q4.24_3", "Q4.24_4")]
survey_q24 <- as.data.frame(lapply(survey_q24, as.numeric))
mean_rank <- colMeans(survey_q24, na.rm = TRUE)

rank_summary_controlled <- data.frame(
  Item = c("0.8%", "2.5%", "3.9%", "6.5%"),
  MeanRank = mean_rank
)
```

combine into one table

```
# main code
q24_prefixes <- list("Q4.24", "Q6.24", "Q8.24", "Q10.24", "Q12.24")
rank_summary <- data.frame(
  Item = c("0.8%", "2.5%", "3.9%", "6.5%")
)
survey_q24 <- data.frame()

for (i in seq_along(group_list)) {
  group_name <- names(group_list)[i]
  df <- group_list[[i]]
  q_cols <- paste0(q24_prefixes[[i]], "_", 1:4)
  temp_table <- df[, c(q_cols, "group", "treatment")]
  # combine the data
  names(temp_table) <- c("Qxx.24_1", "Qxx.24_2", "Qxx.24_3", "Qxx.24_4", "group", "treatment")
  temp_table[, c("Qxx.24_1", "Qxx.24_2", "Qxx.24_3", "Qxx.24_4")] <-
    lapply(temp_table[, c("Qxx.24_1", "Qxx.24_2", "Qxx.24_3", "Qxx.24_4")], as.numeric)
  survey_q24 <- rbind(survey_q24, temp_table)

  # mean rank
  df <- as.data.frame(lapply(df[, q_cols], as.numeric))
  mean_rank <- colMeans(df, na.rm = TRUE)
  rank_summary[[group_name]] <- mean_rank
}

# overall means
overall_means <- rowMeans(rank_summary[, -1], na.rm = TRUE)
rank_summary <- cbind(rank_summary, overall = overall_means)

# correct answer
survey_q24_correct_prop <- survey_q24 %>%
  group_by(group) %>%
  summarise(
    total = n(),
    correct = sum(
      Qxx.24_1 == 3 & Qxx.24_2 == 1 & Qxx.24_3 == 2 & Qxx.24_4 == 4,
      na.rm = TRUE
    ),
    proportion_correct = correct / total,
    .groups = "drop"
  )
```

proportion test

```
prop.test(
  x = survey_q24_correct_prop$correct,
  n = survey_q24_correct_prop$total
)
```

```
##
## 5-sample test for equality of proportions without continuity correction
##
## data: survey_q24_correct_prop$correct out of survey_q24_correct_prop$total
## X-squared = 11.062, df = 4, p-value = 0.02588
## alternative hypothesis: two.sided
## sample estimates:
## prop 1 prop 2 prop 3 prop 4 prop 5
## 0.4032258 0.5084746 0.6557377 0.4696970 0.6229508
```

```
# pairwise proportion test
pairwise.prop.test(
  x = survey_q24_correct_prop$correct,
  n = survey_q24_correct_prop$total,
  p.adjust.method = "bonferroni" # or "holm", "BH"
)
```

```
##
## Pairwise comparisons using Pairwise comparison of proportions
##
## data: survey_q24_correct_prop$correct out of survey_q24_correct_prop$total
##
##   1    2    3    4
## 2 1.000 -    -    -
## 3 0.087 1.000 -    -
## 4 1.000 1.000 0.535 -
## 5 0.240 1.000 1.000 1.000
##
## P value adjustment method: bonferroni
```

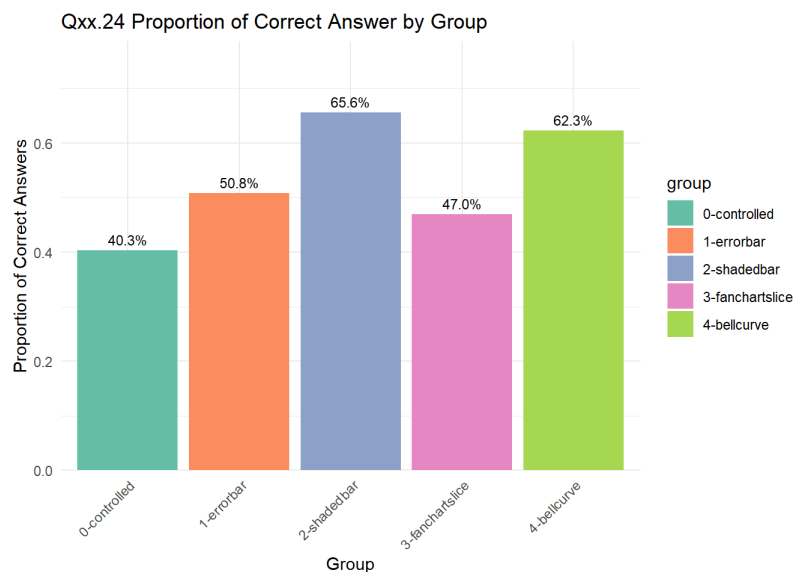
Proportion of correct answer

```
survey_q24 <- controlled_group[, c("Q4.24_1", "Q4.24_2", "Q4.24_3", "Q4.24_4")]
survey_q24 <- as.data.frame(lapply(survey_q24, as.numeric))
survey_q24_correct <- survey_q24[survey_q24$Q4.24_1 == 3 & survey_q24$Q4.24_2 == 1 & survey_q24$Q4.24_3 == 2 & survey_q24$Q4.24_4 == 4, ]
proportion_correct <- nrow(survey_q24_correct) / nrow(survey_q24)
print(proportion_correct)
```

```
## [1] 0.4032258
```

Visualisation Proportion

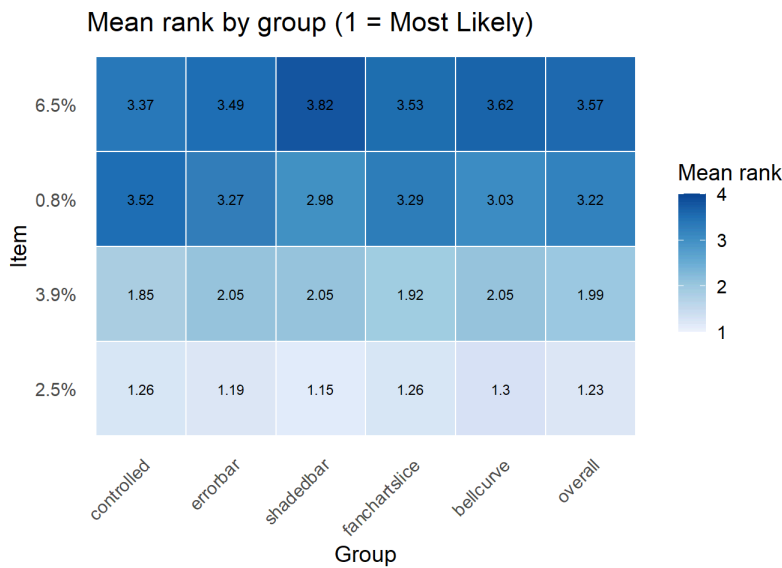
```
ggplot(survey_q24_correct_prop, aes(x = group, y = proportion_correct, fill = group)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = scales::percent(proportion_correct, accuracy = 0.1)),
    vjust = -0.5, size = 3) +
  labs(
    title = "Qxx.24 Proportion of Correct Answer by Group",
    x = "Group",
    y = "Proportion of Correct Answers"
  ) +
  scale_y_continuous(
    expand = expansion(mult = c(0, 0.2))
  ) +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Visualisation Ranking

```
df_long <- rank_summary %>%
  pivot_longer(~Item, names_to = "Group", values_to = "MeanRank") %>%
  mutate(
    # Order items by their overall best rank (smallest = best)
    Item = factor(Item, levels = rank_summary %>% arrange(overall) %>% pull(Item)),
    # Keep groups in the order they appear in the data-frame
    Group = factor(Group, levels = names(rank_summary)[-1])
  )

ggplot(df_long, aes(Group, Item, fill = MeanRank)) +
  geom_tile(colour = "white", linewidth = .4) +
  geom_text(aes(label = round(MeanRank, 2)), size = 3) +
  scale_fill_distiller(palette = "Blues", direction = 1,
    name = "Mean rank", limits = c(1, 4),
    breaks = 1:4) +
  labs(title = "Mean rank by group (1 = Most Likely)") +
  theme_minimal(base_size = 14) +
  theme(panel.grid = element_blank(),
    axis.text.x = element_text(angle = 45, hjust = 1))
```



Qxx.26 Accuracy of central estimate

On a scale of 1 to 10, where 1 means not accurate at all and 10 means very accurate, how accurate do you think the central estimate in the graph is?

1-Not accurate at all; 10-Very accurate

Variable: continuous

Method: ANNOVA test or KW test (depends on whether it's normally distributed)

Result: Fail to reject the null hypothesis.

```
# combine data from different groups
survey_q26 <- data.frame()
survey_q26_list <- list("Q4.26_1", "Q6.26_1", "Q8.26_1", "Q10.26_1", "Q12.26_1")
for (i in seq_along(survey_q26_list)) {
  df <- group_list[[i]]
  col <- survey_q26_list[[i]]
  temp_table <- df[, c(col, "group", "treatment")]
  names(temp_table) <- c("q26_value", "group", "treatment")
  survey_q26 <- rbind(survey_q26, temp_table)
}
survey_q26$q26_value <- as.numeric(survey_q26$q26_value)
summary(survey_q26$q26_value)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1.000   6.000   7.000   6.421   7.000  10.000
```

Summary Table

```

survey_q26_summary <- survey_q26 %>%
  group_by(group) %>%
  summarise(
    mean = mean(q26_value, na.rm = TRUE),
    median = median(q26_value, na.rm = TRUE),
    sd = sd(q26_value, na.rm = TRUE),
    IQR = IQR(q26_value, na.rm = TRUE), # same as Q3 - Q1
    Q1 = quantile(q26_value, 0.25, na.rm = TRUE),
    Q3 = quantile(q26_value, 0.75, na.rm = TRUE)
  )
print(survey_q26_summary)

```

```

## # A tibble: 5 x 7
##   group      mean median    sd   IQR   Q1   Q3
##   <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 0-controlled    6.15     7  1.79     2     5     7
## 2 1-errorbar     6.41     7  1.60     2     5     7
## 3 2-shadedbar     6.64     7  1.60     2     6     8
## 4 3-fanchartslice 6.38     7  1.69     2     5     7
## 5 4-bellcurve     6.54     6  1.60     2     6     8

```

Statistical test

```

# Initial Check
# summary(survey_q26$q26_value)

# check normality --> the data is not normally distributed
# shapiro.test(survey_q26$q26_value)

# run Kruskal-Wallis test
kruskal_test <- kruskal.test(q26_value ~ group, data = survey_q26)
print(kruskal_test)

```

```

##
## Kruskal-Wallis rank sum test
##
## data: q26_value by group
## Kruskal-Wallis chi-squared = 1.3483, df = 4, p-value = 0.8531

```

```

# Effect size
kw_es <- kruskal_effsize(q26_value ~ group, data = survey_q26, ci = TRUE)
print(kw_es)

```

```

## # A tibble: 1 x 7
##   .y.      n effsize conf.low conf.high method magnitude
## * <chr> <int> <dbl> <dbl> <dbl> <chr> <ord>
## 1 q26_value 309 -0.00872 -0.01 0.04 eta2[H] small

```

Combine the treatment groups

```

kruskal_test <- kruskal.test(q26_value ~ treatment, data = survey_q26)
print(kruskal_test)

```

```

##
## Kruskal-Wallis rank sum test
##
## data: q26_value by treatment
## Kruskal-Wallis chi-squared = 0.62468, df = 1, p-value = 0.4293

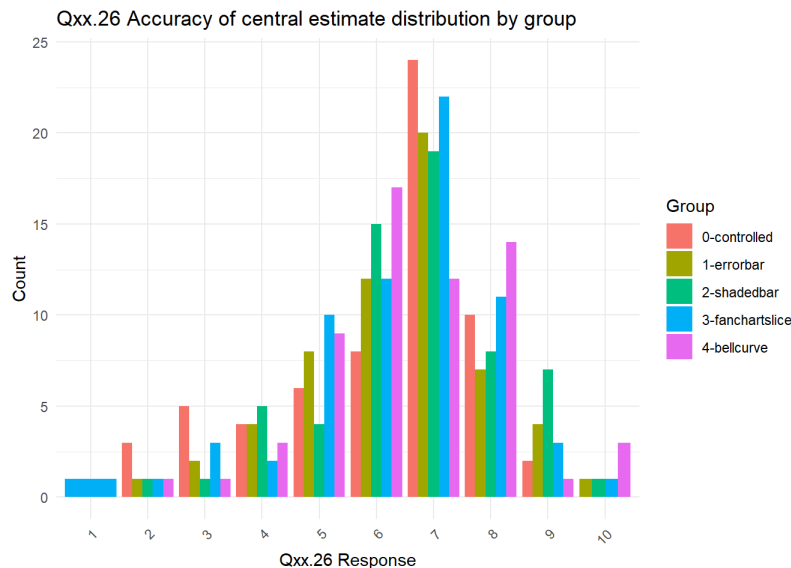
```

Plot the distribution of different groups

```

# Grouped bar-style histogram\
ggplot(survey_q26, aes(x = factor(q26_value), fill = factor(group))) +
  geom_bar(position = "dodge") +
  # facet_wrap(~ group) +
  labs(
    title = "Qxx.26 Accuracy of central estimate distribution by group",
    x = "Qxx.26 Response",
    y = "Count",
    fill = "Group"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```



Qxx.27 Certainty of central estimate

On a scale of 1 to 10, where 1 means very uncertain and 10 means very certain, how certain or uncertain do you think the inflation forecast for 2026 Q2 is?

1-Very uncertain; 10-Very certain

Variable: continuous

Method: ANNOVA test or KW test (depends on whether it's normally distributed)

Result: Fail to reject the null hypothesis.

```
# combine data from different groups
survey_q27 <- data.frame()
survey_q27_list <- list("Q4.27_1", "Q6.27_1", "Q8.27_1", "Q10.27_1", "Q12.27_1")
for (i in seq_along(group_list)) {
  df <- group_list[[i]]
  col <- survey_q27_list[[i]]
  temp_table <- df[, c(col, "group", "treatment")]
  names(temp_table) <- c("q27_value", "group", "treatment")
  survey_q27 <- rbind(survey_q27, temp_table)
}
survey_q27$q27_value <- as.numeric(survey_q27$q27_value)
summary(survey_q27$q27_value)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  1.000   5.000   6.000   5.838   7.000  10.000
```

Statistical test

```
# Initial Check
# summary(survey_q27$q27_value)
# check normality --> the data is not normally distributed
# shapiro.test(survey_q27$q27_value)
# run Kruskal-Wallis test
kruskal_test <- kruskal.test(q27_value ~ group, data = survey_q27)
print(kruskal_test)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  q27_value by group
## Kruskal-Wallis chi-squared = 0.99263, df = 4, p-value = 0.9109
```

```
# Effect size
kw_es <- kruskal_effsize(q27_value ~ group, data = survey_q27, ci = TRUE)
print(kw_es)
```

```
## # A tibble: 1 x 7
##   .y.      n effsize conf.low conf.high method  magnitude
## * <chr>   <int>   <dbl>   <dbl>   <dbl> <chr>   <ord>
## 1 q27_value 309 -0.00989   -0.01    0.03 eta2[H] small
```

Combine the treatment groups

```
kruskal_test <- kruskal.test(q27_value ~ treatment, data = survey_q27)
print(kruskal_test)
```

```
##
## Kruskal-Wallis rank sum test
##
## data: q27_value by treatment
## Kruskal-Wallis chi-squared = 0.056199, df = 1, p-value = 0.8126
```

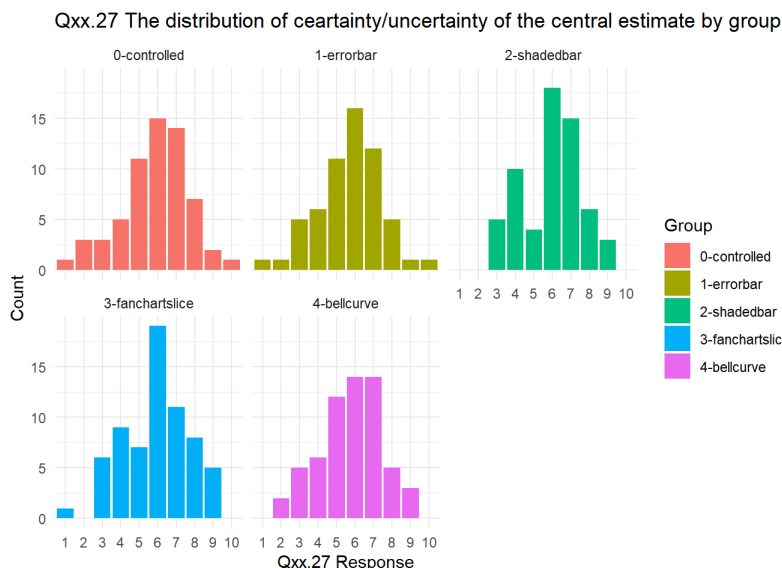
Summary statistics

```
# Summary statistics
survey_q27_summary <- survey_q27 %>%
  group_by(group) %>%
  summarise(
    mean = mean(q27_value, na.rm = TRUE),
    median = median(q27_value, na.rm = TRUE),
    sd = sd(q27_value, na.rm = TRUE),
    IQR = IQR(q27_value, na.rm = TRUE), # same as Q3 - Q1
    Q1 = quantile(q27_value, 0.25, na.rm = TRUE),
    Q3 = quantile(q27_value, 0.75, na.rm = TRUE)
  )
print(survey_q27_summary)
```

```
## # A tibble: 5 x 7
##   group mean median sd IQR Q1 Q3
##   <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 0-controlled 5.85 6 1.84 2 5 7
## 2 1-errorbar 5.69 6 1.72 2 5 7
## 3 2-shadedbar 5.95 6 1.61 2 5 7
## 4 3-fanchartslice 5.91 6 1.80 2 5 7
## 5 4-bellcurve 5.77 6 1.69 2 5 7
```

Plot the distribution of different groups

```
# Grouped bar-style histogram
ggplot(survey_q27, aes(x = factor(q27_value), fill = factor(group))) +
  geom_bar(position = "dodge") +
  facet_wrap(~ group) +
  labs(
    title = "Qxx.27 The distribution of ceartainty/uncertainty of the central estimate by group",
    x = "Qxx.27 Response",
    y = "Count",
    fill = "Group"
  ) +
  theme_minimal()
```



Qxx.28 Rationale for certainty/uncertainty of the forecast

Why do you think that the forecast is more or less uncertain? (Select all that apply)

Variable: categorical

Method: contingency table + Not comparable across group

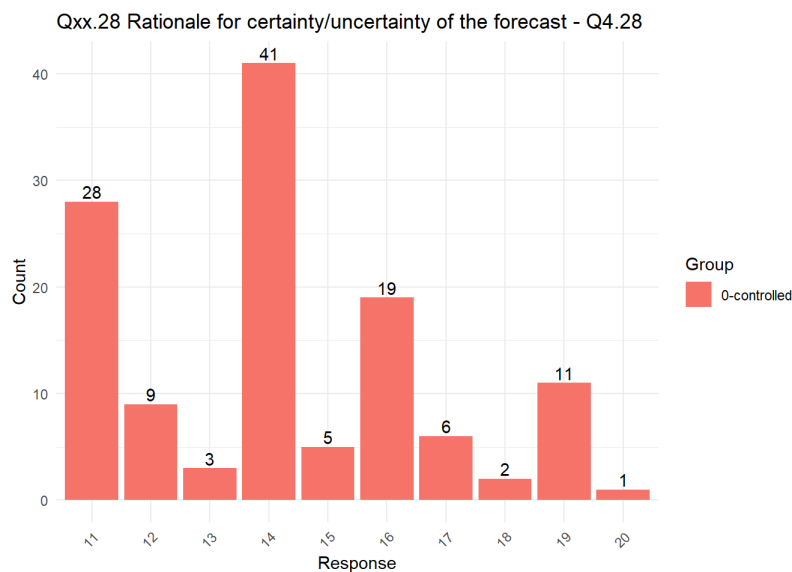
Function:

```

plot_survey_q28 <- function(survey_data, question_header){
  survey_q28 <- survey_data %>%
    # select(all_of(question_header), group) %>%
    mutate(
      !!question_header := lapply(
        as.character(.data[[question_header]]),
        function(x) {
          if (is.na(x) || x == "") return(list())
          nums <- strsplit(x, ",")[[1]]
          as.list(as.numeric(trimws(nums)))
        }
      )
    ) %>%
    unnest(cols = all_of(question_header)) %>%
    group_by(group, value = .data[[question_header]]) %>%
    summarise(count = n(), .groups = "drop") %>%
    mutate(
      value = as.numeric(value)
    )
  # plot the bar chart
  ggplot(survey_q28, aes(x = factor(value), y = count, fill = group)) +
    geom_bar(stat = "identity", position = "dodge") +
    geom_text(aes(label = count), vjust = -0.25, position = position_dodge(0.9)) +
    labs(
      title = paste("Qxx.28 Rationale for certainty/uncertainty of the forecast -", question_header),
      x = "Response",
      y = "Count",
      fill = "Group"
    ) +
    theme_minimal() +
    theme(axis.text.x = element_text(angle = 45, hjust = 1))
  # return(survey_q28)
}

plot_survey_q28(controlled_group, "Q4.28")

```



Qxx.29 Representativeness of the uncertainty

On a scale of 1 to 10, where 1 means not representative at all and 10 means very representative, how representative do you think the uncertainty being communicated is (i.e., how well the given information reflects the true variability or uncertainty of the forecast?)

1-Not representative at all; 10-Very representative

Variable: continuous

Method: ANNOVA test or KW test (depends on whether it's normally distributed)

Result: Fail to reject the null hypothesis.


```
# combine data from different groups
survey_q29 <- data.frame()
survey_q29_list <- list("Q4.29_1", "Q6.29_1", "Q8.29_1", "Q10.29_1", "Q12.29_1")
for (i in seq_along(group_list)) {
  df <- group_list[[i]]
  col <- survey_q29_list[[i]]
  temp_table <- df[, c(col, "group", "treatment")]
  names(temp_table) <- c("q29_value", "group", "treatment")
  survey_q29 <- rbind(survey_q29, temp_table)
}
survey_q29$q29_value <- as.numeric(survey_q29$q29_value)
```

Statistical test

```
# Initial Check
# summary(survey_q29$q29_value)
# check normality --> the data is not normally distributed
# shapiro.test(survey_q29$q29_value)
# run Kruskal-Wallis test
kruskal_test <- kruskal.test(q29_value ~ group, data = survey_q29)
print(kruskal_test)
```

```
##
## Kruskal-Wallis rank sum test
##
## data: q29_value by group
## Kruskal-Wallis chi-squared = 5.1458, df = 4, p-value = 0.2727
```

```
# Effect size
kw_es <- kruskal_effsize(q29_value ~ group, data = survey_q29, ci = TRUE)
print(kw_es)
```

```
## # A tibble: 1 x 7
##   .y.      n effsize conf.low conf.high method magnitude
## * <chr>   <int>   <dbl>   <dbl>   <dbl> <chr>   <ord>
## 1 q29_value 309 0.00377 -0.0085    0.05 eta2[H] small
```

Combine the treatment groups

```
kruskal_test <- kruskal.test(q29_value ~ treatment, data = survey_q29)
print(kruskal_test)
```

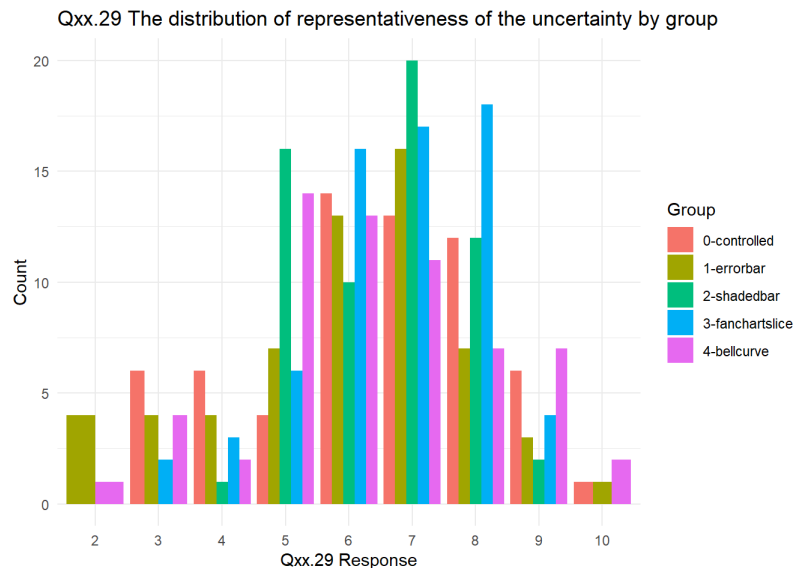
```
##
## Kruskal-Wallis rank sum test
##
## data: q29_value by treatment
## Kruskal-Wallis chi-squared = 0.047402, df = 1, p-value = 0.8276
```

Summary statistics

```
# Summary statistics
survey_q29_summary <- survey_q29 %>%
  group_by(group) %>%
  summarise(
    mean = mean(q29_value, na.rm = TRUE),
    median = median(q29_value, na.rm = TRUE),
    sd = sd(q29_value, na.rm = TRUE),
    IQR = IQR(q29_value, na.rm = TRUE), # same as Q3 - Q1
    Q1 = quantile(q29_value, 0.25, na.rm = TRUE),
    Q3 = quantile(q29_value, 0.75, na.rm = TRUE)
  )
```

Plot the distribution of different groups

```
# Grouped bar-style histogram
ggplot(survey_q29, aes(x = factor(q29_value), fill = factor(group))) +
  geom_bar(position = "dodge") +
  # facet_wrap(~ group) +
  labs(
    title = "Qxx.29 The distribution of representativeness of the uncertainty by group",
    x = "Qxx.29 Response",
    y = "Count",
    fill = "Group"
  ) +
  theme_minimal()
```



! Qxx.30 Readability of the Forecast

On a scale of 1 to 10, where 1 means very difficult and 10 means very easy, how easy or difficult do you find the graph to understand?

1-Very difficult; 10-Very easy

Variable: continuous

Method: ANNOVA test or KW test (depends on whether it's normally distributed)

Result: There are statistical difference between control group vs. fan chart slice group; control group vs. bell curve; error bar group vs. bell curve group.

```
# combine data from different groups
survey_q30 <- data.frame()
survey_q30_list <- list("Q4.30_1", "Q6.30_1", "Q8.30_1", "Q10.30_1", "Q12.30_1")
for (i in seq_along(group_list)) {
  df <- group_list[[i]]
  col <- survey_q30_list[[i]]
  temp_table <- df[, c(col, "group", "treatment")]
  names(temp_table) <- c("q30_value", "group", "treatment")
  survey_q30 <- rbind(survey_q30, temp_table)
}
survey_q30$q30_value <- as.numeric(survey_q30$q30_value)
```

Statistical test

```
# Initial Check
# summary(survey_q30$q30_value)
# check normality --> the data is not normally distributed
# shapiro.test(survey_q30$q30_value)
# run Kruskal-Wallis test
kruskal_test <- kruskal.test(q30_value ~ group, data = survey_q30)
print(kruskal_test)
```

```
##
## Kruskal-Wallis rank sum test
##
## data: q30_value by group
## Kruskal-Wallis chi-squared = 21.14, df = 4, p-value = 0.0002971
```

```
# Effect size
kw_es <- kruskal_effsize(q30_value ~ group, data = survey_q30, ci = TRUE)
print(kw_es)
```

```
## # A tibble: 1 x 7
##   .y.      n effsize conf.low conf.high method magnitude
## * <chr>   <int>   <dbl>   <dbl>   <dbl> <chr>   <ord>
## 1 q30_value 309 0.0564    0.02    0.13 eta2[H] small
```

```
# Post-hoc test
dunnTest(q30_value ~ group, data = survey_q30, method = "bonferroni")
```

```
## Warning: group was coerced to a factor.
```

```
## Dunn (1964) Kruskal-Wallis multiple comparison
```

```
## p-values adjusted with the Bonferroni method.
```

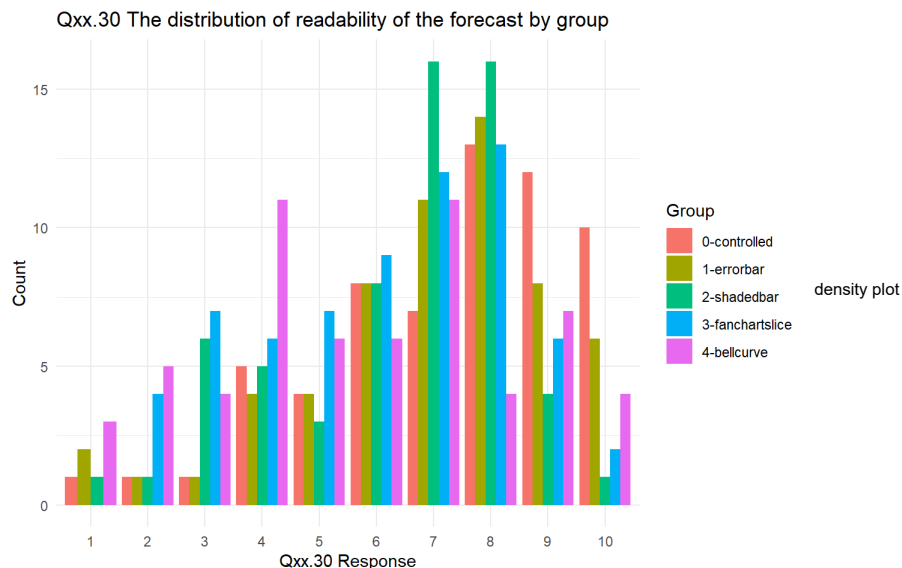
```
##           Comparison      Z      P.unadj      P.adj
## 1  0-controlled - 1-errorbar 0.9304079 3.521599e-01 1.0000000000
## 2  0-controlled - 2-shadedbar 2.5337624 1.128452e-02 0.1128452320
## 3  1-errorbar - 2-shadedbar 1.5756957 1.150960e-01 1.0000000000
## 4  0-controlled - 3-fanchartslice 3.2161670 1.299152e-03 0.0129915217
## 5  1-errorbar - 3-fanchartslice 2.2303493 2.572426e-02 0.2572426134
## 6  2-shadedbar - 3-fanchartslice 0.6299386 5.287348e-01 1.0000000000
## 7  0-controlled - 4-bellcurve 3.9666282 7.289658e-05 0.0007289658
## 8  1-errorbar - 4-bellcurve 2.9908299 2.782205e-03 0.0278220451
## 9  2-shadedbar - 4-bellcurve 1.4270765 1.535579e-01 1.0000000000
## 10 3-fanchartslice - 4-bellcurve 0.8249587 4.093950e-01 1.0000000000
```

Summary statistics

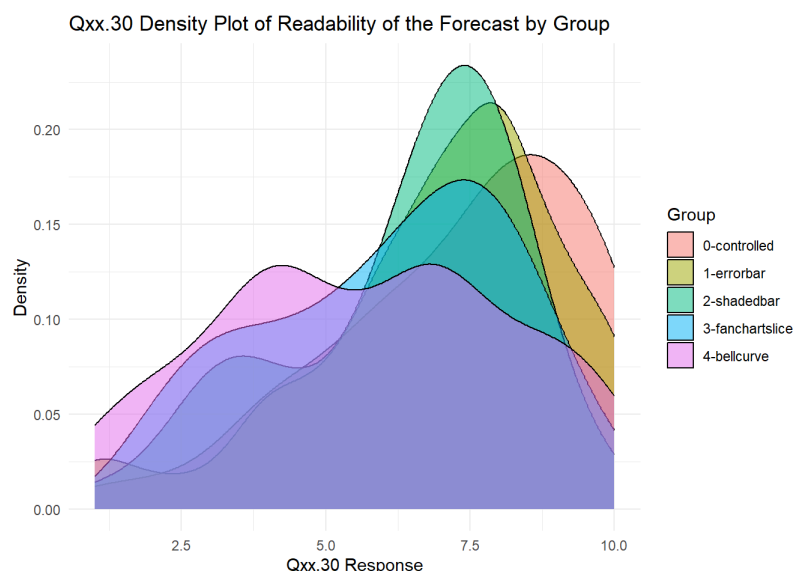
```
# Summary statistics
survey_q30_summary <- survey_q30 %>%
  group_by(group) %>%
  summarise(
    mean = mean(q30_value, na.rm = TRUE),
    median = median(q30_value, na.rm = TRUE),
    sd = sd(q30_value, na.rm = TRUE),
    IQR = IQR(q30_value, na.rm = TRUE), # same as Q3 - Q1
    Q1 = quantile(q30_value, 0.25, na.rm = TRUE),
    Q3 = quantile(q30_value, 0.75, na.rm = TRUE)
  )
```

Plot the distribution of different groups

```
# Grouped bar-style histogram
ggplot(survey_q30, aes(x = factor(q30_value), fill = factor(group))) +
  geom_bar(position = "dodge") +
  # facet_wrap(~ group) +
  labs(
    title = "Qxx.30 The distribution of readability of the forecast by group",
    x = "Qxx.30 Response",
    y = "Count",
    fill = "Group"
  ) +
  theme_minimal()
```



```
# Density plot
ggplot(survey_q30, aes(x = q30_value, fill = group)) +
  geom_density(alpha = 0.5) +
  labs(
    title = "Qxx.30 Density Plot of Readability of the Forecast by Group",
    x = "Qxx.30 Response",
    y = "Density",
    fill = "Group"
  ) +
  theme_minimal()
```



Qxx.31 Trust of the Forecast

On a scale of 1 to 10, where 1 means you don't trust the inflation forecast at all and 10 means you fully trust it, how much do you trust the inflation forecast shown in the graph?

1-Don't trust at all; 10-Fully trust

Variable: continuous

Method: ANNOVA test or KW test (depends on whether it's normally distributed)

Result: Fail to reject the null hypothesis.

```
# combine data from different groups
survey_q31 <- data.frame()
survey_q31_list <- list("Q4.31_1", "Q6.31_1", "Q8.31_1", "Q10.31_1", "Q12.31_1")
for (i in seq_along(group_list)) {
  df <- group_list[[i]]
  col <- survey_q31_list[[i]]
  temp_table <- df[, c(col, "group", "treatment")]
  names(temp_table) <- c("q31_value", "group", "treatment")
  survey_q31 <- rbind(survey_q31, temp_table)
}
survey_q31$q31_value <- as.numeric(survey_q31$q31_value)
summary(survey_q31$q31_value)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.   Max.
##  1.000   5.000   6.000   6.032   7.000  10.000
```

Statistical test

```
# Initial Check
# summary(survey_q31$q31_value)

# check normality --> the data is not normally distributed
# shapiro.test(survey_q31$q31_value)

# run Kruskal-Wallis test
kruskal_test <- kruskal.test(q31_value ~ group, data = survey_q31)
print(kruskal_test)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  q31_value by group
## Kruskal-Wallis chi-squared = 3.7739, df = 4, p-value = 0.4375
```

```
# Effect size
kw_es <- kruskal_effsize(q31_value ~ group, data = survey_q31, ci = TRUE)
print(kw_es)
```

```
## # A tibble: 1 x 7
##   .y.      n  effsize conf.low conf.high method  magnitude
## * <chr>   <int>   <dbl>   <dbl>   <dbl> <chr>   <ord>
## 1 q31_value  309 -0.000744 -0.0097    0.05 eta2[H] small
```

Combine the treatment groups

```
kruskal_test <- kruskal.test(q31_value ~ treatment, data = survey_q31)
print(kruskal_test)
```

```
##
## Kruskal-Wallis rank sum test
##
## data: q31_value by treatment
## Kruskal-Wallis chi-squared = 0.24871, df = 1, p-value = 0.618
```

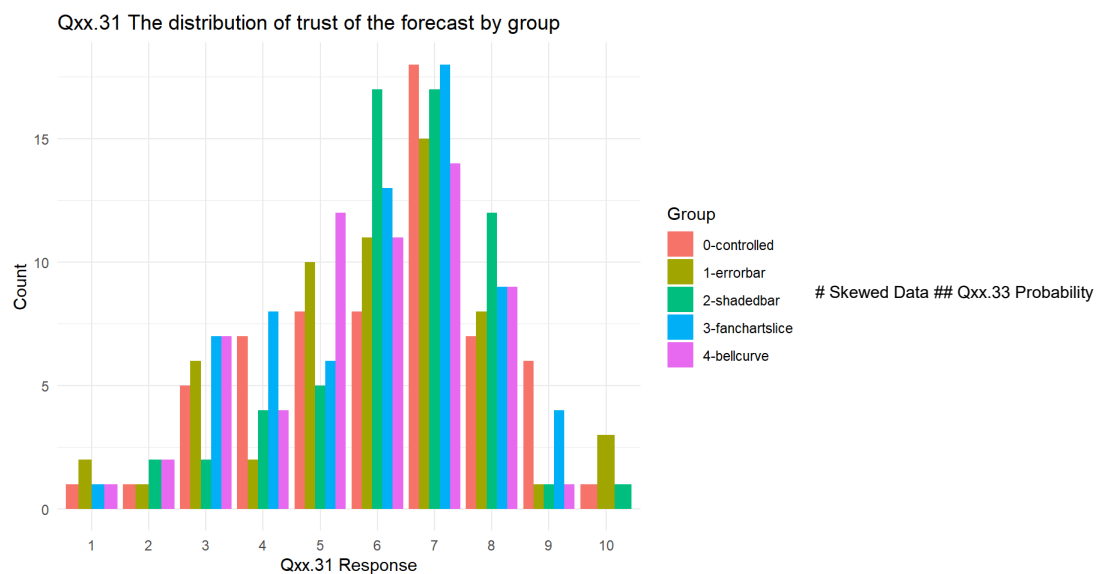
Summary statistics

```
# Summary statistics
survey_q31_summary <- survey_q31 %>%
  group_by(group) %>%
  summarise(
    mean = mean(q31_value, na.rm = TRUE),
    median = median(q31_value, na.rm = TRUE),
    sd = sd(q31_value, na.rm = TRUE),
    IQR = IQR(q31_value, na.rm = TRUE), # same as Q3 - Q1
    Q1 = quantile(q31_value, 0.25, na.rm = TRUE),
    Q3 = quantile(q31_value, 0.75, na.rm = TRUE)
  )
print(survey_q31_summary)
```

```
## # A tibble: 5 x 7
##   group      mean median    sd   IQR    Q1    Q3
##   <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 0-controlled    6.13     7  1.97     2     5     7
## 2 1-errorbar      6      6  2.04     2     5     7
## 3 2-shadedbar    6.34     7  1.60     1     6     7
## 4 3-fanchartslice 6      6  1.82     2     5     7
## 5 4-bellcurve    5.69     6  1.83     2     5     7
```

Plot the distribution of different groups

```
# Grouped bar-style histogram
ggplot(survey_q31, aes(x = factor(q31_value), fill = factor(group))) +
  geom_bar(position = "dodge") +
  # facet_wrap(~ group) +
  labs(
    title = "Qxx.31 The distribution of trust of the forecast by group",
    x = "Qxx.31 Response",
    y = "Count",
    fill = "Group"
  ) +
  theme_minimal()
```



of higher than 3% Based on the inflation forecast provided, what do you think is the probability of inflation ending up higher than 3%?

Definitive Correct Answer - 5

Variable: categorical

Method: contingency table & proportion test

Result: There is no statistical difference across different groups and the largest proportion of participants can understand the probability correctly, but only account for 28% of the entire respondents.

```
# combine data from different groups
survey_q33 <- data.frame()
survey_q33_list <- list("Q6.33", "Q8.33", "Q10.33", "Q12.33")
for (i in seq_along(skewed_group_list)) {
  df <- skewed_group_list[[i]]
  col <- survey_q33_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("q33_value", "group")
  survey_q33 <- rbind(survey_q33, temp_table)
}
survey_q33$q33_value <- as.numeric(survey_q33$q33_value)
```

Test the proportion of correct answer

```
survey_q33_contingency <- table(survey_q33$q33_value, survey_q33$group)
print(survey_q33_contingency)
```

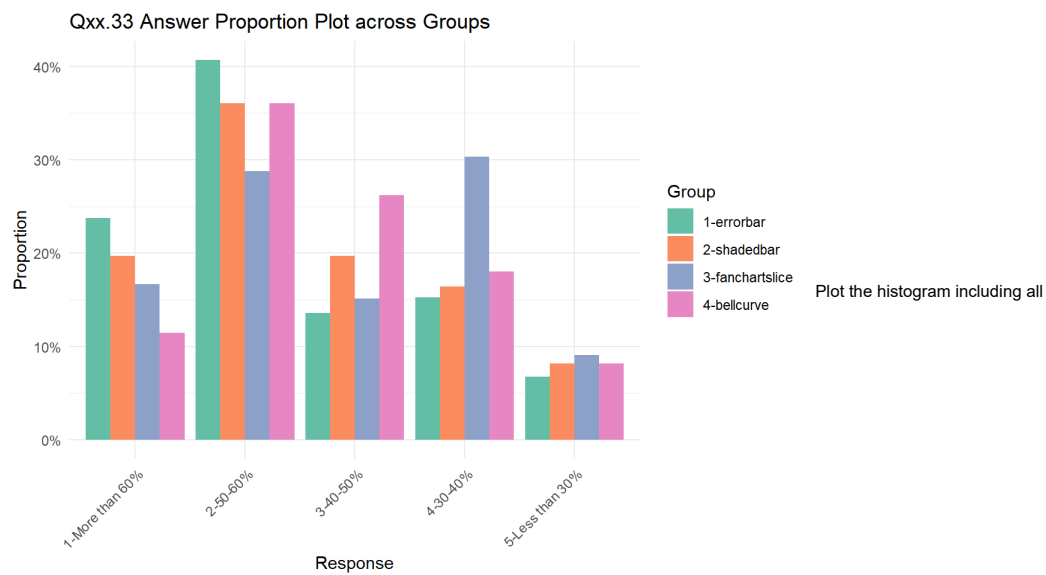
```
##
##      1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1          14          12          11          7
## 2           8          12          10          16
## 3           9          10          20          11
## 4           4           5           6           5
## 5          24          22          19          22
```

```
survey_q33_correct <- survey_q33_contingency[5,]
survey_q33_group_total <- colSums(survey_q33_contingency)
prop.test(survey_q33_correct, survey_q33_group_total)
```

```
##
##      4-sample test for equality of proportions without continuity correction
##
## data: survey_q33_correct out of survey_q33_group_total
## X-squared = 2.0053, df = 3, p-value = 0.5713
## alternative hypothesis: two.sided
## sample estimates:
##      prop 1      prop 2      prop 3      prop 4
## 0.4067797 0.3606557 0.2878788 0.3606557
```

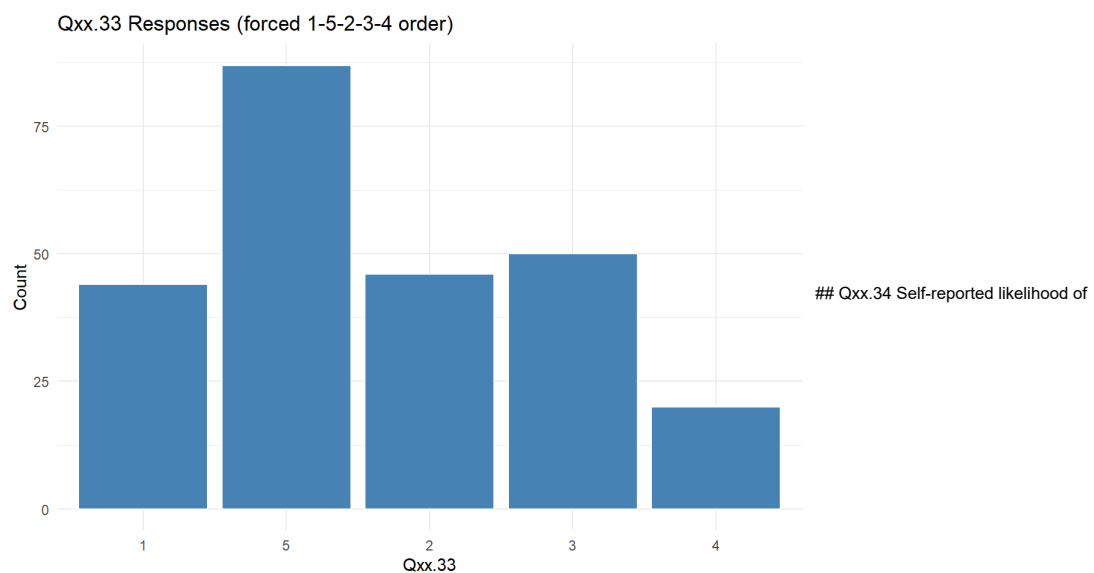
Plot probability of higher than 3% (proportion bar plot across different groups)

```
survey_q33_prop <- as.data.frame(survey_q33_contingency)
# rename columns
names(survey_q33_prop) <- c("Qxx.33", "Group", "Count")
# add a labelled column
survey_q33_prop <- survey_q33_prop %>%
  mutate(Qxx.33_label = case_when(
    Qxx.33 == "1" ~ "1-More than 60%",
    Qxx.33 == "2" ~ "3-40-50%",
    Qxx.33 == "3" ~ "4-30-40%",
    Qxx.33 == "4" ~ "5-Less than 30%",
    Qxx.33 == "5" ~ "2-50-60%"
  ))
# calculate proportions within each group
survey_q33_prop <- survey_q33_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))
ggplot(survey_q33_prop, aes(x = as.factor(`Qxx.33_label`), y = Proportion, fill = Group)) + geom_col(position = "dodge") +
  # facet_wrap(~Group)+
  labs(
    title = "Qxx.33 Answer Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2")+
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal()+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



participants

```
# force order for graph
survey_q33 <- survey_q33 %>%
  mutate(
    q33_value_re = factor(
      q33_value,
      levels = c(1, 5, 2, 3, 4)
    )
  )
# histogram including all participants
ggplot(survey_q33, aes(x = q33_value_re)) +
  geom_bar(fill = "steelblue", color = "white") +
  labs(
    title = "Qxx.33 Responses (forced 1-5-2-3-4 order)",
    x = "Qxx.33",
    y = "Count"
  ) +
  theme_minimal()
```



inflation being higher than 3% How likely do you think it is that the inflation will be higher than 3%?

ascending value-increasing likelihood

Variable: treat as continuous

Method: Kruskal-Wallis test

Result: Fail to reject the null hypothesis. The majority of respondents think it's quite likely.

```
# combine data from different groups
survey_q34 <- data.frame()
survey_q34_list <- list("Q6.34_1", "Q8.34_1", "Q10.34_1", "Q12.34_1")
for (i in seq_along(skewed_group_list)) {
  df <- skewed_group_list[[i]]
  col <- survey_q34_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("q34_value", "group")
  survey_q34 <- rbind(survey_q34, temp_table)
}
survey_q34$q34_value <- as.numeric(survey_q34$q34_value)
```

Summary Table

```
survey_q34_summary <- survey_q34 %>%
  group_by(group) %>%
  summarise(
    median = median(q34_value, na.rm = TRUE),
    IQR = IQR(q34_value, na.rm = TRUE), # same as Q3 - Q1
    Q1 = quantile(q34_value, 0.25, na.rm = TRUE),
    Q3 = quantile(q34_value, 0.75, na.rm = TRUE),
    mean = mean(q34_value, na.rm = TRUE)
  )
print(survey_q34_summary)
```

```
## # A tibble: 4 × 6
##   group      median  IQR  Q1  Q3  mean
##   <chr>      <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 1-errorbar      13     1  12  13  12.7
## 2 2-shadedbar      13     1  12  13  12.8
## 3 3-fanchartslice  12     1  12  13  12.3
## 4 4-bellcurve     12     1  12  13  12.4
```

KW test

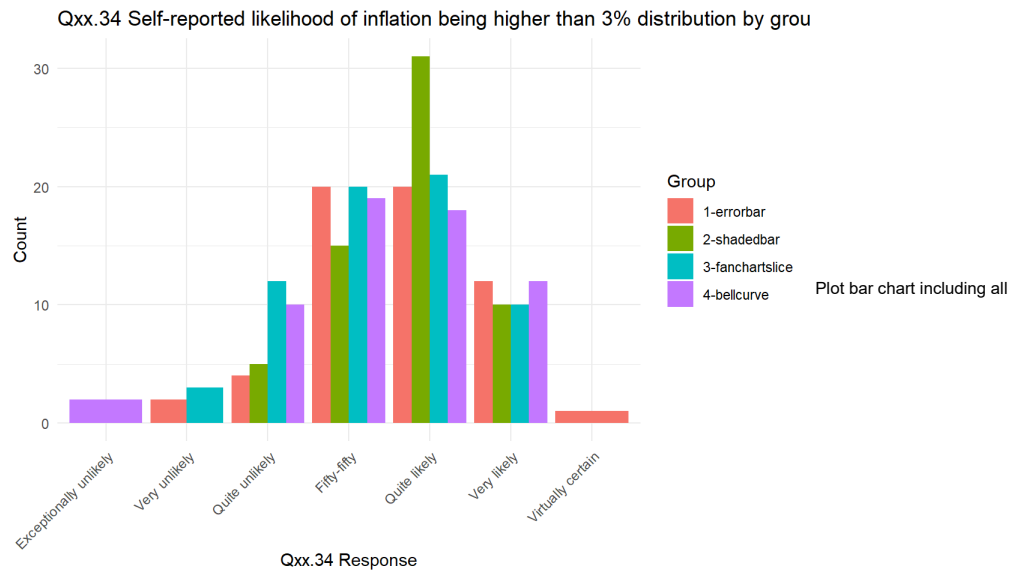
```
# check normality --> the data is not normally distributed
# shapiro.test(survey_q34$q34_value)

# run Kruskal-Wallis test
kruskal_test <- kruskal.test(q34_value ~ group, data = survey_q34)
print(kruskal_test)
```

```
##
## Kruskal-Wallis rank sum test
##
## data: q34_value by group
## Kruskal-Wallis chi-squared = 5.5506, df = 3, p-value = 0.1356
```

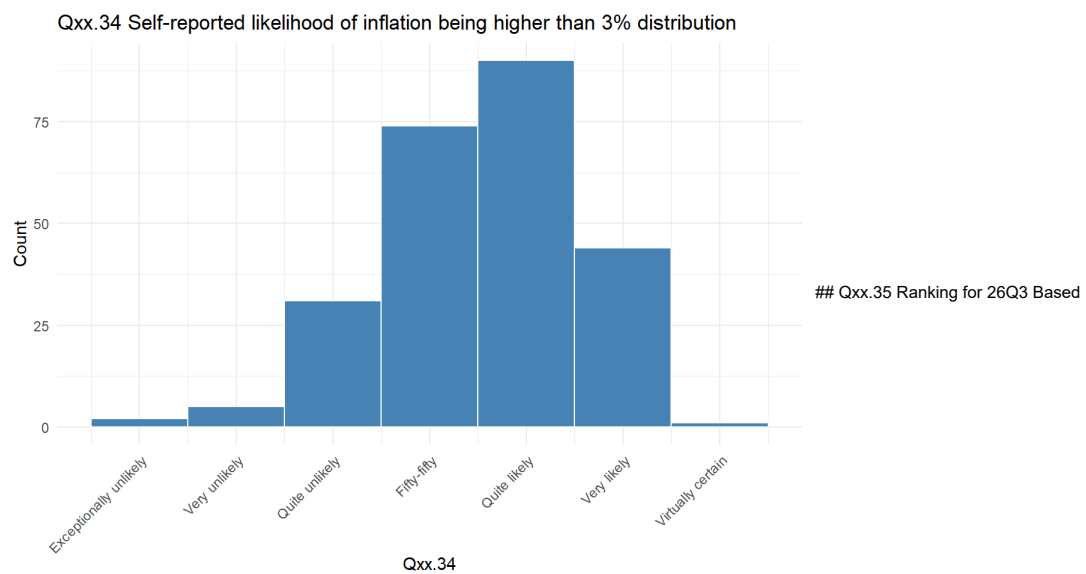
Plot the distribution of different groups

```
# Grouped bar-style histogram
ggplot(survey_q34, aes(x = factor(q34_value), fill = factor(group))) +
  geom_bar(position = "dodge") +
  # facet_wrap(~ group) +
  # add labels
  scale_x_discrete(labels=likelihood_labels) +
  labs(
    title = "Qxx.34 Self-reported likelihood of inflation being higher than 3% distribution by group",
    x = "Qxx.34 Response",
    y = "Count",
    fill = "Group"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

participants

```
# histogram including all participants
ggplot(survey_q34, aes(x = q34_value)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  # facet_wrap(~ group) +
  labs(title = "Qxx.34 Self-reported likelihood of inflation being higher than 3% distribution",
        x = "Qxx.34",
        y = "Count") +
  scale_x_continuous(breaks = as.numeric(names(likelihood_labels)),
                    labels = likelihood_labels) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



on the inflation forecast provided, please rank the following inflation rates from most likely to least likely to occur: 1.1%, 2.6%, 3.3% and 4.1%. - 4.1%

Lower mean rank = more preferred

Method: Mean Rank

```

q35_prefixes <- list("Q6.35", "Q8.35", "Q10.35", "Q12.35")
rank_summary_skewed <- data.frame(
  Item = c("1.1%", "2.6%", "3.3%", "4.1%")
)
survey_q35 <- data.frame()
for (i in seq_along(skewed_group_list)) {
  group_name <- names(skewed_group_list)[i]
  df <- skewed_group_list[[i]]
  q_cols <- paste0(q35_prefixes[[i]], "_", 5:8)
  temp_table <- df[, c(q_cols, "group")]
  # combine the data
  names(temp_table) <- c("Qxx.35_5", "Qxx.35_6", "Qxx.35_7", "Qxx.35_8", "group")
  temp_table[, c("Qxx.35_5", "Qxx.35_6", "Qxx.35_7", "Qxx.35_8")] <-
    lapply(temp_table[, c("Qxx.35_5", "Qxx.35_6", "Qxx.35_7", "Qxx.35_8")], as.numeric)
  survey_q35 <- rbind(survey_q35, temp_table)

  # mean rank
  df <- as.data.frame(lapply(df[,q_cols], as.numeric))
  mean_rank <- colMeans(df, na.rm = TRUE)
  rank_summary_skewed[[group_name]] <- mean_rank
}
# overall means
overall_means <- rowMeans(rank_summary_skewed[, -1], na.rm = TRUE)
rank_summary_skewed <- cbind(rank_summary_skewed, Overall = overall_means)
# correct answer
survey_q35_correct_prop <- survey_q35 %>%
  group_by(group) %>%
  summarise(
    total = n(),
    correct = sum(
      Qxx.35_5 == 4 & Qxx.35_6 == 1 & Qxx.35_7 == 2 & Qxx.35_8 == 3,
      na.rm = TRUE
    ),
    proportion_correct = correct / total,
    .groups = "drop"
  )

```

proportion test

```

prop.test(
  x = survey_q35_correct_prop$correct,
  n = survey_q35_correct_prop$total
)

```

```

##
## 4-sample test for equality of proportions without continuity correction
##
## data: survey_q35_correct_prop$correct out of survey_q35_correct_prop$total
## X-squared = 4.3257, df = 3, p-value = 0.2284
## alternative hypothesis: two.sided
## sample estimates:
##   prop 1   prop 2   prop 3   prop 4
## 0.4237288 0.2622951 0.3939394 0.4098361

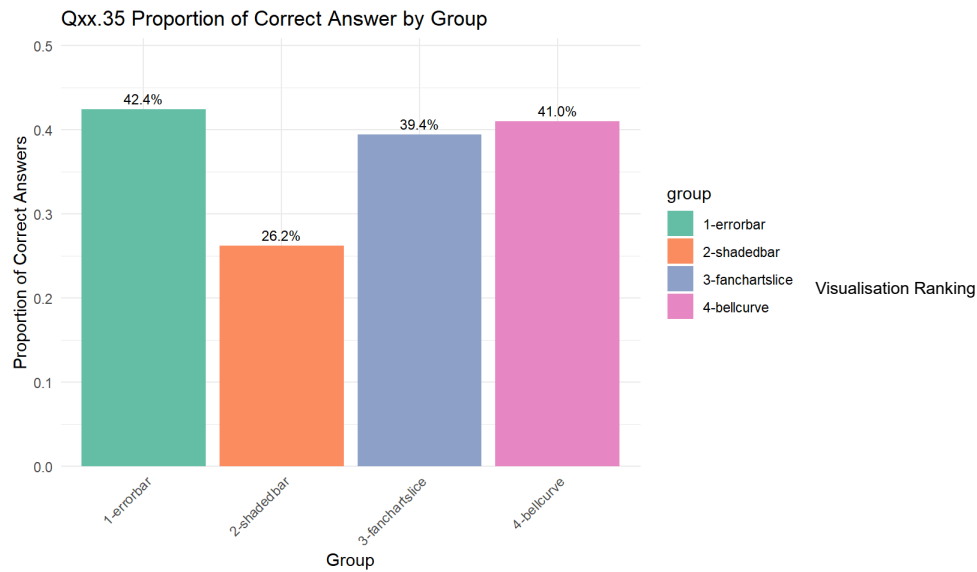
```

Visualisation Proportion

```

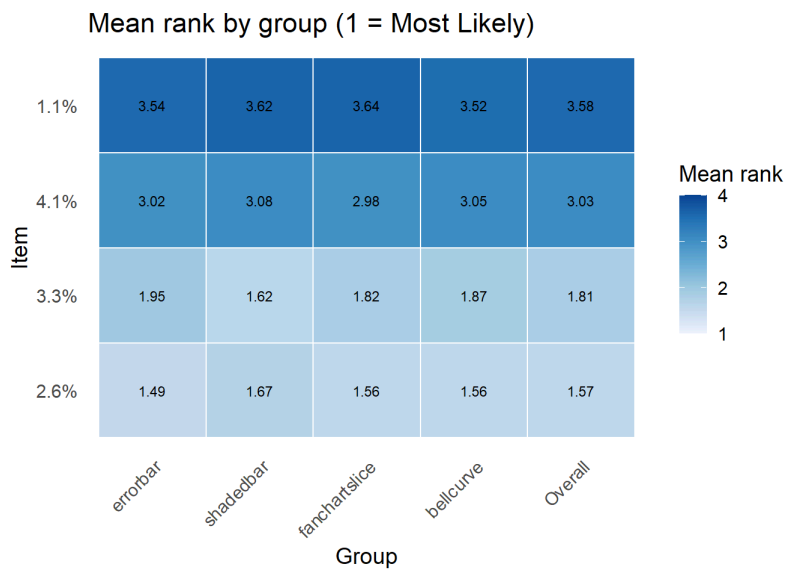
ggplot(survey_q35_correct_prop, aes(x = group, y = proportion_correct, fill = group)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = scales::percent(proportion_correct, accuracy = 0.1)),
    vjust = -0.5, size = 3) +
  labs(
    title = "Qxx.35 Proportion of Correct Answer by Group",
    x = "Group",
    y = "Proportion of Correct Answers"
  ) +
  scale_y_continuous(
    expand = expansion(mult = c(0, 0.2))
  ) +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```



```
df_long_skewed <- rank_summary_skewed %>%
  pivot_longer(-Item, names_to = "Group", values_to = "MeanRank") %>%
  mutate(
    # Order items by their overall best rank (smallest = best)
    Item = factor(Item, levels = rank_summary_skewed %>% arrange(Overall) %>% pull(Item)),
    # Keep groups in the order they appear in the data-frame
    Group = factor(Group, levels = names(rank_summary_skewed)[-1])
  )

ggplot(df_long_skewed, aes(Group, Item, fill = MeanRank)) +
  geom_tile(colour = "white", linewidth = .4) +
  geom_text(aes(label = round(MeanRank, 2)), size = 3) +
  scale_fill_distiller(palette = "Blues", direction = 1,
    name = "Mean rank", limits = c(1, 4),
    breaks = 1:4) +
  labs(title = "Mean rank by group (1 = Most Likely)") +
  theme_minimal(base_size = 14) +
  theme(panel.grid = element_blank(),
    axis.text.x = element_text(angle = 45, hjust = 1))
```



Qxx.36 Deviation from central estimate

Based on the inflation forecast provided, do you think there is a higher risk of inflation exceeding the central estimate or of being less than this central estimate?

Definitive Correct Answer - 1

Variable: categorical

Method: contingency table & proportion test

Result: There is no statistical difference across group. 68% of respondents can understand there is bigger upside risk.

```
# combine data from different groups
survey_q36 <- data.frame()
survey_q36_list <- list("Q6.36", "Q8.36", "Q10.36", "Q12.36")
for (i in seq_along(skewed_group_list)) {
  df <- skewed_group_list[[i]]
  col <- survey_q36_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("q36_value", "group")
  survey_q36 <- rbind(survey_q36, temp_table)
}
survey_q36$q36_value <- as.numeric(survey_q36$q36_value)
```

Contingency table

```
survey_q36_contingency <- table(survey_q36$q36_value, survey_q36$group)
print(survey_q36_contingency)
```

```
##
##      1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1          38          42          47          41
## 2           5           1           7           6
## 3          16          18          12          14
```

```
chisq_result <- chisq.test(survey_q36_contingency)
```

```
## Warning in chisq.test(survey_q36_contingency): Chi-squared approximation may be
## incorrect
```

```
print(chisq_result)
```

```
##
## Pearson's Chi-squared test
##
## data:  survey_q36_contingency
## X-squared = 6.2063, df = 6, p-value = 0.4005
```

Test the proportion of correct answer

```
survey_q36_correct <- survey_q36_contingency[1,]
survey_q36_group_total <- colSums(survey_q36_contingency)
prop.test(survey_q36_correct, survey_q36_group_total)
```

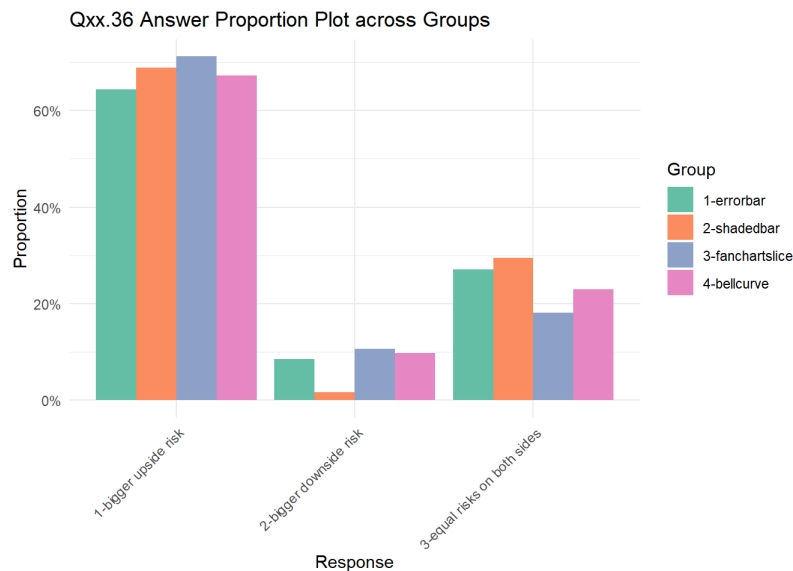
```
##
## 4-sample test for equality of proportions without continuity correction
##
## data:  survey_q36_correct out of survey_q36_group_total
## X-squared = 0.70091, df = 3, p-value = 0.873
## alternative hypothesis: two.sided
## sample estimates:
## prop 1    prop 2    prop 3    prop 4
## 0.6440678 0.6885246 0.7121212 0.6721311
```

Plot proportion bar plot across different groups

```

survey_q36_prop <- as.data.frame(survey_q36_contingency)
# rename columns
names(survey_q36_prop) <- c("Qxx.36", "Group", "Count")
# add a labelled column
survey_q36_prop <- survey_q36_prop %>%
  mutate(Qxx.36_label = case_when(
    Qxx.36 == "1" ~ "1-bigger upside risk",
    Qxx.36 == "2" ~ "2-bigger downside risk",
    Qxx.36 == "3" ~ "3-equal risks on both sides"
  ))
# calculate proportions within each group
survey_q36_prop <- survey_q36_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))
ggplot(survey_q36_prop, aes(x = as.factor(`Qxx.36_label`), y = Proportion, fill = Group)) + geom_col(position = "dodge") +
  # facet_wrap(~Group)+
  labs(
    title = "Qxx.36 Answer Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2")+
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal()+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```

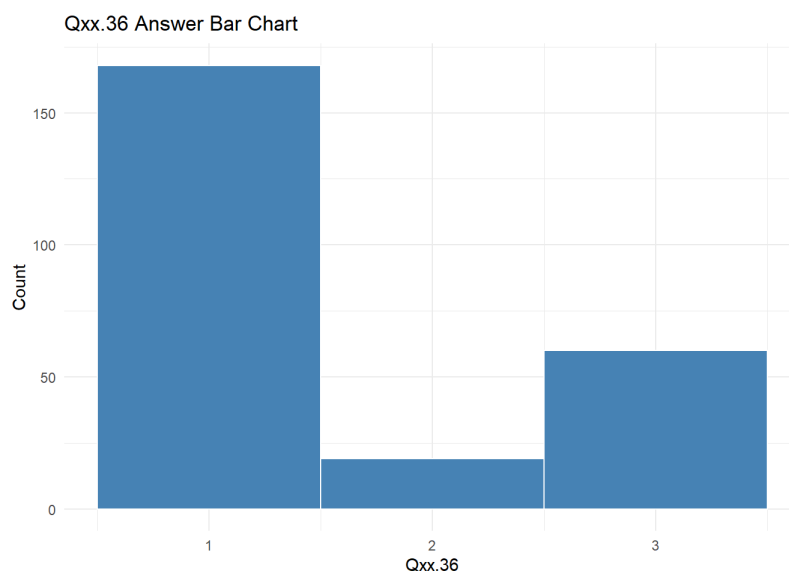


Plot the histogram including all participants

```

# histogram including all participants
ggplot(survey_q36, aes(x = q36_value)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  # facet_wrap(~ group) +
  labs(title = "Qxx.36 Answer Bar Chart",
    x = "Qxx.36",
    y = "Count") +
  scale_x_continuous(breaks = seq(0, max(survey_q36$q36_value, na.rm = TRUE), by = 1)) +
  theme_minimal()

```



? Qxx.37 Risk of ending outside

If the actual inflation ends up outside the black line, which side is more likely, based on the uncertainty depicted in the graph?

Definitive Correct Answer - 1

Variable: categorical

Method: contingency table & proportion test

Result: 63% respondents think if the actual inflation ends up outside, it's more likely to be on the upper side.

```
# combine data from different groups
survey_q37 <- data.frame()
survey_q37_list <- list("Q6.37", "Q8.37", "Q10.37", "Q12.37")
for (i in seq_along(skewed_group_list)) {
  df <- skewed_group_list[[i]]
  col <- survey_q37_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("q37_value", "group")
  survey_q37 <- rbind(survey_q37, temp_table)
}
survey_q37$q37_value <- as.numeric(survey_q37$q37_value)
```

Contingency table

```
survey_q37_contingency <- table(survey_q37$q37_value, survey_q37$group)
print(survey_q37_contingency)
```

```
##
##      1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1          33          35          51          37
## 2          10          11           5           7
## 3          16          15          10          17
```

```
# chisq_result <- chisq.test(survey_q37_contingency)
# print(chisq_result)

# Test the proportion of correct answer
survey_q37_correct <- survey_q37_contingency[1,]
survey_q37_group_total <- colSums(survey_q37_contingency)
prop.test(survey_q37_correct, survey_q37_group_total)
```

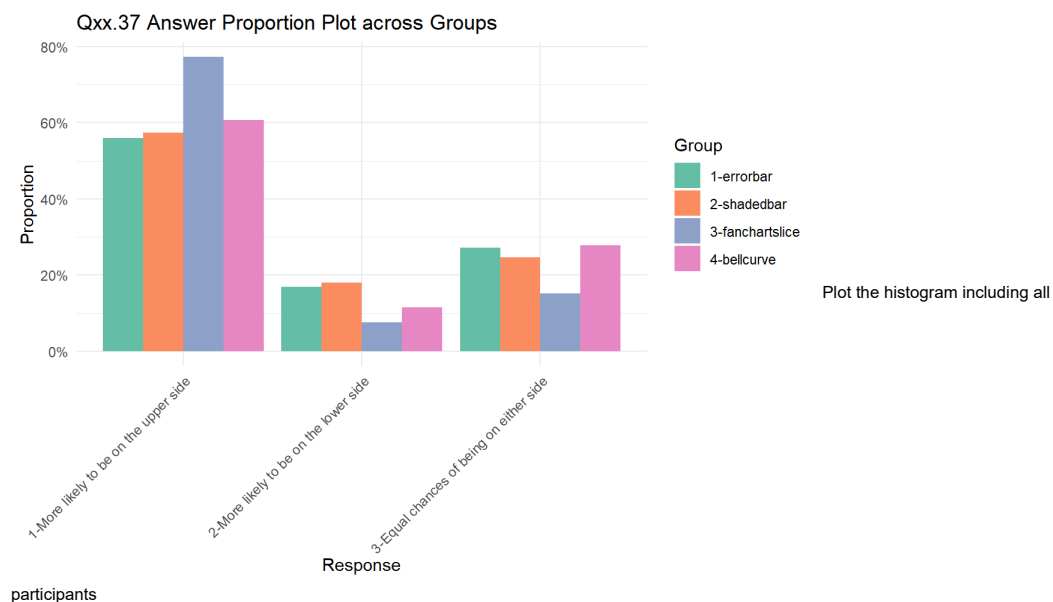
```
##
## 4-sample test for equality of proportions without continuity correction
##
## data: survey_q37_correct out of survey_q37_group_total
## X-squared = 8.015, df = 3, p-value = 0.0457
## alternative hypothesis: two.sided
## sample estimates:
## prop 1 prop 2 prop 3 prop 4
## 0.5593220 0.5737705 0.7727273 0.6065574
```

```
# pairwise proportion test
pairwise.prop.test(
  x = survey_q37_contingency[1,],
  n = colSums(survey_q37_contingency),
  p.adjust.method = "bonferroni" # or "holm", "BH"
)
```

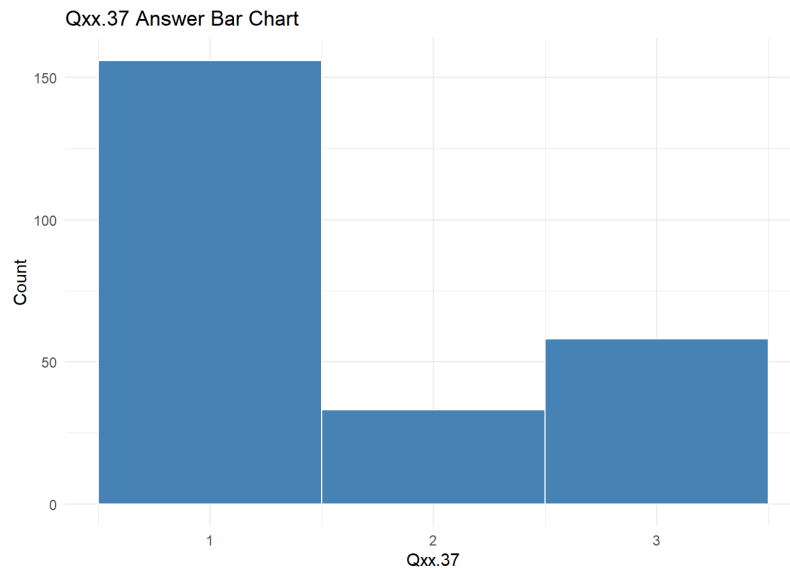
```
##
## Pairwise comparisons using Pairwise comparison of proportions
##
## data: survey_q37_contingency[1, ] out of colSums(survey_q37_contingency)
##
##          1-errorbar 2-shadedbar 3-fanchartslice
## 2-shadedbar    1.00      -          -
## 3-fanchartslice 0.11      0.16      -
## 4-bellcurve    1.00      1.00      0.40
##
## P value adjustment method: bonferroni
```

Plot proportion bar plot across different groups

```
survey_q37_prop <- as.data.frame(survey_q37_contingency)
# rename columns
names(survey_q37_prop) <- c("Qxx.37", "Group", "Count")
# add a labelled column
survey_q37_prop <- survey_q37_prop %>%
  mutate(Qxx.37_label = case_when(
    Qxx.37 == "1" ~ "1-More likely to be on the upper side",
    Qxx.37 == "2" ~ "2-More likely to be on the lower side",
    Qxx.37 == "3" ~ "3-Equal chances of being on either side"
  ))
# calculate proportions within each group
survey_q37_prop <- survey_q37_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))
ggplot(survey_q37_prop, aes(x = as.factor(`Qxx.37_label`), y = Proportion, fill = Group)) + geom_col(position = "dodge") +
  # facet_wrap(~Group)+
  labs(
    title = "Qxx.37 Answer Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2")+
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal()+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
# histogram including all participants
ggplot(survey_q37, aes(x = q37_value)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  # facet_wrap(~ group) +
  labs(title = "Qxx.37 Answer Bar Chart",
        x = "Qxx.37",
        y = "Count") +
  scale_x_continuous(breaks = seq(0, max(survey_q37$q37_value, na.rm = TRUE), by = 1)) +
  theme_minimal()
```



Decision-making

D.3 Investment Decision

Which bond would you choose to invest in?

Variable: categorical

Method: contingency table & proportion test

Result: Fail to reject the null hypothesis.

```
# combine data from different groups
survey_d3 <- data.frame()
survey_d3_list <- list("Q5.3", "Q7.3", "Q9.3", "Q11.3", "Q13.3")
for (i in seq_along(group_list)) {
  df <- group_list[[i]]
  col <- survey_d3_list[[i]]
  temp_table <- df[, c(col, "group", "treatment")]
  names(temp_table) <- c("d3_value", "group", "treatment")
  survey_d3 <- rbind(survey_d3, temp_table)
}
survey_d3$d3_value <- as.numeric(survey_d3$d3_value)
```

Contingency table

```
survey_d3_contingency <- table(survey_d3$d3_value, survey_d3$group)
print(survey_d3_contingency)
```

```
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1          22          17          24          22          17
## 2          40          42          37          44          44
```

```
# chi-squared test
chisq.test(survey_d3_contingency)
```

```
##
## Pearson's Chi-squared test
##
## data: survey_d3_contingency
## X-squared = 2.4805, df = 4, p-value = 0.6481
```



```
# prop.test
# survey_d3_correct <- survey_d3_contingency[1,]
# survey_d3_group_total <- colSums(survey_d3_contingency)
# prop.test(survey_d3_correct, survey_d3_group_total)
```

Combine all treatment groups (p-value = 0.7548)

```
survey_d3_contingency <- table(survey_d3$d3_value, survey_d3$treatment)
print(survey_d3_contingency)
```

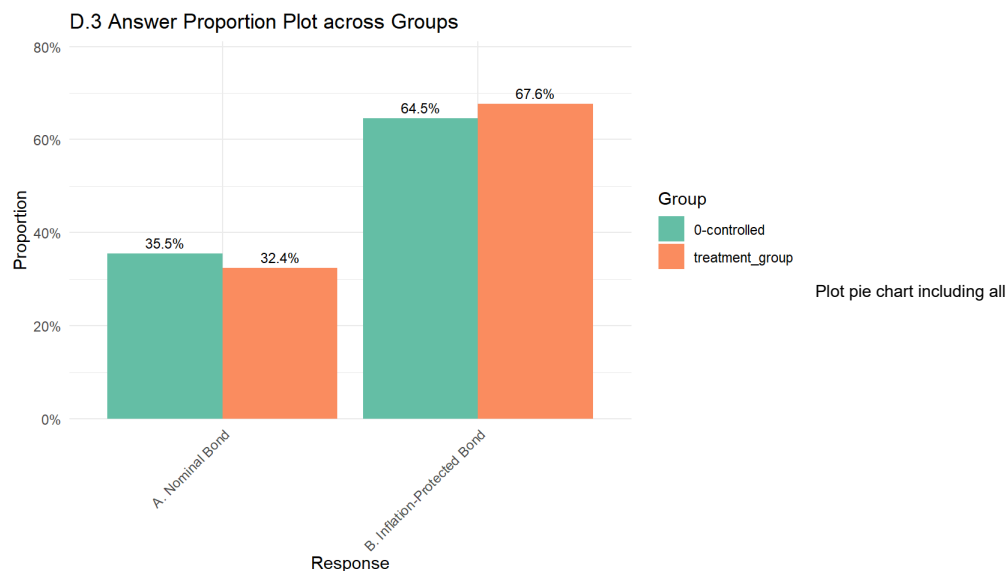
```
##
##      0-controlled treatment_group
##  1      22      80
##  2      40     167
```

```
# chi-squared test
chisq.test(survey_d3_contingency)
```

```
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data: survey_d3_contingency
## X-squared = 0.097553, df = 1, p-value = 0.7548
```

Plot proportion bar plot across different groups

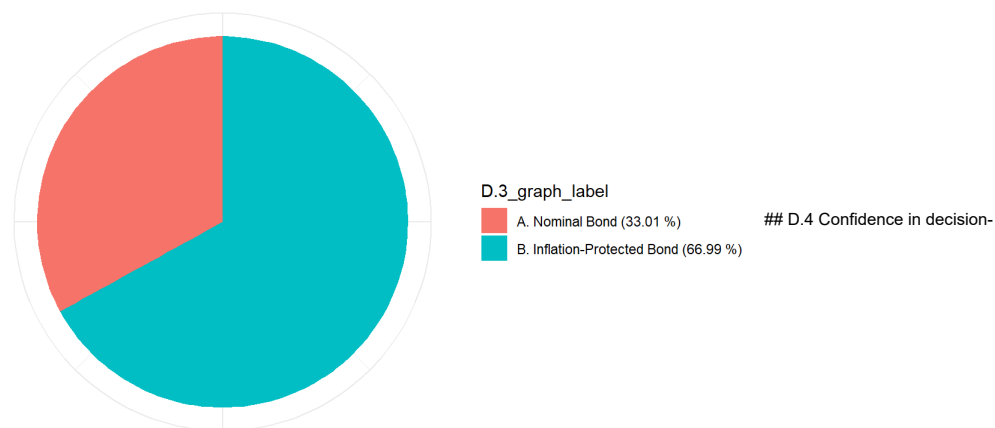
```
survey_d3_prop <- as.data.frame(survey_d3_contingency)
# rename columns
names(survey_d3_prop) <- c("D.3", "Group", "Count")
# add a labelled column
survey_d3_prop <- survey_d3_prop %>%
  mutate(D.3_label = case_when(
    D.3 == "1" ~ "A. Nominal Bond",
    D.3 == "2" ~ "B. Inflation-Protected Bond"
  ))
# calculate proportions within each group
survey_d3_prop <- survey_d3_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))
ggplot(survey_d3_prop, aes(x = as.factor(`D.3_label`), y = Proportion, fill = Group)) + geom_col(position = "dodge") +
  # facet_wrap(~Group)+
  geom_text(aes(label = scales::percent(Proportion, accuracy = 0.1)),
    position = position_dodge(width = 0.9), vjust = -0.5, size = 3) +
  labs(
    title = "D.3 Answer Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2")+
  scale_y_continuous(labels = scales::percent_format(accuracy = 1),
    expand = expansion(mult = c(0, 0.2))
  ) +
  theme_minimal()+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



participants

```
survey_d3_summary <- survey_d3_prop %>%
  group_by(`D.3_label`) %>%
  summarise(total_count = sum(Count), .groups = "drop") %>%
  mutate(proportion = total_count / sum(total_count),
         `D.3_graph_label` = paste0(`D.3_label`, " (", round(proportion * 100, 2), " %)")
ggplot(survey_d3_summary, aes(x = "", y = proportion, fill = `D.3_graph_label`)) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  labs(title = "D.3 Investment Decision Summary Pie Chart",
       x = "",
       y = "") +
  theme_minimal() +
  theme(axis.text.x = element_blank())
```

D.3 Investment Decision Summary Pie Chart



making On a scale of 1 to 10, where 1 means not confident at all and 10 means very confident, how confident are you in making the decision?

1-Not confident at all; 10-Very confident

Variable: continuous

Method: ANNOVA test or KW test (depends on whether it's normally distributed)

Result: Fail to reject the null hypothesis.

```
# combine data from different groups
survey_d4 <- data.frame()
survey_d4_list <- list("Q5.4_1", "Q7.4_1", "Q9.4_1", "Q11.4_1", "Q13.4_1")
for (i in seq_along(group_list)) {
  df <- group_list[[i]]
  col <- survey_d4_list[[i]]
  temp_table <- df[, c(col, "group", "treatment")]
  names(temp_table) <- c("d4_value", "group", "treatment")
  survey_d4 <- rbind(survey_d4, temp_table)
}
survey_d4$d4_value <- as.numeric(survey_d4$d4_value)
summary(survey_d4$d4_value)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  1.000   6.000   7.000   6.896   8.000  10.000
```

Statistical test (p-value = 0.1333)

```
# Initial Check
# summary(survey_d4$d4_value)
# check normality --> the data is not normally distributed
# shapiro.test(survey_d4$d4_value)

# run Kruskal-Wallis test
kruskal_test <- kruskal.test(d4_value ~ group, data = survey_d4)
print(kruskal_test)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  d4_value by group
## Kruskal-Wallis chi-squared = 7.0487, df = 4, p-value = 0.1333
```

```
# Effect size
kw_es <- kruskal_effsize(d4_value ~ group, data = survey_d4, ci = TRUE)
print(kw_es)
```

```
## # A tibble: 1 × 7
##   .y.      n effsize conf.low conf.high method magnitude
## * <chr>   <int>   <dbl>   <dbl>   <dbl> <chr>   <ord>
## 1 d4_value   309  0.0100  -0.0051    0.07 eta2[H] small
```

Combine all treatment groups (p-value = 0.5824)

```
# run Kruskal-Wallis test
kruskal_test <- kruskal.test(d4_value ~ treatment, data = survey_d4)
print(kruskal_test)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  d4_value by treatment
## Kruskal-Wallis chi-squared = 0.30241, df = 1, p-value = 0.5824
```

```
survey_d4_test <- survey_d4 %>%
  group_by(treatment) %>%
  summarise(
    mean = mean(d4_value, na.rm = TRUE))
```

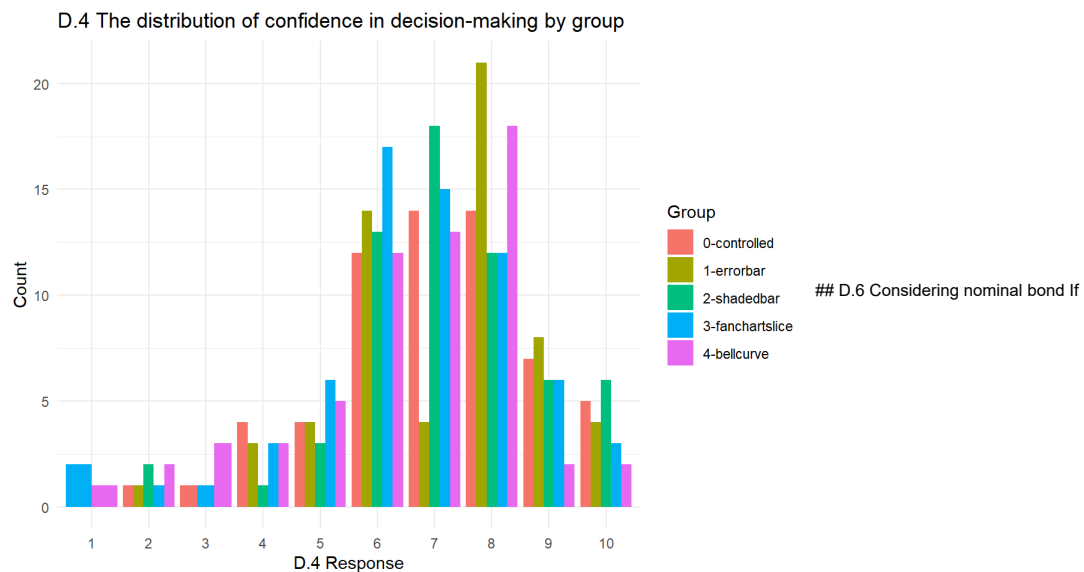
Summary statistics

```
# Summary statistics
survey_d4_summary <- survey_d4 %>%
  group_by(group) %>%
  summarise(
    mean = mean(d4_value, na.rm = TRUE),
    median = median(d4_value, na.rm = TRUE),
    sd = sd(d4_value, na.rm = TRUE),
    IQR = IQR(d4_value, na.rm = TRUE), # same as Q3 - Q1
    Q1 = quantile(d4_value, 0.25, na.rm = TRUE),
    Q3 = quantile(d4_value, 0.75, na.rm = TRUE)
  )
print(survey_d4_summary)
```

```
## # A tibble: 5 x 7
##   group      mean median    sd   IQR    Q1    Q3
##   <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 0-controlled  7.03     7  1.77     2     6     8
## 2 1-errorbar    7.22     8  1.70     2     6     8
## 3 2-shadedbar   7.16     7  1.69     2     6     8
## 4 3-fanchartslice 6.61     7  1.90     2     6     8
## 5 4-bellcurve   6.49     7  1.92     2     6     8
```

Plot the distribution of different groups

```
# Grouped bar-style histogram
ggplot(survey_d4, aes(x = factor(d4_value), fill = factor(group))) +
  geom_bar(position = "dodge") +
  # facet_wrap(~ group) +
  labs(
    title = "D.4 The distribution of confidence in decision-making by group",
    x = "D.4 Response",
    y = "Count",
    fill = "Group"
  ) +
  theme_minimal()
```



the probability of inflation being low ($\leq 3\%$) is greater than the following percentages, I would consider investing in the nominal bond:

Variable: categorical

Method: contingency table & chi-squared test

Result: Fail to reject the null hypothesis.

```
# combine data from different groups
survey_d6 <- data.frame()
survey_d6_list <- list("Q5.6", "Q7.6", "Q9.6", "Q11.6", "Q13.6")
for (i in seq_along(survey_d6_list)) {
  df <- group_list[[i]]
  col <- survey_d6_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("d6_value", "group")
  survey_d6 <- rbind(survey_d6, temp_table)
}
survey_d6$d6_value <- as.numeric(survey_d6$d6_value)
```

Contingency table

```
survey_d6_contingency <- table(survey_d6$d6_value, survey_d6$group)
print(survey_d6_contingency)
```

```
##
##   0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1           0           1           0           3           2
## 2           2           4           1           4           3
## 3           6           5           8           2           2
## 4           2           5           5           6           9
## 5          15          10           8          12          10
## 7           6           8           7          14           8
## 8           9           9           8           3          10
```

```
# run chi-squared test
chisq_result <- chisq.test(survey_d6_contingency)
```

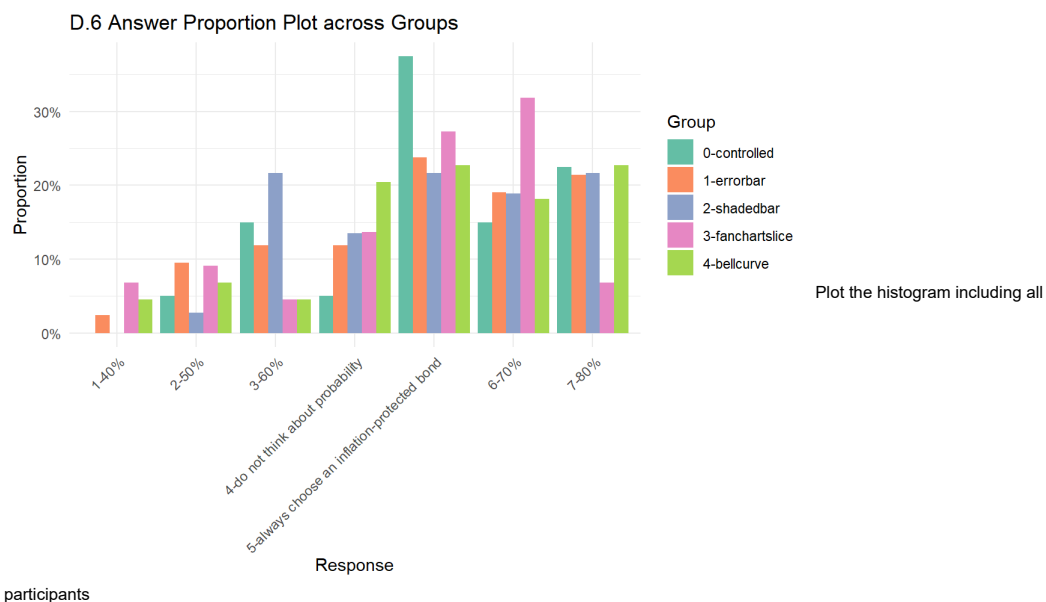
```
## Warning in chisq.test(survey_d6_contingency): Chi-squared approximation may be
## incorrect
```

```
print(chisq_result)
```

```
##
## Pearson's Chi-squared test
##
## data: survey_d6_contingency
## X-squared = 28.8, df = 24, p-value = 0.2277
```

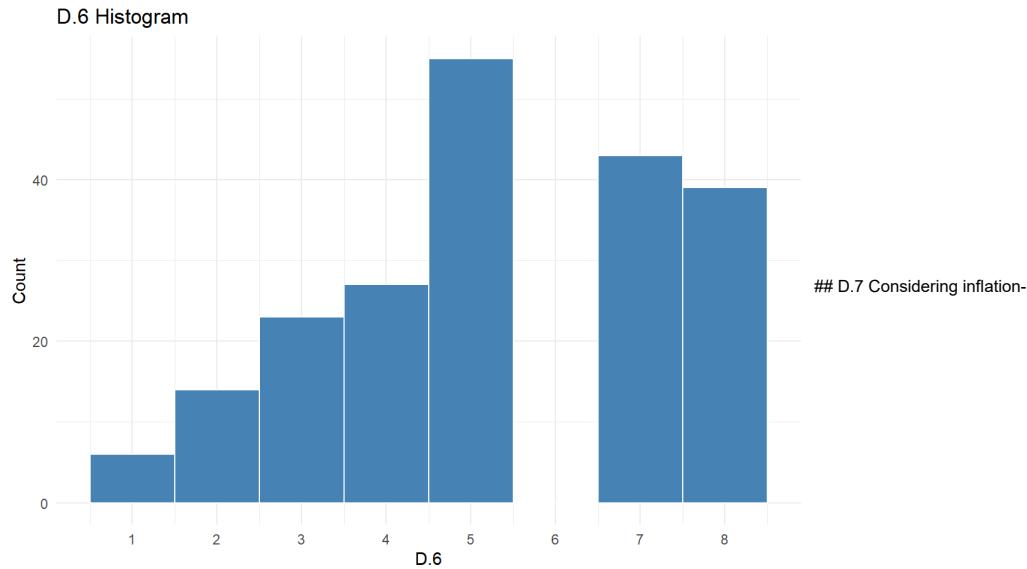
Plot proportion bar plot across different groups

```
survey_d6_prop <- as.data.frame(survey_d6_contingency)
# rename columns
names(survey_d6_prop) <- c("D.6", "Group", "Count")
# add a labelled column
survey_d6_prop <- survey_d6_prop %>%
  mutate(D.6_label = case_when(
    D.6 == "1" ~ "1-40%",
    D.6 == "2" ~ "2-50%",
    D.6 == "3" ~ "3-60%",
    D.6 == "4" ~ "4-do not think about probability",
    D.6 == "5" ~ "5-always choose an inflation-protected bond",
    D.6 == "7" ~ "6-70%",
    D.6 == "8" ~ "7-80%"
  ))
# calculate proportions within each group
survey_d6_prop <- survey_d6_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))
ggplot(survey_d6_prop, aes(x = as.factor(`D.6_label`), y = Proportion, fill = Group)) + geom_col(position = "dodge") +
  # facet_wrap(~Group)+
  labs(
    title = "D.6 Answer Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2")+
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal()+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
# histogram including all participants
ggplot(survey_d6, aes(x = d6_value)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  # facet_wrap(~ group) +
  labs(title = "D.6 Histogram",
        x = "D.6",
        y = "Count") +
  scale_x_continuous(breaks = seq(0, max(survey_d6$d6_value, na.rm = TRUE), by = 1)) +
  theme_minimal()
```

```
## Warning: Removed 102 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



protected bond If the probability of inflation being high (> 3%) is greater than the following percentages, I would consider investing in the inflation-protected bond:

Variable: categorical

Method: contingency table & chi-squared test

*Result: Fail to reject the null hypothesis.**

```
# combine data from different groups
survey_d7 <- data.frame()
survey_d7_list <- list("Q5.7", "Q7.7", "Q9.7", "Q11.7", "Q13.7")
for (i in seq_along(group_list)) {
  df <- group_list[[i]]
  col <- survey_d7_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("d7_value", "group")
  survey_d7 <- rbind(survey_d7, temp_table)
}
survey_d7$d7_value <- as.numeric(survey_d7$d7_value)
```

Contingency table

```
survey_d7_contingency <- table(survey_d7$d7_value, survey_d7$group)
print(survey_d7_contingency)
```

```
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1          2          0          5          1          0
## 2          3          4          0          6          3
## 3          2          4          6          5          5
## 4          6          1          4          3          4
## 5          1          3          1          4          3
## 7          3          4          7          2          2
## 8          5          1          1          1          0
```

```
# run chi-squared test
chisq_result <- chisq.test(survey_d7_contingency)
```

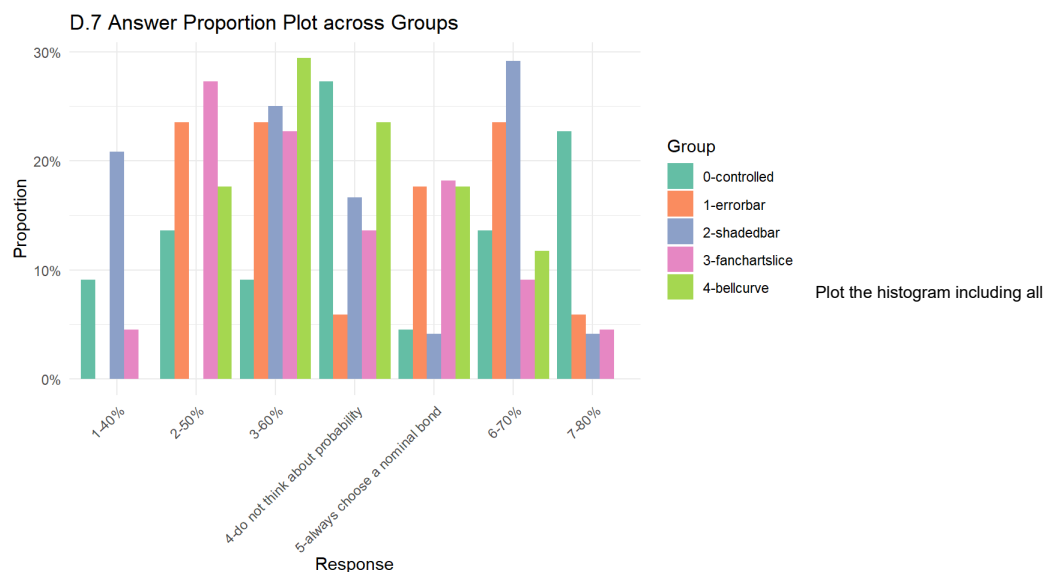
```
## Warning in chisq.test(survey_d7_contingency): Chi-squared approximation may be
## incorrect
```

```
print(chisq_result)
```

```
##
## Pearson's Chi-squared test
##
## data: survey_d7_contingency
## X-squared = 35.733, df = 24, p-value = 0.05824
```

Plot proportion bar plot across different groups

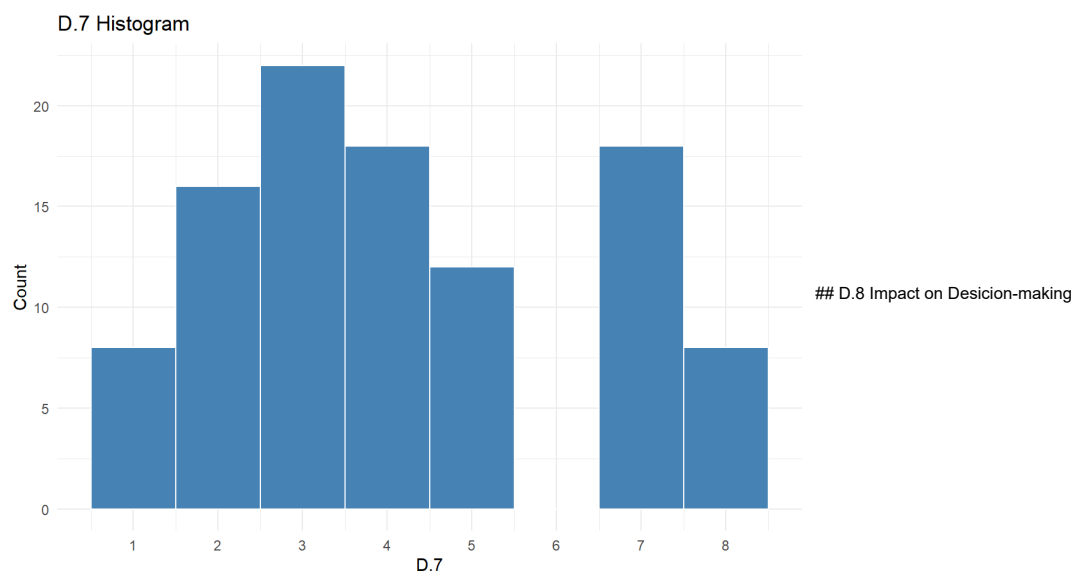
```
survey_d7_prop <- as.data.frame(survey_d7_contingency)
# rename columns
names(survey_d7_prop) <- c("D.7", "Group", "Count")
# add a labelled column
survey_d7_prop <- survey_d7_prop %>%
  mutate(D.7_label = case_when(
    D.7 == "1" ~ "1-40%",
    D.7 == "2" ~ "2-50%",
    D.7 == "3" ~ "3-60%",
    D.7 == "4" ~ "4-do not think about probability",
    D.7 == "5" ~ "5-always choose a nominal bond",
    D.7 == "7" ~ "6-70%",
    D.7 == "8" ~ "7-80%"
  ))
# calculate proportions within each group
survey_d7_prop <- survey_d7_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))
ggplot(survey_d7_prop, aes(x = as.factor(`D.7_label`), y = Proportion, fill = Group)) + geom_col(position = "dodge") +
  # facet_wrap(~Group)+
  labs(
    title = "D.7 Answer Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2")+
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal()+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



participants

```
# histogram including all participants
ggplot(survey_d7, aes(x = d7_value)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  # facet_wrap(~ group) +
  labs(title = "D.7 Histogram",
    x = "D.7",
    y = "Count") +
  scale_x_continuous(breaks = seq(0, max(survey_d7$d7_value, na.rm = TRUE), by = 1)) +
  theme_minimal()
```

```
## Warning: Removed 207 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



Considering the payoff for the Nominal Bond, which of the following has a larger impact on your decision-making?

Variable: categorical

Method: contingency table & chi-squared test

Result: Fail to reject the null hypothesis.

```
# combine data from different groups
survey_d8 <- data.frame()
survey_d8_list <- list("Q5.8", "Q7.8", "Q9.8", "Q11.8", "Q13.8")
for (i in seq_along(group_list)) {
  df <- group_list[[i]]
  col <- survey_d8_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("d8_value", "group")
  survey_d8 <- rbind(survey_d8, temp_table)
}
survey_d8$d8_value <- as.numeric(survey_d8$d8_value)
```

Contingency table

```
survey_d8_contingency <- table(survey_d8$d8_value, survey_d8$group)
print(survey_d8_contingency)
```

```
##
##      0-controlled 1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
##  1           33          28          35          43          40
##  2           16          15          19          12          13
##  3           13          16           7          11           8
```

```
# run chi-squared test
chisq_result <- chisq.test(survey_d8_contingency)
print(chisq_result)
```

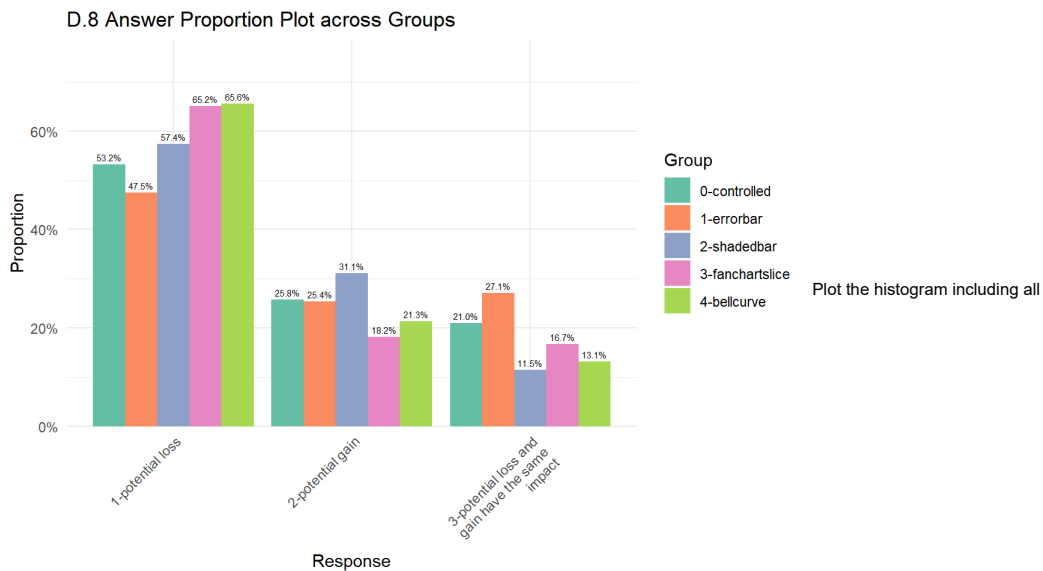
```
##
##  Pearson's Chi-squared test
##
## data:  survey_d8_contingency
## X-squared = 10.474, df = 8, p-value = 0.2333
```

Plot proportion bar plot across different groups


```

library(stringr)
survey_d8_prop <- as.data.frame(survey_d8_contingency)
# rename columns
names(survey_d8_prop) <- c("D.8", "Group", "Count")
# add a labelled column
survey_d8_prop <- survey_d8_prop %>%
  mutate(D.8_label = case_when(
    D.8 == "1" ~ "1-potential loss",
    D.8 == "2" ~ "2-potential gain",
    D.8 == "3" ~ "3-potential loss and gain\ have the same impact"
  ))
# calculate proportions within each group
survey_d8_prop <- survey_d8_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))
ggplot(survey_d8_prop, aes(x = as.factor(`D.8_label`), y = Proportion, fill = Group)) + geom_col(position = "dodge") +
  scale_x_discrete(labels = function(x) str_wrap(x, 20)) +
  geom_text(aes(label = scales::percent(Proportion, accuracy = 0.1)),
    position = position_dodge(width = 0.9), vjust = -0.5, size = 2) +
  # facet_wrap(~Group)+
  labs(
    title = "D.8 Answer Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2")+
  scale_y_continuous(labels = scales::percent_format(accuracy = 1),
    expand = expansion(mult = c(0, 0.2)) ) +
  theme_minimal()+
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```

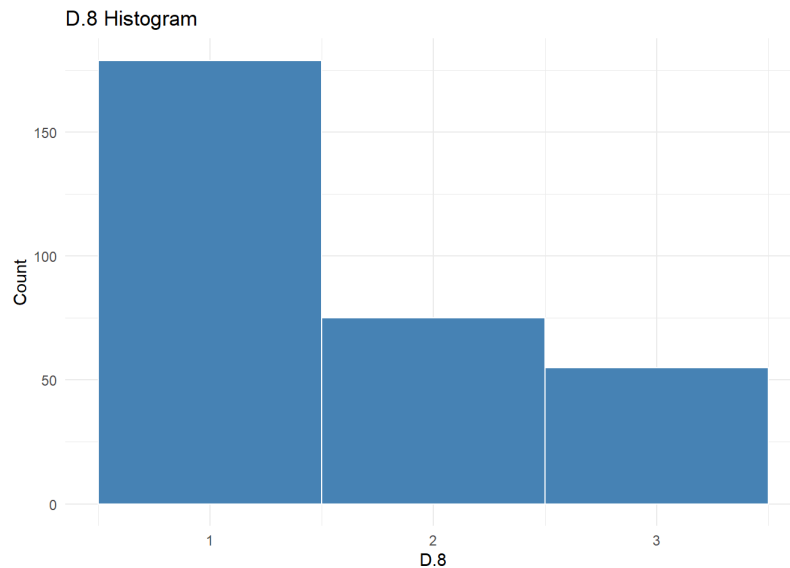


participants

```

# histogram including all participants
ggplot(survey_d8, aes(x = d8_value)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  # facet_wrap(~ group) +
  labs(title = "D.8 Histogram",
    x = "D.8",
    y = "Count") +
  scale_x_continuous(breaks = seq(0, max(survey_d8$d8_value, na.rm = TRUE), by = 1)) +
  theme_minimal()

```



D.9 Effectiveness of the graph for decision-making

On a scale of 1 to 10, where 1 means the forecast was not effective at all and 10 means it was very effective, how effective do you think the forecast was in assisting your decision-making?

1-Not effective at all; 10-Very effective

Variable: continuous

Method: ANNOVA test or KW test (depends on whether it's normally distributed)

Result: At least one group is statistically significant than others.

```
# combine data from different groups
survey_d9 <- data.frame()
survey_d9_list <- list("Q5.9_1", "Q7.9_1", "Q9.9_1", "Q11.9_1", "Q13.9_1")
for (i in seq_along(group_list)) {
  df <- group_list[[i]]
  col <- survey_d9_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("d9_value", "group")
  survey_d9 <- rbind(survey_d9, temp_table)
}
survey_d9$d9_value <- as.numeric(survey_d9$d9_value)
summary(survey_d9$d9_value)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.   Max.
##  1.000   5.000   7.000   6.366   8.000  10.000
```

Statistical test

```
# Initial Check
# summary(survey_d9$d9_value)
# check normality --> the data is not normally distributed
# shapiro.test(survey_d9$d9_value)

# run Kruskal-Wallis test
kruskal_test <- kruskal.test(d9_value ~ group, data = survey_d9)
print(kruskal_test)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  d9_value by group
## Kruskal-Wallis chi-squared = 10.047, df = 4, p-value = 0.03964
```

```
# Effect size
kw_es <- kruskal_effsize(d9_value ~ group, data = survey_d9, ci = TRUE)
print(kw_es)
```

```
## # A tibble: 1 x 7
##   .y.      n effsize conf.low conf.high method magnitude
## *   <chr>   <int>   <dbl>   <dbl>   <dbl>   <chr>   <ord>
## 1 d9_value  309   0.0199 -0.0034    0.08 eta2[H] small
```

```
# Post-hoc test
dunnTest(d9_value ~ group, data = survey_d9, method = "bonferroni")
```

```
## Warning: group was coerced to a factor.
```

```
## Dunn (1964) Kruskal-Wallis multiple comparison
```

```
## p-values adjusted with the Bonferroni method.
```

```
##           Comparison      Z    P.unadj    P.adj
## 1 0-controlled - 1-errorbar 0.3166253 0.75152796 1.0000000
## 2 0-controlled - 2-shadedbar 0.7111562 0.47698744 1.0000000
## 3 1-errorbar - 2-shadedbar 0.3869883 0.69876483 1.0000000
## 4 0-controlled - 3-fanchartslice 2.4870900 0.01287928 0.1287928
## 5 1-errorbar - 3-fanchartslice 2.1337039 0.03286703 0.3286703
## 6 2-shadedbar - 3-fanchartslice 1.7545538 0.07933566 0.7933566
## 7 0-controlled - 4-bellcurve 2.1805856 0.02921408 0.2921408
## 8 1-errorbar - 4-bellcurve 1.8382337 0.06602798 0.6602798
## 9 2-shadedbar - 4-bellcurve 1.4634923 0.14333271 1.0000000
## 10 3-fanchartslice - 4-bellcurve -0.2625307 0.79291233 1.0000000
```

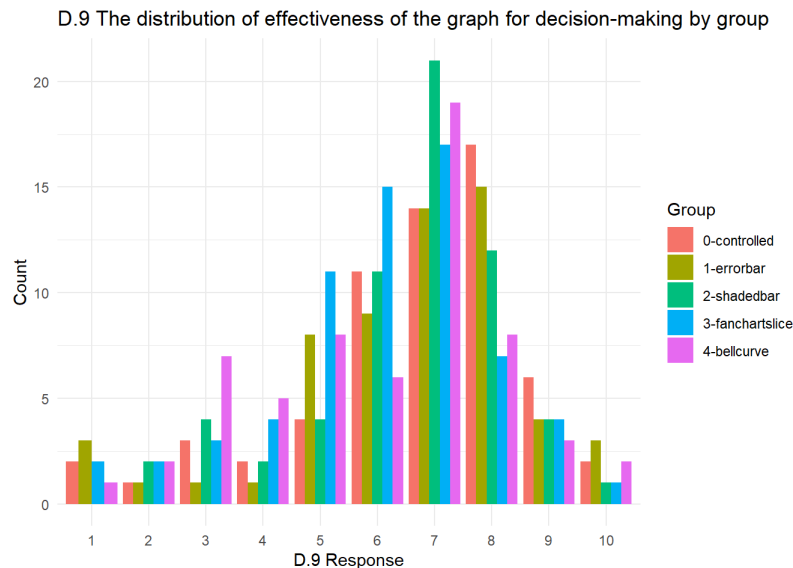
Summary statistics

```
# Summary statistics
survey_d9_summary <- survey_d9 %>%
  group_by(group) %>%
  summarise(
    mean = mean(d9_value, na.rm = TRUE),
    median = median(d9_value, na.rm = TRUE),
    sd = sd(d9_value, na.rm = TRUE),
    IQR = IQR(d9_value, na.rm = TRUE), # same as Q3 - Q1
    Q1 = quantile(d9_value, 0.25, na.rm = TRUE),
    Q3 = quantile(d9_value, 0.75, na.rm = TRUE)
  )
print(survey_d9_summary)
```

```
## # A tibble: 5 x 7
##   group      mean median    sd   IQR    Q1    Q3
##   <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 0-controlled    6.69     7 2.00     2     6     8
## 2 1-errorbar     6.61     7 2.07     2     6     8
## 3 2-shadedbar     6.54     7 1.75     2     6     8
## 4 3-fanchartslice 6.02     6 1.89     2     5     7
## 5 4-bellcurve     6      7 2.07     2     5     7
```

Plot the distribution of different groups

```
# Grouped bar-style histogram
ggplot(survey_d9, aes(x = factor(d9_value), fill = factor(group))) +
  geom_bar(position = "dodge") +
  # facet_wrap(~ group) +
  labs(
    title = "D.9 The distribution of effectiveness of the graph for decision-making by group",
    x = "D.9 Response",
    y = "Count",
    fill = "Group"
  ) +
  theme_minimal()
```



? D.12 Skewed Investment Decision

Which bond would you choose to invest in?

Variable: categorical

Method: contingency table & chi-squared test

Result: There is significant difference in at least one group.

```
# combine data from different groups
survey_d12 <- data.frame()
survey_d12_list <- list("Q7.12", "Q9.12", "Q11.12", "Q13.12")
for (i in seq_along(skewed_group_list)) {
  df <- skewed_group_list[[i]]
  col <- survey_d12_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("d12_value", "group")
  survey_d12 <- rbind(survey_d12, temp_table)
}
survey_d12$d12_value <- as.numeric(survey_d12$d12_value)
```

Contingency table

```
survey_d12_contingency <- table(survey_d12$d12_value, survey_d12$group)
print(survey_d12_contingency)
```

```
##
##      1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1           8          21           25           8
## 2          51          40           41          53
```

```
# run chi-squared test
chisq_result <- chisq.test(survey_d12_contingency)
# print(chisq_result)
std_resid <- chisq_result$stdres
print(std_resid)
```

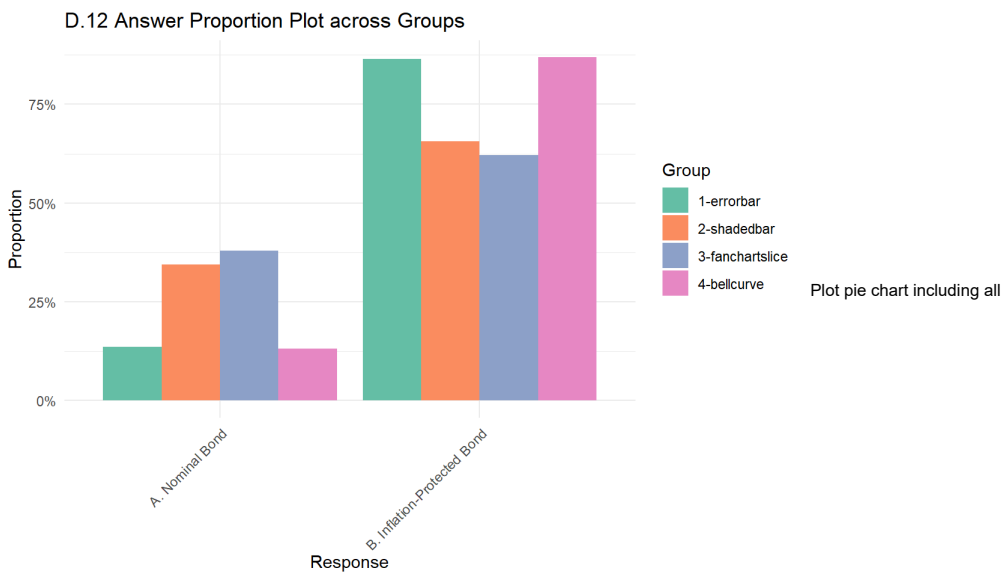
```
##
##      1-errorbar 2-shadedbar 3-fanchartslice 4-bellcurve
## 1 -2.343623    1.935627    2.796693    -2.488073
## 2  2.343623   -1.935627   -2.796693    2.488073
```

```
# pairwise proportion test
pairwise.prop.test(
  x = survey_d12_contingency[2,],
  n = colSums(survey_d12_contingency),
  p.adjust.method = "bonferroni" # or "holm", "BH"
)
```

```
##
## Pairwise comparisons using Pairwise comparison of proportions
##
## data: survey_d12_contingency[2, ] out of colSums(survey_d12_contingency)
##
##          1-errorbar 2-shadedbar 3-fanchartslice
## 2-shadedbar    0.084      -          -
## 3-fanchartslice 0.024    1.000      -
## 4-bellcurve    1.000    0.064    0.017
##
## P value adjustment method: bonferroni
```

Plot proportion bar plot across different groups

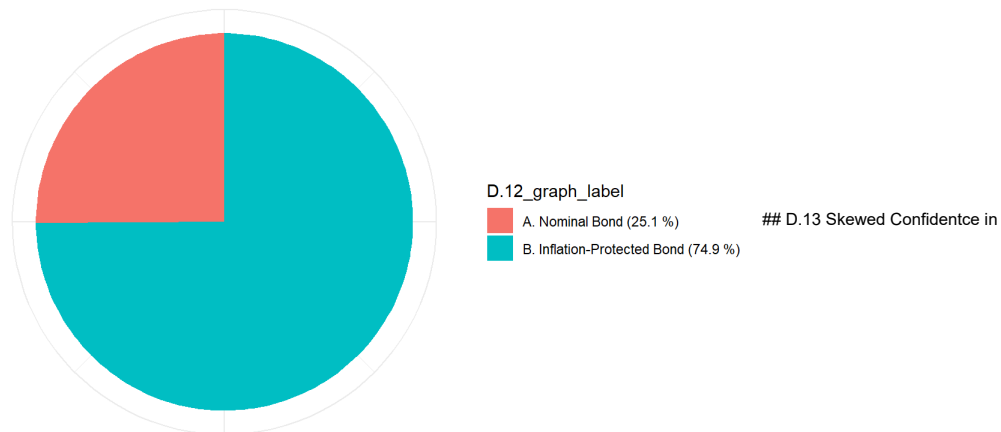
```
survey_d12_prop <- as.data.frame(survey_d12_contingency)
# rename columns
names(survey_d12_prop) <- c("D.12", "Group", "Count")
# add a labelled column
survey_d12_prop <- survey_d12_prop %>%
  mutate(D.12_label = case_when(
    D.12 == "1" ~ "A. Nominal Bond",
    D.12 == "2" ~ "B. Inflation-Protected Bond"
  ))
# calculate proportions within each group
survey_d12_prop <- survey_d12_prop %>%
  group_by(Group) %>%
  mutate(Proportion = Count / sum(Count))
ggplot(survey_d12_prop, aes(x = as.factor(`D.12_label`), y = Proportion, fill = Group)) + geom_col(position = "dodge") +
  # facet_wrap(~Group)+
  labs(
    title = "D.12 Answer Proportion Plot across Groups",
    x = "Response",
    y = "Proportion"
  ) +
  scale_fill_brewer(palette = "Set2") +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



participants

```
survey_d12_summary <- survey_d12_prop %>%
  group_by(`D.12_label`) %>%
  summarise(total_count = sum(Count), .groups = "drop") %>%
  mutate(proportion = total_count / sum(total_count),
    `D.12_graph_label` = paste0(`D.12_label`, " (", round(proportion * 100, 2), "%)"))
ggplot(survey_d12_summary, aes(x = "", y = proportion, fill = `D.12_graph_label`)) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  labs(title = "D.12 Investment Decision Summary Pie Chart",
    x = "",
    y = "") +
  theme_minimal() +
  theme(axis.text.x = element_blank())
```

D.12 Investment Decision Summary Pie Chart



decision-making On a scale of 1 to 10, where 1 means not confident at all and 10 means very confident, how confident are you in making the decision?

1-Not confident at all; 10-Very confident

Variable: continuous

Method: ANNOVA test or KW test (depends on whether it's normally distributed)

Result: Fail to reject the null hypothesis.

```
# combine data from different groups
survey_d13 <- data.frame()
survey_d13_list <- list("Q7.13_1", "Q9.13_1", "Q11.13_1", "Q13.13_1")
for (i in seq_along(skewed_group_list)) {
  df <- skewed_group_list[[i]]
  col <- survey_d13_list[[i]]
  temp_table <- df[, c(col, "group")]
  names(temp_table) <- c("d13_value", "group")
  survey_d13 <- rbind(survey_d13, temp_table)
}
survey_d13$d13_value <- as.numeric(survey_d13$d13_value)
```

Statistical test

```
# Initial Check
# summary(survey_d13$d13_value)
# check normality --> the data is not normally distributed
# shapiro.test(survey_d13$d13_value)

# run Kruskal-Wallis test
kruskal_test <- kruskal.test(d13_value ~ group, data = survey_d13)
print(kruskal_test)
```

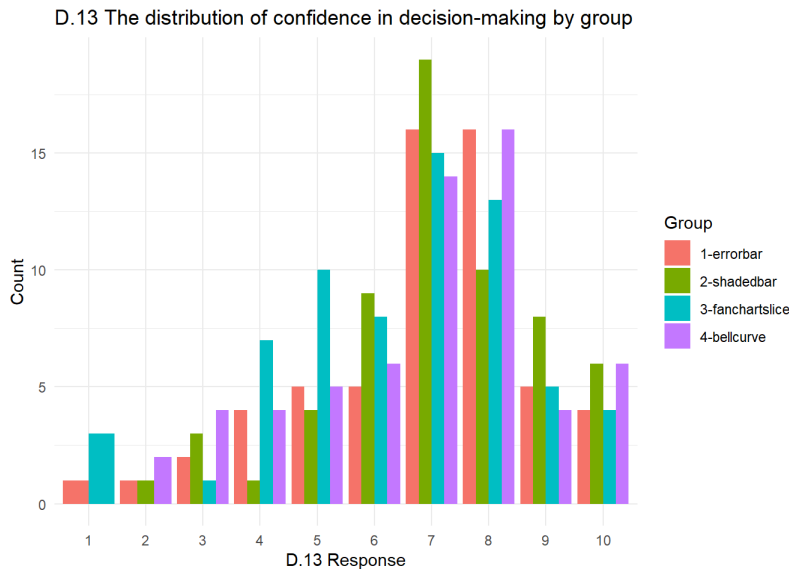
```
##
## Kruskal-Wallis rank sum test
##
## data: d13_value by group
## Kruskal-Wallis chi-squared = 3.1127, df = 3, p-value = 0.3746
```

Summary statistics

```
# Summary statistics
survey_d13_summary <- survey_d13 %>%
  group_by(group) %>%
  summarise(
    mean = mean(d13_value, na.rm = TRUE),
    median = median(d13_value, na.rm = TRUE),
    sd = sd(d13_value, na.rm = TRUE)
  )
```

Plot the distribution of different groups

```
# Grouped bar-style histogram
ggplot(survey_d13, aes(x = factor(d13_value), fill = factor(group))) +
  geom_bar(position = "dodge") +
  # facet_wrap(~ group) +
  labs(
    title = "D.13 The distribution of confidence in decision-making by group",
    x = "D.13 Response",
    y = "Count",
    fill = "Group"
  ) +
  theme_minimal()
```



Cross-tabulation

Current Inflation Knowledge vs. Inflation Forecast Knowledge

Skewed Data vs. Unskewed Data (understanding and decision making)

For people who can correctly understand the probability (? which), what decision do they make?

Does familiarity with the graph type affect understanding and the decision?

Statistical Testing

Basics

Type I error- false positives (saying there's a difference when there isn't) Usually set to 0.05\ Type II error: risk of a false negative (failing to detect a real effect). $\beta = 1 - \text{power} = 0.20$ \ Power = $1 - \beta = 0.80$ (80% chance of detecting a true effect)\ The classical null-hypothesis test (p-value) only tells whether the data are incompatible with zero effect. Pairwise comparison for effect size with confidence interval shows whether the results big enough to matter. – see if the interval include the Cohen's d value we set before. ## Power Calculation Cohen's d / Hedges' g are pairwise numbers, whereas Cohen's f describe omnibus gap across ≥ 3 groups\ f ($\approx d/2$) They are calculated by (effect)/standard deviation. This can be estimated through pilot studies

```
library(pwr)
```

```
## Warning: package 'pwr' was built under R version 4.4.3
```

```
N_total <- 309
sens <- pwr.anova.test(k = 5,
  n = N_total / 5*1.15, # ~10-15 % cushion for KW test
  sig.level = 0.05,
  power = 0.80) # f Left NULL
sens
```

```
##
##      Balanced one-way analysis of variance power calculation
##
##          k = 5
##          n = 71.07
##          f = 0.1845159
##      sig.level = 0.05
##          power = 0.8
##
## NOTE: n is number in each group
```

Chi-squared test

Post-hoc test

1. Standardized residuals

Cells with absolute value > ~2 (or > 1.96 for 95% confidence) indicate contributing most to the overall chi-squared

2. Pairwise proportion tests: can only compare one value

ANOVA test

ANOVA only compares the means of the groups.

Assumptions of ANOVA 1. Normality of residuals within each group: If p-value<0.05, the data is not normally distributed

2. Homogeneity of variances across groups

```
{r}{r eval=FALSE, include=FALSE} # Check the assumption of equal variances (homogeneity of variance) across groups # If p< 0.05, we can
### Post-hoc test If ANOVA is significant, we can perform post-hoc tests to determine which groups are significantly different from each
other.
```

```
```{r}{r eval=FALSE, include=FALSE} # post-hoc test tukey_test <- TukeyHSD(anova_test) print(tukey_test)
```

...

Null hypothesis ( $H_0$ ): All group means are equal

Alternative hypothesis ( $H_1$ ): At least one group mean is different

## Kruskal-Wallis test

(non-parametric test)

when the dependent variable is continuous or ordinal

when the assumptions of one-way ANOVA are not met: normality and equal variances

e.g., Likert-scale responses (ordinal) ### Post-hoc test

## Proportion test -prop.test()

- prop.test(): at least one group differs continuity correction?? The prop.test() function only supports alternative = "less" or "greater" when comparing a single proportion to a hypothesized value — not when comparing multiple groups.
- pairwise.prop.test(): identifying which specific pairs of groups differ significantly in their proportions, choose the p.adjust.method = "holm" or "bonferroni" or "BH"